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SIGNATURES

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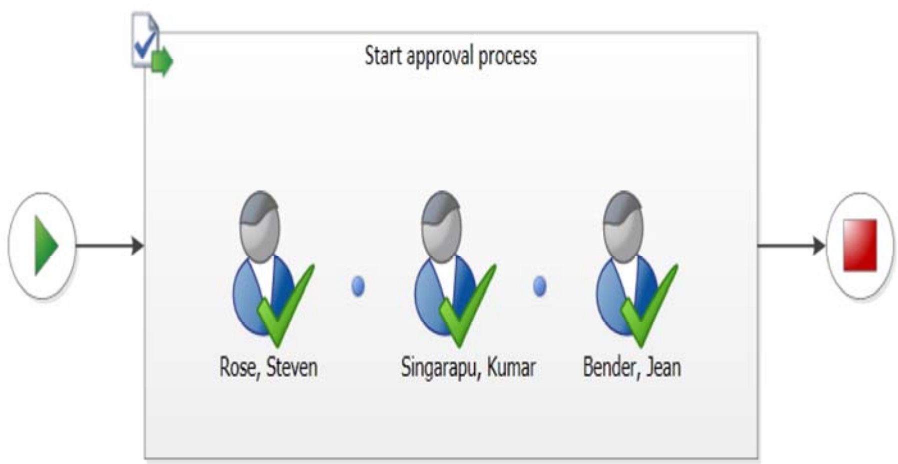

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Tasks

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Assigned To	Document Number	Due Date	Status	Related Content	Outcome
Rose, Steven	Please approve EHIVE03-968320269-617 ED 002 NEW	4/20/2018	Completed	EHIVE03-968320269-617 ED 002	Approved
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Workflow History

The workflow recorded these events.

Date Occurred	Event Type	User ID	Description	Outcome

4/17/2018 7:14 AM	Workflow Initiated	Craighead, Nancy	Approval was started. Participants: Rose, Steven;Singarapu, Kumar;Bender, Jean	
4/17/2018 7:14 AM	Task Created	Craighead, Nancy	Task created for Rose, Steven. Due by: 4/20/2018 12:00:00 AM	
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Report

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1. Summary

Currently, the dissolved carbon dioxide accumulation in the bioreactor is not actively controlled with feedback control based on a carbon dioxide measurement. The accumulation of the carbon dioxide is dependent on the bioreactor configuration and the control strategy used for maintaining the dissolved oxygen and the culture pH to within acceptable values. The model presented in this paper looks at the interaction that carbon dioxide has with both the dissolved oxygen and pH controls in order to effectively design these controllers to achieve a target carbon dioxide accumulation.

There are different methods available for maintaining the dissolved oxygen to set point. The agitation rate, air flow rate, oxygen flow rate, or vessel pressure could be manipulated in response to the on-line dissolved oxygen measurement. The method used to maintain the dissolved oxygen impacts the ability to remove the carbon dioxide produced from cellular respiration. This creates a link between the dissolved oxygen control and carbon dioxide accumulation.

The carbon dioxide is also linked to the pH control loop because it acts as an acid in the bicarbonate or carbonate buffered medium. The pH controller includes carbon dioxide additions to maintain the pH within a specified dead band. If the culture pH exceeds the upper end of the pH controller dead band, then the system will add carbon dioxide. If the culture pH is below the pH controller dead band, then the system will add base.

The dissolved carbon dioxide is measured with off-line samples or with an on-line dissolved carbon dioxide probe. However, these measurements are currently for information only. There are no process actions taken to bring the dissolved carbon dioxide to targeted ranges.

The potential risk with the passive accumulation of carbon dioxide is that the dissolved carbon dioxide levels can impact the cellular biology when it gets either above or below an acceptable range. The carbon dioxide interacts with the cellular biology by impacting the intracellular pH [1] which may then impact the biochemical reactions. The acceptable dissolved carbon dioxide range would be clone dependent and would need to be established empirically. The other potential risk with the passive dissolved carbon dioxide control is that the culture pH within the pH controller dead band is dependent on the accumulation of the carbon dioxide. A system that is more readily able to remove carbon dioxide will tend to run on the alkaline side of the pH dead band. A system that is more readily able to accumulate carbon dioxide will tend to run on the acidic side of the pH dead band. The difference in the culture pH, although within the pH dead band, may impact the cellular metabolism, which may affect culture performance. Therefore, in order to have more predictable performance from processes transferred to different scale, sites, or bioreactor configurations, it would be ideal to have comparable dissolved carbon dioxide and pH profiles.

The objective of this investigation is to describe the factors that contribute to carbon dioxide accumulation in the bioreactor by using first principle equations. The governing first principle systems are the conservation of mass and chemistry equilibria. From the first principle equations purposeful changes in the bioreactor configuration or controls for dissolved oxygen or pH can be tested mathematically before confirming empirically. The goal of this effort is to define a method to rationally specify the bioreactor configuration and control to conserve the dissolve carbon dioxide profile and pH profile for a cell culture process run at different sites, scales, or vessel configuration.

The modeling for the carbon dioxide accumulation and the pH profile in the cell culture bioreactor is performed as a two part system; the mass balance for the gasses and the chemistry equilibrium of the weak acids and bases. This report captures 1) the mathematical derivation of these expressions, 2) the tools to perform parameter fitting of the model, and 3) the tools to evaluate the impact that operating configuration has on the carbon dioxide accumulation and the pH profile in the bioreactor. Each of these three topics listed above are captured in the sections Model Derivation, Model Training, and Model Implementation.

2 Background

The intention of quality by design is to monitor and mitigate the sources of manufacturing variability that can impact the critical product quality attributes that affect patient safety. This ensures that the patient consistently and reliability receives safe and efficacious medicine. To effectively achieve this objective the following need to be performed:

1. the product critical quality attributes are specified,
2. the operating parameters that impact the critical quality attributes are identified,
3. the process operating space that is capable of delivering acceptable product quality is demonstrated,
4. and a robust control strategy of these operating parameters based on equipment design, instrumentation, procedures, methods, and analytics is implemented and maintained.

The outcome of the above efforts are the product specifications, the process design space, and the process control strategy. These three deliverables describe why a parameter needs to be controlled, which parameters are controlled, and how they are controlled (Figure 2-1). The need for control may also include business requirements to ensure the production is profitable, safe, and compliant (Figure 2-1).

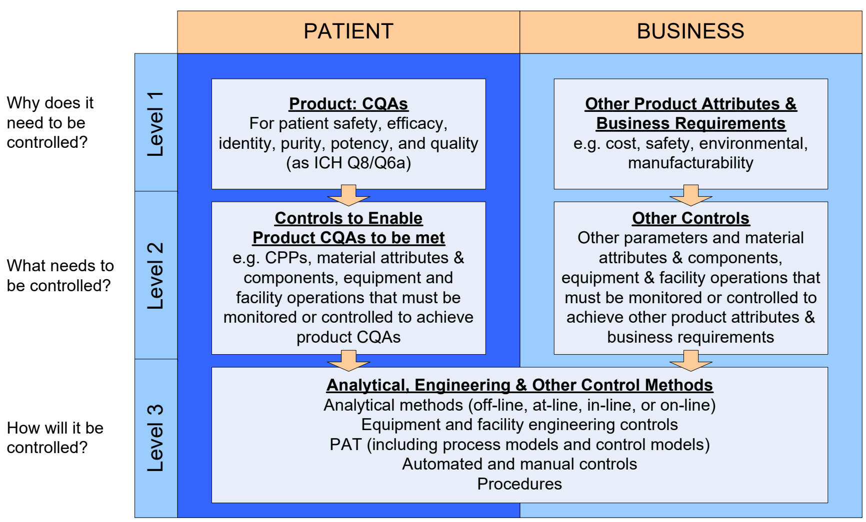

Figure 2-1: ISPE PQLI Control Strategy Model [2].

From the equipment perspective, the configuration of the equipment is defined to perform within the the process design space and equipment specifications to meet the patient and business requirements. There are multiple equipment configurations that can satisfy these requirements as evident with the flexibility of going with different process scales, disposable or fixed vessels, or different bioreactor configurations. It is important to correctly identify which parameters define the process design and operating space to allow for the greatest flexibility during implementation. For example the agitation frequency is specified to achieve adequate mixing with an acceptable level of shear. The appropriate agitation frequency is dependent on impeller design, impeller placement, vessel design, and working volume. If the agitation frequency is incorrectly defined as the process requirement instead of the achieved blend time, energy dissipation, off the bottom suspension energy, or shear rate, then there is a risk that changes in impeller design to improve cleanability, mixing efficiency, or overall mixing, would require respecifying the appropriate agitation frequency and this change would be viewed as a potential regulatory risk. Differentiating the equipment configuration from the process design space offers opportunities to improve the process control capabilities. The capability of the equipment configuration should be understood, and changes in equipment configuration (and the potential impact on the process design space) should be managed.

The dissolved carbon dioxide, the dissolved oxygen, and the culture pH are inherently linked in the bioreactor due to the design of the control strategies for dissolved oxygen and pH. The dissolved oxygen control strategy is affected by the mass transfer coefficient, the total gas flow and the gas composition. These three parameters also affect the dissolved carbon dioxide accumulation. The pH is affected by the amount of weak bases and acids including dissolved carbon dioxide. If the system is running alkaline, the controller will add carbon dioxide. The addition of carbon dioxide would then impact both the total gas flow and the gas composition which would impact the dissolved oxygen control loop. The interactions of the dissolved oxygen, dissolved carbon dioxide, and culture pH are represented in Figure 2-3. In this figure, the parameters that are engineering, biology, or chemistry are color coded yellow, pink, and blue respectively. The modeling will capture these interactions for the purpose to allow for engineering design to enable sufficient control of the carbon dioxide accumulation and pH profiles in different bioreactor scales or configurations without adversely impacting the dissolved oxygen or culture pH controls.

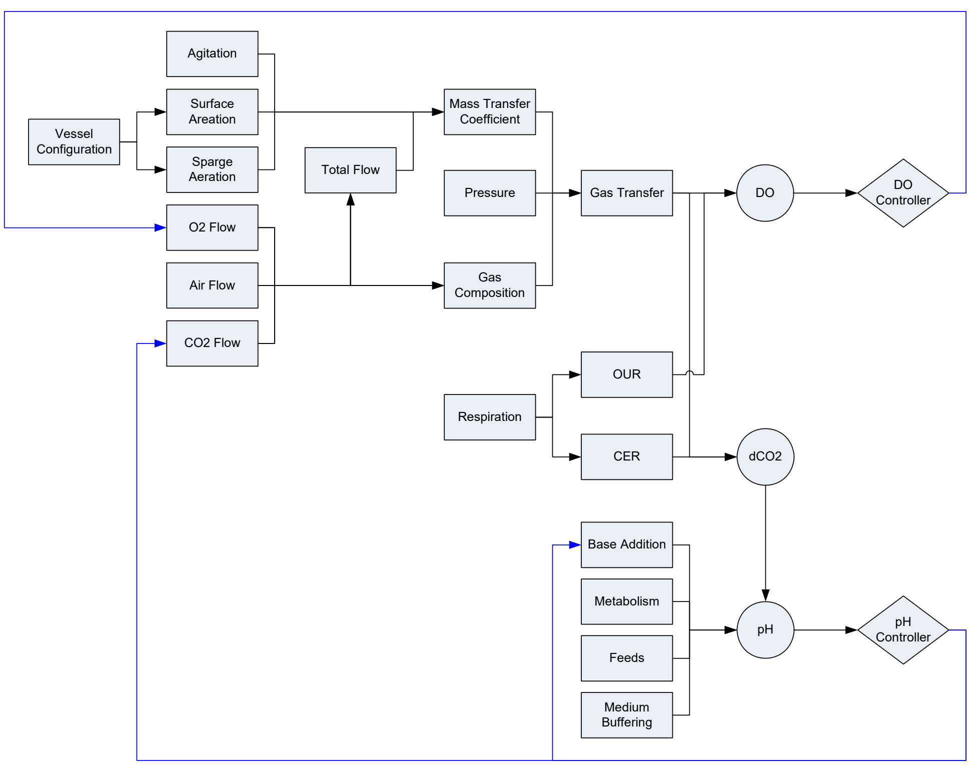

Figure 2-2: Interaction of the Dissolved Oxygen and pH Control Loops.

2.1 List of Parameters

The following list contains the parameters with units and definitions that are used in the equations provided in this report.

A	$1/(hr^*(W/m^3)^a(m/s)^b)$	Coefficient for the Van't Riet gas transfer equation.
A_x		Coefficients for the surface aeration gas transfer equation (x from 0 to 9).
Ala	mol/L	Concentration of alanine
Ala_1	mol/L	Concentration of alanine basic ion.
Ala_2	mol/L	Concentration of alanine dibasic ion.
B_{ala}		Ratio of amount of alanine produced in proportion to lactate.
B_{amm}		Corrective correlation for ammonia concentration due to the production or consumption of other organic acids and bases metabolized in proportion to ammonia.
B_i		Interaction coefficient for the gas transfer coefficients from multiple spargers due to bubble coalescence
B_{lact}		Corrective correlation for lactate concentration due to the production or consumption of other organic acids and bases metabolized in proportion to lactate.
Bunsen	L/L	Gas solubility. Liters of gas at standard temperature and pressure dissolved per liter of solvent under a partial pressure of 1 atm.
C	mol/L	Concentration.
$C_{ambient}$	mol/L	Concentration of in equilibrium with the ambient condition.
C_e	mol/L	Concentration at the electrode.
C_L	mol/L	Concentration in the liquid.
C_{sample}	mol/L	Concentration in the sample.

CER	mmol/(L hr)	Carbon dioxide evolution rate from cellular respiration.
$CO2_{ambient}$	mol/L	The dissolved carbon dioxide concentration in equilibrium with the ambient conditions.
CO2	mol/L	The dissolved carbon dioxide concentration in the liquid.
$CO2_{sample}$	mol/L	The dissolved carbon dioxide concentration in the sample.
$CO2_{spg1\_In}$	mol/L	The dissolved carbon dioxide concentration in equilibrium with the bubble composition of the inlet sparged bubbles resulting from sparger 1.
$CO2_{spg2\_In}$	mol/L	The dissolved carbon dioxide concentration in equilibrium with the bubble composition of the inlet sparged bubbles resulting from sparger 2.
$CO3_i$	mol/L	Initial carbonate concentration.
D_{CO2}	cm^2/s	Carbon dioxide diffusivity coefficient.
D_{O2}	cm^2/s	Oxygen diffusivity coefficient.
D_i	m	Impeller diameter
D_T	m	Vessel diameter
D_x	cm^2/s	Oxygen diffusivity coefficient within the different components x (of the DO probe).
g	m/s^2	Gravitational acceleration
H	mol/L	Hydrogen proton concentration.
H_i	mol/L	Initial hydrogen proton concentration.
H_x	mol/(L * atm)	Henry's coefficient for oxygen solubility within the different components x (of the DO probe).
h_L	m	Liquid height
H_{O2_cp}	mol/(L * atm)	Henry's coefficient for oxygen solubility.

HEPES	mol/L	Initial concentration for HEPES buffer in medium.
$HPO4_i$	mol/L	Initial hydrogen phosphate concentration in the medium formulation.
$H2PO4_i$	mol/L	Initial dihydrogen phosphate concentration in the medium formulation.
$H3PO4_i$	mol/L	Initial phosphoric acid concentration in the medium formulation.
$HCO3_i$	mol/L	Initial bicarbonate concentration.
H_{CO2_cp}	mol/(L * atm)	Henry's coefficient for carbon dioxide solubility.
K_h		Carbon dioxide, carbonic acid equilibrium constant.
K_1	mol/L	Equilibrium constant for the first dissociation in the carbonic acid system.
K_{1Ala}	mol/L	Equilibrium constant for the first dissociation in the alanine system.
K_{1P}	mol/L	Equilibrium constant for the first dissociation in the phosphate system.
K_2	mol/L	Equilibrium constant for the second dissociation in the carbonic acid system.
K_{2Ala}	mol/L	Equilibrium constant for the second dissociation in the alanine system.
K_{2P}	mol/L	Equilibrium constant for the second dissociation in the phosphate system.
K_{3P}	mol/L	Equilibrium constant for the third dissociation in the phosphate system.
$K_{ammonia}$	mol/L	Equilibrium constant for the ammonia, ammonium system.
K_{HEPES}	mol/L	Equilibrium constant for the HEPES buffer.
$K_{Lactate}$	mol/L	Equilibrium constant for the lactic acid, lactate salt system.

$kla_{antifoam}$	1/hr	The mass transfer coefficient adjusted for the impact of antifoam.
$kla_{antifoam\_0}$	1/hr	The mass transfer coefficient with no antifoam.
$kla_{antifoam\_\infty}$	1/hr	The mass transfer coefficient adjusted for the impact of a large volume of antifoam.
kla_{spg1_CO2}	1/hr	The mass transfer coefficient for carbon dioxide between the liquid and the sparged bubbles resulting from sparger 1.
kla_{spg1_O2}	1/hr	The mass transfer coefficient for oxygen between the liquid and the sparged bubbles resulting from sparger 1.
kla_{spg2_CO2}	1/hr	The mass transfer coefficient for carbon dioxide between the liquid and the sparged bubbles resulting from sparger 2.
kla_{spg2_O2}	1/hr	The mass transfer coefficient for oxygen between the liquid and the sparged bubbles resulting from sparger 2.
kla_{surf_CO2}	1/hr	The mass transfer coefficient for carbon dioxide between the liquid and the head space.
kla_{surf_O2}	1/hr	The mass transfer coefficient for oxygen between the liquid and the head space.
LacticAcid	mol/L	Measured lactate concentration.
M_{mOsm}	mOsm	Medium osmolality.
N	1/s	Agitation frequency.
N_p		Impeller power number.
NH3	mol/L	Ammonia concentration.
$O2_{ambient}$	mol/L	The dissolved oxygen concentration in equilibrium with the ambient conditions.
O2	mol/L	The dissolved oxygen concentration in the liquid.
$O2_{feed}$	mol/L	The dissolved oxygen concentration in the nutrient feed.

$O2_{sample}$	mol/L	The dissolved oxygen concentration in the sample.
$O2_{spg1\_In}$	mol/L	The dissolved oxygen concentration in equilibrium with the bubble composition of the inlet sparged bubbles resulting from sparger 1.
$O2_{spg1\_lm\_avg}$	mol/L	The log mean average of the dissolved oxygen concentration in equilibrium with the bubble composition of the sparged bubbles resulting from sparger 1.
$O2_{spg1\_Out}$	mol/L	The dissolved oxygen concentration in equilibrium with the bubble composition of the sparged bubbles resulting from sparger 1 as they are leaving the liquid.
$O2_{spg2\_In}$	mol/L	The dissolved oxygen concentration in equilibrium with the bubble composition of the inlet sparged bubbles resulting from sparger 2.
$O2_{spg2\_lm\_avg}$	mol/L	The log mean average of the dissolved oxygen concentration in equilibrium with the bubble composition of the sparged bubbles resulting from sparger 2.
$O2_{spg2\_Out}$	mol/L	The dissolved oxygen concentration in equilibrium with the bubble composition of the sparged bubbles resulting from sparger 2 as they are leaving the liquid.
OH	mol/L	Hydroxide concentration.
OH_{added}	mol/L	Hydroxide concentration added to bioreactor.
OH_i	mol/L	Initial hydroxide concentration
OUR	mmol/(L hr)	The oxygen uptake rate due to cellular respiration.
p_H	atm	Head space pressure.
pH		Negative log of the hydrogen proton concentration.
p_L	atm	Average liquid pressure.
p_{stp}	atm	Standard pressure (1 atm).
pCO2	atm	Carbon dioxide partial pressure.

pO2	atm	Oxygen partial pressure.
$PO4_i$	mol/L	Initial phosphate ions in the medium formulation.
q_{feed}	L/min	Volumetric flow rate of nutrient feed.
q_{hvst}	L/min	Volumetric flow rate of culture harvest (or sample).
q_{ovly}	L/min	Overlay volumetric flow rate in standard liters per minute.
q_{spg1_In}	L/min	Sparger 1 volumetric flow rate in standard liters per minute.
q_{spg2_In}	L/min	Sparger 2 volumetric flow rate in standard liters per minute.
$r_{antifoam}$	mol/s	Antifoam degradation constant.
R_{gas}	(L atm)/(mol K)	The gas constant (0.082 (L atm)/(mol K)).
respiration	mmol/(L hr)	CER or OUR.
RQ	mol CO_2/mol O_2	Respiration quotient.
$T_{ambient}$	K	Ambient temperature.
T_H	K	Head space temperature.
T_L	K	Liquid temperature.
T_{q_ref}	K	Reference temperature of flow meter.
T_{stp}	K	Standard temperature (288.15 K).
V_H	L	Head space volume.
V_L	L	The culture volume.
V_T	L	Vessel volume.
x_{CO2_hs}		Carbon dioxide mole fraction in the head space.
x_{CO2_ovly}		Carbon dioxide mole fraction in the overlay gas stream

$x_{CO2\_spg1\_In}$		Carbon dioxide mole fraction in the sparger 1 inlet stream.
$x_{CO2\_spg1\_Out}$		Carbon dioxide mole fraction in the sparger 1 bubbles leaving the liquid.
$x_{CO2\_spg2\_In}$		Oxygen mole fraction in the sparger 2 inlet stream.
$x_{CO2\_spg2\_Out}$		Carbon dioxide mole fraction in the sparger 2 bubbles leaving the liquid.
x_{O2_hs}		Oxygen mole fraction in the head space.
x_{O2_ovly}		Oxygen mole fraction in the overlay gas stream
$x_{O2\_spg1\_In}$		Oxygen mole fraction in the sparger 1 inlet stream.
$x_{O2\_spg1\_Out}$		Oxygen mole fraction in the sparger 1 bubbles leaving the liquid.
$x_{O2\_spg2\_In}$		Oxygen mole fraction in the sparger 2 inlet stream.
$x_{O2\_spg2\_Out}$		Oxygen mole fraction in the sparger 2 bubbles leaving the liquid.
$x_{salinity}$	gm/gm	Weight ratio salinity.
z		Exponential decay constant in kla value due to antifoam addition.
a		Power per unit volume exponent in Van't Riet equation
b		Superficial gas velocity exponent in Van't Riet equation
δ_x	m	Length of component x (of the DO probe)
ΔAla	mol/L	Change in alanine concentration due to medium equilibrium.
ΔAla_1	mol/L	Change in alanine basic ion concentration due to medium equilibrium.
ΔAla_2	mol/L	Change in alanine dibasic ion concentration due to to medium equilibrium.
ΔCO3	mol/L	Change in carbonate concentration due to medium equilibrium.

ΔH	mol/L	Change in hydrogen proton concentration due to medium equilibrium.
$\Delta H_{carbonate}$	mol/L	Change in hydrogen proton concentration due to the carbonate equilibrium system.
ΔH_{other}	mol/L	Change in hydrogen proton concentration due to other weak acids or bases in the medium.
ΔH_{water}	mol/L	Change in hydrogen proton concentration due to the water chemistry equilibrium.
ΔHPO4	mol/L	Change in the hydrogen phosphate concentration due to medium chemistry equilibrium.
ΔH2CO3	mol/L	Change in bicarbonate concentration due to medium chemistry equilibrium.
ΔH2PO4	mol/L	Change in the dihydrogen phosphate concentration due to medium chemistry equilibrium.
ΔH3PO4	mol/L	Change in the phosphoric acid concentration due to medium chemistry equilibrium.
$\Delta HEPES_{salt}$	mol/L	Change in the HEPES salt concentration due to medium chemistry equilibrium.
$\Delta Lactate_{salt}$	mol/L	Change in the Lactate salt concentration due to the medium chemistry equilibrium.
ΔNH4	mol/L	Change in the ammonium concentration due to the medium chemistry equilibrium.
ΔPO4	mol/L	Change in the phosphate ion concentration due to medium chemistry equilibrium.
r	kg/m^3	Liquid density
τ_e	s	Electrode response time.
f_x	m^2	Permeability of gas through different components x (of the DO probe).

2.2 Unit Definition

The following units are defined to be used within the equations that follow.

$$Celsius := K \quad (2.2-1)$$

$$mmHg := \frac{atm}{760} \qquad (2.2-2)$$

$$mmol := 0.001 \cdot mol \qquad (2.2-3)$$

$$psig := psi \qquad (2.2-4)$$

$$p_{stp} := 1atm \qquad (2.2-5)$$

$$sccm := \frac{cm^3}{min} \qquad (2.2-6)$$

$$SLPM := \frac{L}{min} \qquad (2.2-7)$$

2.3 Constants

The following constants are the gas constant, zero degrees Celsius, the reference temperature for the mass flow meters, and the standard temperature used in the Bunsen coefficient for gas solubility.

$$R_{gas} := 0.082056 \cdot \frac{atm \cdot L}{mol \cdot K} \qquad (2.3-1)$$

$$T_{0C} := 273.15K \qquad (2.3-2)$$

$$T_{q\_ref} := 294.261K \qquad (2.3-3)$$

$$T_{stp} := 288.15K \qquad (2.3-4)$$

2.4 Other Notations

The report is written in Mathcad version 15.0. This allows for the equations and graphs to be live. As variable assignments are updated, the report will perform the calculations and update the results or graphical output. The following notations are introduced in the report because of the platform that it was written.

...	The symbol of three dots at the end of a line is a Mathcad notation that the equation continues on the next line

3 Model Derivation

The model presented in this report is based on the first principles for mass balance and chemistry equilibria. There are three main components of the bioprocess; the equipment, the chemistry, and the biology. The model is limited to the equipment and the chemistry. The assumption is that there is insufficient understanding of the biology to have a model that captures how the biology would respond to shifts in the carbon dioxide or pH. Instead, the model assumes that the biology performance is known as demonstrated from a previous run. Then, the model is applied to determine how (or whether) the equipment and chemistry will meet the demands that the biology places on the system. This assumes that the biology is conserved across runs, sites, and scales. The following derivations provide the mass balance calculations for oxygen and carbon dioxide in the system and the chemistry equilibria for the weak acids and bases present in the medium (Figure 3-1).

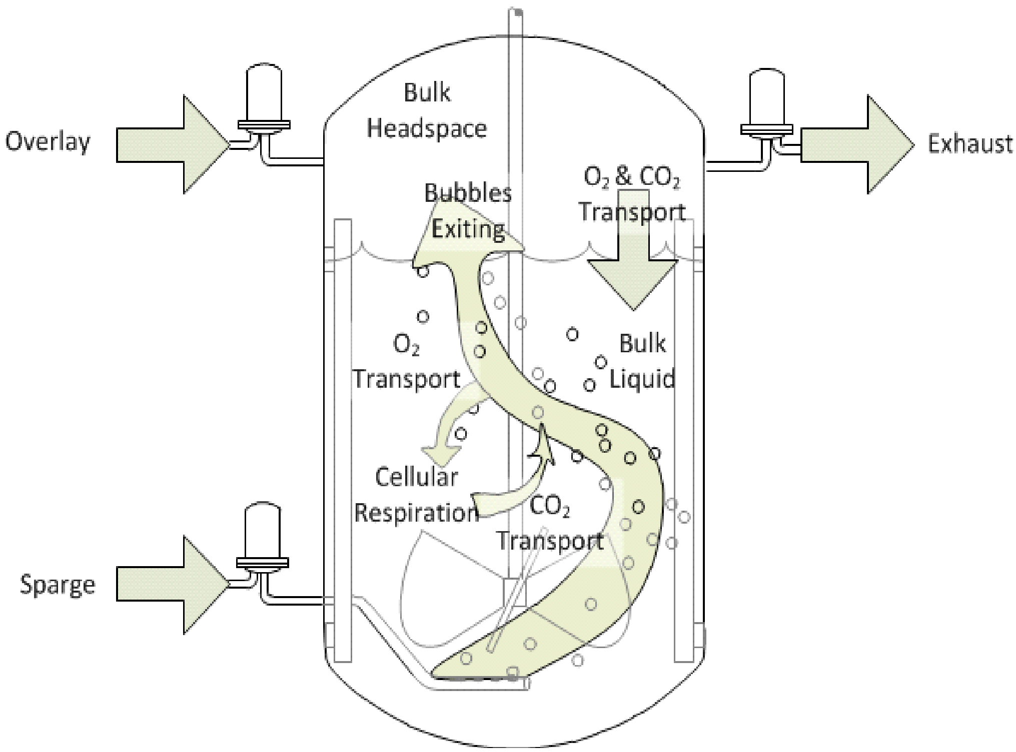

Figure 3.1: Mass transfer of oxygen and carbon dioxide in the bioreactor.

3.1 Conservation of Mass for Oxygen

The purpose of this derivation is to convert the equation for the mole balance of oxygen in the liquid from an expression with potentially unavailable terms to one that has readily available terms by using a first principle analysis. The mole balance of oxygen in the liquid is the sum of oxygen transferred from the sparged bubbles, the oxygen transferred from the surface aeration, the accumulation of oxygen in the liquid, and the consumed oxygen from cellular respiration. This expression contains two terms that may not be readily available; the log mean average of the dissolved oxygen concentration in equilibrium with the bubble composition, and the head space oxygen composition.

$$0 = kla_{spg1\_O2}\cdot\left(O2_{spg1\_lm\_avg} - O2\right)\cdot V_L + kla_{spg2\_O2}\cdot\left(O2_{spg2\_lm\_avg} - O2\right)\cdot V_L \;... \qquad (3.1-1)$$
$$+\, kla_{surf\_O2}\cdot\left(x_{O2\_hs}\cdot p_H\cdot H_{O2\_cp} - O2\right)\cdot V_L + \frac{d}{dt}\left(O2_L\cdot V_L\right) - OUR\cdot V_L$$

The moles of oxygen transferred from the sparge stream can also be determined from the difference of the molar oxygen flow rate in the entering and leaving bubbles. If we assume that the gas hold-up volume is constant and that there is insignificant bubble coalescence, then this is expressed as follows:

$$\left(x_{O2\_spg1\_In} - x_{O2\_spg1\_Out}\right)\cdot\left(\frac{p_{stp}\cdot q_{spg1\_In}}{R_{gas}\cdot T_{q\_ref}}\right)\cdot\frac{60min}{hr} = kla_{spg1\_O2}\cdot\left(O2_{spg1\_lm\_avg} - O2\right)\cdot V_L \qquad (3.1-2)$$

$$\left(x_{O2\_spg2\_In} - x_{O2\_spg2\_Out}\right)\cdot\left(\frac{p_{stp}\cdot q_{spg2\_In}}{R_{gas}\cdot T_{q\_ref}}\right)\cdot\frac{60min}{hr} = kla_{spg2\_O2}\cdot\left(O2_{spg2\_lm\_avg} - O2\right)\cdot V_L \qquad (3.1-3)$$

The log mean average of the dissolved oxygen concentration in equilibrium with the bubble composition (used in equations 3.1-1, 3.1-2, and 3.1-3) is defined from the dissolved oxygen concentration in equilibrium with the sparge inlet oxygen composition, the dissolved oxygen concentration in equilibrium with the sparge exiting oxygen composition, and the bulk liquid oxygen concentration.

$$O2_{spg1\_lm\_avg} = O2_L + \frac{O2_{spg1\_In} - O2_{spg1\_Out}}{\ln\left(\frac{O2_{spg1\_In} - O2}{O2_{spg1\_Out} - O2}\right)} \qquad (3.1-4)$$

$$O2_{spg2\_lm\_avg} = O2_L + \frac{O2_{spg2\_In} - O2_{spg2\_Out}}{\ln\left(\frac{O2_{spg2\_In} - O2}{O2_{spg2\_Out} - O2}\right)} \qquad (3.1-5)$$

Applying Henry's law the dissolved oxygen concentration in equilibrium with the gas streams is converted to Henry's solubility constant and the oxygen partial pressure.

$$O2_{spg1\_lm\_avg} = O2_L + \frac{\left(x_{O2\_spg1\_In} - x_{O2\_spg1\_Out}\right)\cdot H_{O2\_cp}\cdot p_L}{\ln\left(\frac{x_{O2\_spg1\_In}\cdot H_{O2\_cp}\cdot p_L - O2}{x_{O2\_spg1\_Out}\cdot H_{O2\_cp}\cdot p_L - O2}\right)} \qquad (3.1-6)$$

$$O2_{spg2\_lm\_avg} = O2_L + \frac{\left(x_{O2\_spg2\_In} - x_{O2\_spg2\_Out}\right)\cdot H_{O2\_cp}\cdot p_L}{\ln\left(\frac{x_{O2\_spg2\_In}\cdot H_{O2\_cp}\cdot p_L - O2}{x_{O2\_spg2\_Out}\cdot H_{O2\_cp}\cdot p_L - O2}\right)} \qquad (3.1-7)$$

Substituting equation 3.1-6 into equation 3.1-2 and equation 3.1-7 into equation 3.1-3 allows for a solvable expression for the oxygen composition of the bubbles leaving the culture (equations 3.1-10 and 3.1-13).

$$\left(x_{O2\_spg1\_In} - x_{O2\_spg1\_Out}\right)\cdot\left(\frac{p_{stp}\cdot q_{spg1\_In}}{R_{gas}\cdot T_{q\_ref}}\right)\cdot\frac{60min}{hr} = kla_{spg1\_O2}\cdot\left[\frac{\left(\begin{array}{l}x_{O2\_spg1\_In} \cdots \\ + -x_{O2\_spg1\_Out}\end{array}\right)\cdot H_{O2\_cp}\cdot p_L}{\ln\left(\frac{x_{O2\_spg1\_In}\cdot H_{O2\_cp}\cdot p_L - O2}{x_{O2\_spg1\_Out}\cdot H_{O2\_cp}\cdot p_L - O2}\right)}\right]\cdot V_L \qquad (3.1-8)$$

$$\ln\left(\frac{x_{O2\_spg1\_In} \cdot H_{O2\_cp} \cdot p_L - O2}{x_{O2\_spg1\_Out} \cdot H_{O2\_cp} \cdot p_L - O2}\right) = \frac{kla_{spg1\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr}} \qquad (3.1-9)$$

$$x_{O2\_spg1\_Out} \cdot H_{O2\_cp} \cdot p_L = O2 \ldots + \left(x_{O2\_spg1\_In} \cdot H_{O2\_cp} \cdot p_L - O2\right) \cdot \exp\left[-\left(\frac{kla_{spg1\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr}}\right)\right] \qquad (3.1-10)$$

$$\left(x_{O2\_spg2\_In} - x_{O2\_spg2\_Out}\right) \cdot \left(\frac{p_{stp} \cdot q_{spg2\_In}}{R_{gas} \cdot T_{q\_ref}}\right) \cdot \frac{60min}{hr} = kla_{spg2\_O2} \cdot \left[\frac{\left(x_{O2\_spg2\_In} \cdots + -x_{O2\_spg2\_Out}\right) \cdot H_{O2\_cp} \cdot p_L}{\ln\left(\frac{x_{O2\_spg2\_In} \cdot H_{O2\_cp} \cdot p_L - O2}{x_{O2\_spg2\_Out} \cdot H_{O2\_cp} \cdot p_L - O2}\right)}\right] \cdot V_L \qquad (3.1-11)$$

$$\ln\left(\frac{x_{O2\_spg2\_In} \cdot H_{O2\_cp} \cdot p_L - O2}{x_{O2\_spg2\_Out} \cdot H_{O2\_cp} \cdot p_L - O2}\right) = \frac{kla_{spg2\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr}} \qquad (3.1-12)$$

$$x_{O2\_spg2\_Out} \cdot H_{O2\_cp} \cdot p_L = O2 \ldots + \left(x_{O2\_spg2\_In} \cdot H_{O2\_cp} \cdot p_L - O2\right) \cdot \exp\left[-\left(\frac{kla_{spg2\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr}}\right)\right] \qquad (3.1-13)$$

The expression for the log mean average of the oxygen concentration in equilibrium with the bubble composition (equation 3.1-4 and equation 3.1-5) for the different sparge streams is substituted into equation 3.1-1.

$$0 = kla_{spg1\_O2} \cdot \left[\frac{\left(x_{O2\_spg1\_In} - x_{O2\_spg1\_Out}\right) \cdot H_{O2\_cp} \cdot p_L}{\ln\left(\frac{x_{O2\_spg1\_In} \cdot H_{O2\_cp} \cdot p_L - O2}{x_{O2\_spg1\_Out} \cdot H_{O2\_cp} \cdot p_L - O2}\right)}\right] \cdot V_L \ldots + kla_{spg2\_O2} \cdot \left[\frac{\left(x_{O2\_spg2\_In} - x_{O2\_spg2\_Out}\right) \cdot H_{O2\_cp} \cdot p_L}{\ln\left(\frac{x_{O2\_spg2\_In} \cdot H_{O2\_cp} \cdot p_L - O2}{x_{O2\_spg2\_Out} \cdot H_{O2\_cp} \cdot p_L - O2}\right)}\right] \cdot V_L \ldots + kla_{surf\_O2} \cdot \left(x_{O2\_hs} \cdot p_H \cdot H_{O2\_cp} - O2\right) \cdot V_L + -\frac{d}{dt}\left(O2 \cdot V_L\right) - OUR \cdot V_L \qquad (3.1-14)$$

Then, the expressions for the sparge outlet oxygen composition (equations 3.1-10 and 3.1-13) is substituted into equation 3.1-14 to provide a mass balance equation for the dissolved oxygen in the liquid without needing to estimate the log mean average of the oxygen composition of the bubbles.

$$0 = \frac{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr} \cdot \left[x_{O2\_spg1\_In} \cdot \left(H_{O2\_cp} \cdot p_L\right) - O2\right]}{H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla_{spg1\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr}}\right)\right]\right] \cdot V_L \ldots \qquad (3.1-15)$$
$$+ \frac{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr} \cdot \left[x_{O2\_spg2\_In} \cdot \left(H_{O2\_cp} \cdot p_L\right) - O2\right]}{H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla_{spg2\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr}}\right)\right]\right] \cdot V_L \ldots$$
$$+ kla_{surf\_O2} \cdot \left(x_{O2\_hs} \cdot p_H \cdot H_{O2\_cp} - O2\right) \cdot V_L + -\frac{d}{dt}\left(O2 \cdot V_L\right) - OUR \cdot V_L$$

The head space composition could be monitored with an off gas analyzer. However, this measurement may not always be available. To increase the utility of this model, the oxygen mole balance around the head space will also be defined to allow for a numeric calculation of the head space composition. The oxygen mole balance around the head space is determined from the oxygen added to the head space through the overlay, the oxygen added from the bubbles leaving the culture, the oxygen leaving the head space through the exhaust, and the transfer to the culture through surface aeration.

$$0 = x_{O2\_ovly} \cdot \frac{q_{ovly} \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \cdot \frac{60min}{hr} + x_{O2\_spg1\_Out} \cdot \frac{q_{spg1\_Out} \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \cdot \frac{60min}{hr} \ldots \qquad (3.1-16)$$
$$+ x_{O2\_spg2\_Out} \cdot \frac{q_{spg2\_Out} \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \cdot \frac{60min}{hr} \ldots$$
$$+ -x_{O2\_hs} \cdot \frac{\left(q_{ovly} + q_{spg1\_Out} + q_{spg2\_Out}\right) \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \cdot \frac{60min}{hr} \ldots$$
$$+ -kla_{surf\_O2} \cdot \left(x_{O2\_hs} \cdot p_H \cdot H_{O2\_cp} - O2\right) \cdot V_L - \frac{d}{dt}\left(\frac{x_{O2\_hs} \cdot p_H}{R_{gas} \cdot T_H} \cdot V_H\right)$$

Assuming that the head space can be approximated as pseudo steady-state system, then the head space oxygen composition can be approximated as follows.

$$(3.1-17)$$

$$x_{O2\_hs} = \frac{\left[\begin{array}{l} x_{O2\_ovly} \cdot q_{ovly} \cdot \frac{60min}{hr} \ldots \\ + \left[\frac{O2_L}{\left(H_{O2\_cp} \cdot p_L\right)} + \left[x_{O2\_spg1\_In} - \frac{O2}{\left(H_{O2\_cp} \cdot p_L\right)}\right] \cdot \exp\left[-\left(\frac{kla_{spg1\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr}}\right)\right]\right] \cdot q_{spg1\_In} \cdot \frac{60min}{hr} \ldots \\ + \left[\frac{O2_L}{\left(H_{O2\_cp} \cdot p_L\right)} + \left[x_{O2\_spg2\_In} - \frac{O2}{\left(H_{O2\_cp} \cdot p_L\right)}\right] \cdot \exp\left[-\left(\frac{kla_{spg2\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr}}\right)\right]\right] \cdot q_{spg2\_In} \cdot \frac{60min}{hr} \ldots \\ + kla_{surf\_O2} \cdot \frac{R_{gas} \cdot T_{q\_ref}}{p_{stp}} \cdot O2 \cdot V_L \end{array}\right]}{\left[\left(q_{ovly} + q_{spg1\_Out} + q_{spg2\_Out}\right) \cdot \frac{60min}{hr} + kla_{surf\_O2} \cdot H_{O2} \cdot R_{gas} \cdot T_{q\_ref} \cdot \frac{p_H}{p_{stp}} \cdot V_L\right]}$$

Then, by substituting the expression for the head space oxygen composition (equation 3.1-17) into the oxygen mass balance equation (equation 3.1-15) it is possible to solve the oxygen mass balance with readily available terms.

$$
\begin{aligned}
0 = {} & \frac{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr} \cdot \left(x_{O2\_spg1\_In} \cdot H_{O2\_cp} \cdot p_L - O2\right)}{p_L \cdot V_L \cdot H_{O2\_cp} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla_{spg1\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr}}\right)\right]\right] \ldots \\
& + \frac{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr} \cdot \left(x_{O2\_spg2\_In} \cdot H_{O2\_cp} \cdot p_L - O2\right)}{p_L \cdot V_L \cdot H_{O2\_cp} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla_{spg2\_O2} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr}}\right)\right]\right] \ldots \\
& + kla_{surf\_O2} \cdot \left[\frac{\left[\begin{aligned} & x_{O2\_ovly} \cdot \frac{q_{ovly}}{V_L} \cdot H_{O2\_cp} \cdot p_H \cdot \frac{60min}{hr} \ldots \\ & + \left[\begin{aligned} & \left[\begin{aligned} & O2 \cdot \frac{p_H}{p_L} \ldots \\ & + \left(x_{O2\_spg1\_In} \cdot H_{O2\_cp} \cdot p_H \ldots + -O2 \cdot \frac{p_H}{p_L}\right) \cdot \exp\left[-\left(\frac{kla_{spg1\_O2} \cdot H_{O2} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr}}\right)\right] \end{aligned}\right] \cdot \frac{q_{spg1\_In}}{V_L} \cdot \frac{60min}{hr} \ldots \\ & + \left[\begin{aligned} & O2 \cdot \frac{p_H}{p_L} \ldots \\ & + \left(x_{O2\_spg2\_In} \cdot H_{O2\_cp} \cdot p_H \ldots + -O2 \cdot \frac{p_H}{p_L}\right) \cdot \exp\left[-\left(\frac{kla_{spg2\_O2} \cdot H_{O2} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr}}\right)\right] \end{aligned}\right] \cdot \frac{q_{spg2\_In}}{V_L} \cdot \frac{60min}{hr} \end{aligned}\right] \ldots \\ & + kla_{surf\_O2} \cdot \frac{H_{O2\_cp} \cdot R_{gas} \cdot T_{q\_ref} \cdot p_H}{p_{stp}} \cdot O2 \end{aligned}\right]}{\left[\left(\frac{q_{ovly}}{V_L} + \frac{q_{spg1\_In}}{V_L} + \frac{q_{spg2\_In}}{V_L}\right) \cdot \frac{60min}{hr} + kla_{surf\_O2} \cdot H_{O2\_cp} \cdot R_{gas} \cdot T_{q\_ref} \cdot \frac{p_H}{p_{stp}}\right]} - O2_L \right] \ldots \\
& + \frac{-1}{V_L} \cdot \frac{d}{dt}\left(O2 \cdot V_L\right) - OUR
\end{aligned}
\qquad (3.1-18)
$$

If the system cannot be assumed to be under pseudo steady-state, then the time derivatives (equation 3.1-15 and equation 3.1-16) would have to be numerically solved simultaneously.

3.1.1 Oxygen Solubility

The oxygen solubility is affected by the medium temperature and conductivity. There are different forms of Henry's law constant that are reported in the literature. This report is using the form

$$C = H_{cp} \cdot p \qquad (3.1.1-1)$$

where the units for the coefficient are mol/(L*atm). The data used to estimate the oxygen solubility in the cell culture data are the Bunsen coefficient for oxygen at different temperatures and percent salinity (Table 3.1.1-1) [3]. The Bunsen coefficient is the volume of gas at standard temperature (288.15 K) and pressure (1 atm) per unit volume of solvent at the experimental temperature under a partial pressure of the given gas at 1 atm. To convert these values to the Henry's coefficient, the Bunsen coefficient is divided by the ideal gas constant and the standard temperature.

$$\frac{Bunsen}{R_{gas} \cdot T_{stp}} = \frac{C}{p_{stp}} = H_{cp} \qquad (3.1.1-2)$$

Temperature K	Saline 0	0.04	0.08	0.12	0.16	0.2
273.65	0.0489	0.0458	0.0435	0.0414	0.0394	0.0376
278.15	0.0429	N/A	N/A	N/A	N/A	N/A
283.15	0.0380	0.0364	0.0347	0.0332	0.0318	0.0304
288.15	0.0342	N/A	N/A	N/A	N/A	N/A
293.15	0.0310	0.0298	0.0285	0.0273	0.0262	0.0252
298.15	0.0283	0.0273	0.0262	0.0251	0.0242	0.0233
303.15	0.0261	0.0253	0.0243	0.0233	0.0224	0.0216
308.15	0.0245	0.0236	0.0227	0.0218	0.0210	0.0202
313.15	0.0231	N/A	N/A	N/A	N/A	N/A
323.15	0.0209	N/A	N/A	N/A	N/A	N/A
333.15	0.0195	N/A	N/A	N/A	N/A	N/A
343.15	0.0183	N/A	N/A	N/A	N/A	N/A
353.15	0.0176	N/A	N/A	N/A	N/A	N/A
363.15	0.0172	N/A	N/A	N/A	N/A	N/A
373.15	0.0170	N/A	N/A	N/A	N/A	N/A

Table 3.1.1-1: Bunsen coefficients for oxygen at different temperatures with different percent salinity [3].

A fourth order polynomial equation was fitted to the data listed in Table 3.1.1-1 to enable interpolation of the data set. The SAS JMP 8.0 software was used to perform a stepwise regression of mixed direction (0.05 threshold for adding and removing terms) to obtain the following equation for Henry's coefficient.

$$H_{O2\_cp}(x_{salinity}, T_L) := \frac{\left[\begin{array}{l} 0.1398 - 0.02406\cdot x_{salinity} - 0.0003749\cdot\frac{T_L}{K} + 0.0005362(x_{salinity} - 0.08)\cdot\left(\frac{T_L}{K} - 300.55\right) \ldots \\ + 0.009317(x_{salinity} - 0.08)^2 + 0.000007381\cdot\left(\frac{T_L}{K} - 300.55\right)^2 \ldots \\ + -0.0000001294\cdot\left(\frac{T_L}{K} - 300.55\right)^3 \ldots \\ + 0.0000000007597\cdot\left(\frac{T_L}{K} - 300.55\right)^4 - 0.001867\cdot(x_{salinity} - 0.08)^2\cdot\left(\frac{T_L}{K} - 300.55\right) \ldots \\ + -0.00002519\cdot(x_{salinity} - 0.08)\cdot\left(\frac{T_L}{K} - 300.55\right)^2 \end{array}\right]}{R_{gas}\cdot T_{stp}} \qquad (3.1.1-3)$$

The salinity of the medium will be estimated from the osmolality of the medium. Because sodium chloride completely dissociates when dissolved, for every mmol of sodium chloride per liter of medium, the osmolality would increase by two mOsm. If most of the osmolality contributions in the medium are due to salt additions, then we will estimate the salinity equivalence by calculating the amount of sodium chloride that would need to be added to have equivalent osmolality.

$$x_{salinity} = \frac{\frac{M_{mOsm}}{mOsm}}{2\frac{mOsm}{\left(\frac{mmol}{L}\right)}}\cdot 58.44\frac{gm}{mol}\cdot\frac{1L}{1000gm} = \frac{0.000029}{mOsm}\cdot M_{mOsm} \qquad (3.1.1-4)$$

where M_{mOsm} is the medium osmolality.

3.2 Conservation of Mass for Carbon Dioxide

The purpose of this derivation is to solve for the dissolved carbon dioxide in the culture based on the carbon dioxide mole balance. The transfer of carbon dioxide between the bubbles, the head space, and the culture liquid take the same form as that for oxygen. Therefore, we can start with the carbon dioxide equivalence of equation 3.1-18.

$$
\begin{aligned}
0 = {} & \frac{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr} \cdot \left(x_{CO2\_spg1\_In} \cdot H_{CO2\_cp} \cdot p_L - CO2\right)}{p_L \cdot V_L \cdot H_{CO2\_cp} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla_{spg1\_CO2} \cdot H_{CO2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr}}\right)\right]\right] \ldots \qquad (3.2-1) \\
& + \frac{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr} \cdot \left(x_{CO2\_spg2\_In} \cdot H_{CO2\_cp} \cdot p_L - CO2\right)}{p_L \cdot V_L \cdot H_{CO2\_cp} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla_{spg2\_CO2} \cdot H_{CO2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr}}\right)\right]\right] \ldots \\
& + kla_{surf\_CO2} \cdot \left[\frac{
\begin{array}{l}
x_{CO2\_ovly} \cdot \frac{q_{ovly}}{V_L} \cdot H_{CO2\_cp} \cdot p_H \cdot \frac{60min}{hr} \ldots \\
+ \left[\left[CO2 \cdot \frac{p_H}{p_L} \ldots + \left(x_{CO2\_spg1\_In} \cdot H_{CO2\_cp} \cdot p_H \ldots + -CO2 \cdot \frac{p_H}{p_L} \right) \cdot \exp\left[-\left(\frac{kla_{spg1\_CO2} \cdot H_{CO2} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1\_In} \cdot \frac{60min}{hr}}\right)\right] \right] \cdot \frac{q_{spg1\_In}}{V_L} \cdot \frac{60min}{hr} \ldots \right. \\
\quad \left. + \left[CO2 \cdot \frac{p_H}{p_L} \ldots + \left(x_{CO2\_spg2\_In} \cdot H_{CO2\_cp} \cdot p_H \ldots + -CO2 \cdot \frac{p_H}{p_L} \right) \cdot \exp\left[-\left(\frac{kla_{spg2\_CO2} \cdot H_{CO2} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2\_In} \cdot \frac{60min}{hr}}\right)\right] \right] \cdot \frac{q_{spg2\_In}}{V_L} \cdot \frac{60min}{hr} \right] \ldots \\
+ kla_{surf\_CO2} \cdot \frac{H_{CO2\_cp} \cdot R_{gas} \cdot T_{q\_ref} \cdot p_H}{p_{stp}} \cdot CO2
\end{array}
}{
\left[\left(\frac{q_{ovly}}{V_L} + \frac{q_{spg1\_In}}{V_L} + \frac{q_{spg2\_In}}{V_L}\right) \cdot \frac{60min}{hr} + kla_{surf\_CO2} \cdot H_{CO2\_cp} \cdot R_{gas} \cdot T_{q\_ref} \cdot \frac{p_H}{p_{stp}}\right]
} - CO2_L \right] \ldots \\
& + \frac{-1}{V_L} \cdot \frac{d}{dt}\left(CO2 \cdot V_L\right) + CER
\end{aligned}
$$

Assuming the pseudo steady state approximation can be used, the dissolved carbon dioxide concentration is solved as follows, where the term RQ, the respiration quotient, is the moles of carbon dioxide produced per mole of oxygen consumed. Literature suggests that for mammalian cell cultures, assuming an RQ of 1.0 is appropriate [4].

$$
pCO2 = \frac{
\begin{aligned}
&\frac{p_{stp}\cdot q_{CO2\_spg1\_In}\cdot\frac{60min}{hr}}{V_L\cdot R_{gas}\cdot T_{q\_ref}}\cdot\left[1-\exp\left[-\left(\frac{kla_{spg1\_CO2}\cdot H_{CO2\_cp}\cdot p_L\cdot V_L\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1\_In}\cdot\frac{60min}{hr}}\right)\right]\right]\ldots\\
&+\frac{p_{stp}\cdot q_{CO2\_spg2\_In}\cdot\frac{60min}{hr}}{V_L\cdot R_{gas}\cdot T_{q\_ref}}\cdot\left[1-\exp\left[-\left(\frac{kla_{spg2\_CO2}\cdot H_{CO2\_cp}\cdot p_L\cdot V_L\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg2\_In}\cdot\frac{60min}{hr}}\right)\right]\right]\ldots\\
&+kla_{surf\_CO2}\cdot\frac{\left[\begin{aligned}&q_{CO2\_ovly}\cdot\frac{60min}{hr}\ldots\\&+q_{CO2\_spg1\_In}\cdot\frac{60min}{hr}\cdot\exp\left[-\left(\frac{kla_{spg1\_CO2}\cdot H_{CO2\_cp}\cdot p_L\cdot V_L\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1\_In}\cdot\frac{60min}{hr}}\right)\right]\ldots\\&+q_{CO2\_spg2\_In}\cdot\frac{60min}{hr}\cdot\exp\left[-\left(\frac{kla_{spg2\_CO2}\cdot H_{CO2\_cp}\cdot p_L\cdot V_L\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg2\_In}\cdot\frac{60min}{hr}}\right)\right]\end{aligned}\right]}{\left[\left[\frac{q_{ovly}}{\left(H_{CO2\_cp}\cdot p_H\right)}+\frac{q_{spg1\_In}}{\left(H_{CO2\_cp}\cdot p_H\right)}+\frac{q_{spg2\_In}}{\left(H_{CO2\_cp}\cdot p_H\right)}\right]\cdot\frac{60min}{hr}+kla_{surf\_CO2}\cdot\frac{R_{gas}\cdot T_{q\_ref}}{p_{stp}}\cdot V_L\right]}\ldots\\
&+RQ\cdot OUR
\end{aligned}
}{
H_{CO2\_cp}\cdot\left[
\begin{aligned}
&\frac{p_{stp}\cdot q_{spg1\_In}\cdot\frac{60min}{hr}}{p_L\cdot V_L\cdot H_{CO2\_cp}\cdot R_{gas}\cdot T_{q\_ref}}\cdot\left[1-\exp\left[-\left(\frac{kla_{spg1\_CO2}\cdot H_{CO2\_cp}\cdot p_L\cdot V_L\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1\_In}\cdot\frac{60min}{hr}}\right)\right]\right]\ldots\\
&+\frac{p_{stp}\cdot q_{spg2\_In}\cdot\frac{60min}{hr}}{p_L\cdot V_L\cdot H_{CO2\_cp}\cdot R_{gas}\cdot T_{q\_ref}}\cdot\left[1-\exp\left[-\left(\frac{kla_{spg2\_CO2}\cdot H_{CO2\_cp}\cdot p_L\cdot V_L\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg2\_In}\cdot\frac{60min}{hr}}\right)\right]\right]\ldots\\
&+kla_{surf\_CO2}\cdot\frac{\left[\begin{aligned}&\left[\begin{aligned}&-\frac{p_H}{p_L}\ldots\\&+\left(\frac{p_H}{p_L}\right)\cdot\exp\left[-\left(\frac{kla_{spg1\_CO2}\cdot H_{CO2\_cp}\cdot p_L\cdot V_L\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1\_In}\cdot\frac{60min}{hr}}\right)\right]\end{aligned}\right]\cdot q_{spg1\_In}\cdot\frac{60min}{hr}\ldots\\&+\left[\begin{aligned}&-\frac{p_H}{p_L}\ldots\\&+\left(\frac{p_H}{p_L}\right)\cdot\exp\left[-\left(\frac{kla_{spg2\_CO2}\cdot H_{CO2\_cp}\cdot p_L\cdot V_L\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg2\_In}\cdot\frac{60min}{hr}}\right)\right]\end{aligned}\right]\cdot q_{spg2\_In}\cdot\frac{60min}{hr}\ldots\\&+-kla_{surf\_CO2}\cdot\frac{H_{CO2\_cp}\cdot R_{gas}\cdot T_{q\_ref}\cdot p_H}{p_{stp}}\cdot V_L\end{aligned}\right]}{\left[\left(q_{ovly}+q_{spg1\_In}+q_{spg2\_In}\right)\cdot\frac{60min}{hr}+kla_{surf\_CO2}\cdot H_{CO2\_cp}\cdot R_{gas}\cdot T_{q\_ref}\cdot\frac{p_H}{p_{stp}}\cdot V_L\right]}+1
\end{aligned}
\right]
}
\quad (3.2-2)
$$

3.2.1 Carbon Dioxide Solubility

The carbon dioxide solubility is affected by the medium temperature and conductivity. Similar to the oxygen solubility equation (equation 21), the carbon dioxide solubility is obtained from Bunsen coefficient data for carbon dioxide at different temperatures and percent salinity (Table 3.2.1-1) [3]. The Bunsen coefficient is the volume of gas at standard temperature (288.15 K) and pressure (1 atm) per unit volume of solvent at the experimental temperature under a partial pressure of the given gas at 1 atm. Equation 20 is used to convert from the Bunsen coefficient to Henry's coefficient.

Temperature K	Saline 0	0.16	0.18	0.2
273.15	1.7163	1.4781	1.4514	1.4247
277.15	1.4737	1.2755	1.2533	1.2310
281.15	1.2733	1.1108	1.0908	1.0730
283.15	1.1887	1.0374	1.0218	1.0040
285.15	1.1108	0.9728	0.9572	0.9416
287.15	1.0418	0.9149	0.8993	0.8860
289.15	0.9795	0.8615	0.8481	0.8348
291.15	0.9216	0.8147	0.8014	0.7903
293.15	0.8704	0.7702	0.7591	0.7480
295.15	0.8237	0.7324	0.7213	0.7101
297.15	0.7814	0.6945	0.6856	0.6745
299.15	0.7413	0.6612	0.6522	0.6433
301.15	0.7034	0.6300	0.6211	0.6122
303.15	0.6678	0.5988	0.5899	0.5832
313.15	0.5020	N/A	N/A	N/A
323.15	0.3719	N/A	N/A	N/A
333.15	0.2747	N/A	N/A	N/A
343.15	0.1844	N/A	N/A	N/A

Table 3.2.1-1: Bunsen coefficients for carbon dioxide at different temperatures with different percent salinity [3].

A fourth order polynomial equation was fitted to the data listed in Table 3.2.1-1 to enable interpolation of the data set. The SAS JMP 8.0 software was used to perform a stepwise regression of mixed direction (0.05 threshold for adding and removing terms) to obtain the following equation for Henry's coefficient.

$$H_{CO2\_cp}(x_{salinity}, T_L) := \frac{\begin{bmatrix} 7.4436 - 0.6161 \cdot x_{salinity} + 0.2668\left(x_{salinity} - 0.126\right)^2 \ldots \\ + -0.02243 \cdot \frac{T_L}{K} + 0.0005023 \cdot \left(\frac{T_L}{K} - 292.283\right)^2 - 0.00001245 \cdot \left(\frac{T_L}{K} - 292.283\right)^3 \ldots \\ + 0.00000011065 \cdot \left(\frac{T_L}{K} - 292.283\right)^4 + 0.02693 \cdot \left(x_{salinity} - 0.126\right) \cdot \left(\frac{T_L}{K} - 292.283\right) \ldots \\ + -0.0006926 \cdot \left(x_{salinity} - 0.126\right) \cdot \left(\frac{T_L}{K} - 292.283\right)^2 \ldots \\ + -0.02435 \cdot \left(x_{salinity} - 0.126\right)^2 \cdot \left(\frac{T_L}{K} - 292.283\right) \end{bmatrix}}{\left(R_{gas} \cdot T_{stp}\right)} \qquad (3.2.1-1)$$

The salinity of the medium will be estimated from the osmolality of the medium as described in oxygen solubility section 3.1.1.

3.2.2 Carbon Dioxide Mass Transfer Coefficient

The carbon dioxide mass transfer coefficient is proportional to the oxygen mass transfer coefficient with respect to the ratio of the oxygen and carbon dioxide diffusivity raised to the 2/3 power [18].

$$kla_{x\_CO2} = kla_{x\_O2} \cdot \left(\frac{D_{CO2}}{D_{O2}}\right)^{\frac{2}{3}} \qquad (3.2.2-1)$$

The oxygen and carbon dioxide diffusivity are proportional to the ratio of the molecular volumes raised to the 0.6 power [6].

$$D_{O2} = D_{CO2} \cdot \left(\frac{25.6}{34.0}\right)^{0.6} \qquad (3.2.2-2)$$

Combining equations 3.2.2-1 and 3.2.2-2 result in a simple expression for carbon dioxide mass transfer coefficient as a function of the oxygen mass transfer coefficient.

$$kla_{x\_CO2} = 0.893 \cdot kla_{x\_O2} \qquad (3.2.2-3)$$

The expression for the carbon dioxide mass transfer coefficient assumes that the reaction enhancements by the liquid film reactions due to the chemistry equilibria and the cellular respiration are negligible [18].

3.3 Operating Conditions Impact on Gas Transfer Coefficient

When the gas transfer coefficients presented in the mass balance equations are expressed in terms of operating conditions (gas flow rate, agitation, working volume) then it is possible to calculate the impact that changes in the vessel configuration or operation have on the accumulated oxygen or carbon dioxide. There are two types of gas transfer coefficients in this system: 1) bubble aeration and 2) surface aeration. The bubble aeration has been expressed by the Van't Riet equation [7] that relates the average energy dissipation and the sparge rate to the gas transfer coefficient. Multiple spargers may interact with each other. There is no model equation for surface aeration. The surface aeration is affected by the gas boundary layer above the liquid, the extent of surface area relative to working volume, the extent of surface renewal, and whether the mixing is causing any air entrainment. The following two subsections go into more detail on the bubble (sparge) aeration and the surface aeration.

3.3.1 Sparge Aeration

The mass transfer coefficient for sparge aeration follows the Van't Riet equation [7]. The mass transfer coefficient is proportional to the power per unit volume raised to an exponent and the superficial gas velocity raised to an exponent. Both exponents are determined empirically.

$$kla_i = A \cdot \left(\frac{\rho N_p \cdot N^3 \cdot D_i^{\,5}}{V_L}\right)^{\alpha_i} \cdot \left(\frac{4q_{spg\_i}}{\pi \cdot D_T^{\,2}}\right)^{\beta_i} \qquad (3.3.1-1)$$

If there are multiple spargers, there is a possibility that bubble coalescence would result in an interaction between the two spargers. The level of interaction between the spargers could be determined through a design of experiments approach to determine if there is a correction that needs to be applied to one kla based on the gas flow rate through the other sparger.

$$kla_{effective\_i} = kla_i + B_i \cdot f(q_i, q_j) \quad (3.3.1-2)$$

The value for kla_i is the value determined from equation 3.3.1-1 without gas flow through the other sparger(s). If there is no interaction or minimum interaction between the spargers, then the value for B_i would be near zero or the p-value for the B_i term would be insignificant. Pictorially this is represented in Figure 3.3.1-1. If kla2 from trial A is comparable to kla2 from trial C and kla1 from trail B is comparable to kla1 from trial C, then there is no interaction between the spargers. If the kla values determined in trial C are different than those determined in trial A or B, the difference is attributed to the interaction of the spargers.

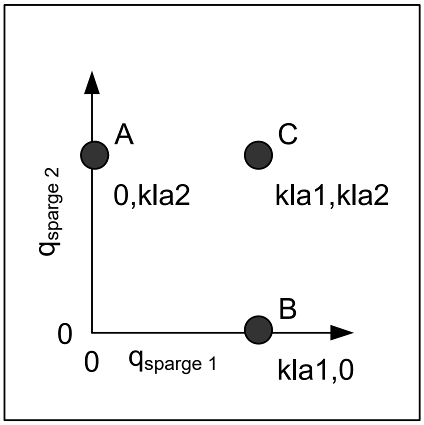

Figure 3.3-1: Independence of gas sparger streams determined through kla evaluations.

3.3.2 Surface Aeration

The surface aeration quantifies the gas exchange between the head space and the liquid at the liquid surface. If the head space is well mixed, then the extent of the surface aeration is not dependent on the overlay gas flow rate. In fact, even when the overlay gas flow rate is zero, surface aeration can occur. This is due to the head space composition being a sum of the overlay and the exiting sparged bubbles. The contribution that the surface aeration has on the overall mass balance model is dependent on the gas boundary layer above the liquid, the extent of surface area relative to working volume, the extent of surface renewal, and whether the mixing is causing any air entrainment.

The gas boundary layer above the liquid describes the extent to which the gas over the liquid represents the bulk head space or the exiting bubble composition. This is the concentration difference in the gas transfer model. As the sparge rate increases, the boundary layer is more representative of the exiting bubbles than the head space. The gas transfer of the boundary would be effectively better represented as an extension of the bubbles than as an exchange with the head space.

The surface area to volume captures the little a term in the kla expression. With increased working volume, the little a term decreases until eventually there is a scale where the surface aeration could be appropriately estimated as zero.

The surface renewal is the movement and mixing of the liquid surface. Surface renewal effectively increases the area for gas exchange. The surface renewal improves with agitation frequency and sparging.

Air entrainment can occur if an impeller is partially submerged or if the liquid height above the impeller is sufficiently small that a vortex forms and comes in contact with the impeller. Air entrainment effectively increases the area for gas transfer.

Currently, there is no universal model to describe how agitation frequency, working volume, or sparge rate correlates to the surface aeration. The surface aeration is determined for each experimental condition. Then a complex polynomial is empirically fitted for the surface aeration as a function of the bioreactor operating conditions. The equation could be simplified to a second order polynomial by using a gate state that if the modeled surface aeration is less than zero, then return zero. The polynomial fit would only need to be performed over the operating conditions where the surface aeration is greater than zero. With limited understanding of the mechanism, the following polynomial is recommended.

$$\begin{aligned} kla_{surf} = {} & A_0 + A_1 \cdot V_L + A_2 \cdot N + A_3 \cdot \left(\sum_i q_{sparge\_i}\right) + A_4 \cdot T \ldots \\ & + A_5 \cdot \left(V_L \cdot \sum_i q_{sparge\_i}\right) + A_6 \cdot (V_L \cdot N) + A_7 \cdot \left(N \cdot \sum_i q_{sparge\_i}\right) \ldots \\ & + A_8 \cdot \left(T \cdot \sum_i q_{sparge\_i}\right) + A_9 \cdot T \cdot N + A_{10} \cdot \left(N \cdot V_L \cdot \sum_i q_{sparge\_i}\right) \ldots \\ & + A_{11} \cdot N^2 + A_{12} \cdot N^3 + A_{13} \cdot \left(\sum_i q_{sparge\_i}\right)^2 + A_{14} \cdot \left(\sum_i q_{sparge\_i}\right)^3 \ldots \\ & + A_{15} \cdot V_L{}^2 \end{aligned} \tag{3.3.2 - 1}$$

3.4 Oxygen and Carbon Dioxide Measurements

There are three different types of measurements that could be taken to monitor either the oxygen, carbon dioxide or both. These would be an offline sample to measure the dissolved gases, an online probe monitoring the dissolved gases, or an at-line or on-line instrument to monitor the composition of the exhaust gas stream. Each of these approaches introduce time constants to the dynamic system. This section discusses the general equations and the factors that could affect the time constants.

3.4.1 Off-line Liquid Sample

The dissolved gas reading could be taken with off-line samples. The sample is pulled from the bioreactor and loaded on a gas analyzer. During the sample handling, there is gas exchange with the ambient and consumption or depletion due to cellular respiration. The mass balance equations for the offline sample is captured in equations 3.4.1-1 and 3.4.1-2.

$$\frac{d}{dt} O2_{sample} = kla_{sample\_O2} \cdot \left(O2_{ambient} - O2_{sample}\right) - OUR \tag{3.4.1 - 1}$$

$$\frac{d}{dt} CO2_{sample} = kla_{sample\_CO2} \cdot \left(CO2_{ambient} - CO2_{sample}\right) + CER \tag{3.4.1 - 2}$$

Due to the differences between oxygen and carbon dioxide solubility in the medium, the off-line dissolved oxygen sample is much more prone to error than the off-line dissolved carbon dioxide sample. Carbon dioxide is approximately 25 times more soluble than oxygen. If the dissolved oxygen is at 80 mm Hg partial pressure, the dissolved carbon dioxide is at 80 mm Hg and the respiration quotient is one, then the dissolved oxygen will change at a rate nearly 25 times faster than the carbon dioxide reading would change. For this reason, while an offline reading for the dissolved carbon dioxide may be acceptable, it is less likely acceptable for the dissolved oxygen reading.

$$\frac{d}{dt} pO2_{sample} = kla_{sample\_O2} \cdot (80mmHg) - \frac{OUR}{H_{O2\_cp}} \tag{3.4.1 - 3}$$

$$\frac{d}{dt}pCO2_{sample} = -0.893 \cdot kla_{sample\_O2} \cdot (80mmHg) + \frac{OUR}{25 \cdot H_{O2\_cp}} \qquad (3.4.1-4)$$

3.4.2 On-line Liquid Reading

The dissolved gas reading could be monitored with an on-line probe in the liquid. There are different types of probe technologies that are capable of doing this task. The probes could be categorized based on whether there is a chemical reaction consuming the analyte at the probe. The Clarke electrodes have a chemical reaction of oxygen at the platinum electrode. Whereas, the pH based or fluorescent based probes for carbon dioxide or oxygen do not have a consuming chemical reaction. The consuming chemical reaction affects the stability of the probe calibration but does not affect the probe response time. The probe response time is affected by the diffusion path length. The diffusion path length includes the electrolyte solution, the membrane, and the liquid boundary layer that is dependent on the mixing of the bulk solution. The impact that the operating conditions have the probe calibration and the probe response time need to be taken into consideration when solving equation 3.4.2-1 or 3.4.2-2.

$$\frac{d}{dt}O2_e = \frac{1}{\tau_{e\_O2}}\left(O2_L - O2_e\right) \qquad (3.4.2-1)$$

$$\frac{d}{dt}CO2_e = \frac{1}{\tau_{e\_CO2}}\left(CO2_L - CO2_e\right) \qquad (3.4.2-2)$$

The impact that the agitation rate has on the probe calibration and response time could be determined from empirical evaluations. This can be done by one of two methods. If the probe uses a consuming reaction to produce a reading, then one can perform step changes in the agitation rate and monitor the probe reading. This is done by running the vessel with a fixed composition sparge until the vessel is at steady-state. The probe is calibrated, then the agitation is stepped through the range that would be evaluated in the study. For each step change in the agitation rate, the vessel is allowed time to reach steady state. Note the change in the online probe reading. The change in the online probe reading is due to the change in the boundary layer thickness [8]. The dissolved liquid concentration does not change because the liquid is in equilibrium with the sparge composition. The response time is determined by fitting a first order equation (equations 3.4.2-1 or 3.4.2-2) to the probe profile. The calibration correction and the response time as a function of the agitation is used in the model data fitting.

For a probe that does not have a consuming chemical reaction, at steady-state, there is no flux through the probe layers. Therefore, changes in the boundary layer would not result in a change in the probe reading. In order to see the probe response difference as a function of operating conditions there would need to be a step change in the dissolved gas readings. This could be accomplished for some vessels by moving the probe from the vessel head space to the liquid while using different agitation rates between trials. If the head space composition and liquid composition are different, then there would be a change in the probe reading once placed in the liquid that would fit equations 3.4.2-1 or 3.4.2-2. For vessels where the probe cannot be moved between the head space and liquid, estimated values from bench studies could be applied as a primer value to a parameter fitting model that tries to fit the kla and probe response time values.

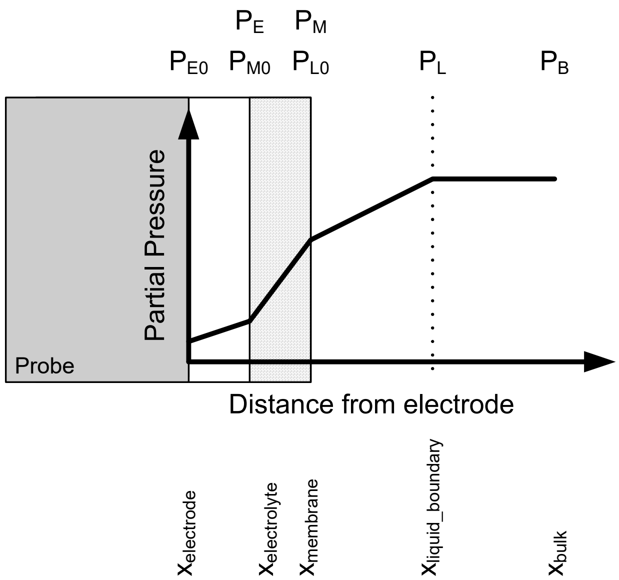

Figure 3.4.2-1: Oxygen diffusion profile for polargraphic probe

The justification for the change in the calibration and response time come from modeling a probe as a three layer diffusion system with reaction kinetics occurring at the electrode (Figure 3.4.2-1, [8]). For a probe that is estimated as the liquid boundary layer, the membrane, and the electrolyte solution, when the system is at steady-state, the molar flux through each of the layers and the reaction rate at the electrode are all equal.

$$J = -D_E \cdot \frac{\Delta C_E}{\delta_E} = -D_M \cdot \frac{\Delta C_M}{\delta_M} = -D_L \cdot \frac{\Delta C_L}{\delta_L} = \frac{r \cdot C_{E0}}{A_{probe}} \qquad (3.4.2-3)$$

where C is the concentration of oxygen or carbon dioxide at the various layers (E is electrolyte, M is membrane, and L is liquid boundary layer. D is the diffusivity through the different layers, and δ is the layer thickness. At each of the junctions of the three layers, the partial pressure of the gases is the same in the different materials. The concentration of the gasses within each of the materials at the material boundary is in proportion to the partition coefficient. The partition coefficient is determined from the ratio of the Henry's coefficient for the different materials.

$$C_E = \frac{H_M}{H_E} \cdot C_{M0} \qquad (3.4.2-4)$$

$$C_M = \frac{H_L}{H_M} \cdot C_{L0} \qquad (3.4.2-5)$$

Then, substituting the concentrations (equations 3.4.2-4 and 3.4.2-5) into the molar flux equilibrium equation (equation 3.4.2-3) allows for us to solve the concentration at each of the material boundaries. This allows us to solve for the reaction rate at the electrode in terms of the partial pressure of the dissolved gas in the bulk solution, the solubility of gas in the different materials, the permeability of gas through the different materials, and the reaction rate constant.

$$\frac{r \cdot C_{E0}}{A_{probe}} = -D_E \cdot \frac{C_{E0} - \frac{H_M}{H_E} \cdot C_{M0}}{\delta_E} = -D_M \cdot \frac{C_{M0} - \frac{H_L}{H_M} \cdot C_{L0}}{\delta_M} = -D_L \cdot \frac{C_{L0} - C_B}{\delta_L} \qquad (3.4.2-6)$$

$$C_{L0} = \frac{\phi_M \cdot C_{M0} + \phi_L \cdot C_B}{\left(\phi_M \cdot \frac{H_L}{H_M} + \phi_L \cdot \phi_M\right)} \qquad (3.4.2-7)$$

$$C_{M0} = \frac{\left[\left(\phi_E \cdot \phi_M \cdot \frac{H_L}{H_M} + \phi_L \cdot \phi_M \cdot \phi_E\right) \cdot C_{E0} + \left(\phi_L \cdot \phi_M \cdot \frac{H_L}{H_M}\right) \cdot C_B\right]}{\left(\phi_L \cdot \phi_M + \phi_E \cdot \phi_M \cdot \frac{H_L}{H_E} + \phi_L \cdot \phi_E \cdot \frac{H_M}{H_E}\right)} \qquad (3.4.2-8)$$

$$C_{E0} = \frac{H_L \cdot C_B}{\frac{r}{A_{probe}} \cdot \left(\frac{H_E}{\phi_E} + \frac{H_M}{\phi_M} + \frac{H_L}{\phi_L}\right) + H_E} \qquad (3.4.2-9)$$

The reaction rate at the probe electrode is proportional to the dissolved gas reading.

$$r \cdot C_{E0} = \frac{1}{\frac{1}{A_{probe}} \cdot \left(\frac{H_E}{\phi_E} + \frac{H_M}{\phi_M} + \frac{H_L}{\phi_L}\right) + \frac{H_E}{r}} \cdot pC_B \qquad (3.4.2-10)$$

The calibration of the probe is an empirical fit that lumps the physical properties of the probe (the solubility of gas in the different materials, the permeability of gas through the different materials, and the reaction rate constant) into one coefficient (equation 3.4.2-11). Temperature would affect the solubility of gas in the different materials, the diffusivity through the different materials, and the reaction rate constant. Most probes have a built-in temperature compensation to account for the impact changes in temperature have on the calibration.

$$pC = C_{cal} \cdot \left(r \cdot C_{E0}\right) = \frac{C_{cal}}{\frac{1}{A} \cdot \left(\frac{H_E}{\phi_E} + \frac{H_M}{\phi_M} + \frac{H_L}{\phi_L}\right) + \frac{H_E}{r}} \cdot pC_B \qquad (3.4.2-11)$$

The probes do not, however, have a compensation for the liquid boundary layer thickness on the probe calibration. The calibration is affected by the liquid boundary layer thickness by affecting the gas permeability through the boundary layer. Increases in the boundary layer would result in a need to increase the probe calibration constant to return the appropriate reading (equation 3.4.2-12). This could be empirically determined as described above.

$$C_{cal2} = C_{cal1}\frac{B1 + B2\cdot\delta_{L2}}{B1 + B2\cdot\delta_{L1}} \qquad (3.4.2-12)$$

The boundary layer thickness could be estimated from the flow past a plate boundary layer equation by using axial flow wall jetting models [9] to determine the velocity. The flow velocity is proportional to the impeller flow number, the impeller agitation frequency, and the impeller diameter cubed. The probe boundary layer is proportional to one over the square root of the flow velocity over the probe. This suggests that the dissolved oxygen probe calibration is related to the agitation frequency as follows:

$$C_{cal2} = C_{cal1}\frac{B1 + \dfrac{B3}{N_2^{\,0.5}}}{B1 + \dfrac{B3}{N_1^{\,0.5}}} \qquad (3.4.2-13)$$

The probe response time is also affected by the liquid boundary layer thickness. The response time of a probe has been characterized as being proportional to the probe path length and the inverse of the effective oxygen permeability through the probe layers [10].

$$\tau_e = \frac{\beta\cdot\delta_{probe}^{\,2}}{D_{probe}} \qquad (3.4.2-14)$$

The effective probe permeability is calculated by deriving equation 3.4.2-11 with a single layer. The probe permeability is described as follows:

$$\phi_{probe} = \frac{\phi_E}{\left(1 + \dfrac{\phi_E}{\phi_M}\cdot\dfrac{H_M}{H_E} + \dfrac{\phi_E}{\phi_L}\cdot\dfrac{H_L}{H_E}\right)} \qquad (3.4.2-15)$$

The probe response time can be described in terms of the thickness of each of the materials.

$$\tau_e = \frac{\beta\cdot\left(1 + \dfrac{\phi_E}{\phi_M}\cdot\dfrac{H_M}{H_E} + \dfrac{\phi_E}{\phi_L}\cdot\dfrac{H_L}{H_E}\right)\cdot\left(\delta_E + \delta_M + \delta_L\right)}{\phi_E} \qquad (3.4.2-16)$$

And this in turn can be described as a function of the impeller frequency.

$$\tau_e = B4 + \frac{B5}{N^{0.5}} + \frac{B6}{N} \qquad (3.4.2-17)$$

Note that if the probe does not have a chemical reaction (such as a fluorescent probe), then the calibration would not be affected by agitation, but the response time would be as described in equations 3.4.2-16 and 3.4.2-17.

3.4.3 On-line Exhaust Gas Reading

The gas levels in the bioreactor head space could also be monitored by measuring the gas composition of the exhaust gas stream. The measurements have a time lag that reflects the gas velocity and line length from the bioreactor head space and the gas monitor.

$$x_{O2\_offgas}(t) = x_{O2\_hs}(t - \tau) \qquad (3.4.3 - 1)$$

$$\tau = \frac{L}{q_{exhaust\_sample}} \qquad (3.4.3 - 2)$$

3.5 Operating Pressure

The operating head space and liquid pressure is incorporated into the equations for the oxygen and carbon dioxide mass balance. The head space pressure is the gauge back pressure plus atmospheric pressure.

$$p_H = p_{gauge} + p_{atm} \qquad (3.5 - 1)$$

The liquid pressure is due to the head space pressure and the hydrostatic pressure. Because the vessel is well mixed with axial impellers, we will assume that the liquid spends equal time at all heights of the vessel. The liquid pressure will then be calculated as the head space pressure plus the average hydrostatic pressure (half the liquid height).

$$p_L = \frac{1}{2} \cdot h_L \cdot \rho_L \cdot g + p_H \qquad (3.5 - 2)$$

Assuming the bioreactor geometry can be estimated as a cylinder, substituting the working volume, tank diameter, and density of water into equation 3.5-2 translates to the following expression.

$$p_L = \frac{1}{2} \cdot \left[\frac{V_L}{\pi \cdot \left(\frac{D}{2}\right)^2} \right] \cdot \left(1 \frac{gm}{cm^3}\right) \cdot g + p_H \qquad (3.5 - 3)$$

3.6 Head Space Temperature

The head space temperature is incorporated in the dynamic equations for the oxygen and carbon dioxide mass balance in the headspace (equations 3.1-16 and 4.1.2-1). The head space temperature is the temperature of the gas in the head space. The temperature is affected by the heat transfer from the liquid, the heat of the gas from the exiting bubbles, the heat of the overlay gas, and the heat transfer to the ambient conditions through the vessel head. The temperature of the head space gas will be a value between the ambient and culture temperature. This could be calculated by using heat balance equations. Or a quick estimate could be used by applying the average of the culture and ambient temperatures.

$$T_H = \frac{T_L + T_{ambient}}{2} \qquad (3.6 - 1)$$

3.7 Antifoam Impact on kla

Antifoam is added to the vessel to manage the accumulation of foam at the liquid surface. Excessive foam can lead to cell entrapment and death, or to fouling of the exhaust filter. The impact that the antifoam has on the gas transfer coefficients would be due to a change in bubble size resulting from a change in the surface tension, or a change in the oxygen diffusivity at the boundary layer. Without fully understanding the mechanism that the antifoam impacts the gas transfer coefficient, we will look at equations that could be empirically determined that help predict the resulting gas transfer coefficient.

The antifoam cannot simply be added to the culture at the time of inoculation and be effective through the course of the run. The antifoam appears to lose its activity over time. We will assume that this can be expressed by first order kinetics.

$$\frac{d}{dt}C_{antifoam} = -r_{antifoam}\cdot C_{antifoam} \qquad (3.7-1)$$

The impact that antifoam has on the kla has been observed to be dependent on the antifoam dosing concentration and the kla. The kla reduction ranges from 5 to 50% under various operating conditions after 3 ppm of simethicone addition. The kinetics of the antifoam impact on kla is under investigation. Until more detailed calculations are available, the following equation could be empirically determined to establish the impact that antifoam concentration has on the gas transfer coefficient.

$$kla_{antifoam} = kla_{antifoam0}\cdot\left[\frac{kla_{antifoam\_\infty}}{kla_{antifoam\_0}} + \left(1 - \frac{kla_{antifoam\_\infty}}{kla_{antifoam\_0}}\right)\cdot e^{-z\cdot C_{antifoam}}\right] \qquad (3.7-2)$$

3.8 Chemistry Equilibrium

The medium is buffered with a number of weak acids and bases. These include the sodium bicarbonate, HEPES, phosphates, amino acids, and other organic acids and bases. The predominate buffering systems in the medium are 1) bicarbonate, carbonate, carbonic acid, and carbon dioxide system, 2) water, 3) phosphate buffering system 4) HEPES buffering system, and 5) lactic acid, alanine, and ammonia accumulation. The first principle model will leverage the available equilibrium expressions for the different buffers. The equilibrium expressions will include an initial concentration and the change in concentration to satisfy the equilibrium. Literature provides examples of simultaneously solving the equilibrium equations to determine the pH of a complex buffering system [11, 12]

3.8.1 Carbonate Buffering System

For sodium bicarbonate system, there are three governing equilibrium equations (equations 3.8.1-1, 3.8.1-2, and 3.8.1-3) [13]. These equilibrium correspond to the dissolved carbon dioxide in equilibrium with carbonic acid, carbon dioxide and carbonic acid in equilibrium with bicarbonate, and bicarbonate in equilibrium with carbonate.

$$K_h = \frac{H2CO3_i + \Delta H2CO3}{\Delta CO2} \qquad (3.8.1-1)$$

$$K_1 = \frac{H\cdot(HCO3_i + \Delta HCO3)}{\Delta CO2 + (H2CO3_i + \Delta H2CO3)} \qquad (3.8.1-2)$$

$$K_2 = \frac{H\cdot(CO3_i + \Delta CO3)}{(HCO3_i + \Delta HCO3)} \qquad (3.8.1-3)$$

where

$$K_h := 10^{-2.59} \qquad (3.8.1-4)$$

$$K_1 := 10^{-6.352} \qquad (3.8.1-5)$$

$$K_2 := 10^{-10.329} \qquad (3.8.1-6)$$

The dissolved carbon dioxide in equilibrium with sodium bicarbonate buffer is determined from equations 3.8.1-1 and 3.8.1-2.

$$\Delta CO2 = \frac{H\cdot(HCO3_i + \Delta HCO3)}{K_1 + K_1\cdot K_h} \qquad (3.8.1-7)$$

In addition, to the chemistry equilibrium, the charge balance also needs to be satisfied.

$$\Delta H_{carbonate} - \Delta HCO3 - 2\cdot\Delta CO3 = 0 \qquad (3.8.1-8)$$

From the charge balance, the change in the carbonate concentration can be expressed by the change in protons and the change in the bicarbonate concentration.

$$\Delta CO3 = \frac{1}{2}\cdot\Delta H_{carbonate} - \frac{1}{2}\cdot\Delta HCO3 \qquad (3.8.1-9)$$

The change in carbonate (equation 3.8.1-9) is substituted into the equilibrium expression for bicarbonate and carbonate (equation 3.8.1-3) to solve for the change in the bicarbonate concentration as a function of the initial bicarbonate concentration and the change in the proton concentration.

$$\Delta HCO3 = \frac{H\cdot(CO3_i + \Delta CO3)}{K_2} - HCO3_i \qquad (3.8.1-10)$$

$$\Delta HCO3 = \frac{H\cdot\left(CO3_i + \frac{1}{2}\cdot\Delta H_{carbonate} - \frac{1}{2}\cdot\Delta HCO3\right)}{K_2} - HCO3_i \qquad (3.8.1-11)$$

$$\Delta HCO3 = \frac{2H}{2K_2 + H}\cdot CO3_i + \frac{H}{2K_2 + H}\cdot\Delta H_{carbonate} - \frac{2K_2}{2K_2 + H}\cdot HCO3_i \qquad (3.8.1-12)$$

The change in the bicarbonate concentration (equation 3.8.1-12) is substituted into the equation for carbon dioxide (equation 3.8.1-7) to provide for an equation for the carbon dioxide in equilibrium with the initial bicarbonate concentration, the pH, and the change in the proton concentration due to other acids and bases.

$$\Delta CO2 = \frac{H^2}{2K_1 \cdot K_2 + 2K_1 \cdot K_2 \cdot K_h + K_1 \cdot H + K_1 \cdot K_h \cdot H} \cdot \left(HCO3_i + 2 \cdot CO3_i + \Delta H_{carbonate}\right) \qquad (3.8.1-13)$$

The change in proton concentration due to other acids and bases is expressed as

$$H = H_i + \Delta H_{carbonate} + \Delta H_{water} + \sum_{other} \Delta H_{other} \qquad (3.8.1-14)$$

$$\Delta H_{carbonate} = H - H_i - \Delta H_{water} - \sum_{other} \Delta H_{other} \qquad (3.8.1-15)$$

The buffering capacity of the bicarbonate system can be determined using the carbon balance of the carbonate system.

$$\Delta CO3 + \Delta HCO3 + \Delta CO2 + \Delta H2CO3 = 0 \qquad (3.8.1-16)$$

Substituting the relationship of carbon acid to carbon dioxide (equation 3.8.1-1) into the carbon balance (equation 3.8.1-16) gives us an expression for the carbonate that can be substituted into equation 3.8.1-10.

$$\Delta CO3 + \Delta HCO3 + \Delta CO2\left(1 + K_h\right) - H2CO3_i = 0 \qquad (3.8.1-17)$$

$$\Delta HCO3 = \frac{H \cdot \left[CO3_i - \left(\Delta HCO3 + \Delta CO2\left(1 + K_h\right) - H2CO3_i\right)\right]}{K_2} - HCO3_i \qquad (3.8.1-18)$$

The change on bicarbonate is simplified as follows

$$\Delta HCO3 = \frac{H}{\left(K_2 + H\right)} \cdot \left[CO3_i - \left(\Delta CO2\left(1 + K_h\right) - H2CO3_i\right)\right] - \frac{K_2}{\left(K_2 + H\right)} \cdot HCO3_i \qquad (3.8.1-19)$$

Substituting the change in bicarbonate (equation 3.8.1-19) into the expression for the change in carbon dioxide as a function of the bicarbonate (equation 3.8.1-7) provides for an expression of the carbon dioxide in terms of the initial concentrations of bicarbonate, carbonate, and carbonic acid.

$$\Delta CO2 = \frac{H^2}{\left[K_1 \cdot \left(K_2 + H\right) + K_1 \cdot K_h \cdot \left(K_2 + H\right) + H^2 \cdot \left(1 + K_h\right)\right]} \cdot \left(HCO3_i + CO3_i + H2CO3_i\right) \qquad (3.8.1-20)$$

Simultaneously solving the change in carbon dioxide with equations 3.8.1-13 and 3.8.1-20 produces an expression for the buffering capacity due to the carbonate system.

$$\begin{aligned}\Delta H\_capacity_{carbonate} = {} & \frac{\left(K_1 \cdot K_2 + K_1 \cdot K_2 \cdot K_h\right) - \left(H^2 + K_h \cdot H^2\right)}{\left[K_1 \cdot \left(K_2 + H\right) + K_1 \cdot K_h \cdot \left(K_2 + H\right) + H^2 \cdot \left(1 + K_h\right)\right]} \cdot HCO3_i \ldots \\ & + \frac{-\left(K_1 \cdot H + K_1 \cdot K_h \cdot H + 2H^2 + 2 \cdot K_h \cdot H^2\right)}{K_1 \cdot \left(K_2 + H\right) + K_1 \cdot K_h \cdot \left(K_2 + H\right) + H^2 \cdot \left(1 + K_h\right)} \cdot CO3_i \ldots \\ & + \frac{\left(2K_1 \cdot K_2 + 2K_1 \cdot K_2 \cdot K_h + K_1 \cdot H + K_1 \cdot K_h \cdot H\right)}{\left[K_1 \cdot \left(K_2 + H\right) + K_1 \cdot K_h \cdot \left(K_2 + H\right) + H^2 \cdot \left(1 + K_h\right)\right]} \cdot H2CO3_i\end{aligned} \qquad (3.8.1-21)$$

The impact of changes on the buffer concentration on the pH equilibrium (equation 3.8.1-13) is account for by taking the ratio of the buffering capacity of the carbonate system (equation 3.8.1-21) relative to the total buffering capacity of new formulation compared to the buffering capacity of the carbonate system (equation 3.8.1-21) relative to the total buffering capacity of the basis formulation.

$$\Delta CO2 = \frac{H^2}{2K_1 \cdot K_2 + 2K_1 \cdot K_2 \cdot K_h + K_1 \cdot H + K_1 \cdot K_h \cdot H} \cdot \left[HCO3_i + 2 \cdot CO3_i + \frac{\left(\dfrac{\Delta H\_capacity_{carbonate}}{\Delta H\_capacity\_basis_{carbonate}}\right)}{\left(\dfrac{\begin{array}{l}\Delta H\_capacity_{carbonate} \\ + \Delta H\_capacity_{PO4} \\ + \Delta H\_capacity_{HEPES} \\ + \Delta H\_capacity_{SaltsA} \\ + \Delta H\_capacity_{SaltsB}\end{array}}{\begin{array}{l}\Delta H\_capacity\_basis_{carbonate} \\ + \Delta H\_capacity\_basis_{PO4} \\ + \Delta H\_capacity\_basis_{HEPES} \\ + \Delta H\_capacity\_basis_{SaltsA} \\ + \Delta H\_capacity\_basis_{SaltsB}\end{array}}\right)} \cdot \left(H - H_i - \Delta H_{water} + -\sum_{other} \Delta H_{other}\right) \right] \quad (3.8.1-22)$$

3.8.2 Water Buffering System

Besides the carbonate buffer system, there are five other key buffering systems that will be calculated out with the first principle approach. These are water, phosphate, HEPES, lactate and ammonia. The other amino acids and components will be lumped into terms for bases and acid equivalents. The change in the proton concentration due to the water equilibrium is determined from the equilibrium that the hydroxide ions have with the hydrogen protons.

$$\left(OH_i + OH_{added} + \Delta OH\right) \cdot (H) = K_{water} \quad (3.8.2-1)$$

$$\Delta OH_{water} = OH_{added} + OH_i - \frac{K_{water}}{H} \quad (3.8.2-2)$$

$$\Delta H_{water} = -\Delta OH_{water} = -OH_{added} - OH_i + \frac{K_{water}}{H} = -OH_{added} - H_i + \frac{K_{water}}{H} \quad (3.8.2-3)$$

3.8.3 Phosphate Buffering System

The phosphate buffering system is complex like the carbonate buffering system because of the number of available dissociations for the phosphates. There are three equilibrium equations for the phosphate ions, the hydrogen phosphates, the dihydrogen phosphates, and the phosphoric acid.

$$K_{1P} = \frac{(H2PO4_i + \Delta H2PO4)\cdot H}{(H3PO4_i + \Delta H3PO4)} \qquad (3.8.3-1)$$

$$K_{2P} = \frac{(HPO4_i + \Delta HPO4)\cdot H}{(H2PO4_i + \Delta H2PO4)} \qquad (3.8.3-2)$$

$$K_{3P} = \frac{(PO4_i + \Delta PO4)\cdot H}{(HPO4_i + \Delta HPO4)} \qquad (3.8.3-3)$$

From the charge balance, the change in the hydrogen protons can be expressed by the change in the phosphate ions, hydrogen phosphate, and dihydrogen phosphate.

$$-3\cdot\Delta PO4 - 2\cdot\Delta HPO4 - \Delta H2PO4 + \Delta H_{PO4} = 0 \qquad (3.8.3-4)$$

There is also an overall balance of the phosphates.

$$\Delta PO4 + \Delta HPO4 + \Delta H2PO4 + \Delta H3PO4 = 0 \qquad (3.8.3-5)$$

With equations 3.8.3-1, 3.8.3-2, 3.8.3-3, 3.8.3-4, and 3.8.3-5, there are five unique equations and five unknowns. Therefore, we should be able to solve for the change in the hydrogen protons due to the initial concentrations of the phosphates. The change in the phosphate ions is represented by the change in the other phosphates by rearranging equation 3.8.3-5.

$$\Delta PO4 = -\Delta HPO4 - \Delta H2PO4 - \Delta H3PO4 \qquad (3.8.3-6)$$

The change in the phosphate ions concentration (equation 3.8.3-6) is substituted into the third equilibrium expression (equation 3.8.3-3). With some rearranging, the change in the hydrogen phosphate concentration is solved in terms of the change in the dihydrogen phosphate concentration and the phosphoric acid concentrations, and the initial phosphate ion and hydrogen phosphate concentrations.

$$\Delta HPO4 = \frac{H}{K_{3P} + H}\cdot(PO4_i - \Delta H2PO4 - \Delta H3PO4) - \frac{K_{3P}}{K_{3P} + H}\cdot HPO4_i \qquad (3.8.3-7)$$

The change in the hydrogen phosphate concentration (equation 3.8.3-7) is substituted into the second equilibrium expression (equation 3.8.3-2). With some rearranging, the change in the dihydrogen phosphate concentration is solved in terms of the change in the phosphoric acid concentration and the initial phosphate ion, hydrogen phosphate, and dihydrogen phosphate concentrations.

$$\Delta H2PO4 = \frac{H^2}{K_{2P}\cdot K_{3P} + K_{2P}\cdot H + H^2}\cdot(PO4_i + HPO4_i - \Delta H3PO4) - \frac{(K_{2P}\cdot K_{3P} + K_{2P}\cdot H)}{(K_{2P}\cdot K_{3P} + K_{2P}\cdot H + H^2)}\cdot H2PO4_i \qquad (3.8.3-8)$$

The change in the dihydrogen phosphate concentration (equation 3.8.3-8) is substituted into the first equilibrium expression (equation 3.8.3-1). With some rearranging, the change in the phosphoric acid concentration is solved in terms of the initial phosphate ion, hydrogen phosphate, dihydrogen phosphate and phosphoric acid concentrations.

$$\begin{aligned}\Delta H3PO4 = {} & \frac{H^3}{K_{1P}\cdot K_{2P}\cdot K_{3P} + K_{1P}\cdot K_{2P}\cdot H + K_{1P}\cdot H^2 + H^3}\cdot\left(PO4_i + HPO4_i + H2PO4_i\right) \ldots \\ & + -\frac{\left(K_{1P}\cdot K_{2P}\cdot K_{3P} + K_{1P}\cdot K_{2P}\cdot H + K_{1P}\cdot H^2\right)}{\left(K_{1P}\cdot K_{2P}\cdot K_{3P} + K_{1P}\cdot K_{2P}\cdot H + K_{1P}\cdot H^2 + H^3\right)}\cdot H3PO4_i\end{aligned} \qquad (3.8.3-9)$$

Then by combining the expressions from equations 3.8.3-4, 3.8.3-5, 3.8.3-6, 3.8.3-7, 3.8.3-8, and 3.8.3-9; it is possible to solve for the change in the hydrogen proton concentration as a function of the initial concentrations of the phosphates.

$$\begin{aligned}\Delta H_{PO4} = {} & \Bigg[\frac{-H}{K_{3P}+H} - \frac{2H^2}{K_{2P}\cdot K_{3P} + K_{2P}\cdot H + H^2} \ldots \\ & + \frac{H}{K_{3P}+H}\cdot\frac{H^2}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)} \ldots \\ & + \left[-3 + \frac{H}{K_{3P}+H} + 2\cdot\frac{H^2}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)} \ldots + -\frac{H}{K_{3P}+H}\cdot\frac{H^2}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)}\right]\cdot\frac{H^3}{\left(K_{1P}\cdot K_{2P}\cdot K_{3P} \ldots + K_{1P}\cdot K_{2P}\cdot H \ldots + K_{1P}\cdot H^2 + H^3\right)}\Bigg]\cdot PO4_i \ldots \\ & + \Bigg[\frac{K_{3P}}{K_{3P}+H} - \frac{2H^2}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)} \ldots \\ & + \frac{H}{K_{3P}+H}\cdot\frac{H^2}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)} \ldots \\ & + \left[-3 + \frac{H}{K_{3P}+H} + 2\cdot\frac{H^2}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)} \ldots + -\frac{H}{K_{3P}+H}\cdot\frac{H^2}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)}\right]\cdot\frac{H^3}{\left(K_{1P}\cdot K_{2P}\cdot K_{3P} \ldots + K_{1P}\cdot K_{2P}\cdot H \ldots + K_{1P}\cdot H^2 + H^3\right)}\Bigg]\cdot HPO4_i \ldots \\ & + \Bigg[2\cdot\frac{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H\right)}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)} - \frac{H}{K_{3P}+H}\cdot\frac{\left(K_{2P}\cdot K_{3P} + K_{2P}\cdot H\right)}{\left(K_{2P}\cdot K_{3P} \ldots + K_{2P}\cdot H + H^2\right)} \ldots \\ & + \left[-3 + \frac{H}{K_{3P}+H} + 2\cdot\frac{H^2}{\left(K_{2P}\cdot K_{3P} \ldots\right.} \ldots\right]\cdot\frac{H^3}{\left(K_{1P}\cdot K_{2P}\cdot K_{3P} \ldots\right.} \Bigg]\cdot H2PO4_i \ldots\end{aligned} \qquad (3.8.3-10)$$

$$\left[\left[\cdots \frac{\cdots}{K_{3P}+H} \cdots \frac{\cdots}{\left(\begin{array}{l} K_{2P}\cdot K_{3P} \ldots \\ +K_{2P}\cdot H+H^2 \end{array}\right)} + -\frac{H}{K_{3P}+H}\cdot\frac{H^2}{\left(\begin{array}{l} K_{2P}\cdot K_{3P} \ldots \\ +K_{2P}\cdot H+H^2 \end{array}\right)}\right] \cdots \frac{\cdots}{\left(\begin{array}{l} K_{1P}\cdot K_{2P}\cdot K_{3P} \ldots \\ +K_{1P}\cdot K_{2P}\cdot H \ldots \\ +K_{1P}\cdot H^2+H^3 \end{array}\right)}\right]$$

$$+\left[\begin{array}{l} 3-\dfrac{H}{K_{3P}+H}-2\cdot\dfrac{H^2}{\left(\begin{array}{l} K_{2P}\cdot K_{3P} \ldots \\ +K_{2P}\cdot H+H^2 \end{array}\right)} \ldots \\ +\dfrac{H}{K_{3P}+H}\cdot\dfrac{H^2}{\left(\begin{array}{l} K_{2P}\cdot K_{3P} \ldots \\ +K_{2P}\cdot H+H^2 \end{array}\right)} \end{array}\right]\frac{\left(\begin{array}{l} K_{1P}\cdot K_{2P}\cdot K_{3P} \ldots \\ +K_{1P}\cdot K_{2P}\cdot H+K_{1P}\cdot H^2 \end{array}\right)}{\left(\begin{array}{l} K_{1P}\cdot K_{2P}\cdot K_{3P} \ldots \\ +K_{1P}\cdot K_{2P}\cdot H \ldots \\ +K_{1P}\cdot H^2+H^3 \end{array}\right)}\cdot H3PO4_i$$

where the equilibrium constants for the phosphate system are the following at 25 degrees Celsius [14].

$$K_{1P} := 10^{-2.12} \qquad (3.8.3-11)$$

$$K_{2P} := 10^{-7.21} \qquad (3.8.3-12)$$

$$K_{3P} := 10^{-12.67} \qquad (3.8.3-13)$$

3.8.4 HEPES Buffering System

The change in the proton concentration due to that addition of the HEPES acid is determined from the equilibrium that the HEPES acid has with the basic salt and proton concentration.

$$K_{HEPES} = \frac{\Delta HEPES_{salt}\cdot H}{HEPES-\Delta HEPES_{salt}} \qquad (3.8.4-1)$$

$$\Delta H_{HEPES} = \Delta HEPES_{salt} = \frac{K_{HEPES}}{K_{HEPES}+H}\cdot HEPES \qquad (3.8.4-2)$$

where the equilibrium constant for HEPES is as follows [15].

$$K_{HEPES} := 10^{-7.562} \qquad (3.8.4-3)$$

3.8.5 Metabolites Buffering System

The change in the proton concentration due to accumulation of lactic acid is determined from the equilibrium of lactic acid with the basic salt and proton concentration.

$$K_{Lactate} = \frac{\Delta Lactate_{salt} \cdot H}{LacticAcid - \Delta Lactate_{salt}} \qquad (3.8.5-1)$$

$$\Delta H_{Lactate} = \Delta Lactate_{salt} = \frac{K_{Lactate}}{K_{Lactate} + H} \cdot Lactate \qquad (3.8.5-2)$$

where the equilibrium constant for lactate is as follows [16].

$$K_{Lactate} := 10^{-3.86} \qquad (3.8.5-3)$$

The change in the proton concentration due to accumulation of ammonia is determined from the equilibrium of ammonia with the basic salt and proton concentration.

$$K_{ammonia} = \frac{(NH3 - \Delta NH4) \cdot H}{\Delta NH4} \qquad (3.8.5-4)$$

$$\Delta H_{ammonia} = -\Delta NH4 = -\frac{H}{K_{ammonia} + H} \cdot NH3 \qquad (3.8.5-5)$$

where the equilibrium constant for ammonia is as follows [16].

$$K_{ammonia} := 10^{-9.255} \qquad (3.8.5-6)$$

Another significant metabolite is alanine. Alanine concentration may be in proportion to the lactate concentration. Alanine buffering system has two dissociations for alanine. There are two equilibrium equations for the α-carboxylic acid and α-amino ion.

$$K_{1Ala} = \frac{\left(Ala\_1_i + \Delta Ala\_1\right) \cdot H}{\left(Ala_i + \Delta Ala\right)} \qquad (3.8.5-7)$$

$$K_{2Ala} = \frac{\left(Ala\_2_i + \Delta Ala\_2\right) \cdot H}{\left(Ala\_1_i + \Delta Ala\_1\right)} \qquad (3.8.5-8)$$

From the charge balance, the change in the hydrogen protons can be expressed by the change in the alanine ions.

$$-2 \cdot \Delta Ala\_2 - \Delta Ala\_1 + \Delta H_{Ala} = 0 \qquad (3.8.5-9)$$

There is also an overall balance of the forms of alanine.

$$\Delta Ala + \Delta Ala\_1 + \Delta Ala\_2 = 0 \qquad (3.8.5-10)$$

With equations 3.8.5-7, 3.8.5-8, 3.8.5-9, and 3.8.5-10, there are four unique equations and four unknowns. Therefore, we should be able to solve for the change in the hydrogen protons due to the initial concentrations of the alanine. The change in the alanine dibasic ions is represented by the change in the other alanines by rearranging equation 3.8.5-10.

$$\Delta Ala\_2 = -\Delta Ala\_1 - \Delta Ala \qquad (3.8.5-11)$$

The change in the alanine dibasic ions concentration (equation 3.8.5-11) is substituted into the second equilibrium expression for alanine (equation 3.8.5-8). With some rearranging, the change in the alanine basic concentration is solved in terms of the change in the alanine basic ion concentrations, the initial alanine basic ion, and the proton concentration.

$$\Delta Ala\_1 = \frac{H}{K_{2Ala} + H} \cdot \left(Ala\_2_i - \Delta Ala\right) - \frac{K_{2Ala}}{K_{2Ala} + H} \cdot Ala\_1_i \qquad (3.8.5-12)$$

The change in the alanine ion concentration (equation 3.8.5-12) is substituted into the first equilibrium expression (equation 3.8.5-7). With some rearranging, the change in the alanine concentration is solved in terms of the initial alanine concentration, initial alanine basic, initial alanine dibasic and proton concentrations.

$$\Delta Ala = \frac{H^2}{\left(K_{1Ala} \cdot K_{2Ala} + K_{1Ala} \cdot H + H^2\right)} \cdot \left(Ala\_1_i + Ala\_2_i\right) - \frac{\left(K_{1Ala} \cdot K_{2Ala} + K_{1Ala} \cdot H\right)}{\left(K_{1Ala} \cdot K_{2Ala} + K_{1Ala} \cdot H + H^2\right)} \cdot Ala_i \qquad (3.8.5-13)$$

Then by combining the expressions from equations 3.8.5-9, 3.8.5-10, 3.8.5-11, 3.8.5-12, and 3.8.5-13, it is possible to solve for the change in the hydrogen proton concentration as a function of the initial concentrations of the phosphates.

$$\begin{aligned}\Delta H_{Ala} = &\left(\frac{2 \cdot K_{1Ala} \cdot K_{2Ala} + K_{1Ala} \cdot H}{K_{1Ala} \cdot K_{2Ala} + K_{1Ala} \cdot H + H^2}\right) \cdot Ala_i \ldots \\ &+ \left(\frac{K_{1Ala} \cdot K_{2Ala} - H^2}{K_{1Ala} \cdot K_{2Ala} + K_{1Ala} \cdot H + H^2}\right) \cdot Ala\_1_i \ldots \\ &+ \left(\frac{K_{1Ala} \cdot H + 2 \cdot H^2}{K_{1Ala} \cdot K_{2Ala} + K_{1Ala} \cdot H + H^2}\right) \cdot Ala\_2_i\end{aligned} \qquad (3.8.5-14)$$

where the equilibrium constants for the phosphate system are the following at 25 degrees Celsius [20].

$$K_{1Ala} := 10^{-2.34} \qquad (3.8.5-15)$$

$$K_{2Ala} := 10^{-9.69} \qquad (3.8.5-16)$$

The change in the proton concentration due to accumulation of dextrose acid is determined from the equilibrium of glucose with the basic salt and proton concentration.

$$K_{Glucose} = \frac{\Delta Glucose_{salt} \cdot H}{Glcuose - \Delta Glucose_{salt}} \qquad (3.8.5-17)$$

$$\Delta H_{Glucose} = \Delta Glucose_{salt} = \frac{K_{Glucose}}{K_{Glucose} + H} \cdot Glucose \qquad (3.8.5-18)$$

where the equilibrium constant for lactate is as follows [16].

$$K_{Glucose} := 10^{-12.28} \qquad (3.8.5-19)$$

3.8.6 Carbon Dioxide Chemistry Equilibrium Equation

Then by combining all of these terms (equations 3.8.1-13, 3.8.1-15, 3.8.2-3, 3.8.3-10, 3.8.4-2, 3.8.5-2, 3.8.5-5, and 3.8.5-14) we get an expression for the carbon dioxide concentration that is in equilibrium with the pH for a given amount of bicarbonate, carbonate, sodium hydroxide, phosphates, HEPES acid, lactic acid, alanine, and ammonia added to the culture.

$$CO2 = \frac{10^{(-2 \cdot pH)}}{\left[\begin{array}{l} 2 \times 10^{K_1+K_2} \cdots \\ + 2 \times 10^{(K_1+K_2+K_h)} \cdots \\ + 10^{(K_1-pH)} \cdots \\ + 10^{(K_1+K_h-pH)} \end{array}\right]} \cdot \left[\begin{array}{l} HCO3_i + 2CO3_i \cdots \\ + \dfrac{\left(\dfrac{\Delta H\_capacity_{carbonate}}{\Delta H\_capacity\_basis_{carbonate}}\right)}{\left(\dfrac{\begin{array}{l}\Delta H\_capacity_{carbonate} \cdots \\ + \Delta H\_capacity_{PO4} \cdots \\ + \Delta H\_capacity_{HEPES} \cdots \\ + \Delta H\_capacity_{SaltsA} \cdots \\ + \Delta H\_capacity_{SaltsB}\end{array}}{\begin{array}{l}\Delta H\_capacity\_basis_{carbonate} \cdots \\ + \Delta H\_capacity\_basis_{PO4} \cdots \\ + \Delta H\_capacity\_basis_{HEPES} \cdots \\ + \Delta H\_capacity\_basis_{SaltsA} \cdots \\ + \Delta H\_capacity\_basis_{SaltsB}\end{array}}\right)} \cdot \left(\begin{array}{l} 10^{-pH} \cdot \dfrac{mole}{L} + OH_i \cdots \\ + -1 \cdot \dfrac{10^{-14}}{10^{-pH}} \cdot \dfrac{mole}{L} \cdots \\ + -\Delta H_{PO4} - \Delta H_{HEPES} \cdots \\ + -\Delta H_{Lactate} + \Delta H_{ammonia} \cdots \\ + -\Delta H_{Glucose} \cdots \\ + \Delta H_{ala} \end{array}\right) \end{array}\right] \qquad (3.8.6-1)$$

There are other organic acids and bases that are produced or consumed as well as the lactate and ammonia. These organic acids and bases may not be monitored during a run, but we will assume that they are produced or consumed in proportion to the lactate and ammonia. As an approximation, we will assume that a multiplier for the measured lactate and ammonia would provide a sufficient approximation. These multipliers would need to be empirically determined for process conditions run. For alanine, we will use the first principle structure for the equilibrium with the different ionic forms of alanine. Since we can not readily measure alanine with standard laboratory instruments, we will assume that it is in proportion to the amount of lactate. In addition, the medium is more complex due to the amino acids, salts, and vitamins. Additional empirical terms for the acidic contributions that these components in the basal medium and in the feeds have on the medium buffering is added to the expression. The model training (section 4) describes a method for determining an appropriate value for these lump sum terms.

$$CO2 = \frac{10^{(-2\cdot pH)}}{\left[\begin{array}{l} 2\times 10^{K_1+K_2} \cdots \\ +2\times 10^{\left(K_1+K_2+K_h\right)} \cdots \\ +10^{\left(K_1-pH\right)} \cdots \\ +10^{\left(K_1+K_h-pH\right)} \end{array}\right]}\cdot\left[\begin{array}{l} HCO3_i + 2CO3_i \cdots \\ + \dfrac{\left(\dfrac{\Delta H\_capacity_{carbonate}}{\Delta H\_capacity\_basis_{carbonate}}\right)}{\left(\dfrac{\begin{array}{l}\Delta H\_capacity_{carbonate} \cdots \\ +\Delta H\_capacity_{PO4} \cdots \\ +\Delta H\_capacity_{HEPES} \cdots \\ +\Delta H\_capacity_{SaltsA} \cdots \\ +\Delta H\_capacity_{SaltsB}\end{array}}{\begin{array}{l}\Delta H\_capacity\_basis_{carbonate} \cdots \\ +\Delta H\_capacity\_basis_{PO4} \cdots \\ +\Delta H\_capacity\_basis_{HEPES} \cdots \\ +\Delta H\_capacity\_basis_{SaltsA} \cdots \\ +\Delta H\_capacity\_basis_{SaltsB}\end{array}}\right)}\cdot\left(\begin{array}{l} 10^{-pH}\cdot\dfrac{mole}{L} + OH_i \cdots \\ + -1\cdot\dfrac{10^{-14}}{10^{-pH}}\cdot\dfrac{mole}{L} \cdots \\ + -\Delta H_{PO4} - \Delta H_{HEPES} \cdots \\ + -B_{lactate}\cdot\Delta H_{Lactate} \cdots \\ + B_{ammonia}\cdot\Delta H_{ammonia} \cdots \\ + B_{ala}\cdot\Delta H_{ala} - \Delta H_{Glucose} \cdots \\ + -1\cdot\dfrac{10^{K_a}}{10^{K_a}+10^{-pH}}\cdot Salts_A\cdot\dfrac{mol}{L} \cdots \\ + \dfrac{10^{-pH}}{10^{K_b}+10^{-pH}}\cdot Salts_B\cdot\dfrac{mol}{L} \cdots \\ + -A_{eq}\cdot\dfrac{mol}{L} \end{array}\right) \end{array}\right] \quad (3.8.6-2)$$

3.8.7 Temperature Impact on the Chemistry Equilibrium

The vessel temperature affects the gas solubility and the pH reading. Because the dissolved gasses are measured by partial pressure (equation 3.4.2-8), changes in the liquid temperature impacts the measurement. This is because for a given concentration of the dissolved gas, the partial pressure that is in equilibrium with that concentration is temperature dependent (equations 3.1.1-1, 3.1.1-3, and 3.2.1-1). Some blood gas analyzers have built in temperature compensation calculations. The Nova BioProfile 400 is a unit that will heat the sample to 37 degrees Celsius with an internal heating block, measure the dissolved gasses partial pressure and liquid pH, and report the values with the option for temperature compensation. The equation for the temperature compensation for oxygen (equation 3.8.7-1), carbon dioxide (equation 3.8.7-2) and pH (equation 3.8.7-3) for the Nova BioProfile 400 are provided below [17].

$$pO2_{Temp} = pO2_{37C}\cdot 10^{\left[\frac{5.49\cdot 10^{-11}\cdot e^{3.88\cdot\ln\left(pO2_{37C}\right)}+0.071}{9.72\cdot 10^{-9}\cdot e^{3.88\cdot\ln\left(pO2_{37C}\right)}+2.30}\cdot(Temp-37)\right]} \quad (3.8.7-1)$$

$$pCO2_{Temp} = pCO2_{37C}\cdot e^{0.0437\cdot(Temp-37)} \quad (3.8.7-2)$$

$$pH_{Temp} = pH_{37C} + \left[-0.0147 + 0.0065\cdot\left(7.40 - pH_{37C}\right)\right]\cdot(Temp - 37) \quad (3.8.7-3)$$

where Temp is the liquid temperature in degrees Celsius.

4 Model Training

The equations captured in section 3 are presented as first principle equations, but in practice they are hybrid models because they contain empirical terms. The bioreactor mass balance model requires an understanding of how the operating conditions impact the gas transfer coefficients. At the time, there are no universal equations for determining the gas transfer coefficients to a sufficient resolution without using empirical evaluations. Model training is performed to obtain the vessel gas transfer coefficients. The medium chemistry models contain lump sum terms to capture the buffering effect of the complex medium components (organic acids and bases) excluded from the mechanistic model. These terms are not mechanistically described because there are either too many or may not be measured (i.e. metabolites). The lump sum expressions simplify the modeling, but requires empirical information that is dependent on medium formulation or process metabolism.

The model training explores the options available for the bioreactor gas models (section 4.1) and the medium chemistry (section 4.2). Both models are under specified. There are fewer equations than parameters. In order to solve these models, there needs to be at least as many equations as unknown parameters. The following sections walk the reader through how to use the empirical data to produce a correctly specified model.

4.1 Bioreactor Gas Models

For the bioreactor gas balance model, there are five general terms: 1) the vessel gas transfer coefficients as a function of operating condition, 2) the cell culture oxygen uptake rate, 3) the necessary gas flow rates, 4) the dissolved oxygen concentration, and 5) the dissolved carbon dioxide concentration. The equations provided above allow for one to solve for two unknowns provided the other three terms are given. This results in ten different combinations for given parameters and solvable parameters (Table 4.1-1). This section provides the necessary programs for all ten different model training combinations.

Model	kla	OUR	gas flows	DO	CO2
1	-	+	+	+	-
2	-	+	+	-	+
3	-	-	+	+	+
4	-	+	-	+	+
5	+	-	-	+	+
6	+	-	+	-	+
7	+	+	-	-	+
8	+	-	+	+	-
9	+	+	-	+	-
10	+	+	+	-	-

Table 4.1-1: Matrix of Correctly Specified Models. The positive signs indicate that the parameter(s) are known. The negative signs indicate that the parameter(s) are not known.

There are seven general dynamic equations for the mass balance of oxygen and carbon dioxide in the bioreactor. Not only do they include the mass balance around the liquid and head space as described in section 3, but also the dynamics around the measuring devices. The equations are listed here for clarity.

4.1.1 Oxygen

Mass Balance of Oxygen in the Head Space (equation 3.1-16)

$$\begin{pmatrix} \frac{V_H \cdot p_H}{R_{gas} \cdot T_H} \cdot \frac{d}{dt} x_{O2\_hs} \cdots \\ + \frac{x_{O2\_hs} \cdot p_H}{R_{gas} \cdot T_H} \cdot \frac{d}{dt} V_H \cdots \\ + \frac{x_{O2\_hs} \cdot V_H}{R_{gas} \cdot T_H} \cdot \frac{d}{dt} p_H \cdots \\ + -\frac{x_{O2\_hs} \cdot p_H \cdot V_H}{R_{gas} \cdot T_H^2} \cdot \frac{d}{dt} T_H \end{pmatrix} = \begin{bmatrix} x_{O2\_ovly} \cdot \frac{q_{ovly} \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \cdots \\ + \sum_{i=1}^{n} \left(x_{O2\_spg\_Out_i} \cdot \frac{q_{spg_i} \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \right) \cdots \\ + -x_{O2\_hs} \cdot \frac{\left[q_{ovly} + \sum_{i=1}^{n} \left(q_{spg_i} \right) \right] \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \cdots \\ + -kla_{surf\_O2} \cdot \left(x_{O2\_hs} \cdot p_H \cdot H_{O2\_cp} - O2 \right) \cdot V_L \end{bmatrix}$$

Mass Balance of Oxygen in the Liquid (equation 3.1-15)

$$\begin{pmatrix} \frac{d}{dt} O2 \cdots \\ + \frac{O2}{V_L} \cdot \frac{d}{dt} V_L \end{pmatrix} = \begin{bmatrix} \sum_{i=1}^{n} \frac{p_{stp} \cdot q_{spg_i} \cdot \left[x_{O2\_spg\_In_i} \cdot \left(H_{O2\_cp} \cdot p_L \right) - O2 \right]}{H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla_{spg\_O2_i} \cdot H_{O2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg_i}} \right) \right] \right] \cdots \\ + kla_{surf\_O2} \cdot \left(x_{O2\_hs} \cdot p_H \cdot H_{O2\_cp} - O2 \right) - OUR + q_{feed} \cdot O2_{feed} - q_{hvst} \cdot O2 \end{bmatrix}$$

4.1.2 Carbon Dioxide

Mass Balance of Carbon Dioxide in the Head Space (equation 4.1.2-1)

$$\begin{pmatrix} \frac{V_H \cdot p_H}{R_{gas} \cdot T_H} \cdot \frac{d}{dt} x_{CO2\_hs} \cdots \\ + \frac{x_{CO2\_hs} \cdot p_H}{R_{gas} \cdot T_H} \cdot \frac{d}{dt} V_H \cdots \\ + \frac{x_{CO2\_hs} \cdot V_H}{R_{gas} \cdot T_H} \cdot \frac{d}{dt} p_H \cdots \\ + -\frac{x_{CO2\_hs} \cdot p_H \cdot V_H}{R_{gas} \cdot T_H^2} \cdot \frac{d}{dt} T_H \end{pmatrix} = \begin{bmatrix} x_{CO2\_ovly} \cdot \frac{q_{ovly} \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \cdots \\ + \sum_{i=1}^{n} \left(x_{CO2\_spg\_Out_i} \cdot \frac{q_{spg_i} \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \right) \cdots \\ + -x_{CO2\_hs} \cdot \frac{\left[q_{ovly} + \sum_{i=1}^{n} \left(q_{spg_i} \right) \right] \cdot p_{stp}}{R_{gas} \cdot T_{q\_ref}} \cdots \\ + -kla_{surf\_CO2} \cdot \left(x_{CO2\_hs} \cdot p_H \cdot H_{CO2\_cp} - CO2 \right) \cdot V_L \end{bmatrix} \qquad (4.1.2-1)$$

Mass Balance of Carbon Dioxide in the Liquid (equation 4.1.2-2)

$$\begin{pmatrix} \frac{d}{dt} CO2 \cdots \\ + \frac{CO2}{V_L} \cdot \frac{d}{dt} V_L \end{pmatrix} = \begin{bmatrix} \sum_{i=1}^{n} \frac{p_{stp} \cdot q_{spg_i} \cdot \begin{bmatrix} x_{CO2\_spg\_In_i} \cdot \left(H_{CO2\_cp} \cdot p_L \right) \cdots \\ + -CO2 \end{bmatrix}}{H_{CO2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla_{spg\_CO2_i} \cdot H_{CO2\_cp} \cdot p_L \cdot V_L \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg_i}} \right) \right] \right] \cdots \\ + kla_{surf\_CO2} \cdot \left(x_{CO2\_hs} \cdot p_H \cdot H_{CO2\_cp} - CO2 \right) - CER + q_{feed} \cdot CO2_{feed} - q_{hvst} \cdot CO2 \end{bmatrix} \qquad (4.1.2-2)$$

4.1.3 Measuring Device

The measuring device could be an offline sample (equation 4.1.3-1), an online probe in the liquid (equation 4.1.3-2), or an at-line measurement of the head space (equation 4.1.3-3).

$$\frac{d}{dt}C_{sample} = kla_{sample\_C} \cdot \left(C_{ambient} - C_{sample}\right) + respiration \qquad (4.1.3-1)$$

$$\frac{d}{dt}C_e = \frac{1}{\tau_{e\_C}} \cdot \left(C_L - C_e\right) \qquad (4.1.3-2)$$

$$x_{C\_offgas}(t) = x_{C\_hs}(t - \tau) \qquad (4.1.3-3)$$

where C is the concentration of oxygen or carbon dioxide.

4.2 Gas Models Considerations

The models presented in Table 4.1-1 differ in only what are the known and unknown parameters. In all cases, the solution is determined by simultaneously solving the seven dynamic equations. The situations that produce the different model scenarios and special considerations when solving the models are presented below.

4.2.1 Gas Model 1: Unknown klas and Dissolved Carbon Dioxide

This model is applied to determine the gas transfer coefficients of the bioreactor when the oxygen uptake rate, the gas flows, and the dissolved oxygen are monitored and known. The oxygen uptake rate is either known from understanding the process load or this evaluation is performed in a cell-free system in which case the oxygen uptake rate is zero. The gas flow rates are monitored with mass flow controllers. The dissolved oxygen is monitored with a dissolved oxygen probe.

In this model there needs to be at least one unique trial per source of gas transfer within an experimental run. For instance, if there are two spargers used in a small vessel, then there are potentially three sources of gas transfer: sparge 1, sparge 2, and surface aeration. An experimental run is defined by the working volume, the agitation frequency, and the total gas flow through each of the spargers. A trial within an experimental run is defined by the gas composition of the spargers or the gas composition of the overlay. The purpose of the trials is to allow for a unique solution for the gas transfer coefficients. The gas transfer coefficients are dependent on the operating conditions that affect the power per unit volume, the superficial gas velocity, or the medium chemistry. Changes in the gas concentration gradient would not impact the gas transfer coefficient, but would result in different dissolved oxygen (or carbon dioxide) temporal profiles. If there are at least as many different dissolved oxygen (or carbon dioxide) profiles that are used in the curve fitting, then a nonlinear optimizer could converge on a unique solution. This is done by finding the kla values that minimize the mean sum of square error of all the trials within an experimental run.

4.2.1.1 Surface Aeration

The surface aeration is determined from each experimental run (with multiple trials that differ by gas composition) to identify the surface mass transfer coefficient for each operating condition. The polynomial equation (equation 3.3.2-1) for the surface aeration as a function of operating conditions is fitted to the data.

4.2.1.2 Sparge Aeration

The sparger(s) gas transfer coefficient is determined from the nonlinear optimization that minimizes the mean sum of square error for the trials within an experimental run. An experimental run is multiple trials with similar temperature, chemistry, working volume, sparge flow rates and agitation rate but with different gas compositions or overlay flow rates. A nonlinear parameter fitting that minimizes the mean square error of the Van't Riet equation (equation 3.3.1-1) for the kla values is determined from all the experimental runs. If necessary, the interaction of the spargers is determined from equation 3.3.1-2.

4.2.2 Gas Model 2: Unknown klas and Dissolved Oxygen

This model is applied to determine the gas transfer coefficients of the bioreactor when the oxygen uptake rate, the gas flows, and the dissolved carbon dioxide are monitored and known. The oxygen uptake rate is either known from understanding the process load or this evaluation is performed in a cell-free system in which case the oxygen uptake rate is zero. The gas flow rates are monitored with mass flow controllers. The dissolved oxygen is monitored with a dissolved carbon dioxide probe or instrumentation to monitor the carbon dioxide composition of the exhaust stream.

This model is very similar to model 1 expect the carbon dioxide is being monitored. In this model there needs to be at least one unique trial per source of gas transfer within an experimental run. For instance, if there are two spargers used in a small vessel, then there are potentially three sources of gas transfer: sparge 1, sparge 2, and surface aeration. An experimental run is defined by the working volume, the agitation frequency, and the total gas flow through each of the spargers. A trial within an experimental run is defined by the gas composition of the spargers or the gas composition of the overlay. The purpose of the trials is to allow for a unique solution for the gas transfer coefficients. The gas transfer coefficients are dependent on the operating conditions that affect the power per unit volume, the superficial gas velocity, or the medium chemistry. Changes in the gas concentration gradient would not impact the gas transfer coefficient, but would result in different dissolved oxygen (or carbon dioxide) temporal profiles. If there are at least as many different dissolved oxygen (or carbon dioxide) profiles that are used in the curve fitting, then a nonlinear optimizer could converge on a unique solution. This is done by finding the kla values that minimize the mean sum of square error of all the trials within an experimental run.

4.2.3 Gas Model 3: Unknown kla and Oxygen Uptake Rate

This model is applied when the kla and the cellular respiration is not known. The dissolved oxygen may be monitored with an online probe. The dissolved carbon dioxide may be monitored with an off-line sample, an on-line dissolved carbon dioxide probe, or an at-line gas composition instrument.

This model is very similar to model 1 and 2 in that for each source of gas transfer there would need to be a unique trial within each experimental run. For instance, if there are two spargers used in a small vessel, then there are potentially three sources of gas transfer: sparge 1, sparge 2, and surface aeration. An experimental run is defined by the working volume, the agitation frequency, and the total gas flow through each of the spargers. A trial within an experimental run is defined by the gas composition of the spargers or the gas composition of the overlay. The purpose of the trials is to allow for a unique solution for the gas transfer coefficients. The gas transfer coefficients are dependent on the operating conditions that affect the power per unit volume, the superficial gas velocity, or the medium chemistry. Changes in the gas concentration gradient would not impact the gas transfer coefficient, but would result in different dissolved oxygen and carbon dioxide temporal profiles. If there are at least as many different dissolved oxygen and carbon dioxide profiles that are used in the curve fitting, then a nonlinear optimizer could converge on a unique solution. This is done by finding the kla values that minimize the mean sum of square error of all the trials within an experimental run.

4.2.4 Gas Model 4: Unknown kla and Gas Flow Rates

This model is applied when the kla and the gas flow rates are not known. The gas flow rates may not be known because the system is using rotometers with poor resolution, or the system has a very active controller, and it is difficult to know the delivered gas flow rates. This model would be rarely applied because one would need to have a trial for each source of gas transfer plus one for each gas stream. For the model to be solvable, the composition of the gas streams would need to be known and the dissolved oxygen and dissolved carbon dioxide would need to be monitored.

4.2.5 Gas Model 5: Unknown Oxygen Uptake Rate and Gas Flow Rates

This model is applied when the vessel kla has been well characterized (by previously applying models 1, 2, 3, or 4), there is data available for the dissolved gases, but the respiration and gas flows may not be known. This can arise when a process is run in characterized equipment that is using rotometers and the cellular respiration is not understood. The model is capable of finding a unique solution from one experimental trial. There is no need to have multiple trials to decouple parameters.

4.2.6 Gas Model 6: Unknown Oxygen Uptake Rate and Dissolved Oxygen

This model is applied when the vessel kla have been well characterized (by previously applying models 1, 2, 3, or 4), there is data available for the gas flow rates and dissolved carbon dioxide, but there is no information on the cellular respiration or dissolved oxygen. This can arise when using disposable reactors that are not fitted with a dissolved oxygen probe. There is limited confidence in the off-line dissolved oxygen reading. Due to the higher solubility of carbon dioxide, there is confidence in the off-line dissolved carbon dioxide reading. The model is capable of finding a unique solution from one experimental trial. There is no need to have multiple trials to decouple parameters.

4.2.7 Gas Model 7: Unknown Gas Flow Rates and Dissolved Oxygen

This model is applied when the vessel kla have been well characterized (by previously applying models 1, 2, 3, or 4), there is data available for the cellular respiration, but there is no information on the gas flow rates or dissolved oxygen. This can arise when using disposable reactors that are not fitted with a dissolved oxygen probe and use rotometers to deliver the gases. There is limited confidence in the off-line dissolved oxygen reading. Due to the higher solubility of carbon dioxide, there is confidence in the off-line dissolved carbon dioxide reading. The model is capable of finding a unique solution from one experimental trial. There is no need to have multiple trials to decouple parameters.

4.2.8 Gas Model 8: Unknown Oxygen Uptake Rate and Dissolved Carbon Dioxide

This model is the one applied to most process monitoring situations. The vessel has been well characterized to provide the gas transfer coefficients (by previously applying models 1, 2, 3, or 4). The dissolved oxygen is monitored with an on-line probe. The vessel operating conditions are monitored and available with on-line values. There may be a predicted cellular respiration but due to inherent process variability it may differ from run to run. Likewise, the dissolved carbon dioxide may fall within a predicted value, but the inherent process variableness will also cause this to differ between runs. This model is capable of finding a unique solution from one experimental trial. There is no need to have multiple trials to decouple parameters.

4.2.9 Gas Model 9: Unknown Gas Flow Rates and Dissolved Carbon Dioxide

This model is predominantly used for predicting process performance. If the cellular respiration is known and the dissolved oxygen is controlled to a set point value, then one could calculate the necessary gas flow rate and the resulting dissolved carbon dioxide for a specific well characterized vessel or vessel configuration (by previously applying models 1, 2, 3, or 4). This model is capable of finding a unique solution from one experimental trial. There is no need to have multiple trials to decouple parameters.

4.2.10 Gas Model 10: Unknown Dissolved Oxygen and Dissolved Carbon Dioxide

This model evaluates the resulting dissolved oxygen and dissolved carbon dioxide for a given cellular respiration and operating condition in a well characterized vessel (by previously applying models 1, 2, 3, or 4). This model is applied to understanding what sort of nonconformance could be expected in the dissolved oxygen and dissolved carbon dioxide if a flow meter were to either have an excursion, have zero flow, or becomes max out. This model could be used to predict the impact of limits in equipment capability or to model the impact that a nonconformance had on the dissolved gasses for a process. This model is capable of finding a unique solution from one experimental trial. There is no need to have multiple trials to decouple parameters.

4.3 Medium Chemistry Models

The medium chemistry models are fitted with empirical data to determine the value for lumped term coefficients. The lumped term coefficients are required because not all of the sources of weak acids and weak bases are detailed in the mechanistic model. The amino acids in the basal and feed medium are not individually detailed. Not all of the metabolic intermediates are monitored either. These are assumed to be produced in proportion to the lactate or ammonia. The empirical data allows for these terms to be fitted so a hybrid (first principle plus empirical) model could be used to predict the impact that the medium chemistry has on the carbon dioxide and pH equilibrium. Alanine is produced in larger amounts than the other amino acids and will therefore be treated separate from the other amino acids. The first principle equilibrium equations will be applied to express the proportions of the different ionic alanine forms. In the absence of direct measurements, the alanine will be assumed to be produced in proportion to lactate. There are three sets of experiments performed in the listed order that are required to determine the empirical terms: 1) basal medium pH, pCO_2 equilibrium curve, 2) basal medium plus feed and base pH, pCO_2 equilibrium curve, and 3) process lactate and ammonia data mining parameter fitting. The details of these experiments are described below.

4.3.1 Basal Medium pH, pCO_2 Equilibrium Curve

The purpose of this empirical study to produce a curve of the carbon dioxide and pH that would allow for parameter fitting to describe the weak acids and bases present in the medium. This is done by taking a sample of basal medium and gassing it with carbon dioxide. Over a period of time, the carbon dioxide will off-gas from the sample. During this period of time, the dissolved carbon dioxide and pH are monitored to produce a curve that demonstrates the relationship of pH and carbon dioxide over the pH or pCO_2 range that would be observed by the process. A nonlinear optimization is performed to minimize the mean square error between the calculated value for pCO_2 (equation 3.8.6-2) with the measured value for pCO_2 over the different pH values by varying the values for K_a, K_b, $Salts_A$, and $Salts_B$. Because the system is only basal medium, the values for $\Delta H_{Lactate}$, $\Delta H_{ammonia}$, $\Delta H_{Alanine}$, and A_{eq} are set to zero. The terms for HCO_{3i}, CO_{3i}, ΔH_{PO4}, ΔH_{HEPES} are calculated from the medium formulation using equations 3.8.3-10 and 3.8.4-2.

In practice, this study can be done with a small volume (~ 60mL) of medium. The sample can be bubbled with carbon dioxide. This can be an open operation because the test would be on the order of 45 minutes and the medium would be discarded after the test is complete. The gassed medium can then be transferred to a syringe to allow loading on a blood gas analyzer. Between instrument loadings, the sample can be mixed to promote carbon dioxide off-gassing. To ensure a good fit for the four parameters, one would want to collect at least 8 readings within the pCO_2 range expected for the process. The pCO_2 range may be 20 - 160 mm Hg.

4.3.2 Basal Medium pH, pCO_2 plus Feed and Base Equilibrium Curve

After there is an understanding for the basal medium empirical coefficients, it is possible to characterize the impact that the feed and base additions have on the equilibrium curves. These terms are expressed as acid or base equivalents. The study described in section 4.3.1 is repeated with fixed volumes of feed or fixed volumes of base added to the sample. The pH, pCO_2 curve is generated similar to section 4.3.1. The values for varying the values for K_a, K_b, $Salts_A$, and $Salts_B$ are applied as determined in section 4.3.1. The nonlinear optimization is performed to minimize the mean square error between the calculated value for pCO_2 (equation 3.8.6-2) with the measured value for pCO_2 over the different pH values by varying the values for A_{eq} and OH_i. The term A_{eq} is adjusted when the evaluation is looking for the feed effect. The term OH_i is adjusted when the evaluation is looking for the base addition effect. The values for $\Delta H_{Lactate}$, $\Delta H_{ammonia}$, and $\Delta H_{Alanine}$ are set to zero. The terms for $HCO3_i$, $CO3_i$, ΔH_{PO4}, and ΔH_{HEPES} are calculated from the medium formulation using equations 3.8.3-10 and 3.8.4-2.

4.3.3 Process Data Mining Parameter Fitting

After the values for K_a, K_b, $Salts_A$, and $Salts_B$ have been determined from section 4.3.1 and the values for A_{eq} and OH_i have been determined from section 4.3.2, it is possible to determine the values for $B_{lactate}$ and $B_{ammonia}$ from the process data. To mine the process data, the values for lactate concentration, ammonia concentration, volume of feed added, volume of base added, culture pH, dissolved carbon dioxide, and temperature need to be available from multiple runs. A nonlinear optimization is performed to minimize the mean square error between the calculated value for pCO_2 (equation 3.8.6-2) with the measured value for pCO_2 over the different pH values by varying the values for $B_{lactate}$, $B_{ammonia}$, and $B_{alanine}$. The values for $B_{lactate}$, $B_{ammonia}$, and $B_{alanine}$ reflect the proportion of other weak acid or base metabolites that produced in proportion to the production or consumption of lactate or ammonia. This model assumes a simple linear relationship that is dependent on the clone and process conditions. This simplification may not be the most representative of the process, but it may provide sufficient predictive capabilities for the pH and pCO_2 relationship. If the process data suggests a more sophisticated relationship is required, then the model should reflect the simplest expression that provides the needed predictive capabilities.

5 Model Implementations

There are four general model implementations that are captured in the appendix of this report. These implementations include predictive process modeling (section 5.1), dynamic systems modeling (section 5.2), defining the tuning loop parameters (section 5.3), and predicting the process control loop performance (section 5.4). The predictive process modeling is performed to determine how scale and operating conditions impact the dissolved oxygen accumulation and the pH profile. This model assumes that the process controls are sufficiently stable to allow pseudo steady-state approximations to be appropriate. The dynamic systems modeling is performed in instances where due to a loss of control the pseudo steady-state approximation is not correct. The dynamic model presents the resulting dissolved oxygen, dissolved carbon dioxide, and pH due to the loss of control. The third model is a direct synthesis method to obtaining the tuning parameters for the dissolved oxygen and pH control loops. The forth model is a simulation for how the process controller will behave to process disturbances. This report was written in Mathcad 15.0 to allow for the analysis to be performed by either populating the fields in the appendix or by linking this report to raw data files.

5.1 Predictive Process Modeling

The modeling is applied to predict the impact that operating conditions have on the accumulated carbon dioxide and pH profile. The model assumes that the biology is understood, the vessel and medium are well characterized, and system can be approximated as being in pseudo steady-state.. The biology is established from empirical runs with similar operating conditions. The empirical runs establish the oxygen uptake rate, the lactate temporal profile, the ammonia temporal profile, and the values for $B_{lactate}$, $B_{ammonia}$, and $B_{alanine}$ that reflect the proportion of other weak acid or base metabolites that produced in proportion to the production or consumption of lactate or ammonia.

The first step to the modeling is to characterize the gas transfer coefficients for the bioreactors that were used to generate the process data and the bioreactors will be used and require the evaluation to ensure similar performance of future runs (section 9.1). Models 1, 2, 3, or 4 from Section 4.2 could be applied to determine the gas transfer coefficients for the bioreactor. An experimental run is performed per operating condition per source for gas transfer. If there are two spargers and the surface aeration is significant, then for each agitation frequency, working volume, and sparge rate, there would be three experimental trials that use different compositions of oxygen (relative to each other) in the three gas streams (sparge 1, sparge 2, and head space). This would allow for the nonlinear optimization to identify a unique solution for the three gas transfer coefficients. This evaluation is performed with enough operating conditions to span the typical operating space and enable for equations 3.3.1-1 and 3.3.2-1 to be solved.

The second step to the modeling is to determine the medium chemistry properties (section 9.2). This is performed as described in section 4.3. The outcome of the medium characterization studies would be the coefficients to solve equation 3.8.6-2 as a function of the volume of base added to the culture, the volume of feed added to the culture, and the amount of lactate or ammonia accumulated in the culture.

The third step to the modeling is to identify the representative run(s) (section 9.3). This is the cell culture run that represents the target cell growth and performance. From this run, we obtain the time profile for the oxygen uptake rate, the lactate accumulation, and the ammonia accumulation. We assume that these three profiles are conserved (with some small inherent variability) between the vessel site, scale, or configurations. The oxygen uptake from the representative run(s) is determined by solving either model 3, 5, 6, or 8 using the available data from that run (typically model 8 is applied). Sixth order polynomials are then fitted to the oxygen uptake rate, the lactate accumulation, and ammonia accumulation rates with respect to time to provide continuous functions that can interpolate between the discrete time points that have available data. Time dependent functions are defined for the nutrient feed schedule.

The fourth step to the modeling is to select the vessel that will run the process (section 9.4). The model is populated with 1) the gas transfer coefficients for the bioreactor, 2) the medium equilibrium coefficients, 3) the time dependent profiles for oxygen uptake rate, lactate accumulation, and ammonia accumulation and 4) the nutrient feed schedule. The user then selects operating conditions and methods for controlling the dissolved oxygen to set point. The model will numerically solve for how the equipment will satisfy the dissolved oxygen set point and the culture pH by stepping through the information flow tree depicted in Figure 5-1. At each interval of time, the model calculates the gas flow required to satisfy the oxygen uptake rate. This is done iteratively because changes in the gas flow rates impacts the gas transfer coefficient. When the model converges on a solution for the gas flow necessary for dissolved oxygen control, the model calculates the resulting dissolved carbon dioxide. The dissolved carbon dioxide is then incorporated into the medium chemistry calculation to determine the culture pH. If the culture pH is above the control range, then then carbon dioxide is added to the gas flow. The amount of air or oxygen that is added for dissolved oxygen control is recalculated and this loop is repeated until the model converges on a value that will meet the dissolved oxygen and pH requirements. If the pH is below the control range, then the model calculates the amount of base that is required to bring the pH within range. The amount of base used through the calculation is totalized and incorporated into the medium chemistry equations for later time points. Once a solution has been determined, the model advances through the process to the next time point. Decisions for the gas flow schedule or carbon dioxide enrichment are based on the predicted process environment and the capability to implement the solution. Given the number of parameters that can be adjusted, it has been the experience that there are multiple viable solutions. The user is able to select the more appropriate solution based on ease of implementation, safety, equipment capability, as well as the comparison of carbon dioxide and pH profile between scales, sites, or vessels.

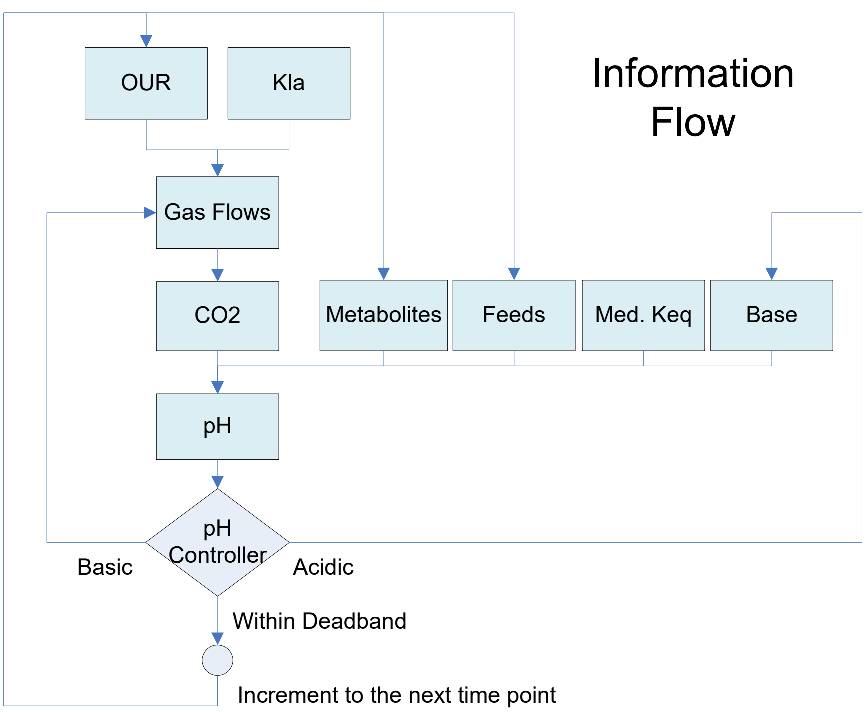

Figure 5-1: Information flow for the predictive process modeling.

5.2 Dynamic Model

The predictive process modeling assumes that the system is at pseudo steady state. The dynamic model calculates the impact that non stable systems would have on accumulation of the dissolved gases and pH. The dynamic model is solved by using differential equations for the gas balances in the liquid (3.1-15), the headspace (3.1-16), and the chemistry equilibrium equation (3.8.6-1) with discrete step changes in the inputs. At each discrete step change of any of the process inputs, the dynamic equations are solved with the initial conditions determined from the last solved dynamic equations and the inputs parameters are the new stepped values. The elapsed time is calculated from the time of the step change to the next step change. This is repeated for the duration of time that is being evaluated with the dynamic model.

5.3 Determining Process Tuning Parameters

The tuning parameters can be determined with the direct synthesis method. The direct syntheses method calculates how a step change in the controlling parameter impacts the steady-state value of the output parameter [19]. The ratio of the response to the output parameter to the change of the controlling parameter is the gain of the system. The response time of the system is the inverse of the gas transfer coefficient. The time lag of the system is due to response time of the valves and probe, and the length of piping between the mass flow controllers and the vessel.

For dissolved oxygen control, typically there is sufficient noise in the dissolved oxygen measurement, that a derivative term in the controller should not be used. An appropriate controller is a gain with integral. Therefore, using the first-order plus time-delay model we can solve for the controller gain (equation 5.3-1) and the controller integral (equation 5.3-2).

$$K_c = \frac{1}{K} \cdot \frac{\tau}{\theta + \tau_c} \qquad (5.3-1)$$

where K is the gain of the system (for example, the change in dissolved oxygen with respect to change in gas flow rate), τ is the inverse of the gas transfer coefficient, θ is the system time lag (probe response time), and τ_c is the closed loop controller time constant.

$$\tau_I = \tau = \frac{1}{kla} \qquad (5.3-2)$$

The tuning parameters obtained by the direct synthesis approach have been noted to be designed for set-point changes and are sluggish to correcting for disturbances. A modification to this approach is to presented in the literature based on disturbance rejection [21]. The disturbance rejection controller gain and controller integral are calculated in equation 5.3-3 and equation 5.3-4 respectively [21].

$$K_c = \frac{1}{K} \cdot \frac{\tau^2 + \theta \cdot \tau - (\tau_c - \tau)^2}{(\theta + \tau_c)^2} \qquad (5.3-4)$$

$$\tau i = \frac{\tau^2 + \theta \cdot \tau - (\tau_c - \tau)^2}{(\theta + \tau)} \qquad (5.3-5)$$

where K is the gain of the system (for example, the change in dissolved oxygen with respect to change in gas flow rate), τ is the inverse of the gas transfer coefficient, θ is the system time lag (probe response time), and τ_c is the closed loop controller time constant.

6 Conclusion

This report provides the first principle calculations for the oxygen and carbon dioxide mass balance in the cell culture bioreactor. Because the equations are not limited to a specific bioreactor configuration or method for dissolved oxygen control, these equations can be viewed as universally appropriate for any bioreactor scale, geometric configuration, or process control method. The carbon dioxide also contributes to the cell culture pH. This interaction is captured in first principle models with lump sum terms. By understanding how the carbon dioxide is involved in both the methods for controlling the dissolved oxygen and the methods for controlling the culture pH, it is possible to forecast the accumulation of the dissolved carbon dioxide based on operating conditions. It is through this predictive capability that we can select operating conditions and define the process controls that meet target dissolved carbon dioxide levels and pH profiles.

7 Acknowledgment

Thank you to Kumar Singarapu for performing the first principle modeling of alanine.

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9 Appendix

The appendix contains the working model for carbon dioxide accumulation and pH profile described in the report above. The implementation of the model is separated into six parts. The first part is used for vessel characterization to determine the gas transfer coefficients. The second part is used for medium characterization to determine the medium coefficients. The third part is used for establishing the representative run standard for oxygen uptake rate demand, lactate profile, and ammonia profile. The fourth part is used for predicting the impact that run decisions have on the accumulation of carbon dioxide, the dissolved oxygen profile, and the culture pH. The fifth part is used for dynamic modeling of operating disturbances impact on the dissolved oxygen, dissolved carbon dioxide, and pH. The sixth part is a direct synthesis approach to determining PID tuning parameters from the gas transport and medium equilibrium models. Each of these parts of the model can be selectively disable to speed the calculation time.

9.1 Part 1. Vessel Characterization

The vessel characterization is performed to determine the gas transfer coefficients for the surface aeration and multiple spargers as a function of operating conditions. If the box DO NOT Run Part 1 is selected, it will skip the calculations in Section 9.1 to conserve on *CPU* time.

Utilize Model Part 1	DO NOT Run Part 1 Run Part 1

Table 9.1-1: Decision to run 9.1 Part 1 for vessel characterization

BIOREACTOR PARAMETERS

Enter the vessel total volume and the vessel inner diameter.

Vessel Volume		L
Vessel Diameter		m

Table 9.1-2: Vessel volume and diameter

Enter the impeller characteristics of power number and diameter, and the working volume at which the impeller is submerged and is contributing to the mixing of the liquid in the tank.

Impeller Characteristics

"Power Number"	"Diameter"	"Volume"
0	0	0
0	0	0

Table 9.1-3: Impeller characteristics and volume when submerged

Select the number of spargers that are within the vessel (or are active in this evaluation). Enter an estimate for the gas transfer coefficients for the relative spargers (equation 3.3.1-1). The units for the superficial gas *velocity is m/s. Enter the estimated coefficients for the surface aeration gas transfer coefficient.*

Number of Spargers

Zero
One
Two

Spargers Gas Transfer Coefficients

"Coefficient"	"alpha"	"beta"
0	0	0
0	0	0

Surface Aeration Gas Transfer Coefficients

"Int"	1
"V (L)"	0
"N (1/s)"	0
"S (slpm)"	0
"VN (L/s)"	0
"VS (L*slpm)"	0
"NS (slpm/s)"	0
"V2 (L2)"	0
"N2 (1/s2)"	0
"S2 (slpm2)"	0

Table 9.1-4: Number of spargers, sparger gas transfer coefficients, and surface aeration gas transfer coefficients

Enter the estimated values for the delay time, the calibration adjustment, the surface kla, the sparge 1 kla, and the sparge 2 kla for each of the experimental studies.

Experimental Run Number

	0	1	2	3	4	
0	0	0	0	0	0	Surface Kla (1/hr)
1	0	0	0	0	0	Sparge1 Kla (1/hr)
2	0	0	0	0	0	Sparge2 Kla (1/hr)
3	0	0	0	0	0	Delay Time Trial 1 (hours)
4	0	0	0	0	0	Calibration Value Trial 1
5	0	0	0	0	0	Delay Time Trial 2 (hours)
6	0	0	0	0	...	Calibration Value Trial 2

Table 9.1-5: Estimated values for start time, probe calibration, and kla coefficients for the experimental runs.

Assume a respiration quotient (moles of carbon dioxide produced per mole of oxygen consumed) of 1.0. Assume a probe response time of 10 seconds. Enter the atmospheric pressure.

$RQ := 1$ (9.1 − 1)

$\tau(Nrpm) := 10s$ (9.1 − 2)

$p_{atm} := 12.2 \cdot psi$ (9.1 − 2.1)

Enter the saline equivalence of the medium (equation 3.1.1-4) to determine the solubility constants for the dissolve oxygen and dissolved carbon dioxide.

Saline	0.011

Table 9.1-6: Saline equivalent of medium

Enter the file name for the location of the raw data. That contains the information in the column order listed below. Each worksheet should be labeled "Exp x" where x is the experimental number ranging from 1 to n.

Experimental Data file := "C:\WorkingFolder\"

Number of Experiments	30

Column Order

"Elapsed Time"	"Volume"	"Temperature"	...

Table 9.1-7: The experimental data file location, number of experiments, and the proper column order for the excel data file from which the data is pulled.

Load Experimental Data Section

This program loads the experimental data into a matrix called Exp_Data. The program is only active if the user selects the option to activate section 9.1.

```
Pages := | for i ∈ 0.. str2num(Exp) − 1
         |    | num ← num2str(i + 1)
         |    | result_{i,0} ← concat("Exp " , num)
         | result
```
(9.1 − 3)

ReadExcel(file, Page) := [ ] (file Page) (9.1 − 4)

$fp(x, y) := 0$ (9.1 − 5)

```
Exp_Data := | for x ∈ 0.. str2num(Exp) − 1                                   if Part1 = 1
            |    | M_x ← ReadExcel(file, Pages_x)
            |    | M_x ← augment(M_x^T, 0·submatrix(M_x, 0, 0, 0, cols(M_x) − 1)^T)^T
            | M_0 ← matrix(2, cols(Columns), fp)   otherwise
            | M
```
(9.1 − 6)

Load Experimental Data Section

Experiment number(s) *(The values can be entered in this box similar to the print pages command by using a combination of commas for discrete values or dashes for continuous ranges.)*

Table 9.1-8: The experimental trials to use in the analysis.

Data Vectors and Mass Balance Differential Equations

This program builds a vector containing all the experimental numbers that are to be used for the evaluation.

(9.1 – 7)

```
eNum := | strCount ← 1
        | prevSymbol ← 0
        | arrayCount ← 0
        | prevCount ← 0
        | while strCount < strlen(ExpNum)
        |   | if substr(ExpNum, strCount, 1) = " " ∨ substr(ExpNum, strCount, 1) = "," ∨ substr(ExpNum, strCount, 1) = "-"
        |   |   | if strCount − prevCount > 0
        |   |   |   | arrayValue ← str2num(substr(ExpNum, prevCount, strCount − prevCount))
        |   |   |   | if prevSymbol = 1
        |   |   |   |   | for i ∈ prevValue + 1 .. arrayValue
        |   |   |   |   |   | result_arrayCount ← i
        |   |   |   |   |   | arrayCount ← arrayCount + 1
        |   |   |   |   | prevSymbol ← 0
        |   |   |   | otherwise
        |   |   |   |   | result_arrayCount ← arrayValue
        |   |   |   |   | arrayCount ← arrayCount + 1
        |   |   | prevCount ← strCount + 1
        |   |   | if substr(ExpNum, strCount, 1) = "-"
        |   |   |   | prevSymbol ← 1
        |   |   |   | prevValue ← arrayValue
        |   | strCount ← strCount + 1
        | if strCount − prevCount > 0 ∧ strlen(ExpNum) > 0
        |   | arrayValue ← str2num(substr(ExpNum, prevCount, strCount − prevCount))
        |   | if prevSymbol = 1
        |   |   | for i ∈ prevValue + 1 .. arrayValue
        |   |   |   | result_arrayCount ← i
        |   |   |   | arrayCount ← arrayCount + 1
        |   |   | prevSymbol ← 0
        |   | otherwise
        |   |   | result_arrayCount ← arrayValue
        |   |   | arrayCount ← arrayCount + 1
        | result
```

Prime the Input vector.

$$\text{Inputs} := \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \qquad (9.1-8)$$

These equations load the vessel diameter and total tank volume.

$$D := \text{str2num}(D)\cdot m \qquad (9.1-9)$$

$$V_T := \text{str2num}(V_T)\cdot L \qquad (9.1-10)$$

These equations load the vectors for the monitored parameters from each experimental number and trial within the experiments.

$$\text{ElapsedTime}(\text{ExpData}, \text{deltaT}, \text{Trial}) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (\text{ExpData})_{i,1+(\text{Trial}-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} \text{result}_i \leftarrow (\text{ExpData})_{i,0+(\text{Trial}-1)\cdot 20} - \text{deltaT} \\ \text{result}_i \leftarrow 0 \ \text{ if } \ \text{result}_i < 0 \\ i \leftarrow i+1 \end{array} \right. \\ \text{result} \end{array} \right. \qquad (9.1-11)$$

$$V_L(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 1+(Trial-1)\cdot 20} \cdot L \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \qquad (9.1-12)$$

$$T_L(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 2+(Trial-1)\cdot 20} \cdot Celsius + T_{0C} \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \qquad (9.1-13)$$

$$p_H(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 3+(Trial-1)\cdot 20} \cdot psig + p_{atm} \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \qquad (9.1-14)$$

$$Nrpm(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 4+(Trial-1)\cdot 20} \cdot \frac{1}{min} \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \qquad (9.1-15)$$

$$DO(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 5+(Trial-1)\cdot 20} \text{ if } (ExpData)_{i, 5+(Trial-1)\cdot 20} > 0 \\ result_i \leftarrow 0 \text{ otherwise} \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \qquad (9.1-16)$$

$$CO2(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 6+(Trial-1)\cdot 20} \cdot mmHg \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \qquad (9.1-17)$$

$$\text{qair\_ovly}(\text{ExpData},\text{Trial}) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (\text{ExpData})_{i,1+(\text{Trial}-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} \text{result}_i \leftarrow (\text{ExpData})_{i,7+(\text{Trial}-1)\cdot 20}\cdot \text{SLPM} \\ i \leftarrow i+1 \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.1 – 18}$$

$$\text{qO2\_ovly}(\text{ExpData},\text{Trial}) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (\text{ExpData})_{i,1+(\text{Trial}-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} \text{result}_i \leftarrow (\text{ExpData})_{i,8+(\text{Trial}-1)\cdot 20}\cdot \text{SLPM} \\ i \leftarrow i+1 \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.1 – 19}$$

$$\text{qCO2\_ovly}(\text{ExpData},\text{Trial}) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (\text{ExpData})_{i,1+(\text{Trial}-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} \text{result}_i \leftarrow (\text{ExpData})_{i,9+(\text{Trial}-1)\cdot 20}\cdot \text{SLPM} \\ i \leftarrow i+1 \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.1 – 20}$$

$$\text{qN2\_ovly}(\text{ExpData},\text{Trial}) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (\text{ExpData})_{i,1+(\text{Trial}-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} \text{result}_i \leftarrow (\text{ExpData})_{i,10+(\text{Trial}-1)\cdot 20}\cdot \text{SLPM} \\ i \leftarrow i+1 \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.1 – 21}$$

$$\text{qair\_1}(\text{ExpData},\text{Trial}) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (\text{ExpData})_{i,1+(\text{Trial}-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} \text{result}_i \leftarrow (\text{ExpData})_{i,11+(\text{Trial}-1)\cdot 20}\cdot \text{SLPM} \\ i \leftarrow i+1 \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.1 – 22}$$

$$\text{qO2\_1}(\text{ExpData},\text{Trial}) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (\text{ExpData})_{i,1+(\text{Trial}-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} \text{result}_i \leftarrow (\text{ExpData})_{i,12+(\text{Trial}-1)\cdot 20}\cdot \text{SLPM} \\ i \leftarrow i+1 \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.1 – 23}$$

$$qCO2\_1(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 13+(Trial-1)\cdot 20} \cdot SLPM \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \tag{9.1 – 24}$$

$$qN2\_1(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 14+(Trial-1)\cdot 20} \cdot SLPM \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \tag{9.1 – 25}$$

$$qair\_2(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 15+(Trial-1)\cdot 20} \cdot SLPM \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \tag{9.1 – 26}$$

$$qO2\_2(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 16+(Trial-1)\cdot 20} \cdot SLPM \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \tag{9.1 – 27}$$

$$qCO2\_2(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 17+(Trial-1)\cdot 20} \cdot SLPM \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \tag{9.1 – 28}$$

$$qN2\_2(ExpData, Trial) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (ExpData)_{i, 1+(Trial-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} result_i \leftarrow (ExpData)_{i, 18+(Trial-1)\cdot 20} \cdot SLPM \\ i \leftarrow i + 1 \end{array} \right. \\ result \end{array} \right. \tag{9.1 – 29}$$

$$\mathrm{OUR}(\mathrm{ExpData},\mathrm{Trial}) := \left| \begin{array}{l} i \leftarrow 0 \\ \text{while } (\mathrm{ExpData})_{i,1+(\mathrm{Trial}-1)\cdot 20} > 0 \\ \quad \left| \begin{array}{l} \mathrm{result}_i \leftarrow (\mathrm{ExpData})_{i,19+(\mathrm{Trial}-1)\cdot 20}\cdot \dfrac{\mathrm{mmol}}{\mathrm{L}\cdot\mathrm{hr}} \\ i \leftarrow i+1 \end{array} \right. \\ \mathrm{result} \end{array} \right. \tag{9.1 - 30}$$

This program calculates the average value for a vector.

$$\mathrm{Average}(\mathrm{vector}) := \left| \begin{array}{l} \mathrm{sum} \leftarrow 0 \\ \text{for } i \in 0..\mathrm{rows}(\mathrm{vector})-1 \\ \quad \mathrm{sum} \leftarrow \mathrm{sum} + \mathrm{vector}_i \\ \mathrm{result} \leftarrow \dfrac{\mathrm{sum}}{\mathrm{rows}(\mathrm{vector})} \end{array} \right. \tag{9.1 - 31}$$

This program converts discrete data values into a continuous function. The method is not interpolating based on slope, but rather, using a stair step approach. A value is held constant until a new value is recorded.

$$\mathrm{Value}(X,Y,t) := \left| \begin{array}{l} i \leftarrow 0 \\ \mathrm{result}(t) \leftarrow Y_{\mathrm{rows}(X)-1} \ \text{ if } t \ge X_{\mathrm{rows}(X)-1} \\ \text{otherwise} \\ \quad \left| \begin{array}{l} \text{while } X_i < t \\ \quad i \leftarrow i+1 \\ i \leftarrow i-1 \\ i \leftarrow 0 \ \text{ if } i < 0 \\ \mathrm{result}(t) \leftarrow Y_i \end{array} \right. \\ \mathrm{result}(t) \end{array} \right. \tag{9.1 - 32}$$

The average liquid pressure due to the headspace pressure and hydrodynamic pressure.

$$p_L(V_L, D, p_H) := \frac{1}{2}\cdot\frac{4\cdot V_L}{\pi\cdot(D)^2}\cdot 1\frac{\mathrm{gm}}{\mathrm{cm}^3}\cdot g + p_H \tag{9.1 - 33}$$

The volume of the headspace.

$$V_H(V_T, V_L) := V_T - V_L \tag{9.1 - 34}$$

The total gas flow rate.

$$q_{total}(\mathrm{qair}, \mathrm{qO2}, \mathrm{qCO2}, \mathrm{qN2}) := \mathrm{qair} + \mathrm{qO2} + \mathrm{qCO2} + \mathrm{qN2} \tag{9.1 - 35}$$

The percent oxygen within a gas stream.

$$x_{O2}(qair, qO2, qCO2, qN2) := \frac{0.21 \cdot qair + qO2}{q_{total}(qair, qO2, qCO2, qN2) + 0.000000000000001\ \mathrm{SIUnitsOf}(qair)} \qquad (9.1-36)$$

The percent carbon dioxide within a gas stream.

$$x_{CO2}(qair, qO2, qCO2, qN2) := \frac{qCO2}{q_{total}(qair, qO2, qCO2, qN2) + 0.000000000000001\ \mathrm{SIUnitsOf}(qair)} \qquad (9.1-37)$$

The headspace temperature estimated to be the mid value between the liquid and ambient temperatures.

$$T_H(T_{Liq}) := \frac{T_{Liq}}{2} + \frac{T_{0C} + 20\mathrm{Celsius}}{2} \qquad (9.1-38)$$

This program builds the matrix of average values used during the experiments.

$$\begin{aligned}
AvgMtx := \; & \text{for } ExpNum \in 0..\,rows(Exp_Data) - 1 \qquad \text{if } Part1 = 1 \qquad (9.1-39)\\
& \quad \text{for } Trial \in 1..\,\frac{cols(Exp_Data_{ExpNum})}{20}\\
& \qquad V_{Liq} \leftarrow \frac{Average(V_L(Exp_Data_{ExpNum,0}, Trial))}{L}\\
& \qquad T_{Liq} \leftarrow \frac{Average(T_L(Exp_Data_{ExpNum,0}, Trial))}{K}\\
& \qquad p_{HS} \leftarrow \frac{Average(p_H(Exp_Data_{ExpNum,0}, Trial))}{psi}\\
& \qquad N_{rpm} \leftarrow Average(Nrpm(Exp_Data_{ExpNum,0}, Trial)) \cdot min\\
& \qquad qair_ovly \leftarrow \frac{Average(qair_ovly(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qO2_ovly \leftarrow \frac{Average(qO2_ovly(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qCO2_ovly \leftarrow \frac{Average(qCO2_ovly(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qN2_ovly \leftarrow \frac{Average(qN2_ovly(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qair_1 \leftarrow \frac{Average(qair_1(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qO2_1 \leftarrow \frac{Average(qO2_1(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qCO2_1 \leftarrow \frac{Average(qCO2_1(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qN2_1 \leftarrow \frac{Average(qN2_1(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qair_2 \leftarrow \frac{Average(qair_2(Exp_Data_{ExpNum,0}, Trial))}{SLPM}\\
& \qquad qO2_2 \leftarrow \frac{Average(qO2_2(Exp_Data_{ExpNum,0}, Trial))}{SLPM}
\end{aligned}$$

SLPM

$$qCO2\_2 \leftarrow \frac{\text{Average}\left(qCO2\_2\left(Exp\_Data_{ExpNum,0}, Trial\right)\right)}{SLPM}$$

$$qN2\_2 \leftarrow \frac{\text{Average}\left(qN2\_2\left(Exp\_Data_{ExpNum,0}, Trial\right)\right)}{SLPM}$$

$$OUR \leftarrow \frac{\text{Average}\left(OUR\left(Exp\_Data_{ExpNum,0}, Trial\right)\right)}{\frac{mmol}{L \cdot hr}}$$

$$result_{ExpNum,Trial} \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \end{pmatrix}$$

$result \leftarrow 0$ otherwise

result

This program builds the coefficients that are used in equations 3.1-15, 3.1-16, 4.1.2-1, 4.1.2-2, and 4.1.3-2.

$$\text{Coefficient(Inputs)} := \left| \begin{array}{l}
V_{Liq} \leftarrow \text{Inputs}_0 \cdot L \\
T_{Liq} \leftarrow \text{Inputs}_1 \cdot K \\
p_{HS} \leftarrow \text{Inputs}_2 \cdot psi \\
N_{rpm} \leftarrow \text{Inputs}_3 \cdot \dfrac{1}{min} \\
\text{qair_ovly} \leftarrow \text{Inputs}_4 \cdot SLPM \\
\text{qO2_ovly} \leftarrow \text{Inputs}_5 \cdot SLPM \\
\text{qCO2_ovly} \leftarrow \text{Inputs}_6 \cdot SLPM \\
\text{qN2_ovly} \leftarrow \text{Inputs}_7 \cdot SLPM \\
\text{qair_1} \leftarrow \text{Inputs}_8 \cdot SLPM \\
\text{qO2_1} \leftarrow \text{Inputs}_9 \cdot SLPM \\
\text{qCO2_1} \leftarrow \text{Inputs}_{10} \cdot SLPM \\
\text{qN2_1} \leftarrow \text{Inputs}_{11} \cdot SLPM \\
\text{qair_2} \leftarrow \text{Inputs}_{12} \cdot SLPM \\
\text{qO2_2} \leftarrow \text{Inputs}_{13} \cdot SLPM \\
\text{qCO2_2} \leftarrow \text{Inputs}_{14} \cdot SLPM \\
\text{qN2_2} \leftarrow \text{Inputs}_{15} \cdot SLPM \\
OUR \leftarrow \text{Inputs}_{16} \cdot \dfrac{mmol}{L \cdot hr} \\
kla_{surf} \leftarrow \text{Inputs}_{18} \cdot \dfrac{1}{hr} \\
kla1 \leftarrow \text{Inputs}_{19} \cdot \dfrac{1}{hr} \\
kla2 \leftarrow \text{Inputs}_{20} \cdot \dfrac{1}{hr} \\
T_{HS} \leftarrow T_H\left(T_{Liq}\right) \\
V_{HS} \leftarrow V_H\left(V_T, V_{Liq}\right) \\
p_{Liq} \leftarrow p_L\left(V_{Liq}, D, p_{HS}\right) \\
q_{ovly} \leftarrow q_{total}(\text{qair_ovly}, \text{qO2_ovly}, \text{qCO2_ovly}, \text{qN2_ovly}) + 0.000000000000001 \cdot \text{SIUnitsOf(qair_ovly)} \\
q_{spg1} \leftarrow q_{total}(\text{qair_1}, \text{qO2_1}, \text{qCO2_1}, \text{qN2_1}) + 0.000000000000001 \cdot \text{SIUnitsOf(qair_1)} \\
q_{spg2} \leftarrow q_{total}(\text{qair_2}, \text{qO2_2}, \text{qCO2_2}, \text{qN2_2}) + 0.000000000000001 \cdot \text{SIUnitsOf(qair_2)} \\
\text{xO2_ovly} \leftarrow x_{O2}(\text{qair_ovly}, \text{qO2_ovly}, \text{qCO2_ovly}, \text{qN2_ovly}) \\
\text{xO2spg_1} \leftarrow x_{O2}(\text{qair_1}, \text{qO2_1}, \text{qCO2_1}, \text{qN2_1}) \\
\text{xO2spg_2} \leftarrow x_{O2}(\text{qair_2}, \text{qO2_2}, \text{qCO2_2}, \text{qN2_2}) \\
\text{xCO2_ovly} \leftarrow x_{CO2}(\text{qair_ovly}, \text{qO2_ovly}, \text{qCO2_ovly}, \text{qN2_ovly}) \\
\text{xCO2spg_1} \leftarrow x_{CO2}(\text{qair_1}, \text{qO2_1}, \text{qCO2_1}, \text{qN2_1}) \\
\text{xCO2spg_2} \leftarrow x_{CO2}(\text{qair_2}, \text{qO2_2}, \text{qCO2_2}, \text{qN2_2}) \\
H_{O2_} \leftarrow H_{O2_cp}\left(\text{str2num(saline)}, T_{Liq}\right) \\
H_{CO2_} \leftarrow H_{CO2_cp}\left(\text{str2num(saline)}, T_{Liq}\right)
\end{array} \right. \qquad (9.1-40)$$

$$result_0 \leftarrow xO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_1 \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_2 \leftarrow xO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_3 \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$$result_4 \leftarrow kla_{surf} \cdot hr$$

$$result_5 \leftarrow p_{HS} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_6 \leftarrow OUR \cdot \frac{L \cdot hr}{mmol}$$

$$result_7 \leftarrow xO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_8 \leftarrow \frac{1}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_9 \leftarrow \left(xO2spg\_1 \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot hr$$

$$result_{10} \leftarrow \left[\frac{1}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{11} \leftarrow \left(xO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{12} \leftarrow \frac{\left(q_{ovly} + q_{spg1} + q_{spg2}\right) \cdot T_{HS} \cdot p_{stp}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{13} \leftarrow 0$$

$$result_{14} \leftarrow \frac{R_{gas} \cdot T_{HS} \cdot V_{Liq}}{p_{HS} \cdot V_{HS}} \cdot \frac{mol}{L}$$

$$result_{15} \leftarrow xCO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{16} \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_{17} \leftarrow xCO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{18} \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$$result_{19} \leftarrow 0.893 \cdot kla_{surf} \cdot hr$$

$$result_{20} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{21} \leftarrow xCO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result \leftarrow \frac{1}{} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{}\right)\right]\right] \cdot \frac{mol}{} \cdot hr$$

$$result_{22} \leftarrow \left[\frac{1}{H_{CO2\_}\cdot p_{Liq}}\cdot\frac{q_{spg1}\cdot p_{stp}\cdot T_{HS}}{V_{HS}\cdot T_{q\_ref}\cdot p_{HS}}\cdot\left[1-\exp\left[-\left(\frac{0.893\cdot kla1\cdot H_{CO2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1}}\right)\right]\right]\right]\cdot\frac{mol}{L}\cdot hr$$

$$result_{23} \leftarrow \left(xCO2spg\_1\cdot\frac{q_{spg1}\cdot p_{stp}\cdot T_{HS}}{V_{HS}\cdot T_{q\_ref}\cdot p_{HS}}\right)\cdot\exp\left[-\left(\frac{0.893\cdot kla1\cdot H_{CO2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1}}\right)\right]\cdot hr$$

$$result_{24} \leftarrow \left[\frac{1}{H_{CO2\_}\cdot p_{Liq}}\cdot\frac{q_{spg2}\cdot p_{stp}\cdot T_{HS}}{V_{HS}\cdot T_{q\_ref}\cdot p_{HS}}\cdot\left[1-\exp\left[-\left(\frac{0.893\cdot kla2\cdot H_{CO2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg2}}\right)\right]\right]\right]\cdot\frac{mol}{L}\cdot hr$$

$$result_{25} \leftarrow \left(xCO2spg\_2\cdot\frac{q_{spg2}\cdot p_{stp}\cdot T_{HS}}{V_{HS}\cdot T_{q\_ref}\cdot p_{HS}}\right)\cdot\exp\left[-\left(\frac{0.893\cdot kla2\cdot H_{CO2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg2}}\right)\right]\cdot hr$$

$$result_{26} \leftarrow 0$$

$$result_{27} \leftarrow p_{Liq}\cdot H_{O2\_}\cdot\frac{L}{mol}$$

$$result$$

The differential equations that are solved to obtain the time dependent curves for the dissolved oxygen measurement.

Given

$$O2(0) = Initial_0 \qquad (9.1-41)$$

$$x_{O2\_hs}(0) = Initial_1 \qquad (9.1-42)$$

$$O_e(0) = Initial_2 \qquad (9.1-43)$$

$$CO2(0) = Initial_3 \qquad (9.1-44)$$

$$x_{CO2\_hs}(0) = Initial_4 \qquad (9.1-45)$$

$$\frac{d}{dt}O2(t) = \left[\begin{aligned}&Coefficient(Inputs)_0\cdot\left(\frac{mol}{L\cdot hr}\right) - O2(t)\cdot\frac{mol}{L}\cdot Coefficient(Inputs)_1\cdot\frac{1}{hr}\ \ldots\\&+Coefficient(Inputs)_2\cdot\left(\frac{mol}{L\cdot hr}\right) - O2(t)\cdot\frac{mol}{L}\cdot Coefficient(Inputs)_3\cdot\frac{1}{hr}\ \ldots\\&+\frac{Coefficient(Inputs)_4}{hr}\cdot\left(x_{O2\_hs}(t)\cdot Coefficient(Inputs)_5\cdot\frac{mol}{L} - O2(t)\cdot\frac{mol}{L}\right)\ldots\\&+-Coefficient(Inputs)_6\cdot\frac{mmol}{L\cdot hr}\end{aligned}\right]\cdot\frac{hr\cdot L}{mol} \qquad (9.1-46)$$

$$\frac{d}{dt}CO2(t) = \left[\begin{aligned}&Coefficient(Inputs)_{15}\cdot\frac{mol}{L\cdot hr} - CO2(t)\cdot\frac{mol}{L}\cdot Coefficient(Inputs)_{16}\cdot\frac{1}{hr}\ \ldots\\&+Coefficient(Inputs)_{17}\cdot\frac{mol}{L\cdot hr} - CO2(t)\cdot\frac{mol}{L}\cdot Coefficient(Inputs)_{18}\cdot\frac{1}{hr}\ \ldots\\&+\frac{Coefficient(Inputs)_{19}}{hr}\cdot\left(x_{CO2\_hs}(t)\cdot Coefficient(Inputs)_{20}\cdot\frac{mol}{L} - CO2(t)\cdot\frac{mol}{L}\right)\ldots\\&+RQ\,Coefficient(Inputs)_6\cdot\frac{mmol}{L\cdot hr}\end{aligned}\right]\cdot\frac{hr\cdot L}{mol} \qquad (9.1-47)$$

$$\frac{d}{dt}x_{O2\_hs}(t) = \left[\begin{array}{l} \text{Coefficient(Inputs)}_7\cdot\frac{1}{hr}\;... \\ +\text{Coefficient(Inputs)}_8\cdot\frac{L}{hr\cdot mol}\cdot\left(O2(t)\cdot\frac{mol}{L}\right)+\text{Coefficient(Inputs)}_9\cdot\frac{1}{hr}\;... \\ +\text{Coefficient(Inputs)}_{10}\cdot\frac{L}{hr\cdot mol}\cdot\left(O2(t)\cdot\frac{mol}{L}\right)+\text{Coefficient(Inputs)}_{11}\cdot\frac{1}{hr}\;... \\ +-x_{O2\_hs}(t)\cdot\text{Coefficient(Inputs)}_{12}\cdot\frac{1}{hr}\;... \\ +-\frac{\text{Coefficient(Inputs)}_4}{hr}\cdot\left(x_{O2\_hs}(t)\cdot\text{Coefficient(Inputs)}_5\cdot\frac{mol}{L}-O2(t)\cdot\frac{mol}{L}\right)\cdot\text{Coefficient(Inputs)}_{14}\cdot\frac{L}{mol} \end{array}\right]\cdot hr \qquad (9.1-48)$$

(9.1 – 49)

$$\frac{d}{dt}x_{CO2\_hs}(t) = \left[\begin{array}{l} \text{Coefficient(Inputs)}_{21}\cdot\frac{1}{hr}\;... \\ +\text{Coefficient(Inputs)}_{22}\cdot\frac{L}{mol\cdot hr}\cdot\left(CO2(t)\cdot\frac{mol}{L}\right)+\text{Coefficient(Inputs)}_{23}\cdot\frac{1}{hr}\;... \\ +\text{Coefficient(Inputs)}_{24}\cdot\frac{L}{mol\cdot hr}\cdot\left(CO2(t)\cdot\frac{mol}{L}\right)+\text{Coefficient(Inputs)}_{25}\cdot\frac{1}{hr}\;... \\ +-x_{CO2\_hs}(t)\cdot\text{Coefficient(Inputs)}_{12}\cdot\frac{1}{hr}\;... \\ +\frac{-\text{Coefficient(Inputs)}_4\cdot 0.893}{hr}\cdot\left(x_{CO2\_hs}(t)\cdot\text{Coefficient(Inputs)}_{20}\cdot\frac{mol}{L}-CO2(t)\cdot\frac{mol}{L}\right)\cdot\text{Coefficient(Inputs)}_{14}\cdot\frac{L}{mol} \end{array}\right]\cdot hr$$

$$\frac{d}{dt}O_e(t) = \frac{hr}{\tau(Nrpm)}\cdot\left(\frac{\frac{100}{0.21}\cdot O2(t)\cdot\frac{mol}{L}}{\text{Coefficient(Inputs)}_{27}\cdot\frac{mol}{L}}-O_e(t)\right) \qquad (9.1-50)$$

$$\text{Result(Inputs, Initial, DataPoints)} := \text{Odesolve}\left[\begin{pmatrix} O2 \\ x_{O2\_hs} \\ O_e \\ CO2 \\ x_{CO2\_hs} \end{pmatrix}, t, 48, \text{DataPoints}\right] \qquad (9.1-51)$$

Data Vectors and Mass Balance Differential Equations

Select the model form used to fit the data. For a dynamic step change, cell-free system select item 1.

Basis for Characterization

0: No Model
1: OUR, Gas Flows, DO
2: OUR, Gas Flows, CO2
3: Gas Flows, DO, CO2
4: OUR, DO, CO2

Table 9.1-9: Model used for kla characterization.

Select which spargers were active during the evaluation.

Data for Sparger

Surface Only
Sparger 1 and Surface
Sparger 2 and Surface
Sparger 1, 2, and Surface
Sparger 1 Only

Table 9.1-10: Experimental data set on which sparger.

Primer Matrix

User Provided
Calculate Estimate

Table 9.1-11: Primer kla matrix. *Select whether to use the kla matrix provided or to use a calculated estimate to initiate the nonlinear fitting. The user provided one is listed in Table 9.1-5.*

Resolution

Low
Medium
High

Table 9.1-12: Nonlinear fitting resolution. *Select the resolution for the nonlinear curve fitting. The lower resolution would have a faster computation time.*

Estimate kla Values

This program provides an estimate for the kla values based on the time required to achieve 63% of the step change. The program is only active when the user selects Calculate Estimate.

(9.1 – 52)

```
klaMatrix(klaValues, eNum, Primer) := | result ← klaValues
                                      | for x ∈ 0.. rows(Pages) − 1                                    if Part1 = 1 ∧ Primer = 1
                                      |   for i ∈ 0.. rows(eNum) − 1
                                      |     if eNum_i = x
                                      |       sum_x ← 0
                                      |       Data_x ← Exp_Data_x
                                      |       for Trial ∈ 1.. cols(Data_x)/20
                                      |         deltaT ← klaValues_(Trial·2+1, x)
                                      |         eTime ← ElapsedTime(Data_x, deltaT, Trial)
                                      |         DO(t) ← Value(eTime, DO(Data_x, Trial), t)
                                      |         j ← rows(eTime) − 1
                                      |         i ← 0
                                      |         while DO(eTime_i) < 0.63·(DO(eTime_j) ...
                                      |                                  + −DO(eTime_0)) ... ∧ i < j
                                      |                         + DO(eTime_0)
                                      |           i ← i + 1
                                      |         kla_result_(x, Trial) ← 1/eTime_(i−1)
                                      |       sum_x ← sum_x + kla_result_(x, Trial)
                                      |     result_(1, x) ← sum_x/Trial
                                      | result
```

klaValues := klaMatrix(klaValues_, eNum, Primer) (9.1 − 53)

Estimate kla Values

The delay times, probe calibration corrections and kla values that are used as the primer for the fitting.

Experimental Run Number

klaValues =

	0	1	2	3	
0	0	0	0	0	Surface Kla (1/hr)
1	0	0	0	0	Sparge1 Kla (1/hr)
2	0	0	0	0	Sparge2 Kla (1/hr)
3	0	0	0	0	Delay Time Trial 1 (hours)
4	0	0	0	0	Calibration Value Trial 1
5	0	0	0	0	Delay Time Trial 2 (hours)
6	0	0	0	...	Calibration Value Trial 2

Table 9.1-13: Kla primer matrix used for evaluation.

Select whether a nonlinear solver or a contour map would be used to determine the initial kla value(s) of best fit. The nonlinear solver converges on the minimum sum of square error, and the graphical solution plots to sum of square error as a function of the fitted parameters.

Kla Solver Method

Nonlinear Solver
Graphical

Table 9.1-14: Kla solver method.

Nonlinear Parameter Fitting

This program calculates the mean sum of square error for each of the selected experimental runs based on the difference between the observed dissolved oxygen reading and the calculated dissolved oxygen reading, or the observed dissolved carbon dioxide and the calculated dissolved carbon dioxide. This program will work for models 1, 2, 3, or 4.

(9.1 – 54)

```
SSE_M(InMx, exp_num, M, Points) := | count ← 0
                                   | SSE ← 0
                                   | for Trial ∈ 1 .. cols(Exp_Data_exp_num)/20                 if Part1 = 1 ∧ exp_num ≥ 0
                                   |   | eTime ← ElapsedTime(Exp_Data_exp_num,0, InMx_2Trial+1, Trial)
                                   |   | V_Liq ← (AvgMtx_exp_num,Trial)_0
                                   |   | T_Liq ← (AvgMtx_exp_num,Trial)_1
                                   |   | p_HS ← (AvgMtx_exp_num,Trial)_2
                                   |   | N_rpm ← (AvgMtx_exp_num,Trial)_3
                                   |   | DO(t) ← Value(eTime, DO(Exp_Data_exp_num,0, Trial), t)
                                   |   | CO2(t) ← Value(eTime, CO2(Exp_Data_exp_num,0, Trial), t)
                                   |   | qair_ovly ← (AvgMtx_exp_num,Trial)_4
                                   |   | qO2_ovly ← (AvgMtx_exp_num,Trial)_5
                                   |   | qCO2_ovly ← (AvgMtx_exp_num,Trial)_6
                                   |   | qN2_ovly ← (AvgMtx_exp_num,Trial)_7
                                   |   | qair_1 ← (AvgMtx_exp_num,Trial)_8
                                   |   | qO2_1 ← (AvgMtx_exp_num,Trial)_9
                                   |   | qCO2_1 ← (AvgMtx_exp_num,Trial)_10
                                   |   | qN2_1 ← (AvgMtx_exp_num,Trial)_11
                                   |   | qair_2 ← (AvgMtx_exp_num,Trial)_12
                                   |   | qO2_2 ← (AvgMtx_exp_num,Trial)_13
                                   |   | qCO2_2 ← (AvgMtx_exp_num,Trial)_14
                                   |   | qN2_2 ← (AvgMtx_exp_num,Trial)_15
                                   |   | OUR ← (AvgMtx_exp_num,Trial)_16
                                   |   | if M = 1 ∨ M = 2 ∨ M = 4
                                   |   |   |   ( V_Liq
                                   |   |   |     T_Liq
                                   |   |   |     p_HS
                                   |   |   |     N_rpm
                                   |   |   |     qair_ovly
                                   |   |   |     qO2_ovly
                                   |   |   |     qCO2_ovly
                                   |   |   |     qN2_ovly
                                   |   |   |     qair_1
```

$$InVtr \leftarrow \begin{pmatrix} \vdots \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMx_{2\cdot Trial+1} \\ InMx_{0} \\ 0 \\ 0 \end{pmatrix} \quad \text{if } Sparger = 0$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMx_{2\cdot Trial+1} \\ InMx_{0} \\ InMx_{1} \\ 0 \end{pmatrix} \quad \text{if } Sparger = 1$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ \vdots \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMx_{2 \cdot Trial+1} \\ InMx_0 \\ 0 \\ InMx_2 \end{pmatrix} \quad \text{if } Sparger = 2$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMx_{2Trial+1} \\ InMx_0 \\ InMx_1 \\ InMx_2 \end{pmatrix} \quad \text{if } Sparger = 3$$

$$\begin{pmatrix} V_{Liq} \end{pmatrix}$$

$$
InVtr \leftarrow \begin{pmatrix} T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMx_{2Trial+1} \\ 0 \\ InMx_{1} \\ 0 \end{pmatrix} \quad \text{if } Sparger = 4
$$

$$
InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMx_{2Trial+1} \\ 0 \end{pmatrix} \quad \text{if } Sparger = 5
$$

$$\begin{pmatrix} 0 \\ 0 \\ InMx_2 \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMx_{2Trial+1} \\ 0 \\ InMx_1 \\ InMx_2 \end{pmatrix} \text{ if } Sparger = 6$$

$$p_{Liq} \leftarrow p_L(V_{Liq}\cdot L, D, p_{HS}\cdot psig)$$

$$H_{O2\_} \leftarrow H_{O2\_cp}(str2num(saline), T_{Liq}\cdot K)$$

$$Initial_0 \leftarrow \frac{DO(0.0001)}{100}\cdot 0.21\cdot H_{O2\_}\cdot p_{Liq}\cdot \frac{L}{mol}\cdot \frac{100}{InMx_{2Trial+2}}$$

$$Initial_1 \leftarrow \frac{DO(0.0001)}{100}\cdot \frac{100}{InMx_{2Trial+2}}\cdot 0.21$$

$$Initial_2 \leftarrow DO(0.0001)\cdot \frac{100}{InMx_{2Trial+2}}$$

$$H_{CO2\_} \leftarrow H_{CO2\_cp}(str2num(saline), T_{Liq}\cdot K)$$

$$Initial_3 \leftarrow CO2(0.0001)\cdot H_{CO2\_}\cdot \frac{L}{mol}$$

$$Initial_4 \leftarrow \frac{CO2(0.0001)}{760\cdot mmHg}\cdot \frac{p_{stp}}{p_{HS}\cdot psi}$$

$$DOcalc \leftarrow Result(InVtr, Initial, Points)_2$$

$$CO2calc \leftarrow Result(InVtr, Initial, Points)_3$$

$$\text{for } i \in 0 .. rows(eTime) - 1$$

$$SSE \leftarrow SSE + \left(DO(eTime_i)\cdot\frac{100}{InMx_{2Trial+2}} + -DOcalc(eTime_i)\right)^2 \quad \text{if } M = 1$$

$$T2 \leftarrow \frac{CO2calc(eTime_i)\cdot\frac{mol}{L}}{H_{CO2\_}}$$

$$SSE \leftarrow SSE + \left(\frac{CO2(eTime_i) - T2}{mmHg}\right)^2 \quad \text{if } M = 2$$

$$SSE \leftarrow SSE + \left(\frac{CO2(eTime_i) - T2}{mmHg}\right)^2 + \left(DO(eTime_i) + -DOcalc(eTime_i)\right)^2 \quad \text{if } M = 3 \vee M = 4$$

$$count \leftarrow count + 1$$

otherwise

$$SSE \leftarrow 0$$

$$count \leftarrow 1$$

$$MSSE \leftarrow \frac{SSE}{count + 1}$$

$$temp \leftarrow trace(exp\_num)$$

$$temp \leftarrow trace(InMx_0)$$

$$temp \leftarrow trace(InMx_1)$$

$$temp \leftarrow trace(InMx_2)$$

$$temp \leftarrow trace(InMx_3)$$

$$temp \leftarrow trace(InMx_4)$$

$$temp \leftarrow trace(InMx_5)$$

$$temp \leftarrow trace(InMx_6)$$

$$temp \leftarrow trace(MSSE)$$

$$MSSE$$

This program produces a matrix of the calculated dissolved oxygen with respect to elapsed time.

(9.1 – 55)

DOmodel(InMtx, exp_num, GasModel, Trial, Points) := if Part1 = 1

$eTime \leftarrow ElapsedTime\left(Exp\_Data_{exp\_num,0}, InMtx_{2Trial+1}, Trial\right)$

$V_{Liq} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_0$

$T_{Liq} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_1$

$p_{HS} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_2$

$N_{rpm} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_3$

$DO(t) \leftarrow Value\left(eTime, DO\left(Exp\_Data_{exp\_num,0}, Trial\right), t\right)$

$CO2(t) \leftarrow Value\left(eTime, CO2\left(Exp\_Data_{exp\_num,0}, Trial\right), t\right)$

$qair\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_4$

$qO2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_5$

$qCO2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_6$

$qN2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_7$

$qair\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_8$

$qO2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_9$

$qCO2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{10}$

$qN2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{11}$

$qair\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{12}$

$qO2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{13}$

$qCO2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{14}$

$qN2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{15}$

$OUR \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{16}$

if GasModel = 1 ∨ GasModel = 2 ∨ GasModel = 4

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \end{pmatrix} \text{ if } Sparger = 0$$

$$\begin{pmatrix} \text{qair\_2} \\ \text{qO2\_2} \\ \text{qCO2\_2} \\ \text{qN2\_2} \\ \text{OUR} \\ \text{InMtx}_{2\text{Trial}+1} \\ \text{InMtx}_{0} \\ 0 \\ 0 \end{pmatrix}$$

$$\text{InVtr} \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ \text{qair\_ovly} \\ \text{qO2\_ovly} \\ \text{qCO2\_ovly} \\ \text{qN2\_ovly} \\ \text{qair\_1} \\ \text{qO2\_1} \\ \text{qCO2\_1} \\ \text{qN2\_1} \\ \text{qair\_2} \\ \text{qO2\_2} \\ \text{qCO2\_2} \\ \text{qN2\_2} \\ \text{OUR} \\ \text{InMtx}_{2\text{Trial}+1} \\ \text{InMtx}_{0} \\ \text{InMtx}_{1} \\ 0 \end{pmatrix} \quad \text{if Sparger} = 1$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ \text{qair\_ovly} \\ \text{qO2\_ovly} \\ \text{qCO2\_ovly} \\ \text{qN2\_ovly} \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_0 \\ 0 \\ InMtx_2 \end{pmatrix} \quad \text{if } Sparger = 2$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_0 \\ InMtx_1 \\ InMtx_2 \end{pmatrix} \quad \text{if } Sparger = 3$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ InMtx_{1} \\ 0 \end{pmatrix} \quad \text{if } Sparger = 4$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ 0 \\ InMtx_{2} \end{pmatrix} \quad \text{if } Sparger = 5$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ InMtx_{1} \\ InMtx_{2} \end{pmatrix} \quad \text{if } Sparger = 6$$

$$p_{Liq} \leftarrow p_L\left(V_{Liq} \cdot L, D, p_{HS} \cdot psig\right)$$

$$Initial_0 \leftarrow \frac{DO(0.0001)}{100} \cdot \frac{100}{InMtx_{2Trial+2}} \cdot 0.21 \cdot H_{O2\_cp}\left(str2num(saline), T_{Liq} \cdot K\right) \cdot \frac{p_{Liq} \cdot L}{mol}$$

$$Initial_1 \leftarrow \frac{DO(0.0001)}{100} \cdot \frac{100}{InMtx_{2Trial+2}} \cdot 0.21$$

$$Initial_2 \leftarrow DO(00.0001) \cdot \frac{100}{InMtx_{2Trial+2}}$$

$$Initial_3 \leftarrow CO2(0.0001) \cdot H_{CO2\_cp}\left(str2num(saline), T_{Liq} \cdot K\right) \cdot \frac{L}{mol}$$

$$Initial_4 \leftarrow \frac{CO2(0.0001)}{760 \cdot mmHg} \cdot \frac{p_{stp}}{p_{HS} \cdot psi}$$

$$DOcalc \leftarrow Result(InVtr, Initial, Points)_2$$

$$\text{for } i \in 0..rows(eTime) - 1$$

$$result_i \leftarrow DOcalc\left(eTime_i\right)$$

$$result$$

The mean sum of square error value for each experimental run is calculated and provided in the vector MSSE_Exp.

$$MSSE\_Exp := \left| \begin{array}{l} \text{for } i \in 0..\,rows(eNum)-1 \\ \quad Result_{0,eNum_i} \leftarrow SSE\_M\left(klaValues^{\langle eNum_i \rangle}, eNum_i, GasModel, Points\right) \\ Result \end{array} \right. \text{ if } Part1 = 1 \qquad (9.1-56)$$

$$MSSE\_Exp = \blacksquare \qquad (9.1-57)$$

If the user selects to solve for the kla values by using the nonlinear solver, the following block of code is active.

$$exp\_num := \left| \begin{array}{ll} -1 & \text{if } klaSolver = 0 \\ eNum_0 & \text{otherwise} \end{array} \right. \qquad (9.1-58)$$

$$klaInput := \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 100 \\ 0 \\ 100 \\ 0 \\ 100 \\ 0 \\ 100 \\ 0 \\ 100 \end{pmatrix} \qquad (9.1-59)$$

$$TOL := \frac{1}{Points} \qquad (9.1-60)$$

$$CTOL := \frac{1}{Points} \qquad (9.1-61)$$

The following constraints are to help the solver not get caught in a local minimum that does not make physcial sense.

Given

$$\text{klaInput} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 95 \\ 0 \\ 95 \\ 0 \\ 95 \\ 0 \\ 95 \\ 0 \\ 95 \end{pmatrix} \qquad (9.1-62)$$

$$\text{klaInput} \leq \begin{pmatrix} 20 \\ 20 \\ 20 \\ 0.05 \\ 105 \\ 0.05 \\ 105 \\ 0.05 \\ 105 \\ 0.05 \\ 105 \\ 0.05 \\ 105 \end{pmatrix} \qquad (9.1-63)$$

$$\left| \begin{array}{l} 0 \ \text{if} \ \text{exp\_num} = -1 \\ \text{SSE\_M(klaInput, exp\_num, GasModel, Points)} \ \ \text{if} \ \ \text{klaSolver} = 0 \\ 0 \ \ \text{otherwise} \end{array} \right. = 0 \qquad (9.1-64)$$

$$\text{kla\_result(klaInput, exp\_num)} := \text{Minerr(klaInput)} \qquad (9.1-65)$$

$$\text{kla\_} := \left| \begin{array}{l} \text{result} \leftarrow \text{klaValues} \\ \text{for} \ \ i \in 0..\text{rows(eNum)} - 1 \\ \quad \left| \begin{array}{l} \text{klaValues}_{0,\text{eNum}_i} \leftarrow 0 \ \ \text{if} \ \ \text{Sparger} = 4 \vee \text{Sparger} = 5 \vee \text{Sparger} = 6 \\ \text{result}^{\langle \text{eNum}_i \rangle} \leftarrow \text{kla\_result}\left(\text{klaValues}^{\langle \text{eNum}_i \rangle}, \text{eNum}_i\right) \ \ \text{if} \ \ \text{klaSolver} = 0 \end{array} \right. \\ \text{result} \end{array} \right. \qquad (9.1-66)$$

The calculated dissolved oxygen (equation 9.1-67), head space composition (equation 9.1-68), and the observed dissolved oxygen (equation 9.1-69) are calculated with the determined kla values.

(9.1 – 67)

$O2\_liquid(InMtx, exp\_num, GasModel, Trial, Points) :=$ if $Part1 = 1$

$eTime \leftarrow ElapsedTime\left(Exp\_Data_{exp\_num,0}, InMtx_{2Trial+1}, Trial\right)$

$V_{Liq} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_0$

$T_{Liq} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_1$

$p_{HS} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_2$

$N_{rpm} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_3$

$DO(t) \leftarrow Value\left(eTime, DO\left(Exp\_Data_{exp\_num,0}, Trial\right), t\right)$

$CO2(t) \leftarrow Value\left(eTime, CO2\left(Exp\_Data_{exp\_num,0}, Trial\right), t\right)$

$qair\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_4$

$qO2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_5$

$qCO2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_6$

$qN2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_7$

$qair\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_8$

$qO2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_9$

$qCO2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{10}$

$qN2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{11}$

$qair\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{12}$

$qO2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{13}$

$qCO2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{14}$

$qN2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{15}$

$OUR \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{16}$

if $GasModel = 1 \lor GasModel = 2 \lor GasModel = 4$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \end{pmatrix} \text{ if } Sparger = 0$$

$$\begin{pmatrix} qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_{0} \\ 0 \\ 0 \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ P_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_{0} \\ InMtx_{1} \\ 0 \end{pmatrix} \text{ if } Sparger = 1$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ P_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_0 \\ 0 \\ InMtx_2 \end{pmatrix} \text{ if } Sparger = 2$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ P_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_0 \\ InMtx_1 \\ InMtx_2 \end{pmatrix} \text{ if } Sparger = 3$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ P_{HS} \\ N_{rpm} \end{pmatrix}$$

$$\mathrm{InVtr} \leftarrow \begin{pmatrix} \mathrm{qair\_ovly} \\ \mathrm{qO2\_ovly} \\ \mathrm{qCO2\_ovly} \\ \mathrm{qN2\_ovly} \\ \mathrm{qair\_1} \\ \mathrm{qO2\_1} \\ \mathrm{qCO2\_1} \\ \mathrm{qN2\_1} \\ \mathrm{qair\_2} \\ \mathrm{qO2\_2} \\ \mathrm{qCO2\_2} \\ \mathrm{qN2\_2} \\ \mathrm{OUR} \\ \mathrm{InMtx}_{2\mathrm{Trial}+1} \\ 0 \\ \mathrm{InMtx}_{1} \\ 0 \end{pmatrix} \quad \text{if } \mathrm{Sparger} = 4$$

$$\mathrm{InVtr} \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ \mathrm{qair\_ovly} \\ \mathrm{qO2\_ovly} \\ \mathrm{qCO2\_ovly} \\ \mathrm{qN2\_ovly} \\ \mathrm{qair\_1} \\ \mathrm{qO2\_1} \\ \mathrm{qCO2\_1} \\ \mathrm{qN2\_1} \\ \mathrm{qair\_2} \\ \mathrm{qO2\_2} \\ \mathrm{qCO2\_2} \\ \mathrm{qN2\_2} \\ \mathrm{OUR} \\ \mathrm{InMtx}_{2\mathrm{Trial}+1} \\ 0 \\ 0 \\ \mathrm{InMtx}_{2} \end{pmatrix} \quad \text{if } \mathrm{Sparger} = 5$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ InMtx_{1} \\ InMtx_{2} \end{pmatrix} \quad \text{if } Sparger = 6$$

$$p_{Liq} \leftarrow p_L\left(V_{Liq} \cdot L, D, p_{HS} \cdot psig\right)$$

$$Initial_0 \leftarrow \frac{DO(0.0001)}{100} \cdot \frac{100}{InMtx_{2Trial+2}} \cdot 0.21 \cdot H_{O2\_cp}\left(str2num(saline), T_{Liq} \cdot K\right) \cdot p_{Liq} \cdot \frac{L}{mol}$$

$$Initial_1 \leftarrow \frac{DO(0.0001)}{100} \cdot \frac{100}{InMtx_{2Trial+2}} \cdot 0.21$$

$$Initial_2 \leftarrow DO(0.0001) \cdot \frac{100}{InMtx_{2Trial+2}}$$

$$Initial_3 \leftarrow CO2(0.0001) \cdot H_{CO2\_cp}\left(str2num(saline), T_{Liq} \cdot K\right) \cdot \frac{L}{mol}$$

$$Initial_4 \leftarrow \frac{CO2(0.0001)}{760 \cdot mmHg} \cdot \frac{p_{stp}}{p_{HS} \cdot psi}$$

$$DOcalc \leftarrow Result(InVtr, Initial, Points)_0$$

$$\text{for } i \in 0..rows(eTime) - 1$$

$$result_i \leftarrow \frac{DOcalc\left(eTime_i\right) \cdot 100}{\left(0.21 \cdot H_{O2\_cp}\left(str2num(saline), T_{Liq} \cdot K\right) \cdot p_{Liq} \cdot psig \cdot \frac{L}{mol}\right)}$$

$$result$$

(9.1 – 68)

```
xO2_HS(InMtx, exp_num, GasModel, Trial, Points) := | if Part1 = 1
    | eTime ← ElapsedTime(Exp_Data_{exp_num,0}, InMtx_{2Trial+1}, Trial)
    | V_Liq ← (AvgMtx_{exp_num,Trial})_0
    | T_Liq ← (AvgMtx_{exp_num,Trial})_1
    | p_HS ← (AvgMtx_{exp_num,Trial})_2
    | N_rpm ← (AvgMtx_{exp_num,Trial})_3
    | DO(t) ← Value(eTime, DO(Exp_Data_{exp_num,0}, Trial), t)
    | CO2(t) ← Value(eTime, CO2(Exp_Data_{exp_num,0}, Trial), t)
    | qair_ovly ← (AvgMtx_{exp_num,Trial})_4
    | qO2_ovly ← (AvgMtx_{exp_num,Trial})_5
    | qCO2_ovly ← (AvgMtx_{exp_num,Trial})_6
    | qN2_ovly ← (AvgMtx_{exp_num,Trial})_7
    | qair_1 ← (AvgMtx_{exp_num,Trial})_8
    | qO2_1 ← (AvgMtx_{exp_num,Trial})_9
    | qCO2_1 ← (AvgMtx_{exp_num,Trial})_10
    | qN2_1 ← (AvgMtx_{exp_num,Trial})_11
    | qair_2 ← (AvgMtx_{exp_num,Trial})_12
    | qO2_2 ← (AvgMtx_{exp_num,Trial})_13
    | qCO2_2 ← (AvgMtx_{exp_num,Trial})_14
    | qN2_2 ← (AvgMtx_{exp_num,Trial})_15
    | OUR ← (AvgMtx_{exp_num,Trial})_16
    | if GasModel = 1 ∨ GasModel = 2 ∨ GasModel = 4
    |   | InVtr ← ( V_Liq
    |   |           T_Liq
    |   |           p_HS
    |   |           N_rpm
    |   |           qair_ovly
    |   |           qO2_ovly
    |   |           qCO2_ovly
    |   |           qN2_ovly
    |   |           qair_1
    |   |           qO2_1
    |   |           qCO2_1     if Sparger = 0
    |   |           qN2_1
```

$$\begin{pmatrix} qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_{0} \\ 0 \\ 0 \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_{0} \\ InMtx_{1} \\ 0 \end{pmatrix} \quad \text{if } Sparger = 1$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_0 \\ 0 \\ InMtx_2 \end{pmatrix} \text{ if Sparger} = 2$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_0 \\ InMtx_1 \\ InMtx_2 \end{pmatrix} \text{ if Sparger} = 3$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} \ldots \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ InMtx_{1} \\ 0 \end{pmatrix} \quad \text{if } Sparger = 4$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ 0 \\ InMtx_{2} \end{pmatrix} \quad \text{if } Sparger = 5$$

$$
InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ InMtx_1 \\ InMtx_2 \end{pmatrix} \quad \text{if } Sparger = 6
$$

$$p_{Liq} \leftarrow p_L\left(V_{Liq}\cdot L, D, p_{HS}\cdot psig\right)$$

$$Initial_0 \leftarrow \frac{DO(0.0001)}{100}\cdot\frac{100}{InMtx_{2Trial+2}}\cdot 0.21\cdot H_{O2\_cp}\left(str2num(saline), T_{Liq}\cdot K\right)\cdot p_{Liq}\cdot\frac{L}{mol}$$

$$Initial_1 \leftarrow \frac{DO(0.0001)}{100}\cdot\frac{100}{InMtx_{2Trial+2}}\cdot 0.21$$

$$Initial_2 \leftarrow DO(0.0001)\cdot\frac{100}{InMtx_{2Trial+2}}$$

$$Initial_3 \leftarrow CO2(0.0001)\cdot H_{CO2\_cp}\left(str2num(saline), T_{Liq}\cdot K\right)\cdot\frac{L}{mol}$$

$$Initial_4 \leftarrow \frac{CO2(0.0001)}{760\cdot mmHg}\cdot\frac{p_{stp}}{p_{HS}\cdot psi}$$

$$DOcalc \leftarrow Result(InVtr, Initial, Points)_1$$

$$\text{for } i \in 0..rows(eTime)-1$$

$$result_i \leftarrow \frac{DOcalc\left(eTime_i\right)\cdot 100}{0.21}$$

$$result$$

(9.1 – 69)

DO_probe(InMtx, exp_num, GasModel, Trial, Points) := if Part1 = 1

$eTime \leftarrow ElapsedTime\left(Exp\_Data_{exp\_num,0}, InMtx_{2Trial+1}, Trial\right)$

$V_{Liq} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_0$

$T_{Liq} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_1$

$p_{HS} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_2$

$N_{rpm} \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_3$

$DO(t) \leftarrow Value\left(eTime, DO\left(Exp\_Data_{exp\_num,0}, Trial\right), t\right)$

$CO2(t) \leftarrow Value\left(eTime, CO2\left(Exp\_Data_{exp\_num,0}, Trial\right), t\right)$

$qair\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_4$

$qO2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_5$

$qCO2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_6$

$qN2\_ovly \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_7$

$qair\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_8$

$qO2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_9$

$qCO2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{10}$

$qN2\_1 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{11}$

$qair\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{12}$

$qO2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{13}$

$qCO2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{14}$

$qN2\_2 \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{15}$

$OUR \leftarrow \left(AvgMtx_{exp\_num,Trial}\right)_{16}$

if GasModel = 1 ∨ GasModel = 2 ∨ GasModel = 4

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \end{pmatrix} \text{ if } Sparger = 0$$

$$\begin{pmatrix} \text{qair\_2} \\ \text{qO2\_2} \\ \text{qCO2\_2} \\ \text{qN2\_2} \\ \text{OUR} \\ \text{InMtx}_{2\text{Trial}+1} \\ \text{InMtx}_{0} \\ 0 \\ 0 \end{pmatrix}$$

$$\text{InVtr} \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ \text{qair\_ovly} \\ \text{qO2\_ovly} \\ \text{qCO2\_ovly} \\ \text{qN2\_ovly} \\ \text{qair\_1} \\ \text{qO2\_1} \\ \text{qCO2\_1} \\ \text{qN2\_1} \\ \text{qair\_2} \\ \text{qO2\_2} \\ \text{qCO2\_2} \\ \text{qN2\_2} \\ \text{OUR} \\ \text{InMtx}_{2\text{Trial}+1} \\ \text{InMtx}_{0} \\ \text{InMtx}_{1} \\ 0 \end{pmatrix} \quad \text{if Sparger} = 1$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ \text{qair\_ovly} \\ \text{qO2\_ovly} \\ \text{qCO2\_ovly} \\ \text{qN2\_ovly} \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_{0} \\ 0 \\ InMtx_{2} \end{pmatrix} \text{ if } Sparger = 2$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ InMtx_{0} \\ InMtx_{1} \\ InMtx_{2} \end{pmatrix} \text{ if } Sparger = 3$$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \end{pmatrix}$$

$$InVtr \leftarrow \begin{pmatrix} \vdots \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ InMtx_1 \\ 0 \end{pmatrix} \quad \text{if } Sparger = 4$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ 0 \\ InMtx_2 \end{pmatrix} \quad \text{if } Sparger = 5$$

$$InVtr \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ InMtx_{2Trial+1} \\ 0 \\ InMtx_1 \\ InMtx_2 \end{pmatrix} \text{ if } Sparger = 6$$

$$p_{Liq} \leftarrow p_L(V_{Liq} \cdot L, D, p_{HS} \cdot psig)$$

$$Initial_0 \leftarrow \frac{DO(0.0001)}{100} \cdot \frac{100}{InMtx_{2Trial+2}} \cdot 0.21 \cdot H_{O2\_cp}(str2num(saline), T_{Liq} \cdot K) \cdot \frac{p_{Liq} \cdot L}{mol}$$

$$Initial_1 \leftarrow \frac{DO(0.0001)}{100} \cdot \frac{100}{InMtx_{2Trial+2}} \cdot 0.21$$

$$Initial_2 \leftarrow DO(0.0001) \cdot \frac{100}{InMtx_{2Trial+2}}$$

$$Initial_3 \leftarrow CO2(0.0001) \cdot H_{CO2\_cp}(str2num(saline), T_{Liq} \cdot K) \cdot \frac{L}{mol}$$

$$Initial_4 \leftarrow \frac{CO2(0.0001)}{760 \cdot mmHg} \cdot \frac{p_{stp}}{p_{HS} \cdot psi}$$

$$DOcalc \leftarrow Result(InVtr, Initial, Points)_2$$

$$\text{for } i \in 0..rows(eTime) - 1$$
$$result_i \leftarrow DOcalc(eTime_i)$$

$$result$$

These expressions produce an array for the graph below.

$$\mathrm{DOprobe}(ExpNum, Trial) := \begin{cases} \mathrm{DO\_probe}\left(kla\_^{\langle ExpNum\rangle}, ExpNum, GasModel, Trial, Points\right) & \text{if } Part1 = 1 \\ 0 & \text{otherwise} \end{cases} \quad (9.1-70)$$

$$\mathrm{DOliquid}(ExpNum, Trial) := \begin{cases} \mathrm{O2\_liquid}\left(kla\_^{\langle ExpNum\rangle}, ExpNum, GasModel, Trial, Points\right) & \text{if } Part1 = 1 \\ 0 & \text{otherwise} \end{cases} \quad (9.1-71)$$

$$\mathrm{DO\_HS}(ExpNum, Trial) := \begin{cases} \mathrm{xO2\_HS}\left(kla\_^{\langle ExpNum\rangle}, ExpNum, GasModel, Trial, Points\right) & \text{if } Part1 = 1 \\ 0 & \text{otherwise} \end{cases} \quad (9.1-72)$$

$$\mathrm{DOobs}(ExpNum, deltaT, cal, Trial) := \left| \begin{array}{l} \text{if } Part1 = 1 \\ \quad \left| \begin{array}{l} eTime \leftarrow \mathrm{ElapsedTime}\left(Exp\_Data_{ExpNum,0}, deltaT, Trial\right) \\ DO(t) \leftarrow \mathrm{Value}\left(eTime, DO\left(Exp\_Data_{ExpNum,0}, Trial\right), t\right) \\ \text{for } i \in 0..\mathrm{rows}(eTime) - 1 \\ \quad result_i \leftarrow DO\left(eTime_i\right) \cdot \frac{100}{cal} \\ result \end{array} \right. \\ 0 \quad \text{otherwise} \end{array} \right. \quad (9.1-73)$$

$$\mathrm{e\_Time}(ExpNum, deltaT, Trial) := \begin{cases} \mathrm{ElapsedTime}\left(Exp\_Data_{ExpNum,0}, deltaT, Trial\right) & \text{if } Part1 = 1 \\ 0 & \text{otherwise} \end{cases} \quad (9.1-74)$$

The mean square error is calculated for each experimental trial.

$$MSSE\_Exp := \left| \begin{array}{l} \text{for } i \in 0..\mathrm{rows}(eNum) - 1 \\ \quad Result_{0, eNum_i} \leftarrow \mathrm{SSE\_M}\left(kla\_^{\langle eNum_i \rangle}, eNum_i, GasModel, Points\right) \\ Result \end{array} \right. \quad \text{if } Part1 = 1 \quad (9.1-75)$$

Nonlinear Parameter Fitting

Experimental Run Number

Surface Kla (1/hr)
Sparge1 Kla (1/hr)
Sparge2 Kla (1/hr)
Delay Time Trial 1 (hours)
Calibration Value Trial 1

$kla\_ = ▮$

Delay Time Trial 2 (hours)

Calibration Value Trial 2

Table 9.1-14: Nonlinear fitting for delay time, calibration values, and gas transfer coefficients for the experimental runs.

The resulting mean square error for the nonlinear parameter fitting. Note if the kla parameter fitting is done by the graphical contour map, then the value returned in equation 9.1-76 will be zero.

$$MSSE\_Exp = \blacksquare \tag{9.1 - 76}$$

Enter the value here for what would be plotted in Figure 9.1-1 below.

Plot Experiment

Table 9.1-15: Experimental trial to plot in Figure 9.1-1.

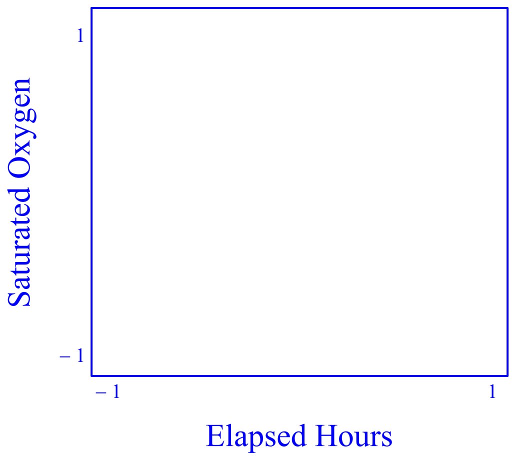
Experiment Plotting

The following program primes the klaPlot vector to contain the kla values for the specified experimental run.

$$ExNum := str2num(ExNum) \tag{9.1 - 77}$$

$$klaPlot(kla\_surf, kla1, kla2, deltaT, cal, ExNum) := \left| \begin{array}{l} result_0 \leftarrow kla\_surf \\ result_1 \leftarrow kla1 \\ result_2 \leftarrow kla2 \\ result_3 \leftarrow deltaT \\ result_4 \leftarrow cal \\ \text{for } i \in 5..rows(kla\_) - 1 \\ \quad result_i \leftarrow kla\_{i, ExNum} \\ result \end{array} \right. \tag{9.1 - 78}$$

Experiment Plotting



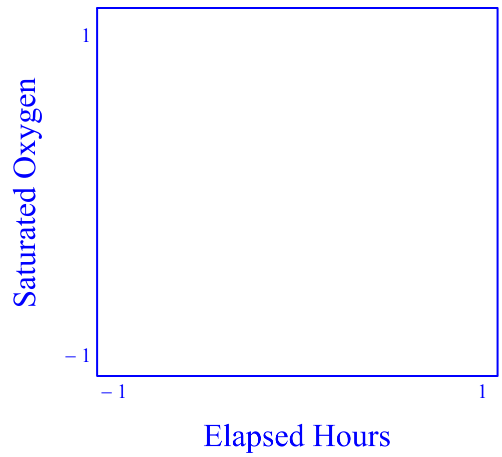


Figure 9.1-1: The observed probe profile, the fitted probe profile, and the calculated dissolved oxygen and headspace compositions for the study listed in table 9.1-15.

Error Surface

The mean sum of square error is calculated as a function of the kla values.

$$\text{ErrorSurf}(\text{kla\_surf}, \text{kla1}, \text{kla2}, \text{deltaT}, \text{cal}, \text{ExpNum}) := \left| \begin{array}{l} \text{temp} \leftarrow \text{ExpNum}_0 \\ \text{SSE\_M}(\text{klaPlot}(\text{kla\_surf}, \text{kla1}, \text{kla2}, \text{deltaT}, \text{cal}, \text{temp}), \text{temp}, \text{GasModel}, \text{Points}) \end{array} \right. \qquad (9.1-79)$$

The ExNum scalar value is converted to a matrix to be consistent with the expectations of the programs.

$$\text{ExN} := \left| \begin{array}{l} \text{result}_0 \leftarrow \text{ExNum} \\ \text{result} \end{array} \right. \qquad (9.1-80)$$

Error Surface

$$\text{SSE\_M}\left(\text{kla\_}^{\langle \text{ExN}_0 \rangle}, \text{ExN}_0, \text{GasModel}, \text{Points}\right) = \blacksquare \qquad (9.1-81)$$

Select the x and y axis to be used in the error contour plot. Visually identify the minimum error in the contour plot and adjust the values in the table where the estimated values for the delay time, the calibration adjustment, the surface kla, the sparge 1 kla, and the sparge 2 kla for each of the experimental studies (this is above equation 9.1-1).

X axis		Y axis	
	kla Surface		kla Surface
	kla Sparge 1		kla Sparge 1
	kla Sparge 2		kla Sparge 2
	Start Time		Start Time
	Calibration		Calibration

Table 9.1-15: X and Y axis to perform contour plot of mean square error in Figure 9.1-2.

Contour Plot Parameters

The following program builds the response surface for the mean sum of square error based on the kla types selected for the x and y axes. The program returns the value zero if the two kla types selected are the same.

(9.1 – 82)

$$
\begin{array}{l}
\mathrm{errorMesh1}(x,y) := \\
\quad \mathrm{result} \leftarrow 0 \\
\quad \text{if } \mathrm{Part1} = 1 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[x, \left(kla\_^{\langle ExN_0\rangle}\right)_1, \left(kla\_^{\langle ExN_0\rangle}\right)_2, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 0 \wedge \mathrm{Yvalue} = 0 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[x, y, \left(kla\_^{\langle ExN_0\rangle}\right)_2, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 0 \wedge \mathrm{Yvalue} = 1 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[x, \left(kla\_^{\langle ExN_0\rangle}\right)_1, y, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 0 \wedge \mathrm{Yvalue} = 2 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[x, \left(kla\_^{\langle ExN_0\rangle}\right)_1, \left(kla\_^{\langle ExN_0\rangle}\right)_2, y, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 0 \wedge \mathrm{Yvalue} = 3 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[x, \left(kla\_^{\langle ExN_0\rangle}\right)_1, \left(kla\_^{\langle ExN_0\rangle}\right)_2, \left(kla\_^{\langle ExN_0\rangle}\right)_3, y, ExN\right] \text{ if } \mathrm{Xvalue} = 0 \wedge \mathrm{Yvalue} = 4 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[y, x, \left(kla\_^{\langle ExN_0\rangle}\right)_2, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 1 \wedge \mathrm{Yvalue} = 0 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, x, \left(kla\_^{\langle ExN_0\rangle}\right)_2, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 1 \wedge \mathrm{Yvalue} = 1 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, x, y, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 1 \wedge \mathrm{Yvalue} = 2 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, x, \left(kla\_^{\langle ExN_0\rangle}\right)_2, y, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 1 \wedge \mathrm{Yvalue} = 3 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, x, \left(kla\_^{\langle ExN_0\rangle}\right)_2, \left(kla\_^{\langle ExN_0\rangle}\right)_3, y, ExN\right] \text{ if } \mathrm{Xvalue} = 1 \wedge \mathrm{Yvalue} = 4 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[y, \left(kla\_^{\langle ExN_0\rangle}\right)_1, x, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 2 \wedge \mathrm{Yvalue} = 0 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, y, x, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 2 \wedge \mathrm{Yvalue} = 1 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, \left(kla\_^{\langle ExN_0\rangle}\right)_1, x, \left(kla\_^{\langle ExN_0\rangle}\right)_3, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 2 \wedge \mathrm{Yvalue} = 2 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, \left(kla\_^{\langle ExN_0\rangle}\right)_1, x, y, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 2 \wedge \mathrm{Yvalue} = 3 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, \left(kla\_^{\langle ExN_0\rangle}\right)_1, x, \left(kla\_^{\langle ExN_0\rangle}\right)_3, y, ExN\right] \text{ if } \mathrm{Xvalue} = 2 \wedge \mathrm{Yvalue} = 4 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, \left(kla\_^{\langle ExN_0\rangle}\right)_1, \left(kla\_^{\langle ExN_0\rangle}\right)_2, x, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 3 \wedge \mathrm{Yvalue} = 3 \\
\qquad \mathrm{result} \leftarrow \mathrm{ErrorSurf}\left[y, \left(kla\_^{\langle ExN_0\rangle}\right)_1, \left(kla\_^{\langle ExN_0\rangle}\right)_2, x, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } \mathrm{Xvalue} = 3 \wedge \mathrm{Yvalue} = 0
\end{array}
$$

$$\begin{array}{l} \quad \text{result} \leftarrow \text{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, y, \left(kla\_^{\langle ExN_0\rangle}\right)_2, x, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } Xvalue = 3 \wedge Yvalue = 1 \\ \quad \text{result} \leftarrow \text{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, \left(kla\_^{\langle ExN_0\rangle}\right)_1, y, x, \left(kla\_^{\langle ExN_0\rangle}\right)_4, ExN\right] \text{ if } Xvalue = 3 \wedge Yvalue = 2 \\ \quad \text{result} \leftarrow \text{ErrorSurf}\left[\left(kla\_^{\langle ExN_0\rangle}\right)_0, \left(kla\_^{\langle ExN_0\rangle}\right)_1, \left(kla\_^{\langle ExN_0\rangle}\right)_2, x, y, ExN\right] \text{ if } Xvalue = 3 \wedge Yvalue = 4 \\ \text{result} \end{array}$$

$$minX := \begin{array}{l} \text{result} \leftarrow 0 \\ \text{if } Part1 = 1 \\ \quad \text{result} \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_0 \text{ if } Xvalue = 0 \\ \quad \text{result} \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_1 \text{ if } Xvalue = 1 \\ \quad \text{result} \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_2 \text{ if } Xvalue = 2 \\ \quad \text{result} \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_3 \text{ if } Xvalue = 3 \\ \quad \text{result} \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_4 \text{ if } Xvalue = 4 \\ \text{result} \end{array} \qquad (9.1-83)$$

$$maxX := \begin{array}{l} \text{result} \leftarrow 0 \\ \text{if } Part1 = 1 \\ \quad \text{result} \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_0 \text{ if } Xvalue = 0 \\ \quad \text{result} \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_1 \text{ if } Xvalue = 1 \\ \quad \text{result} \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_2 \text{ if } Xvalue = 2 \\ \quad \text{result} \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_3 \text{ if } Xvalue = 3 \\ \quad \text{result} \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0\rangle}\right)_4 \text{ if } Xvalue = 4 \\ \text{result} \end{array} \qquad (9.1-84)$$

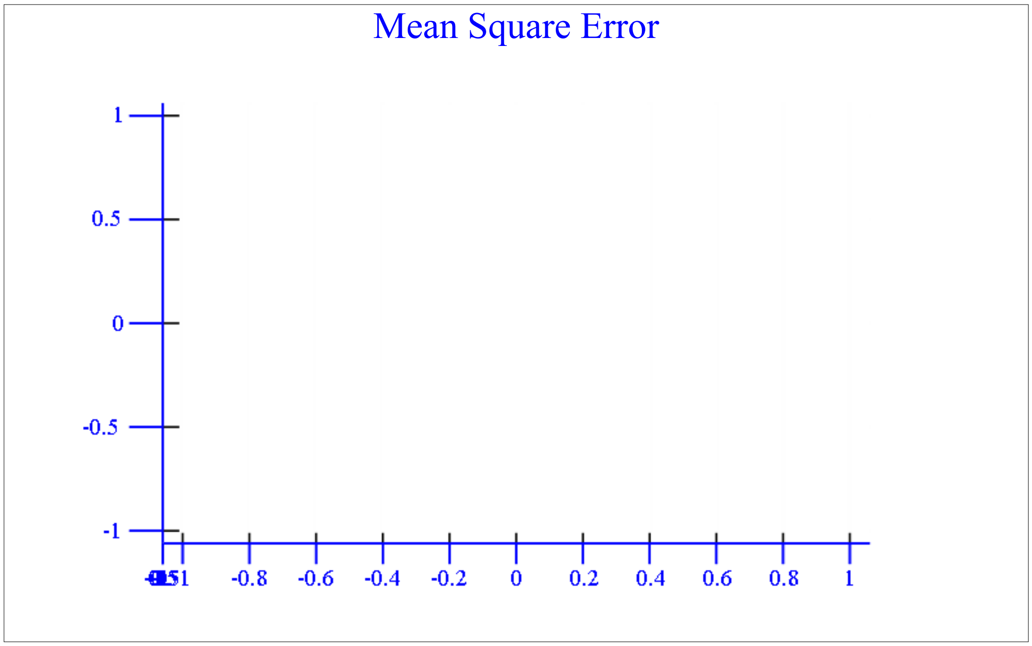
$$
minY := \left| \begin{array}{l} result \leftarrow 0 \\ \text{if } Part1 = 1 \\ \quad \left| \begin{array}{l} result \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_0 \text{ if } Yvalue = 0 \\ result \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_1 \text{ if } Yvalue = 1 \\ result \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_2 \text{ if } Yvalue = 2 \\ result \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_3 \text{ if } Yvalue = 3 \\ result \leftarrow 0.6 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_4 \text{ if } Yvalue = 4 \end{array} \right. \\ result \end{array} \right. \qquad (9.1-85)
$$

$$
maxY := \left| \begin{array}{l} result \leftarrow 0 \\ \text{if } Part1 = 1 \\ \quad \left| \begin{array}{l} result \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_0 \text{ if } Yvalue = 0 \\ result \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_1 \text{ if } Yvalue = 1 \\ result \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_2 \text{ if } Yvalue = 2 \\ result \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_3 \text{ if } Yvalue = 3 \\ result \leftarrow 1.4 \cdot \left(kla\_^{\langle ExN_0 \rangle}\right)_4 \text{ if } Yvalue = 4 \end{array} \right. \\ result \end{array} \right. \qquad (9.1-86)
$$

$$
errorPlot := CreateMesh(errorMesh1, minX, maxX, minY, maxY, 5) \qquad (9.1-87)
$$

Contour Plot Parameters

kla Surface



kla Surface

Figure 9.1-2: Contour plot of mean square error of the observed dissolved oxygen probe reading and the fitted oxygen probe values with respect to the X and Y parameters selected in table 9.1-16.

Kla Fitting

Surface
Runs to Include

Sparge
Runs to Include

(The values can be entered in these boxes similar to the print pages command by using a combination of commas for discrete values or dashes for continuous ranges.)

Weight :=

	0	1	2	3	4	5	6	7	8	9	10	11	12
0	0	0	0	0	0	0	0	0	0	0	0	0	...

Table 9.1-16: The experimental runs used to fit the kla surface aeration polynomial and the kla spargers coefficients on the Van't Riet equation.

Kla Parameter Fitting

The following programs build the vector for which experimental runs to include in determining the function for the sparge and surface klas.

(9.1 – 88)

```
kla_spg_Num := | strCount ← 1
               | prevSymbol ← 0
               | arrayCount ← 0
               | prevCount ← 0
               | while strCount < strlen(eNum_spg)
               |   if substr(eNum_spg, strCount, 1) = " " ∨ substr(eNum_spg, strCount, 1) = "," ∨ substr(eNum_spg, strCount, 1) = "-"
               |     if strCount − prevCount > 0
               |       arrayValue ← str2num(substr(eNum_spg, prevCount, strCount − prevCount))
               |       if prevSymbol = 1
               |         for i ∈ prevValue + 1 .. arrayValue
               |           result_arrayCount ← i
               |           arrayCount ← arrayCount + 1
               |         prevSymbol ← 0
               |       otherwise
               |         result_arrayCount ← arrayValue
               |         arrayCount ← arrayCount + 1
               |     prevCount ← strCount + 1
               |     if substr(eNum_spg, strCount, 1) = "-"
               |       prevSymbol ← 1
               |       prevValue ← arrayValue
               |   strCount ← strCount + 1
               | if strCount − prevCount > 0 ∧ strlen(eNum_spg) > 0
               |   arrayValue ← str2num(substr(eNum_spg, prevCount, strCount − prevCount))
               |   if prevSymbol = 1
               |     for i ∈ prevValue + 1 .. arrayValue
               |       result_arrayCount ← i
               |       arrayCount ← arrayCount + 1
               |     prevSymbol ← 0
               |   otherwise
               |     result_arrayCount ← arrayValue
               |     arrayCount ← arrayCount + 1
               | result
```

(9.1 − 89)

```
kla_surf_Num := | strCount ← 1
                | prevSymbol ← 0
                | arrayCount ← 0
                | prevCount ← 0
                | while strCount < strlen(eNum_surf)
                |   | if substr(eNum_surf, strCount, 1) = " " ∨ substr(eNum_surf, strCount, 1) = "," ∨ substr(eNum_surf, strCount, 1) = "-"
                |   |   | if strCount − prevCount > 0
                |   |   |   | arrayValue ← str2num(substr(eNum_surf, prevCount, strCount − prevCount))
                |   |   |   | if prevSymbol = 1
                |   |   |   |   | for i ∈ prevValue + 1 .. arrayValue
                |   |   |   |   |   | result_arrayCount ← i
                |   |   |   |   |   | arrayCount ← arrayCount + 1
                |   |   |   |   | prevSymbol ← 0
                |   |   |   | otherwise
                |   |   |   |   | result_arrayCount ← arrayValue
                |   |   |   |   | arrayCount ← arrayCount + 1
                |   |   | prevCount ← strCount + 1
                |   |   | if substr(eNum_surf, strCount, 1) = "-"
                |   |   |   | prevSymbol ← 1
                |   |   |   | prevValue ← arrayValue
                |   | strCount ← strCount + 1
                | if strCount − prevCount > 0 ∧ strlen(eNum_surf) > 0
                |   | arrayValue ← str2num(substr(eNum_surf, prevCount, strCount − prevCount))
                |   | if prevSymbol = 1
                |   |   | for i ∈ prevValue + 1 .. arrayValue
                |   |   |   | result_arrayCount ← i
                |   |   |   | arrayCount ← arrayCount + 1
                |   |   | prevSymbol ← 0
                |   | otherwise
                |   |   | result_arrayCount ← arrayValue
                |   |   | arrayCount ← arrayCount + 1
                | result
```

This program will fit a third order polynomial for the kla surface data to volume, agitation frequency, total sparge flow rate, and temperature.

$$\text{MSE\_klasurf}(\text{Surface},\text{Part}) := \qquad (9.1-90)$$

```
count ← 0
SSE ← 0
for exp_num ∈ 0.. rows(Pages) − 1                                   if Part = 1
    Match ← 0
    for i ∈ 0.. rows(kla_surf_Num) − 1
        Match ← 1  if kla_surf_Num_i = exp_num
    Trial ← 1
    if Match = 1
        V_Liq ← (AvgMtx_exp_num,Trial)_0
        T_Liq ← (AvgMtx_exp_num,Trial)_1
        P_HS ← (AvgMtx_exp_num,Trial)_2
        N_rpm ← (AvgMtx_exp_num,Trial)_3
        qair_ovly ← (AvgMtx_exp_num,Trial)_4
        qO2_ovly ← (AvgMtx_exp_num,Trial)_5
        qCO2_ovly ← (AvgMtx_exp_num,Trial)_6
        qN2_ovly ← (AvgMtx_exp_num,Trial)_7
        qair_1 ← (AvgMtx_exp_num,Trial)_8
        qO2_1 ← (AvgMtx_exp_num,Trial)_9
        qCO2_1 ← (AvgMtx_exp_num,Trial)_10
        qN2_1 ← (AvgMtx_exp_num,Trial)_11
        qair_2 ← (AvgMtx_exp_num,Trial)_12
        qO2_2 ← (AvgMtx_exp_num,Trial)_13
        qCO2_2 ← (AvgMtx_exp_num,Trial)_14
        qN2_2 ← (AvgMtx_exp_num,Trial)_15
        qtotalov ← q_total(qair_ovly, qO2_ovly, qCO2_ovly, qN2_ovly)
        qtotal1 ← q_total(qair_1, qO2_1, qCO2_1, qN2_1)
        qtotal2 ← q_total(qair_2, qO2_2, qCO2_2, qN2_2)
        kla_surf ← Surface_0,1 + Surface_1,1·V_Liq ...
                   + Surface_2,1·N_rpm·(min/s) ...
                   + Surface_3,1(qtotal1 + qtotal2) ...
                   + Surface_4,1·(V_Liq·N_rpm·(min/s)) ...
                   + Surface_5,1·[V_Liq·(qtotal1 + qtotal2)] ...
                   + Surface_6,1·[N_rpm·(min/s)·(qtotal1 + qtotal2)] ...
                   + Surface_8,1·(N_rpm·(min/s))^2 ...
```

$$+ \text{Surface}_{9,1} \cdot (\text{qtotal1} + \text{qtotal2})^2 \ldots$$
$$+ \text{Surface}_{7,1} \cdot V_{Liq}^{\;2}$$

$$\text{kla\_surf} \leftarrow \text{kla\_surf} \;\; \text{if} \;\; \text{kla\_surf} < 0$$

$$\text{SSE} \leftarrow \text{SSE} + \text{Weight}_{0,\text{exp\_num}} \cdot \left(\text{kla\_surf} - \text{kla}_{0,\text{exp\_num}}\right)^2$$

$$\text{count} \leftarrow \text{count} + 1$$

$$\text{MSSE} \leftarrow \frac{\text{SSE}}{\text{count}}$$

$$\text{MSSE}$$

$$\text{MSE\_klasurf}(\text{Surface}, \text{Part1}) = \blacksquare \tag{9.1 - 91}$$

$$\text{Surface\_solve} := \text{augment}(\text{submatrix}(\text{Surface}, 0, 9, 1, 1), \text{submatrix}(\text{Surface}, 0, 9, 1, 1)) \tag{9.1 - 92}$$

$$\text{TOL} := 10^{-9} \tag{9.1 - 92.1}$$

$$\text{CTOL} := 10^{-9} \tag{9.1 - 92.2}$$

The numerical solver block for the surface aeration polynomial.

Given

$$\text{MSE\_klasurf}(\text{Surface\_solve}, \text{Part1}) = 0 \tag{9.1 - 93}$$

$$\text{Surface\_solve} := \text{Minerr}(\text{Surface\_solve}) \tag{9.1 - 94}$$

$$\text{Surface} := \text{augment}(\text{submatrix}(\text{Surface}, 0, 9, 0, 0), \text{submatrix}(\text{Surface\_solve}, 0, 9, 1, 1)) \tag{9.1 - 95}$$

This program will fit the sparge kla values to the terms to the Van't Riet equation.

(9.1 – 96)

$$
\begin{aligned}
&\text{MSE\_klaspg}(\text{Sparge},\text{Part}) := \\
&\quad \text{count} \leftarrow 0 \\
&\quad \text{SSE} \leftarrow 0 \\
&\quad \text{if Part} = 1 \\
&\qquad \text{count} \leftarrow 0 \\
&\qquad \text{SSE} \leftarrow 0 \\
&\qquad \text{for exp\_num} \in 0..\text{rows}(\text{Pages}) - 1 \\
&\qquad\quad \text{Match} \leftarrow 0 \\
&\qquad\quad \text{Trial} \leftarrow 1 \\
&\qquad\quad \text{for } i \in 0..\text{rows}(\text{kla\_spg\_Num}) - 1 \\
&\qquad\qquad \text{Match} \leftarrow 1 \text{ if kla\_spg\_Num}_i = \text{exp\_num} \\
&\qquad\quad \text{if Match} = 1 \\
&\qquad\qquad V_{Liq} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_0 \\
&\qquad\qquad T_{Liq} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_1 \\
&\qquad\qquad p_{HS} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_2 \\
&\qquad\qquad N_{rpm} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_3 \\
&\qquad\qquad \text{qair\_ovly} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_4 \\
&\qquad\qquad \text{qO2\_ovly} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_5 \\
&\qquad\qquad \text{qCO2\_ovly} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_6 \\
&\qquad\qquad \text{qN2\_ovly} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_7 \\
&\qquad\qquad \text{qair\_1} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_8 \\
&\qquad\qquad \text{qO2\_1} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_9 \\
&\qquad\qquad \text{qCO2\_1} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_{10} \\
&\qquad\qquad \text{qN2\_1} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_{11} \\
&\qquad\qquad \text{qair\_2} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_{12} \\
&\qquad\qquad \text{qO2\_2} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_{13} \\
&\qquad\qquad \text{qCO2\_2} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_{14} \\
&\qquad\qquad \text{qN2\_2} \leftarrow \left(\text{AvgMtx}_{\text{exp\_num},\text{Trial}}\right)_{15} \\
&\qquad\qquad \text{Power} \leftarrow 0 \cdot W \\
&\qquad\qquad \text{for } i \in 1..\text{rows}(\text{Impeller}) - 1 \\
&\qquad\qquad\quad \text{Power} \leftarrow \text{Power} + \text{Impeller}_{i,0} \cdot \left(\frac{N_{rpm}}{\text{min}}\right)^3 \cdot \left(\text{Impeller}_{i,1} \cdot m\right)^5 \frac{gm}{cm^3} \text{ if Impeller}_{i,2} \cdot L < V_{Liq} \cdot L \\
&\qquad\qquad \text{P\_V} \leftarrow \frac{\text{Power}}{V_{Liq} \cdot L} \cdot \frac{m^3}{W} \text{ if } V_{Liq} \cdot L > 0 \cdot L \\
&\qquad\qquad \text{q\_sup1} \leftarrow \frac{q_{total}(\text{qair\_1},\text{qO2\_1},\text{qCO2\_1},\text{qN2\_1}) \cdot \text{SLPM}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}
\end{aligned}
$$

$$A1 \leftarrow Sparge_{1,0}$$
$$alpha1 \leftarrow Sparge_{1,1}$$
$$beta1 \leftarrow Sparge_{1,2}$$
$$kla1 \leftarrow A1 \cdot P\_V^{alpha1} \cdot q\_sup1^{beta1}$$
$$q\_sup2 \leftarrow \frac{q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) \cdot SLPM}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$
$$A2 \leftarrow Sparge_{2,0}$$
$$alpha2 \leftarrow Sparge_{2,1}$$
$$beta2 \leftarrow Sparge_{2,2}$$
$$kla2 \leftarrow A2 \cdot P\_V^{alpha2} \cdot q\_sup2^{beta2}$$
$$SSE \leftarrow SSE + Weight_{0,exp\_num} \cdot \left(\begin{array}{ll} kla1 - kla\_{1,exp\_num} & \text{if } kla\_{1,exp\_num} = 0 \\ \frac{kla1 - kla\_{1,exp\_num}}{kla\_{1,exp\_num}} & \text{otherwise} \end{array} \right)^2 \ldots$$
$$+ Weight_{0,exp\_num} \cdot \left(\begin{array}{ll} kla2 - kla\_{2,exp\_num} & \text{if } kla\_{2,exp\_num} = 0 \\ \frac{kla2 - kla\_{2,exp\_num}}{kla\_{2,exp\_num}} & \text{otherwise} \end{array} \right)^2$$
$$count \leftarrow count + 1$$
$$MSSE \leftarrow \frac{SSE}{count + 1}$$
$$MSSE$$

$$MSE\_klaspg(Sparge, Part1) = \blacksquare \quad (9.1-97)$$

$$Sparge\_solve := augment\left(submatrix(Sparge,1,1,0,2)^T, submatrix(Sparge,1,2,0,2)^T\right)^T \quad (9.1-98)$$

The numerical solver block for the sparger Van't Reit equation.

Given

$$MSE\_klaspg(Sparge\_solve, Part1) = 0 \quad (9.1-99)$$

$$Sparge\_solve := Minerr(Sparge\_solve) \quad (9.1-100)$$

$$Sparge := augment\left(submatrix(Sparge,0,0,0,2)^T, submatrix(Sparge\_solve,1,2,0,2)^T\right)^T \quad (9.1-101)$$

Kla Parameter Fitting

The coefficients for the surface aeration contribution determined by the least mean square error were determined to be as follows:

Surface = ▮

Table 9.1-17: The fitted coefficients for the surface aeration polynomial.

The mean square error resulting from the surface aeration fitted parameters is as follows:

MSE_klasurf(Surface_solve, Part1) = ▮ (9.1 − 102)

The Van't Riet coefficients for the sparge aeration that minimizes the mean square error are as follows:

Sparge = ▮

Table 9.1-18: The fitted coefficients for the Van't Riet equations for sparge aeration.

The mean square error for the sparge aeration fitted parameters is as follows:

MSE_klaspg(Sparge_solve, Part1) = ▮ (9.1 − 103)

9.2 Part 2. Medium Characterization

The medium characterization studies are performed to establish the empirical parameters for the medium equations. The empirical parameters include the lump sum terms for the acids and bases in the medium, the acid contributions of the feeds, and the proportion of organic acids and bases produced relative to the lactate and ammonia. The medium characterization is performed in three steps. First the basal medium is characterized by establishing the constants for the carbon dioxide, pH equilibrium. Second, the contributions for the feeds and base are determined. Third the contributions of the metabolites correlated to lactate and ammonia are determined.

9.2.1 Basal Medium Characterization

The following section is performed by taking basal medium and sparging a sample with carbon dioxide. The sample is loaded on a blood gas analyzer to determine the dissolved carbon dioxide and pH. While the instrument is processing the injection, the original sample is mixed to promote some carbon dioxide off-gassing. When the blood gas analyzer is available for another injection, the sample is loaded on the instrument. This process of sample injection and sample off-gassing is repeated until a range of dissolved carbon dioxide or pH values are obtained that covers the region of interest.

Utilize Model Part 2

DO NOT Run Part 2
Run Part 2

Table 9.2.1-1: Decision to run 9.2 Part 2 for medium characterization.

pK values

"K1"	-6.352
"K2"	-10.323
"Kh"	-2.59
"K1P"	-2.12
"K2P"	-7.21
"K3P"	-12.67
"K_HEPES"	-7.55
"K_glucose"	-9.41
"K_Lactate"	-3.86
"K1Ala"	-2.34
"K2Ala"	-9.69
"K_Ammonia"	-9.255
"K_Water"	-14
"Ka"	0
"Kb"	0

Table 9.2.1-2: pKa values for the carbonate, phosphates, hepes, lactate, ammonia, alanine, water, and lump sum terms of medium.

Concentrations of Characterized Medium (mol/L)

"CO3"	0
"HCO3"	0
"H2CO3"	0
"PO4"	0
"HPO4"	0
"H2PO4"	0
"H3PO4"	0
"HEPES"	0
"Glucose"	0
"Lactate"	0
"Ammonia"	0
"Ala"	0
"OH"	0
"A_salts"	0
"B_salts"	0
"A_eq"	0

Table 9.2.1-3: Concentration of medium components in mol/L.

Saline	0.011	
Temperature	37	C

Table 9.2.1-4: Medium salinity equivalence and temperature.

Adjustable Coefficients

"B_ammonia"	0
"B_feed"	0
"B_lactate"	0
"B_alanine"	0
"B_base"	0

Table 9.2.1-5: Estimates for terms to be fitted in nonlinear medium model.

Basal Medium First Principle Equations

$$\begin{pmatrix} CO3_i \\ HCO3_i \\ H2CO3_i \\ PO4_i \\ HPO4_i \\ H2PO4_i \\ H3PO4_i \\ HEPES \\ Gluc \\ Lact \\ Amm \\ Ala_i \\ OH_i \\ Salts_A \\ Salts_B \\ A_{eq} \end{pmatrix} := Conc^{\langle 1 \rangle} \qquad (9.2.1-1)$$

$$\begin{pmatrix} K_1 \\ K_2 \\ K_h \\ K_{1P} \\ K_{2P} \\ K_{3P} \\ K_{HEPES} \\ K_{Gluc} \\ K_{lact} \\ K_{1Ala} \\ K_{2Ala} \\ K_{amm} \\ K_w \\ K_a \\ K_b \end{pmatrix} := Kvalues^{\langle 1 \rangle} \qquad (9.2.1-2)$$

$$\begin{pmatrix} B_{amm} \\ B_{feed} \\ B_{lact} \\ B_{ala} \\ B_{base} \end{pmatrix} := Coeff^{\langle 1 \rangle} \qquad (9.2.1-3)$$

$$\mathrm{PO4\_i} := \left(\mathrm{PO4}_i \;\; \mathrm{HPO4}_i \;\; \mathrm{H2PO4}_i \;\; \mathrm{H3PO4}_i\right) \tag{9.2.1 – 4}$$

$$\mathrm{carbonate\_i} := \left(\mathrm{CO3}_i \;\; \mathrm{HCO3}_i \;\; \mathrm{H2CO3}_i\right) \tag{9.2.1 – 5}$$

$$\begin{pmatrix} \mathrm{CO3}_{basis} \\ \mathrm{HCO3}_{basis} \\ \mathrm{H2CO3}_{basis} \\ \mathrm{PO4}_{basis} \\ \mathrm{HPO4}_{basis} \\ \mathrm{H2PO4}_{basis} \\ \mathrm{H3PO4}_{basis} \\ \mathrm{HEPES}_{basis} \\ \mathrm{Gluc}_{basis} \\ \mathrm{Lact} \\ \mathrm{Amm} \\ \mathrm{Ala}_{basis} \\ \mathrm{OH}_{basis} \\ \mathrm{Salts}_{A\_basis} \\ \mathrm{Salts}_{B\_basis} \\ \mathrm{A}_{eq\_basis} \end{pmatrix} := \begin{pmatrix} \mathrm{CO3}_i \\ \mathrm{HCO3}_i \\ \mathrm{H2CO3}_i \\ \mathrm{PO4}_i \\ \mathrm{HPO4}_i \\ \mathrm{H2PO4}_i \\ \mathrm{H3PO4}_i \\ \mathrm{HEPES} \\ \mathrm{Gluc} \\ \mathrm{Lact} \\ \mathrm{Amm} \\ \mathrm{Ala}_i \\ \mathrm{OH}_i \\ \mathrm{Salts}_A \\ \mathrm{Salts}_B \\ \mathrm{A}_{eq} \end{pmatrix} \tag{9.2.1 – 6}$$

$$\mathrm{PO4\_basis} := \mathrm{PO4\_i} \tag{9.2.1 – 7}$$

$$\mathrm{carbonate\_basis} := \mathrm{carbonate\_i} \tag{9.2.1 – 8}$$

$$\mathrm{HEPES\_basis} := \mathrm{HEPES} \tag{8.2.1 – 9}$$

The proton contribution due to the carbonate buffer is calculated as described in equation 3.8.1-21.

$$\begin{aligned} \Phi\_\mathrm{carbonate}(\mathrm{pH}, \mathrm{carbonate\_i}) := {} & \frac{\left[10^{(K_1+K_2)} + 10^{(K_1+K_2+K_h)}\right] - \left(10^{-2\mathrm{pH}} + 10^{K_h}\cdot 10^{-2\mathrm{pH}}\right)}{\left[10^{K_1}\cdot\left(10^{K_2} + 10^{-\mathrm{pH}}\right) + 10^{K_1}\cdot 10^{K_h}\cdot\left(10^{K_2} + 10^{-\mathrm{pH}}\right) + 10^{-2\mathrm{pH}}\cdot\left(1 + 10^{K_h}\right)\right]}\cdot \mathrm{carbonate\_i}_{0,1}\cdot\frac{\mathrm{mol}}{\mathrm{L}} \; \ldots \\ & + \frac{\left(10^{K_1}\cdot 10^{-\mathrm{pH}} + 10^{K_1}\cdot 10^{K_h}\cdot 10^{-\mathrm{pH}} + 2\times 10^{-2\mathrm{pH}} + 2\cdot 10^{K_h}\cdot 10^{-2\mathrm{pH}}\right)}{10^{K_1}\cdot\left(10^{K_2} + 10^{-\mathrm{pH}}\right) + 10^{K_1}\cdot 10^{K_h}\cdot\left(10^{K_2} + 10^{-\mathrm{pH}}\right) + 10^{-2\mathrm{pH}}\cdot\left(1 + 10^{K_h}\right)}\cdot \mathrm{carbonate\_i}_{0,0}\cdot\frac{\mathrm{mol}}{\mathrm{L}} \; \ldots \\ & + \frac{-\left(2\times 10^{K_1}\cdot 10^{K_2} + 2\times 10^{K_1}\cdot 10^{K_2}\cdot 10^{K_h} + 10^{K_1}\cdot 10^{-\mathrm{pH}} + 10^{K_1}\cdot 10^{K_h}\cdot 10^{-\mathrm{pH}}\right)}{\left[10^{K_1}\cdot\left(10^{K_2} + 10^{-\mathrm{pH}}\right) + 10^{K_1}\cdot 10^{K_h}\cdot\left(10^{K_2} + 10^{-\mathrm{pH}}\right) + 10^{-2\mathrm{pH}}\cdot\left(1 + 10^{K_h}\right)\right]}\cdot \mathrm{carbonate\_i}_{0,2}\cdot\frac{\mathrm{mol}}{\mathrm{L}} \end{aligned} \tag{9.2.1 – 10}$$

The proton contribution due to the phosphate buffering is calculated as described in equation 3.8.3-10.

(9.2.1 – 11)

$$
\begin{aligned}
\Delta H\_PO4(pH, PO4\_i) :=\ & \Bigg[\frac{-10^{-pH}}{10^{K_{3P}} + 10^{-pH}} - \frac{2 \times 10^{-2pH}}{10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}} + \frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \ldots \\
& + \Bigg[-3 + \frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} + 2 \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \ldots \\
& + -\frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \Bigg] \cdot \frac{10^{-3pH}}{\left(10^{K_{1P}+K_{2P}+K_{3P}} + 10^{K_{1P}+K_{2P}-pH} + 10^{K_{1P}-2 \cdot pH} + 10^{-3pH}\right)} \Bigg] \cdot PO4\_i_{0,0} \cdot \frac{mol}{L} \ldots \\
& + \Bigg[\frac{10^{K_{3P}}}{10^{K_{3P}} + 10^{-pH}} - \frac{2 \times 10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} + \frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \ldots \\
& + \Bigg[-3 + \frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} + 2 \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \ldots \\
& + -\frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \Bigg] \cdot \frac{10^{-3pH}}{\left(10^{K_{1P}+K_{2P}+K_{3P}} + 10^{K_{1P}+K_{2P}-pH} + 10^{K_{1P}-2 \cdot pH} + 10^{-3pH}\right)} \Bigg] \cdot PO4\_i_{0,1} \cdot \frac{mol}{L} \ldots \\
& + \Bigg[2 \cdot \frac{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH}\right)}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} - \frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} \cdot \frac{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH}\right)}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \ldots \\
& + \Bigg[-3 + \frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} + 2 \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \ldots \\
& + -\frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \Bigg] \cdot \frac{10^{-3pH}}{\left(10^{K_{1P}+K_{2P}+K_{3P}} + 10^{K_{1P}+K_{2P}-pH} + 10^{K_{1P}-2 \cdot pH} + 10^{-3pH}\right)} \Bigg] \cdot PO4\_i_{0,2} \cdot \frac{mol}{L} \ldots \\
& + \Bigg[3 - \frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} - 2 \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \ldots \\
& + \frac{10^{-pH}}{10^{K_{3P}} + 10^{-pH}} \cdot \frac{10^{-2pH}}{\left(10^{K_{2P}+K_{3P}} + 10^{K_{2P}-pH} + 10^{-2pH}\right)} \Bigg] \cdot \frac{\left(10^{K_{1P}+K_{2P}+K_{3P}} + 10^{K_{1P}+K_{2P}-pH} + 10^{K_{1P}-2 \cdot pH}\right)}{\left(10^{K_{1P}+K_{2P}+K_{3P}} + 10^{K_{1P}+K_{2P}-pH} + 10^{K_{1P}-2 \cdot pH} + 10^{-3pH}\right)} \cdot PO4\_i_{0,3} \cdot \frac{mol}{L}
\end{aligned}
$$

The total proton contribution from all the buffers is calculated as follows

(9.2.1 – 12)

$$\Phi\_total\left(pH, carbonate\_i, PO4\_i, HEPES, Salts_A, Salts_B\right) := \Phi\_carbonate(pH, carbonate\_i) + \Delta H\_PO4(pH, PO4\_i) \ldots$$
$$+ \frac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-pH}} \cdot HEPES \cdot \frac{mol}{L} \ldots$$
$$+ \frac{10^{K_a}}{10^{K_a} + 10^{-pH}} \cdot Salts_A \cdot \frac{mol}{L} \ldots$$
$$+ \frac{10^{-pH}}{10^{K_b} + 10^{-pH}} \cdot Salts_B \cdot \frac{mol}{L}$$

The dissolved carbon dioxide in equilibrium with a pH for a basal medium is calculated as described in equation 3.8.6-2 with the lactate, ammonia, alanine and base addition set to zero.

(9.2.1 – 13)

$$pCO2(pH) := \left| \begin{array}{l} K1 \leftarrow \dfrac{\Phi\_carbonate(pH, carbonate\_i)}{\Phi\_carbonate(pH, carbonate\_basis)} \cdot \dfrac{\Phi\_total\left(pH, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_{A\_basis}, Salts_{B\_basis}\right)}{\Phi\_total\left(pH, carbonate\_i, PO4\_i, HEPES, Salts_A, Salts_B\right)} \\ \dfrac{\left[\dfrac{10^{(-2 \cdot pH)}}{\begin{array}{l} 2 \times 10^{K_1+K_2} + 2 \times 10^{\left(K_1+K_2+K_h\right)} \ldots \\ + 10^{\left(K_1-pH\right)} + 10^{\left(K_1+K_h-pH\right)} \end{array}} \cdot \left[\begin{array}{l} HCO3_i \cdot \frac{mol}{L} \ldots \\ + 2CO3_i \cdot \frac{mol}{L} \ldots \\ + K1 \cdot \left(\begin{array}{l} 10^{-pH} \cdot \frac{mole}{L} \ldots \\ + OH_i \cdot \frac{mol}{L} - \frac{10^{K_w}}{10^{-pH}} \cdot \frac{mole}{L} \ldots \\ + -\Delta H\_PO4(pH, PO4\_i) \ldots \\ + -\frac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-pH}} \cdot HEPES \cdot \frac{mol}{L} \ldots \\ + -\frac{10^{K_a}}{10^{K_a} + 10^{-pH}} \cdot Salts_A \cdot \frac{mol}{L} \ldots \\ + \frac{10^{-pH}}{10^{K_b} + 10^{-pH}} \cdot Salts_B \cdot \frac{mol}{L} + -A_{eq} \cdot \frac{mol}{L} \ldots \\ + \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-pH}} \cdot Gluc \cdot \frac{mol}{L} \end{array}\right) \end{array}\right]\right]}{H_{CO2\_cp}\left(str2num(saline), str2num(Temp)\,Celsius + T_{0C}\right)} \end{array}\right.$$

Basal Medium First Principle Equations

Basal Medium with Base Data

"pH"	"pCO2"	"base"

Table 9.2.1-6: Data for pH, pCO2 (mm Hg), and amount of base added (mol/L) for the basal medium.

Data Columns

pHvalue := 0 CO2value := 1 Basevalue := 2

Basal Medium Parameter Fitting

This program calculates the mean sum of square error between the observed dissolved carbon dioxide and the calculated dissolved carbon dioxide.

(9.2.1 − 14)

$$
\begin{aligned}
&MSE(K_a, Salts_A, K_b, Salts_B, A_{eq}, B_b) := \\
&\quad Result \leftarrow 0 \\
&\quad temp \leftarrow 0 \\
&\quad \text{for } i \in 1..rows(Data) - 1 \qquad\qquad \text{if } Part2 = 1 \\
&\qquad carbonate\_i \leftarrow \left(CO3_i + B_b \cdot Data_{i,Basevalue} \quad HCO3_i \quad H2CO3_i\right) \\
&\qquad K1 \leftarrow \frac{\Phi\_carbonate\left(Data_{i,pHvalue}, carbonate\_i\right)}{\Phi\_carbonate\left(Data_{i,pHvalue}, carbonate\_basis\right)} \\
&\qquad K2 \leftarrow \frac{\Phi\_total\left(Data_{i,pHvalue}, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_A, Salts_B\right)}{\Phi\_total\left(Data_{i,pHvalue}, carbonate\_i, PO4\_i, HEPES, Salts_A, Salts_B\right)} \\
&\qquad A1 \leftarrow HCO3_i \cdot \frac{mol}{L} + 2\left(CO3_i + B_b \cdot Data_{i,Basevalue}\right) \cdot \frac{mol}{L} \; ... \\
&\qquad\qquad + K1 \cdot K2 \cdot \Bigg(10^{-Data_{i,pHvalue}} \cdot \frac{mole}{L} \; ... \\
&\qquad\qquad\qquad + OH_i \cdot \frac{mol}{L} \; ... \\
&\qquad\qquad\qquad + -\frac{10^{-14}}{10^{-Data_{i,pHvalue}}} \cdot \frac{mole}{L} \; ... \\
&\qquad\qquad\qquad + -\Delta H\_PO4\left(Data_{i,pHvalue}, PO4\_i\right) \; ... \\
&\qquad\qquad\qquad + -\frac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-Data_{i,pHvalue}}} \cdot HEPES \cdot \frac{mol}{L} \; ... \\
&\qquad\qquad\qquad + -\frac{10^{K_a}}{10^{K_a} + 10^{-Data_{i,pHvalue}}} \cdot Salts_A \cdot \frac{mol}{L} \; ... \\
&\qquad\qquad\qquad + \frac{10^{-Data_{i,pHvalue}}}{10^{K_b} + 10^{-Data_{i,pHvalue}}} \cdot Salts_B \cdot \frac{mol}{L} \; ... \\
&\qquad\qquad\qquad + \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-Data_{i,pHvalue}}} \cdot Gluc \cdot \frac{mol}{L} \; ...
\end{aligned}
$$

$$+ -A_{eq} \cdot \frac{mol}{L}$$

$$B2 \leftarrow H_{CO2\_cp}(\text{str2num(saline)}, \text{str2num(Temp)} \cdot \text{Celsius} + T_{0C})$$

$$\text{temp} \leftarrow \text{temp} \ldots + \left[\text{Data}_{i,\text{CO2value}} \ldots + \left[\frac{-10^{(-2 \cdot \text{Data}_{i,\text{pHvalue}})}}{\left[2 \times 10^{K_1+K_2} \ldots + 2 \times 10^{(K_1+K_2+K_h)} \ldots + 10^{(K_1 - \text{Data}_{i,\text{pHvalue}})} \ldots + 10^{(K_1+K_h-\text{Data}_{i,\text{pHvalue}})} \right]} \cdot A1}{B2 \cdot \text{mmHg}} \right] \right]^2$$

$$\text{count} \leftarrow \text{rows(Data)} - 1$$

$$\text{temp} \leftarrow \frac{\text{temp}}{\text{count}}$$

$$\text{Result} \leftarrow \text{temp}$$

$$\text{Result}$$

$$CTOL := 10^{-20} \qquad TOL := 10^{-20} \qquad (9.2.1-15)$$

$$MSE(K_a, Salts_A, K_b, Salts_B, A_{eq}, B_{base}) = \blacksquare \qquad (9.2.1-16)$$

The solver block solves for the lump sum pKa values and concentrations for the acid and basic salts of the medium.

Given

$$0 = MSE(K_a, Salts_A, K_b, Salts_B, A_{eq}, B_{base}) \qquad (9.2.1-17)$$

$$\begin{pmatrix} K_a \\ K_b \end{pmatrix} > -14 \qquad \begin{pmatrix} K_a \\ K_b \end{pmatrix} < 0$$

$$\begin{pmatrix} Salts_A \\ Salts_B \\ A_{eq} \end{pmatrix} < 0.15 \qquad \begin{pmatrix} Salts_A \\ Salts_B \\ A_{eq} \end{pmatrix} > -0.15$$

$$\begin{pmatrix} K_a \\ Salts_A \\ K_b \\ Salts_B \\ A_{eq} \\ B_{base} \end{pmatrix} := Minerr(K_a, Salts_A, K_b, Salts_B, A_{eq}, B_{base}) \qquad (9.2.1-18)$$

$$\begin{pmatrix} K_a \\ Salts_A \\ K_b \\ Salts_B \\ A_{eq} \\ B_{base} \end{pmatrix} = \blacksquare \qquad (9.2.1-19)$$

$$MSE\left(K_a, Salts_A, K_b, Salts_B, A_{eq}, B_{base}\right) = \blacksquare \qquad (9.2.1-20)$$

This program produces a matrix of the calculated dissolved carbon dioxide with the new fitted parameters with respect to sample order.

(9.2.1 – 21)

$$\begin{array}{l}
pCO2_calc := \\
\quad result \leftarrow 0 \\
\quad \text{for } i \in 1..rows(Data)-1 \\
\quad\quad carbonate_i \leftarrow \left(CO3_i + B_{base}\cdot Data_{i,Basevalue} \quad HCO3_i \quad H2CO3_i\right) \\
\quad\quad K1 \leftarrow \dfrac{\Phi_carbonate\left(Data_{i,pHvalue}, carbonate_i\right)}{\Phi_carbonate\left(Data_{i,pHvalue}, carbonate_basis\right)} \\
\quad\quad K2 \leftarrow \dfrac{\Phi_total\left(Data_{i,pHvalue}, carbonate_basis, PO4_basis, HEPES_basis, Salts_A, Salts_B\right)}{\Phi_total\left(Data_{i,pHvalue}, carbonate_i, PO4_i, HEPES, Salts_A, Salts_B\right)} \\
\quad\quad result_i \leftarrow \left[\dfrac{\dfrac{10^{-2\cdot Data_{i,pHvalue}}}{\left[\begin{array}{l} 2\times 10^{K_1+K_2} \ldots \\ +2\times 10^{\left(K_1+K_2+K_h\right)} \ldots \\ +10^{\left(K_1-Data_{i,pHvalue}\right)} \ldots \\ +10^{\left(K_1+K_h-Data_{i,pHvalue}\right)} \end{array}\right]}\cdot\left[\begin{array}{l} HCO3_i\cdot\dfrac{mol}{L} + 2\left(CO3_i + B_{base}\,Data_{i,Basevalue}\right)\cdot\dfrac{mol}{L} \ldots \\ +K1\cdot K2\cdot\left[\begin{array}{l} 10^{-Data_{i,pHvalue}}\cdot\dfrac{mole}{L} \ldots \\ +OH_i\cdot\dfrac{mol}{L} \ldots \\ +-\dfrac{10^{-14}}{10^{-Data_{i,pHvalue}}}\cdot\dfrac{mole}{L} \ldots \\ +-\Delta H_PO4\left(Data_{i,pHvalue}, PO4_i\right) \ldots \\ +-\dfrac{10^{K_{HEPES}}}{10^{K_{HEPES}}+10^{-Data_{i,pHvalue}}}\cdot HEPES\cdot\dfrac{mol}{L} \ldots \\ +-\dfrac{10^{K_a}}{10^{K_a}+10^{-Data_{i,pHvalue}}}\cdot Salts_A\cdot\dfrac{mol}{L} \ldots \\ +\dfrac{10^{-Data_{i,pHvalue}}}{10^{K_b}+10^{-Data_{i,pHvalue}}}\cdot Salts_B\cdot\dfrac{mol}{L} \ldots \\ +-A_{eq}\cdot\dfrac{mol}{L} \ldots \\ +\dfrac{-10^{K_{Gluc}}}{10^{K_{Gluc}}+10^{-Data_{i,pHvalue}}}\cdot(Gluc)\cdot\dfrac{mol}{L} \end{array}\right] \end{array}\right]}{H_{CO2_cp}\left(str2num(saline), str2num(Temp)\cdot Celsius + T_{0C}\right)\cdot mmHg}\right] \\
\quad result_0 \leftarrow result_1 \\
\quad result
\end{array}$$

The following programs produce a matrix for the observed dissolved carbon dioxide, the observed pH, the residual between the observed and calculated dissolved carbon dioxide as a function of sample order. The last program calculates the overall standard deviation in the dissolved carbon dioxide prediction.

```
pCO2_data := | result ← 0
             | result_0 ← Data_{1,CO2value}
             | for i ∈ 1..rows(Data) − 1
             |   result_i ← Data_{i,CO2value}
             | result
```
(9.2.1 – 22)

```
pH_data := | result ← 0
           | result_0 ← Data_{1,pHvalue}
           | for i ∈ 1..rows(Data) − 1
           |   result_i ← Data_{i,pHvalue}
           | result
```
(9.2.1 – 23)

```
resid := | result ← 0
         | for i ∈ 1..rows(Data) − 1
         |   result_i ← pCO2_data_i − pCO2_calc_i
         | result
```
(9.2.1 – 24)

```
std := | result ← 0
       | for i ∈ 1..rows(Data) − 1
       |   result_i ← pCO2_data_i − pCO2_calc_i
       | Result ← Stdev(result)
```
(9.2.1 – 25)

Basal Medium Parameter Fitting

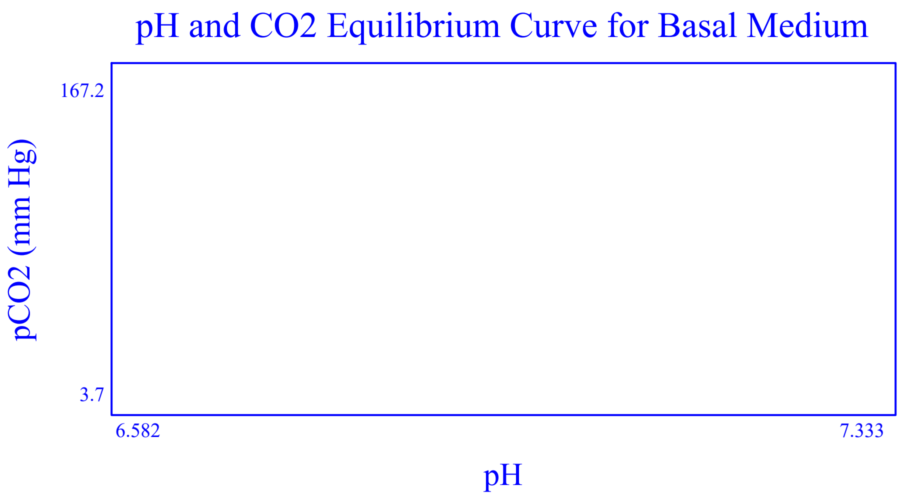


Figure 9.2.1-1: pCO2 (mm Hg) versus pH curve for the observed and fitted curves.

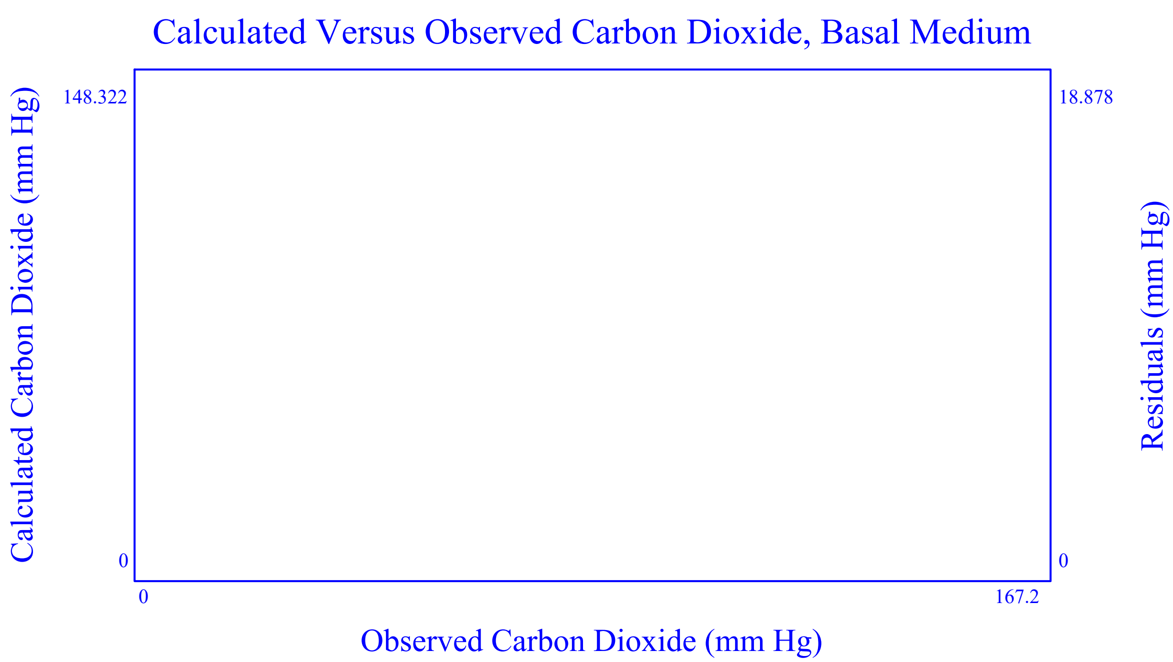


Figure 9.2.1-2: Calculated pCO2 (mm Hg) versus observed pCO2 (mm Hg) and the residuals between the calculated and observed pCO2 (mm Hg) versus observed pCO2 (mm Hg) for the basal medium.

The nonlinear regression of the medium was performed to identify a lump term expression for the contribution that the amino acids and the other weak acids and bases have on the medium dissolved carbon dioxide and pH equilibrium curve. The terms that were adjusted were the acid salts equilibrium constant, the acid salts concentration, the base salts equilibrium concentration, the base salts concentration, and the acid equivalent concentration. If base was added, the coefficient for the molar equivalence of the base was also fitted.

$$\begin{pmatrix} K_a \\ Salts_A \\ K_b \\ Salts_B \\ A_{eq} \\ B_{base} \end{pmatrix} = $$

The standard deviation for the basal medium dissolved carbon dioxide and pH equilibrium model was found to be as follows: $std =$.

Table 9.2.1-7: Fitted values for the pKa, concentration of A salts (mol/L), the pKb, the concentration of the B salts, the concentration of acid equivalence and the base coefficient.

$$Salts_{A\_basis} := Salts_A \tag{9.2.1 – 26}$$

$$Salts_{B\_basis} := Salts_B \tag{9.2.1 – 27}$$

$$A_{eq\_basis} := A_{eq} \tag{9.2.1 – 28}$$

9.2.2 Basal Medium with Feeds Characterization

The following section is performed by taking basal medium with different volumes of feed added and sparging a sample with carbon dioxide. The sample is loaded on a blood gas analyzer to determine the dissolved carbon dioxide and pH. While the instrument is processing the injection, the original sample is mixed to promote some carbon dioxide off-gassing. When the blood gas analyzer is available for another injection, the sample is loaded on the instrument. This process of sample injection and sample off-gassing is repeated until a range of dissolved carbon dioxide or pH values are obtained that covers the region of interest. The value for the volume of feed added should be in volume percent to the initial medium volume.

Basal Medium with Feed Data

"pH"	"pCO2"	"Glucose"	"feed"	"base"

Table 9.2.2-1: Experimental data for basal medium with feed. Data contains the pH, the pCO2 (mm Hg), glucose concentration (g/L), the ratio of total feed volume to medium volume and the base concentration (mol /L).

Data Columns

pHvalue := 0 CO2value := 1 GLUCvalue := 2 FEEDvalue := 3 Basevalue := 4

Medium Feed Parameter Fitting

This program calculates the mean square error of the observed dissolved carbon dioxide versus the calculated dissolved carbon dioxide as a function of the pH and the volume of feed added to the basal medium determined above. This program is only active if the user selects to run section 9.2.

(9.2.2 − 1)

$$MSE\_feed(B_{feed}) := \begin{array}{l} Result \leftarrow 0 \\ temp \leftarrow 0 \\ \text{for } i \in 1 .. rows(feed) - 1 \\ \quad carbonate\_i \leftarrow \left(CO3_i + B_{base} \cdot feed_{i,Basevalue} \;\; HCO3_i \;\; H2CO3_i\right) \\ \quad K1 \leftarrow \dfrac{\Phi\_carbonate\left(feed_{i,pHvalue}, carbonate\_i\right)}{\Phi\_carbonate\left(feed_{i,pHvalue}, carbonate\_basis\right)} \\ \quad K2 \leftarrow \dfrac{\Phi\_total\left(feed_{i,pHvalue}, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_A, Salts_B\right)}{\Phi\_total\left(feed_{i,pHvalue}, carbonate\_i, PO4\_i, HEPES, Salts_A, Salts_B\right)} \\ \quad A1 \leftarrow \dfrac{-10^{\left(-2 \cdot feed_{i,pHvalue}\right)}}{\begin{bmatrix} 2 \times 10^{K_1+K_2} \ldots \\ + 2 \times 10^{\left(K_1+K_2+K_h\right)} \ldots \\ + 10^{\left(K_1 - feed_{i,pHvalue}\right)} \ldots \\ + 10^{\left(K_1+K_h-feed_{i,pHvalue}\right)} \end{bmatrix}} \cdot \left[\begin{array}{l} HCO3_i \cdot \frac{mol}{L} + 2\left(CO3_i + B_{base} \cdot feed_{i,Basevalue}\right) \cdot \frac{mol}{L} \ldots \\ + K1 \cdot K2 \cdot \left(\begin{array}{l} 10^{-feed_{i,pHvalue}} \cdot \frac{mole}{L} \ldots \\ + OH_i \cdot \frac{mol}{L} \ldots \\ + -\frac{10^{-14}}{10^{-feed_{i,pHvalue}}} \cdot \frac{mole}{L} \ldots \\ + -\Delta H\_PO4\left(feed_{i,pHvalue}, PO4\_i\right) \ldots \\ + -\frac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-feed_{i,pHvalue}}} \cdot HEPES \cdot \frac{mol}{L} \ldots \\ + \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-feed_{i,pHvalue}}} \cdot \frac{feed_{i,GLUCvalue}}{180} \cdot \frac{mol}{L} \ldots \\ + -\frac{10^{K_a}}{10^{K_a} + 10^{-feed_{i,pHvalue}}} \cdot Salts_A \cdot \frac{mol}{L} \ldots \\ \quad 10^{-feed_{i,pHvalue}} \qquad mol \end{array} \right. \end{array} \right. \end{array} \quad \text{if } Part2 = 1$$

$$+\frac{}{10^{K_b}+10^{-feed_{i,pHvalue}}}\cdot Salts_B\cdot\frac{}{L}\ \ldots$$
$$+-A_{eq}\cdot\frac{mol}{L}+B_{feed}\cdot feed_{i,FEEDvalue}\cdot\frac{mol}{L}$$

$$temp \leftarrow temp\ \ldots$$
$$+\left[feed_{i,CO2value}\ \cdots + \left(\frac{A1}{H_{CO2\_cp}(str2num(saline), str2num(Temp)\cdot Celsius + T_{0C})\cdot mmHg}\right)\right]^2$$

$$count \leftarrow rows(feed) - 1$$
$$temp \leftarrow \frac{temp}{count}$$
$$Result \leftarrow temp$$
$$Result$$

$$CTOL \equiv 10^{-20} \qquad (9.2.2-2)$$

$$MSE\_feed(B_{feed}) = \blacksquare \qquad (9.2.2-3)$$

The following solver block will determine the feed coefficient that minimizes the mean square error.

Given

$$0 = MSE\_feed(B_{feed}) \qquad (9.2.2-4)$$

$$B_{feed} := Minerr(B_{feed}) \qquad (9.2.2-5)$$

$$B_{feed} = \blacksquare \qquad (9.2.2-6)$$

$$MSE\_feed(B_{feed}) = \blacksquare \qquad (9.2.2-7)$$

This program produces a matrix of the calculated dissolved carbon dioxide with the new fitted parameters with respect to sample order.

(9.2.2 – 8)

$$
\begin{aligned}
\text{pCO2\_calc} := \;& \text{result} \leftarrow 0 \\
& \text{for } i \in 1..\text{rows(feed)} - 1 \\
& \quad \text{carbonate\_i} \leftarrow \left(CO3_i + B_{base} \cdot feed_{i,Basevalue} \quad HCO3_i \quad H2CO3_i \right) \\
& \quad K1 \leftarrow \frac{\Phi\_\text{carbonate}\left(feed_{i,pHvalue}, \text{carbonate\_i}\right)}{\Phi\_\text{carbonate}\left(feed_{i,pHvalue}, \text{carbonate\_basis}\right)} \\
& \quad K2 \leftarrow \frac{\Phi\_\text{total}\left(feed_{i,pHvalue}, \text{carbonate\_basis}, \text{PO4\_basis}, \text{HEPES\_basis}, Salts_A, Salts_B\right)}{\Phi\_\text{total}\left(feed_{i,pHvalue}, \text{carbonate\_i}, \text{PO4\_i}, \text{HEPES}, Salts_A, Salts_B\right)} \\
& \quad \text{result}_i \leftarrow \left[\frac{ \dfrac{10^{-2 \cdot feed_{i,pHvalue}}}{\left[2 \times 10^{K_1+K_2} + 2 \times 10^{\left(K_1+K_2+K_h\right)} + 10^{\left(K_1 - feed_{i,pHvalue}\right)} + 10^{\left(K_1+K_h-feed_{i,pHvalue}\right)} \right]} \cdot \left[\begin{aligned} & HCO3_i \cdot \frac{mol}{L} + 2\left(CO3_i + B_{base} \cdot feed_{i,Basevalue}\right) \cdot \frac{mol}{L} \; ... \\ & + K1 \cdot K2 \cdot \left(\begin{aligned} & 10^{-feed_{i,pHvalue}} \cdot \frac{mole}{L} \; ... \\ & + OH_i \cdot \frac{mol}{L} \; ... \\ & + -\frac{10^{-14}}{10^{-feed_{i,pHvalue}}} \cdot \frac{mole}{L} \; ... \\ & + -\Delta H\_PO4\left(feed_{i,pHvalue}, \text{PO4\_i}\right) \; ... \\ & + -\frac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-feed_{i,pHvalue}}} \cdot HEPES \cdot \frac{mol}{L} \; ... \\ & + -\frac{10^{K_a}}{10^{K_a} + 10^{-feed_{i,pHvalue}}} \cdot Salts_A \cdot \frac{mol}{L} \; ... \\ & + \frac{10^{-feed_{i,pHvalue}}}{10^{K_b} + 10^{-feed_{i,pHvalue}}} \cdot Salts_B \cdot \frac{mol}{L} \; ... \\ & + -A_{eq} \cdot \frac{mol}{L} + B_{feed} \cdot feed_{i,FEEDvalue} \cdot \frac{mol}{L} \; ... \\ & + \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-feed_{i,pHvalue}}} \cdot \frac{feed_{i,GLUCvalue}}{180} \cdot \frac{mol}{L} \end{aligned} \right) \end{aligned} \right] }{ H_{CO2\_cp}\left(\text{str2num(saline)}, \text{str2num(Temp)} \cdot \text{Celsius} + T_{0C}\right) \cdot \text{mmHg} } \right] \\
& \text{result}_0 \leftarrow \text{result}_1 \\
& \text{result}
\end{aligned}
$$

The following programs produce a matrix for the observed dissolved carbon dioxide, the observed pH, the residual between the observed and calculated dissolved carbon dioxide as a function of sample order. The last program calculates the overall standard deviation in the dissolved carbon dioxide prediction.

$$
\begin{aligned}
\text{pCO2\_data} := \;& \text{result} \leftarrow 0 \\
& \text{result}_0 \leftarrow feed_{1,CO2value} \\
& \text{for } i \in 1..\text{rows(feed)} - 1 \\
& \quad \text{result}_i \leftarrow feed_{i,CO2value} \\
& \text{result}
\end{aligned}
\qquad (9.2.2 - 9)
$$

```
pH_data := | result ← 0
           | result_0 ← feed_1,pHvalue
           | for  i ∈ 1 .. rows(feed) − 1
           |   result_i ← feed_i,pHvalue
           | result
```
(9.2.2 – 10)

```
resid := | result ← 0
         | for  i ∈ 1 .. rows(feed) − 1
         |   result_i ← pCO2_data_i − pCO2_calc_i
         | result
```
(9.2.2 – 11)

```
std := | result ← 0
       | for  i ∈ 1 .. rows(feed) − 1
       |   result_i ← pCO2_data_i − pCO2_calc_i
       | Result ← Stdev(result)
```
(9.2.2 – 12)

Medium Feed Parameter Fitting

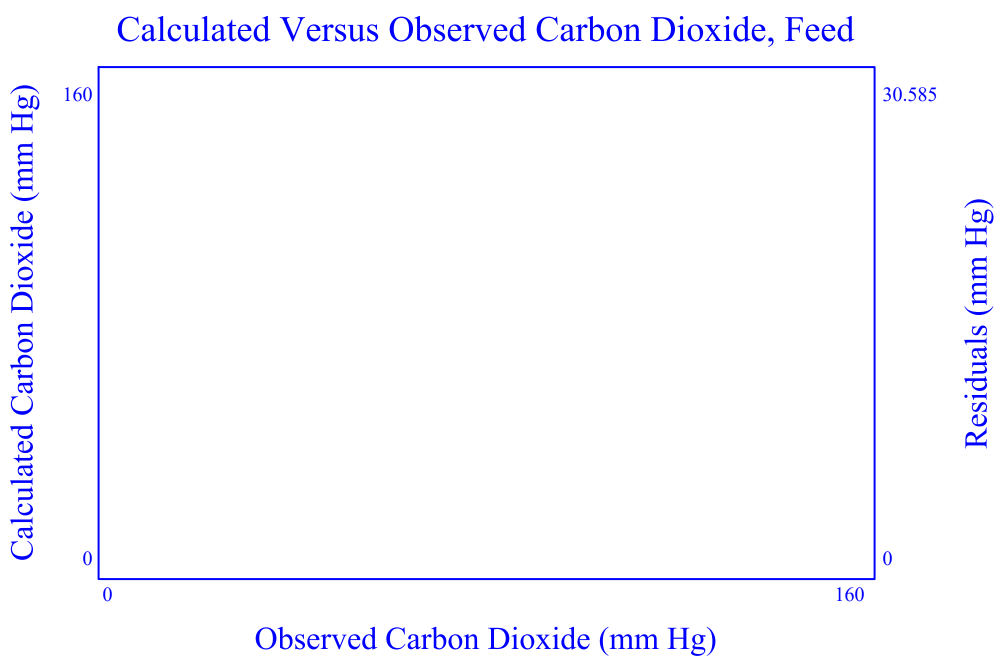


Figure 9.2.2-1: Calculated pCO2 (mm Hg) versus observed pCO2 (mm Hg) and the residuals between the calculated and observed pCO2 (mm Hg) versus observed pCO2 (mm Hg) for the basal medium with feed.

The nonlinear regression of the medium was performed to identify a effective acid contribution of the feeds. The other terms of the model were conserved.

$B_{feed} = \blacksquare$

The standard deviation for the basal medium dissolved carbon dioxide and pH equilibrium model was found to be as follows: $std = \blacksquare$.

9.2.3 Process Characterization

The following section is performed by taking process data where the amount of feed and base addition is known. The lactate, ammonia, dissolved carbon dioxide, and pH are also known. A nonlinear regression is performed to fit the proportional constants for lactate and ammonia. The alanine is fitted to be in proportion to the lactate concentration.

Process Data

SAMPLE NAME"	"DATE TIME"	LAPSED DAYS"	"VCC"	"VIABILITY"	...

Table 9.2.3-1: Experimental data from processes. Data contains the pH, the pCO2 (mm Hg), the ratio of total feed volume to medium volume, glucose concentration (g/L), the base concentration (mol /L), the lactate concentration (g/L), and the ammonia concentration (mmol/L).

Data Columns

TempValue := 6 pHvalue := 7 CO2value := 8 GlucValue := 9

LactValue := 10 AmmValue := 11 FEEDvalue := 12 Basevalue := 13

Metabolites Medium Equilibrium Parameter Fitting

This program calculates the mean square error of the observed dissolved carbon dioxide versus the calculated dissolved carbon dioxide as a function of the pH and the metabolites accumulated from the process data. This program is only active if the user selects to run section 9.2.

(9.2.3 − 1)

$$MSE\_proc(B_{a1}, B_{l1}, B_b, B_{ala}, B_{feed}) := \begin{array}{l} Result \leftarrow 0 \\ temp \leftarrow 0 \\ \text{for } i \in 1..rows(proc)-1 \qquad\qquad \text{if } Part2 = 1 \end{array}$$

$$carbonate\_i \leftarrow \left(CO3_i + B_b proc_{i,Basevalue} \quad HCO3_i \quad H2CO3_i\right)$$

$$K1 \leftarrow \frac{\Phi\_carbonate\left(proc_{i,pHvalue}, carbonate\_i\right)}{\Phi\_carbonate\left(proc_{i,pHvalue}, carbonate\_basis\right)}$$

$$K2 \leftarrow \frac{\Phi\_total\left(proc_{i,pHvalue}, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_A, Salts_B\right)}{\Phi\_total\left(proc_{i,pHvalue}, carbonate\_i, PO4\_i, HEPES, Salts_A, Salts_B\right)}$$

$$\begin{aligned} A1 \leftarrow\ & HCO3_i \cdot \frac{mol}{L} + 2\left(CO3_i + B_b proc_{i,Basevalue}\right)\cdot\frac{mol}{L} \ldots \\ & + K1 \cdot K2 \cdot \Bigg[10^{-proc_{i,pHvalue}} \cdot \frac{mole}{L} + OH_i \cdot \frac{mol}{L} \ldots \\ & + \left[B_{feed}\cdot\left(proc_{i,FEEDvalue}\right)\cdot\frac{mol}{L}\right] \ldots \\ & + -\frac{10^{-14}}{10^{-proc_{i,pHvalue}}}\cdot\frac{mole}{L} \ldots \\ & + -\Delta H\_PO4\left(proc_{i,pHvalue}, PO4\_i\right) \ldots \\ & + -\frac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-proc_{i,pHvalue}}}\cdot HEPES\cdot\frac{mol}{L} \ldots \\ & + -\frac{10^{K_a}}{10^{K_a} + 10^{-proc_{i,pHvalue}}}\cdot Salts_A\cdot\frac{mol}{L} \ldots \\ & + \frac{10^{-proc_{i,pHvalue}}}{10^{K_b} + 10^{-proc_{i,pHvalue}}}\cdot Salts_B\cdot\frac{mol}{L} \ldots \\ & + -A_{eq}\cdot\frac{mol}{L} \ldots \\ & + \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-proc_{i,pHvalue}}}\cdot\left(\frac{proc_{i,GlucValue}}{180}\right)\cdot\frac{mol}{L} \ldots \\ & + \frac{-10^{K_{lact}}}{10^{K_{lact}} + 10^{-proc_{i,pHvalue}}}\cdot\left(\frac{B_{l1}\,proc_{i,LactValue}}{90.08}\right)\cdot\frac{mol}{L} \ldots \\ & + \frac{10^{-proc_{i,pHvalue}}}{10^{K_{amm}} + 10^{-proc_{i,pHvalue}}}\cdot\left(\frac{B_{a1}\,proc_{i,AmmValue}}{1000}\right)\cdot\frac{mol}{L} \ldots \\ & + -\left(\frac{2\cdot 10^{K_{1Ala}+K_{2Ala}} \ldots + 10^{K_{1Ala}-proc_{i,pHvalue}}}{10^{K_{1Ala}+K_{2Ala}} \ldots + 10^{K_{1Ala}-proc_{i,pHvalue}} \ldots + 10^{-2proc_{i,pHvalue}}}\right)\cdot\left(\frac{B_{ala}\,proc_{i,LactValue}}{90.08}\right)\cdot\frac{mol}{L} \Bigg] \end{aligned}$$

$$temp \leftarrow temp \ldots$$

$$+ \left[proc_{i,CO2value} \ldots \left[\frac{-10^{\left(-2\cdot proc_{i,pHvalue}\right)}}{\left[K_1 + K_2 \ldots\right.} \cdot (A1) \right.\right]^2$$

$$+\frac{\begin{bmatrix} 2\times 10^{\cdots} \ldots \\ +2\times 10^{(K_1+K_2+K_h)} \ldots \\ +10^{(K_1-\mathrm{proc}_{i,\mathrm{pHvalue}})} \ldots \\ +10^{(K_1+K_h-\mathrm{proc}_{i,\mathrm{pHvalue}})} \end{bmatrix}}{H_{CO2\_cp}(\mathrm{str2num(saline)},\mathrm{proc}_{i,\mathrm{TempValue}}\cdot\mathrm{Celsius}+T_{0C})\cdot\mathrm{mmHg}}$$

$\mathrm{count} \leftarrow \mathrm{rows(proc)} - 1$

$\mathrm{Result} \leftarrow \dfrac{\mathrm{temp}}{\mathrm{count}}$

Result

$$\mathrm{CTOL} \equiv 10^{-20} \qquad (9.2.3-2)$$

$$\mathrm{MSE\_proc}(B_{amm}, B_{lact}, B_{base}, B_{ala}, B_{feed}) = \blacksquare \qquad (9.2.3-3)$$

The following solver block determines the coefficients for the lactate, ammonia, alanine, and base.

Given

$$0 = \mathrm{MSE\_proc}(B_{amm}, B_{lact}, B_{base}, B_{ala}, B_{feed}) \qquad (9.2.3-4)$$

$$\begin{pmatrix} B_{amm} \\ B_{lact} \\ B_{base} \\ B_{ala} \end{pmatrix} > -1 \qquad \begin{pmatrix} B_{amm} \\ B_{lact} \\ B_{base} \\ B_{ala} \end{pmatrix} < 1$$

Only one of these equations should be active.

$$\begin{pmatrix} B_{amm} \\ B_{lact} \\ B_{base} \\ B_{ala} \end{pmatrix} := \mathrm{Minerr}(B_{amm}, B_{lact}, B_{base}, B_{ala})$$

Disable evaluation if the feed term is solved in section 9.2.2. Activate if the feed terms is not solved here.

(9.2.3 – 5)

$$\begin{pmatrix} B_{amm} \\ B_{lact} \\ B_{base} \\ B_{ala} \\ B_{feed} \end{pmatrix} := \mathrm{Minerr}(B_{amm}, B_{lact}, B_{base}, B_{ala}, B_{feed})$$

Disable evaluation if the feed term is solved in section 9.2.3. Activate if the feed term is solved here.

$$\begin{pmatrix} B_{amm} \\ B_{lact} \\ B_{base} \\ B_{ala} \\ B_{feed} \end{pmatrix} = \blacksquare \qquad (9.2.3-6)$$

$$\mathrm{MSE\_proc}\left(B_{amm}, B_{lact}, B_{base}, B_{ala}, B_{feed}\right) = \blacksquare \qquad (9.2.3-7)$$

This program produces a matrix of the calculated dissolved carbon dioxide with the new fitted parameters with respect to sample order.

(9.2.3 – 8)

$$
\begin{aligned}
&\mathrm{pCO2\_calc} := \;\; \mathrm{result} \leftarrow 0\\
&\quad \text{for } i \in 1..\mathrm{rows(proc)} - 1\\
&\qquad \mathrm{carbonate\_i} \leftarrow \left(CO3_i + B_{base}\,\mathrm{proc}_{i,\mathrm{Basevalue}} \quad HCO3_i \quad H2CO3_i\right)\\
&\qquad G1 \leftarrow \frac{\Phi\_\mathrm{carbonate}\left(\mathrm{proc}_{i,\mathrm{pHvalue}}, \mathrm{carbonate\_i}\right)}{\Phi\_\mathrm{carbonate}\left(\mathrm{proc}_{i,\mathrm{pHvalue}}, \mathrm{carbonate\_basis}\right)}\\
&\qquad G2 \leftarrow \frac{\Phi\_\mathrm{total}\left(\mathrm{proc}_{i,\mathrm{pHvalue}}, \mathrm{carbonate\_basis}, \mathrm{PO4\_basis}, \mathrm{HEPES\_basis}, \mathrm{Salts}_A, \mathrm{Salts}_B\right)}{\Phi\_\mathrm{total}\left(\mathrm{proc}_{i,\mathrm{pHvalue}}, \mathrm{carbonate\_i}, \mathrm{PO4\_i}, \mathrm{HEPES}, \mathrm{Salts}_A, \mathrm{Salts}_B\right)}\\
&\qquad \frac{10^{-2\cdot \mathrm{proc}_{i,\mathrm{pHvalue}}}}{\begin{bmatrix} 2\times 10^{K_1+K_2} \ldots \\ +\,2\times 10^{\left(K_1+K_2+K_h\right)} \ldots \\ +\,10^{\left(K_1-\mathrm{proc}_{i,\mathrm{pHvalue}}\right)} \ldots \\ +\,10^{\left(K_1+K_h-\mathrm{proc}_{i,\mathrm{pHvalue}}\right)} \end{bmatrix}} \cdot \Bigg[HCO3_i\cdot\frac{mol}{L} + 2\left(CO3_i + B_{base}\,\mathrm{proc}_{i,\mathrm{Basevalue}}\right)\cdot\frac{mol}{L} \ldots\\
&\qquad\quad + G1\cdot G2\cdot \Bigg[10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}\cdot\frac{mole}{L} \ldots\\
&\qquad\qquad + OH_i\cdot\frac{mol}{L} \ldots\\
&\qquad\qquad + \left[B_{feed}\cdot\left(\mathrm{proc}_{i,\mathrm{FEEDvalue}}\right)\cdot\frac{mol}{L}\right] \ldots\\
&\qquad\qquad + -\frac{10^{-14}}{10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}\cdot\frac{mole}{L} \ldots\\
&\qquad\qquad + -\Delta H\_\mathrm{PO4}\left(\mathrm{proc}_{i,\mathrm{pHvalue}}, \mathrm{PO4\_i}\right) \ldots\\
&\qquad\qquad + -\frac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}\cdot \mathrm{HEPES}\cdot\frac{mol}{L} \ldots\\
&\qquad\qquad + -\frac{10^{K_a}}{10^{K_a} + 10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}\cdot \mathrm{Salts}_A\cdot\frac{mol}{L} \ldots\\
&\qquad\qquad + \frac{10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}{10^{K_b} + 10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}\cdot \mathrm{Salts}_B\cdot\frac{mol}{L} \ldots\\
&\qquad\qquad + -A_{eq}\cdot\frac{mol}{L} \ldots\\
&\qquad\qquad + \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}\cdot\left(\frac{\mathrm{proc}_{i,\mathrm{GlucValue}}}{180}\right)\cdot\frac{mol}{L} \ldots\\
&\qquad\qquad + \frac{-10^{K_{lact}}}{10^{K_{lact}} + 10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}\cdot\left(\frac{B_{lact}\,\mathrm{proc}_{i,\mathrm{LactValue}}}{90.08}\right)\cdot\frac{mol}{L} \ldots\\
&\qquad\qquad + \frac{10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}{10^{K_{amm}} + 10^{-\mathrm{proc}_{i,\mathrm{pHvalue}}}}\cdot\left(B_{amm}\cdot\frac{\mathrm{proc}_{i,\mathrm{AmmValue}}}{1000}\right)\cdot\frac{mol}{L} \ldots\\
&\qquad\qquad + -\frac{\begin{pmatrix} 2\cdot 10^{K_{1Ala}+K_{2Ala}} \ldots \\ +\,10^{K_{1Ala}-\mathrm{proc}_{i,\mathrm{pHvalue}}} \end{pmatrix}}{\ldots}\cdot\left(\frac{B_{ala}\,\mathrm{proc}_{i,\mathrm{LactValue}}}{90.08}\right)\cdot\frac{mol}{L}
\end{aligned}
$$

$$10^{K_{1Ala}+K_{2Ala}} \ldots + 10^{K_{1Ala}-proc_{i,pHvalue}} \ldots + 10^{-2 proc_{i,pHvalue}}$$

$$result_i \leftarrow \frac{\ldots}{H_{CO2\_cp}\left(str2num(saline), proc_{i,TempValue} \cdot Celsius + T_{0C}\right) \cdot mmHg}$$

$$result_0 \leftarrow result_1$$

$$result$$

The following programs produce a matrix for the observed dissolved carbon dioxide, the observed pH, the residual between the observed and calculated dissolved carbon dioxide as a function of sample order. The last program calculates the overall standard deviation in the dissolved carbon dioxide prediction.

$$pCO2\_data := \begin{array}{|l} result \leftarrow 0 \\ result_0 \leftarrow proc_{1,CO2value} \\ \text{for } i \in 1..rows(proc) - 1 \\ \quad result_i \leftarrow proc_{i,CO2value} \\ result \end{array} \qquad (9.2.3-9)$$

$$pH\_data := \begin{array}{|l} result \leftarrow 0 \\ result_0 \leftarrow proc_{1,pHvalue} \\ \text{for } i \in 1..rows(proc) - 1 \\ \quad result_i \leftarrow proc_{i,pHvalue} \\ result \end{array} \qquad (9.2.3-10)$$

$$resid := \begin{array}{|l} result \leftarrow 0 \\ \text{for } i \in 1..rows(proc) - 1 \\ \quad result_i \leftarrow pCO2\_data_i - pCO2\_calc_i \\ result \end{array} \qquad (9.2.3-11)$$

$$std := MSE\_proc\left(B_{amm}, B_{lact}, B_{base}, B_{ala}, B_{feed}\right)^{0.5} \qquad (9.2.3-12)$$

Metabolites Medium Equilibrium Parameter Fitting

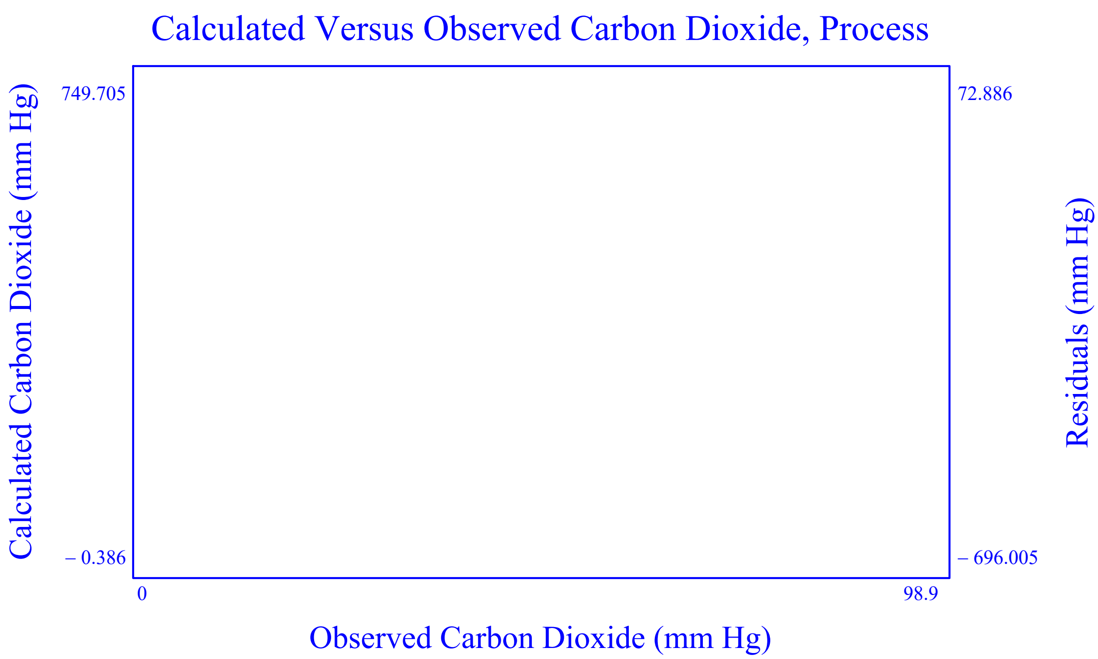


Figure 9.2.3-1: Calculated pCO2 (mm Hg) versus observed pCO2 (mm Hg) and the residuals between the calculated and observed pCO2 (mm Hg) versus observed pCO2 (mm Hg) for the process data.

The nonlinear regression of the medium was performed to identify a effective acid contribution of the feeds. The other terms of the model were conserved.

$$\begin{pmatrix} B_{amm} \\ B_{lact} \\ B_{ala} \\ B_{base} \\ B_{feed} \end{pmatrix} =$$

The standard deviation for the basal medium dissolved carbon dioxide and pH equilibrium model was found to be as follows $std =$.

Table 9.2.3-2: Coefficients for ammonia, lactate, and base from the empirical data.

The overall properties of the medium were determined to be as follows (Table 9.2.3-3):

Medium Properties Matrix

$$\text{MediumProperties} := \left| \begin{array}{l} \text{Labels} \leftarrow \text{augment}\left(\text{Conc}^{\langle 0 \rangle^T}, \text{Kvalues}^{\langle 0 \rangle^T}, \text{Coeff}^{\langle 0 \rangle^T}\right)^T \\ \text{Values} \leftarrow \text{augment}\left[\begin{pmatrix} CO3_i \\ HCO3_i \\ H2CO3_i \\ PO4_i \\ HPO4_i \\ H2PO4_i \\ H3PO4_i \\ HEPES \\ Gluc \\ Lact \\ Amm \\ Ala_i \\ OH_i \\ Salts_A \\ Salts_B \\ A_{eq} \end{pmatrix}^T, \begin{pmatrix} K_1 \\ K_2 \\ K_h \\ K_{1P} \\ K_{2P} \\ K_{3P} \\ K_{HEPES} \\ K_{Gluc} \\ K_{lact} \\ K_{1Ala} \\ K_{2Ala} \\ K_{amm} \\ K_w \\ K_a \\ K_b \end{pmatrix}^T, \begin{pmatrix} B_{amm} \\ B_{feed} \\ B_{lact} \\ B_{ala} \\ B_{base} \end{pmatrix}^T\right]^T \\ \text{result} \leftarrow \text{stack}\left(\text{Labels}^T, \text{Values}^T\right)^T \end{array} \right. \qquad (9.2.3-13)$$

Medium Properties Matrix

MediumProperties = ∎

Table 9.2.3-3: Coefficients for the medium from the empirical data.

9.2.4 Changes to Medium Buffer Design

The following section calculates the impact that changes in the buffer concentrations have on the equilibrium curves. The proposed concentrations of the components are captured in the table below. The values for the original medium and the proposed medium need to be entered to properly calculate the adjustment.

Proposed Concentrations (mol/L)

"CO3"	0
"HCO3"	0
"H2CO3"	0
"PO4"	0
"HPO4"	0
"H2PO4"	0
"H3PO4"	0
"HEPES"	0
"Glucose"	0

"Lactate"	0
"Ammonia"	0
"Ala"	0
"OH"	0

Table 9.2.4-1: Concentrations (mol/L) of buffers for proposed medium.

Proposed Buffer Medium Equilibrium

The values from the proposed concentrations are used to assign values for the buffer concentrations.

$$\begin{pmatrix} CO3_i \\ HCO3_i \\ H2CO3_i \\ PO4_i \\ HPO4_i \\ H2PO4_i \\ H3PO4_i \\ HEPES \\ Gluc \\ Lact \\ Amm \\ Ala_i \\ OH_i \end{pmatrix} := Conc2^{\langle 1 \rangle} \qquad (9.2.4-1)$$

$$PO4\_i := \left(PO4_i \;\; HPO4_i \;\; H2PO4_i \;\; H3PO4_i \right) \qquad (9.2.4-2)$$

$$carbonate\_i := \left(CO3_i \;\; HCO3_i \;\; H2CO3_i \right) \qquad (9.2.4-3)$$

The carbon dioxide and pH equilibrium curve with the new buffer concentrations is calculated by using equation 3.8.6-2.

$$
\text{pCO2\_new} := \left|
\begin{array}{l}
\text{result} \leftarrow 0 \\
\text{for } i \in 650..750 \\
\quad \left|
\begin{array}{l}
G1 \leftarrow \dfrac{\Phi\_\text{carbonate}\left(\dfrac{i}{100}, \text{carbonate\_i}\right)}{\Phi\_\text{carbonate}\left(\dfrac{i}{100}, \text{carbonate\_basis}\right)} \\
G2 \leftarrow \dfrac{\Phi\_\text{total}\left(\dfrac{i}{100}, \text{carbonate\_basis}, \text{PO4\_basis}, \text{HEPES\_basis}, \text{Salts}_A, \text{Salts}_B\right)}{\Phi\_\text{total}\left(\dfrac{i}{100}, \text{carbonate\_i}, \text{PO4\_i}, \text{HEPES}, \text{Salts}_A, \text{Salts}_B\right)} \\
\text{result}_i \leftarrow \dfrac{\left[\dfrac{10^{-2\cdot\frac{i}{100}}}{\left[\begin{array}{l} 2\times 10^{K_1+K_2} \ldots \\ +\, 2\times 10^{\left(K_1+K_2+K_h\right)} \ldots \\ +\, 10^{\left(K_1-\frac{i}{100}\right)} \ldots \\ +\, 10^{\left(K_1+K_h-\frac{i}{100}\right)} \end{array}\right]} \cdot \left[\begin{array}{l} \text{HCO3}_i\cdot\dfrac{mol}{L} + 2\text{CO3}_i\cdot\dfrac{mol}{L} \ldots \\ +\, G1\cdot G2\cdot\left(\begin{array}{l} 10^{-\frac{i}{100}}\cdot\dfrac{mole}{L} \ldots \\ +\, \text{OH}_i\cdot\dfrac{mol}{L} \ldots \\ +\, -\dfrac{10^{-14}}{10^{-\frac{i}{100}}}\cdot\dfrac{mole}{L} \ldots \\ +\, -\Delta\text{H\_PO4}\left(\dfrac{i}{100}, \text{PO4\_i}\right) \ldots \\ +\, -\dfrac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-\frac{i}{100}}}\cdot\text{HEPES}\cdot\dfrac{mol}{L} \ldots \\ +\, -\dfrac{10^{K_a}}{10^{K_a} + 10^{-\frac{i}{100}}}\cdot\text{Salts}_A\cdot\dfrac{mol}{L} \ldots \\ +\, \dfrac{10^{-\frac{i}{100}}}{10^{K_b} + 10^{-\frac{i}{100}}}\cdot\text{Salts}_B\cdot\dfrac{mol}{L} \ldots \\ +\, \dfrac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{\frac{-i}{100}}}\cdot\text{Gluc}\cdot\dfrac{mol}{L} \ldots \\ +\, -A_{eq}\cdot\dfrac{mol}{L} \end{array}\right) \end{array}\right]\right]}{H_{CO2\_cp}\left(\text{str2num(saline)}, \text{str2num(Temp)}\cdot\text{Celsius} + T_{0C}\right)\cdot\text{mmHg}}
\end{array}\right. \\
\text{result}_0 \leftarrow \text{result}_1 \\
\text{result}
\end{array}\right.
\qquad (9.2.4-4)
$$

The calculation for the basis carbon dioxide and pH equilibrium curve is determined by having the ratio of the proton capacity versus basis in equation 3.8.6-2 to be set to one.

$$
pCO2\_basis := \left|
\begin{array}{l}
result \leftarrow 0 \\
\text{for } i \in 650..750 \\
\quad \left|
\begin{array}{l}
K1 \leftarrow 1 \\
K2 \leftarrow 1 \\
result_i \leftarrow \dfrac{
\left[\dfrac{10^{-2\cdot\frac{i}{100}}}{\left[\begin{array}{l} 2\times 10^{K_1+K_2} \ldots \\ +\,2\times 10^{\left(K_1+K_2+K_h\right)} \ldots \\ +\,10^{\left(K_1-\frac{i}{100}\right)} \ldots \\ +\,10^{\left(K_1+K_h-\frac{i}{100}\right)} \end{array}\right]} \cdot
\left[\begin{array}{l}
HCO3_{basis}\cdot\dfrac{mol}{L} + 2CO3_{basis}\cdot\dfrac{mol}{L} \ldots \\
+\,K1\cdot K2\cdot\left(\begin{array}{l}
10^{-\frac{i}{100}}\cdot\dfrac{mole}{L} \ldots \\
+\,OH_{basis}\cdot\dfrac{mol}{L} \ldots \\
+\,-\dfrac{10^{-14}}{10^{-\frac{i}{100}}}\cdot\dfrac{mole}{L} \ldots \\
+\,-\Delta H\_PO4\left(\dfrac{i}{100}, PO4\_basis\right) \ldots \\
+\,-\dfrac{10^{K_{HEPES}}}{10^{K_{HEPES}}+10^{-\frac{i}{100}}}\cdot HEPES_{basis}\cdot\dfrac{mol}{L} \ldots \\
+\,-\dfrac{10^{K_a}}{10^{K_a}+10^{-\frac{i}{100}}}\cdot Salts_A\cdot\dfrac{mol}{L} \ldots \\
+\,\dfrac{10^{-\frac{i}{100}}}{10^{K_b}+10^{-\frac{i}{100}}}\cdot Salts_B\cdot\dfrac{mol}{L} \ldots \\
+\,\dfrac{-10^{K_{Gluc}}}{10^{K_{Gluc}}+10^{\frac{-i}{100}}}\cdot Gluc_{basis}\cdot\dfrac{mol}{L} \ldots \\
+\,-A_{eq}\cdot\dfrac{mol}{L}
\end{array}\right)
\end{array}\right]
\right]
}{H_{CO2\_cp}\left(str2num(saline), str2num(Temp)\cdot Celsius + T_{0C}\right)\cdot mmHg}
\end{array}\right. \\
result_0 \leftarrow result_1 \\
result
\end{array}\right.
\qquad (9.2.4-5)
$$

$$
pH\_values := \left|
\begin{array}{l}
\text{for } i \in 650..750 \\
\quad result_i \leftarrow \dfrac{i}{100} \\
result
\end{array}\right.
\qquad (9.2.4-6)
$$

Proposed Buffer Medium Equilibrium

The carbon dioxide versus pH equilibrium curves are provided in the following plot.

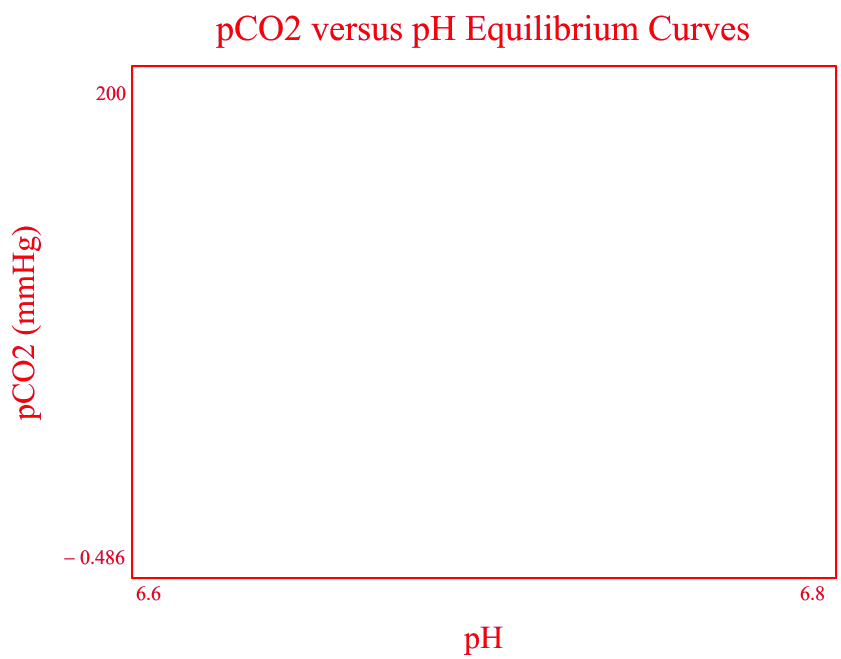


Figure 9.2.4-1: pCO2 (mm Hg) versus pH for the characterized medium and the proposed medium.

9.3 Part 3. Representative Run

A representative run is selected to establish the expected oxygen uptake rate, lactate profile, and ammonia profile. To complete this section, at a minimum the online data (volume, temperature, agitation rate, gas flow rates, dissolved oxygen and pressure) and the offline data (lactate and ammonia) are provided from a run that was conducted in a characterized vessel (known kla). From this information the temporal oxygen uptake rate and metabolite concentrations are estimated.

Utilize Model Part 3

DO NOT Run Part 3
Run Part 3

Table 9.3-1: Decision to run 9.3 Part 3 for capturing the oxygen uptake rate, the lactate, and ammonia profile of a representative run.

BIOREACTOR PARAMETERS

Vessel Volume ______ L

Vessel Diameter ______ m

Impeller Characteristics

"Power Number"	"Diameter"	"Volume"
0	0	0
0	0	0

Number of Spargers

Zero
One
Two

Spargers Gas Transfer Coefficients

"Coefficient"	"alpha"	"beta"
0	0	0
0	0	0

Surface Aeration Gas Transfer Coefficients

"Int"	0
"V"	0
"N"	0
"S"	0
"VN"	0
"VS"	0
"NS"	0
"V2"	0
"N2"	0
"S2"	0

$P_{atm} := 14.7psi$

Table 9.3-2: Properties of the bioreactor used to run the representative run.

Enter the saline equivalence of the representative run.

Saline 0.011

Table 9.3-3: Saline equivalence of medium. Used to determine the oxygen and carbon dioxide gas solubility.

Process Data

Data :=

"Hours"	"Temp"	"Agitation"	"Pressure"	"DO"	...

Table 9.3-3: Process data from a representative run. Column order is run duration (hours), temperature (degrees Celsius), agitation frequency (rpm), back pressure (psig), dissolved oxygen (percent saturation), volume (L), pH, air overlay flow rate (SLPM), oxygen overlay flow rate (SLPM), carbon dioxide overlay flow rate (SLPM), nitrogen overlay flow rate (SLMP), air sparge 1 flow rate (SLPM), oxygen sparge 1 flow rate (SLPM), carbon dioxide sparge 1 flow rate (SLPM), nitrogen sparge 1 flow rate (SLPM), air sparge 2 flow rate (SLPM), oxygen sparge 2 flow rate (SLPM), carbon dioxide sparge 2 flow rate (SLPM), nitrogen sparge 2 flow rate (SLPM), antifoam added (L), feed A added (kg), feed B added (kg), base added (kg), glucose feed (kg), viable cell density (10^5 cells/mL), pCO2 (mm Hg), lactate (g/L), ammonia (mmol/L),

Representative Run Calculation

$$D := \mathrm{str2num}(D)\cdot m \qquad (9.3-1)$$

$$V_T := \mathrm{str2num}(V_T)\cdot L \qquad (9.3-2)$$

The coefficient matrix for equations 3.1-15, 3.1-16, 4.1.2-1, 4.1.2-2, and 4.1.3-2.is rebuilt to account for values that may be different in the gold standard run than were used for characterizing the vessel.

$$\mathrm{Coefficient(Inputs)} := \left| \begin{aligned}
& V_{Liq} \leftarrow Inputs_0 \cdot L \\
& T_{Liq} \leftarrow Inputs_1 \cdot K \\
& p_{HS} \leftarrow Inputs_2 \cdot psi \\
& N_{rpm} \leftarrow Inputs_3 \cdot \frac{1}{min} \\
& qair_ovly \leftarrow Inputs_4 \cdot SLPM \\
& qO2_ovly \leftarrow Inputs_5 \cdot SLPM \\
& qCO2_ovly \leftarrow Inputs_6 \cdot SLPM \\
& qN2_ovly \leftarrow Inputs_7 \cdot SLPM \\
& qair_1 \leftarrow Inputs_8 \cdot SLPM \\
& qO2_1 \leftarrow Inputs_9 \cdot SLPM \\
& qCO2_1 \leftarrow Inputs_{10} \cdot SLPM \\
& qN2_1 \leftarrow Inputs_{11} \cdot SLPM \\
& qair_2 \leftarrow Inputs_{12} \cdot SLPM \\
& qO2_2 \leftarrow Inputs_{13} \cdot SLPM \\
& qCO2_2 \leftarrow Inputs_{14} \cdot SLPM \\
& qN2_2 \leftarrow Inputs_{15} \cdot SLPM \\
& OUR \leftarrow Inputs_{1\ldots} \cdot mmol
\end{aligned} \right. \qquad (9.3-3)$$

$$\cdot \; 16 \quad L\cdot hr$$

$$kla_{surf} \leftarrow Inputs_{17}\cdot\frac{1}{hr}$$

$$kla1 \leftarrow Inputs_{18}\cdot\frac{1}{hr}$$

$$kla2 \leftarrow Inputs_{19}\cdot\frac{1}{hr}$$

$$saline \leftarrow Inputs_{20}$$

$$T_{HS} \leftarrow T_H(T_{Liq})$$

$$V_{HS} \leftarrow V_H(V_T, V_{Liq})$$

$$p_{Liq} \leftarrow p_L(V_{Liq}, D, p_{HS})$$

$$q_{ovly} \leftarrow q_{total}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly) + 0.000000000000000001\cdot SLPM$$

$$q_{spg1} \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.000000000000000001\cdot SLPM$$

$$q_{spg2} \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.000000000000000001\cdot SLPM$$

$$xO2\_ovly \leftarrow x_{O2}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$$

$$xO2spg\_1 \leftarrow x_{O2}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$$

$$xO2spg\_2 \leftarrow x_{O2}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$$

$$xCO2\_ovly \leftarrow x_{CO2}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$$

$$xCO2spg\_1 \leftarrow x_{CO2}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$$

$$xCO2spg\_2 \leftarrow x_{CO2}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$$

$$H_{O2\_} \leftarrow H_{O2\_cp}(saline, T_{Liq})$$

$$H_{CO2\_} \leftarrow H_{CO2\_cp}(saline, T_{Liq})$$

$$result_0 \leftarrow xO2spg\_1\cdot\frac{p_{stp}\cdot q_{spg1}}{V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}\cdot\left[1-\exp\left[-\left(\frac{kla1\cdot H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1}}\right)\right]\right]\cdot\frac{L\cdot hr}{mol}$$

$$result_1 \leftarrow \frac{p_{stp}\cdot q_{spg1}}{H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}\cdot\left[1-\exp\left[-\left(\frac{kla1\cdot H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1}}\right)\right]\right]\cdot hr$$

$$result_2 \leftarrow xO2spg\_2\cdot\frac{p_{stp}\cdot q_{spg2}}{V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}\cdot\left[1-\exp\left[-\left(\frac{kla2\cdot H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg2}}\right)\right]\right]\cdot\frac{L\cdot hr}{mol}$$

$$result_3 \leftarrow \frac{p_{stp}\cdot q_{spg2}}{H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}\cdot\left[1-\exp\left[-\left(\frac{kla2\cdot H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg2}}\right)\right]\right]\cdot hr$$

$$result_4 \leftarrow kla_{surf}\cdot hr$$

$$result_5 \leftarrow p_{HS}\cdot H_{O2\_}\cdot\frac{L}{mol}$$

$$result_6 \leftarrow OUR\cdot\frac{L\cdot hr}{mmol}$$

$$result_7 \leftarrow xO2\_ovly\cdot\frac{q_{ovly}\cdot p_{stp}\cdot T_{HS}}{V_{HS}\cdot T_{q\_ref}\cdot p_{HS}}\cdot hr$$

$$result_8 \leftarrow \frac{1}{H_{O2\_}\cdot p_{Liq}}\cdot\frac{q_{spg1}\cdot p_{stp}\cdot T_{HS}}{V_{HS}\cdot T_{q\_ref}\cdot p_{HS}}\cdot\left[1-\exp\left[-\left(\frac{kla1\cdot H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1}}\right)\right]\right]\cdot\frac{mol}{L}\cdot hr$$

$$result_9 \leftarrow \left(xO2spg\_1\cdot\frac{q_{spg1}\cdot p_{stp}\cdot T_{HS}}{V_{HS}\cdot T_{q\_ref}\cdot p_{HS}}\right)\cdot\exp\left[-\left(\frac{kla1\cdot H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{p_{stp}\cdot q_{spg1}}\right)\right]\cdot hr$$

$$result \leftarrow \left[\frac{1}{\quad}\cdot\frac{q_{spg2}\cdot p_{stp}\cdot T_{HS}}{\quad}\cdot\left[1-\exp\left[-\left(\frac{kla2\cdot H_{O2\_}\cdot p_{Liq}\cdot V_{Liq}\cdot R_{gas}\cdot T_{q\_ref}}{\quad}\right)\right]\right]\right]\cdot\frac{mol}{\quad}\cdot hr$$

$$result_{10} \leftarrow \left[\frac{\quad}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{\quad}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{\quad}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{\quad}{L} \cdot hr$$

$$result_{11} \leftarrow \left(xO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{12} \leftarrow \frac{\left(q_{ovly} + q_{spg1} + q_{spg2}\right) \cdot T_{HS} \cdot p_{stp}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{13} \leftarrow p_{HS} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_{14} \leftarrow \frac{R_{gas} \cdot T_{HS} \cdot V_{Liq}}{p_{HS} \cdot V_{HS}} \cdot \frac{mol}{L}$$

$$result_{15} \leftarrow xCO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{16} \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_{17} \leftarrow xCO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{18} \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$$result_{19} \leftarrow 0.893 \cdot kla_{surf} \cdot hr$$

$$result_{20} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{21} \leftarrow xCO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{22} \leftarrow \frac{1}{H_{CO2\_} \cdot p_{Liq}} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{23} \leftarrow \left(xCO2spg\_1 \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot hr$$

$$result_{24} \leftarrow \left[\frac{1}{H_{CO2\_} \cdot p_{Liq}} \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{25} \leftarrow \left(xCO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{26} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{27} \leftarrow p_{Liq} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_{28} \leftarrow OUR \cdot \frac{L \cdot hr}{mmol}$$

$$result$$

The following program uploads and calculates the process data from the gold standard run. The values as a function of run duration are loaded into matrix M3. This program is only active in section 9.3 is selected.

(9.3 – 4)

$M3(Input) :=$ for $i \in 1..rows(Input) - 1$ if $Part3 = 1$

$eTime \leftarrow Input_{i,0}$

$V_{Liq} \leftarrow Input_{i,5}$

$T_{Liq} \leftarrow Input_{i,1} + \frac{T_{0C}}{K}$

$DO \leftarrow Input_{i,4}$

$p_{HS} \leftarrow Input_{i,3} + \frac{p_{atm}}{psi}$

$N_{rpm} \leftarrow Input_{i,2}$

$qair\_ovly \leftarrow Input_{i,7}$

$qO2\_ovly \leftarrow Input_{i,8}$

$qCO2\_ovly \leftarrow Input_{i,9}$

$qN2\_ovly \leftarrow Input_{i,10}$

$qair\_1 \leftarrow Input_{i,11}$

$qO2\_1 \leftarrow Input_{i,12}$

$qCO2\_1 \leftarrow Input_{i,13}$

$qN2\_1 \leftarrow Input_{i,14}$

$qair\_2 \leftarrow Input_{i,15}$

$qO2\_2 \leftarrow Input_{i,16}$

$qCO2\_2 \leftarrow Input_{i,17}$

$qN2\_2 \leftarrow Input_{i,18}$

$OUR \leftarrow 1$

$qtotalovly \leftarrow q_{total}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$

$qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$

$qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$

$$
\begin{aligned}
kla\_surf \leftarrow\ & Surface_{0,1} + Surface_{1,1} \cdot V_{Liq} \ldots \\
& + Surface_{2,1} \cdot N_{rpm} \cdot \frac{min}{s} \ldots \\
& + Surface_{3,1}(qtotal1 + qtotal2) \ldots \\
& + Surface_{4,1} \cdot \left(V_{Liq} \cdot N_{rpm} \cdot \frac{min}{s}\right) \ldots \\
& + Surface_{5,1} \cdot \left[V_{Liq} \cdot (qtotal1 + qtotal2)\right] \ldots \\
& + Surface_{6,1} \cdot \left[N_{rpm} \cdot \frac{min}{s} \cdot (qtotal1 + qtotal2)\right] \ldots \\
& + Surface_{8,1} \cdot \left(N_{rpm} \cdot \frac{min}{s}\right)^2 \ldots \\
& + Surface_{9,1} \cdot (qtotal1 + qtotal2)^2 \ldots \\
& + Surface_{7,1} \cdot V_{Liq}^{\ 2}
\end{aligned}
$$

$kla\_surf \leftarrow 0$ if $kla\_surf < 0$

$Power \leftarrow 0 \cdot W$

for $k \in 1..rows(Impeller) - 1$

$\text{Power} \leftarrow \text{Power} + \text{Impeller}_{k,0} \cdot \left(\frac{N_{rpm}}{\text{min}}\right)^3 \cdot \left(\text{Impeller}_{k,1} \cdot \text{m}\right)^5 \frac{\text{gm}}{\text{cm}^3} \quad \text{if } \text{Impeller}_{k,2} \cdot L < V_{Liq} \cdot L$

$P\_V \leftarrow \frac{\text{Power}}{V_{Liq} \cdot L} \cdot \frac{\text{m}^3}{\text{W}} \quad \text{if } V_{Liq} \cdot L > 0 \cdot L$

$P\_V \leftarrow 0 \quad \text{otherwise}$

$q\_sup1 \leftarrow \frac{\text{qtotal1} \cdot \text{SLPM}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{\text{s}}{\text{m}}$

$A1 \leftarrow \text{Sparge}_{1,0}$

$\text{alpha1} \leftarrow \text{Sparge}_{1,1}$

$\text{beta1} \leftarrow \text{Sparge}_{1,2}$

$\text{kla1} \leftarrow A1\, P\_V^{\text{alpha1}} \cdot q\_sup1^{\text{beta1}}$

$q\_sup2 \leftarrow \frac{\text{qtotal2} \cdot \text{SLPM}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{\text{s}}{\text{m}}$

$A2 \leftarrow \text{Sparge}_{2,0}$

$\text{alpha2} \leftarrow \text{Sparge}_{2,1}$

$\text{beta2} \leftarrow \text{Sparge}_{2,2}$

$\text{kla2} \leftarrow A2 \cdot P\_V^{\text{alpha2}} \cdot q\_sup2^{\text{beta2}}$

$$\text{Inputs} \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ \text{qair\_ovly} \\ \text{qO2\_ovly} \\ \text{qCO2\_ovly} \\ \text{qN2\_ovly} \\ \text{qair\_1} \\ \text{qO2\_1} \\ \text{qCO2\_1} \\ \text{qN2\_1} \\ \text{qair\_2} \\ \text{qO2\_2} \\ \text{qCO2\_2} \\ \text{qN2\_2} \\ \text{OUR} \\ \text{kla\_surf} \\ \text{kla1} \end{pmatrix}$$

$$\begin{pmatrix} \text{kla2} \\ 0.011 \end{pmatrix}$$

$\text{temp1} \leftarrow \text{kla1}$

$$T1 \leftarrow \text{Coefficient(Inputs)}_0 \cdot \left(\frac{\text{mol}}{\text{L}\cdot\text{hr}}\right) - \frac{\text{DO}}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} \cdot \frac{\text{mol}}{\text{L}} \cdot \text{Coefficient(Inputs)}_1 \cdot \frac{1}{\text{hr}}$$

$$T2 \leftarrow \text{Coefficient(Inputs)}_2 \cdot \left(\frac{\text{mol}}{\text{L}\cdot\text{hr}}\right) - \frac{\text{DO}}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} \cdot \frac{\text{mol}}{\text{L}} \cdot \text{Coefficient(Inputs)}_3 \cdot \frac{1}{\text{hr}}$$

$$T3 \leftarrow \frac{\left(\begin{array}{l} \text{Coefficient(Inputs)}_7 \ldots \\ + \text{Coefficient(Inputs)}_8 \cdot \frac{\text{DO}}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} + \text{Coefficient(Inputs)}_9 \ldots \\ + \text{Coefficient(Inputs)}_{10} \cdot \frac{\text{DO}}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} + \text{Coefficient(Inputs)}_{11} \ldots \\ + \text{Coefficient(Inputs)}_4 \cdot \frac{\text{DO}}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} \cdot \text{Coefficient(Inputs)}_{14} \end{array}\right)}{\begin{array}{l} \text{Coefficient(Inputs)}_{12} \ldots \\ + \text{Coefficient(Inputs)}_{13} \cdot \text{Coefficient(Inputs)}_4 \cdot \text{Coefficient(Inputs)}_{14} \end{array}} \cdot \text{Coefficient(Inputs)}_5 \cdot \frac{\text{mol}}{\text{L}}$$

$$T4 \leftarrow T3 - \frac{\text{DO}}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} \cdot \frac{\text{mol}}{\text{L}}$$

$$T5 \leftarrow \text{Coefficient(Inputs)}_{15} \cdot \frac{\text{mol}}{\text{L}} + \text{Coefficient(Inputs)}_{17} \cdot \frac{\text{mol}}{\text{L}}$$

$$T6 \leftarrow \text{Coefficient(Inputs)}_{19} \cdot \left[\frac{\left(\begin{array}{l} \text{Coefficient(Inputs)}_{21} \ldots \\ + \text{Coefficient(Inputs)}_{23} \ldots \\ + \text{Coefficient(Inputs)}_{25} \end{array}\right) \cdot \left(\text{Coefficient(Inputs)}_{20} \cdot \frac{\text{mol}}{\text{L}}\right)}{\begin{array}{l} \text{Coefficient(Inputs)}_{12} \ldots \\ + \text{Coefficient(Inputs)}_4 \cdot 0.893 \cdot \text{Coefficient(Inputs)}_{26} \cdot \text{Coefficient(Inputs)}_{14} \end{array}}\right]$$

$$T7 \leftarrow \frac{\left(\begin{array}{l} \text{Coefficient(Inputs)}_{22} \ldots \\ + \text{Coefficient(Inputs)}_{24} \ldots \\ + \text{Coefficient(Inputs)}_4 \cdot 0.893 \cdot \text{Coefficient(Inputs)}_{14} \end{array}\right) \cdot \text{Coefficient(Inputs)}_{20}}{\text{Coefficient(Inputs)}_{12} + \text{Coefficient(Inputs)}_4 \cdot 0.893 \cdot \text{Coefficient(Inputs)}_{26} \cdot \text{Coefficient(Inputs)}_{14}} - 1$$

$$\text{OUR1} \leftarrow T1 + T2 \ldots + \frac{\text{Coefficient(Inputs)}_4}{\text{hr}} \cdot T4 - \left| \begin{array}{l} \text{dO2\_dt} \leftarrow \frac{\text{Input}_{i,4} - \text{Input}_{i-1,4}}{\left(\text{Input}_{i,0} - \text{Input}_{i-1,0}\right) \cdot \text{hr}} \cdot \frac{0.21}{100} \cdot \text{Coefficient(Inputs)}_{27} \cdot \frac{\text{mol}}{\text{L}} \quad \text{if } i > 1 \\ \text{dO2\_dt} \leftarrow 0 \quad \text{otherwise} \end{array}\right.$$

$$\text{pCO21} \leftarrow \frac{(T5 + T6 + \text{RQ} \cdot \text{OUR1} \cdot \text{hr})}{H_{\text{CO2\_cp}}\left(0.011, T_{\text{Liq}} \cdot K\right) \cdot \text{mmHg} \cdot \left[\begin{array}{l} \text{Coefficient(Inputs)}_{16} \ldots \\ + \text{Coefficient(Inputs)}_{18} - \text{Coefficient(Inputs)}_{19} \cdot (T7) \end{array}\right]}$$

for $j \in 1..60$

$$\begin{pmatrix} V_{\text{Liq}} \\ T_{\text{Liq}} \\ p_{\text{HS}} \\ N_{\text{rpm}} \\ \text{qair\_ovly} \\ \text{qO2\_ovly} \\ \text{qCO2\_ovly} \end{pmatrix}$$

$$\mathrm{Inputs} \leftarrow \begin{pmatrix} \mathrm{qN2\_ovly} \\ \mathrm{qair\_1} \\ \mathrm{qO2\_1} \\ \mathrm{qCO2\_1} \\ \mathrm{qN2\_1} \\ \mathrm{qair\_2} \\ \mathrm{qO2\_2} \\ \mathrm{qCO2\_2} \\ \mathrm{qN2\_2} \\ \mathrm{OUR} \\ \mathrm{kla\_surf} \\ \mathrm{kla1} \\ \mathrm{kla2} \\ 0.011 \end{pmatrix}$$

$$R1 \leftarrow \mathrm{Coefficient(Inputs)}_0 \cdot \left(\frac{mol}{L \cdot hr}\right) - \frac{DO}{100} \cdot 0.21 \cdot \mathrm{Coefficient(Inputs)}_{27} \cdot \frac{mol}{L} \cdot \mathrm{Coefficient(Inputs)}_1 \cdot \frac{1}{hr}$$

$$R2 \leftarrow \mathrm{Coefficient(Inputs)}_2 \cdot \left(\frac{mol}{L \cdot hr}\right) - \frac{DO}{100} \cdot 0.21 \cdot \mathrm{Coefficient(Inputs)}_{27} \cdot \frac{mol}{L} \cdot \mathrm{Coefficient(Inputs)}_3 \cdot \frac{1}{hr}$$

$$R3 \leftarrow \left[\frac{\begin{pmatrix} \mathrm{Coefficient(Inputs)}_7 \ldots \\ + \mathrm{Coefficient(Inputs)}_8 \cdot \frac{DO}{100} \cdot 0.21 \cdot \mathrm{Coefficient(Inputs)}_{27} + \mathrm{Coefficient(Inputs)}_9 \ldots \\ + \mathrm{Coefficient(Inputs)}_{10} \cdot \frac{DO}{100} \cdot 0.21 \cdot \mathrm{Coefficient(Inputs)}_{27} + \mathrm{Coefficient(Inputs)}_{11} \ldots \\ + \mathrm{Coefficient(Inputs)}_4 \cdot \frac{DO}{100} \cdot 0.21 \cdot \mathrm{Coefficient(Inputs)}_{27} \cdot \mathrm{Coefficient(Inputs)}_{14} \end{pmatrix}}{\mathrm{Coefficient(Inputs)}_{12} + \mathrm{Coefficient(Inputs)}_{13} \cdot \mathrm{Coefficient(Inputs)}_4 \cdot \mathrm{Coefficient(Inputs)}_{14}}\right]$$

$$R4 \leftarrow \left(R3 \cdot \mathrm{Coefficient(Inputs)}_5 \cdot \frac{mol}{L} - \frac{DO}{100} \cdot 0.21 \cdot \mathrm{Coefficient(Inputs)}_{27} \cdot \frac{mol}{L}\right)$$

$$R5 \leftarrow \mathrm{Coefficient(Inputs)}_{15} \cdot \frac{mol}{L} + \mathrm{Coefficient(Inputs)}_{17} \cdot \frac{mol}{L}$$

$$R6 \leftarrow \mathrm{Coefficient(Inputs)}_{19} \cdot \frac{\begin{pmatrix} \mathrm{Coefficient(Inputs)}_{21} \ldots \\ + \mathrm{Coefficient(Inputs)}_{23} \ldots \\ + \mathrm{Coefficient(Inputs)}_{25} \end{pmatrix} \cdot \left(\mathrm{Coefficient(Inputs)}_{20} \cdot \frac{mol}{L}\right)}{\begin{matrix} \mathrm{Coefficient(Inputs)}_{12} \ldots \\ + \mathrm{Coefficient(Inputs)}_4 \cdot 0.893 \cdot \mathrm{Coefficient(Inputs)}_{26} \cdot \mathrm{Coefficient(Inputs)}_{14} \end{matrix}}$$

$$R7 \leftarrow \frac{\begin{pmatrix} \mathrm{Coefficient(Inputs)}_{22} \ldots \\ + \mathrm{Coefficient(Inputs)}_{24} \ldots \\ + \mathrm{Coefficient(Inputs)}_4 \cdot 0.893 \cdot \mathrm{Coefficient(Inputs)}_{14} \end{pmatrix} \cdot \mathrm{Coefficient(Inputs)}_{20}}{\mathrm{Coefficient(Inputs)}_{12} + \mathrm{Coefficient(Inputs)}_4 \cdot 0.893 \cdot \mathrm{Coefficient(Inputs)}_{26} \cdot \mathrm{Coefficient(Inputs)}_{14}} - 1$$

$$OUR2 \leftarrow \Bigg[R1 + R2 \ldots \\ + \frac{\mathrm{Coefficient(Inputs)}_4}{hr} \cdot (R4) \ldots \\ + -1 \cdot \Big| dO2\_dt \leftarrow \frac{\mathrm{Input}_{i,4} - \mathrm{Input}_{i-1,4}}{(\mathrm{Input} \quad \mathrm{Input}}\Big) hr \cdot \frac{0.21}{100} \cdot \mathrm{Coefficient(Inputs)}_{27} \cdot \frac{mol}{L} \quad \text{if } i > 1$$

$dO2\_dt \leftarrow 0$ otherwise

$$pCO22 \leftarrow \frac{(R5 + R6 + RQ\,OUR2 \cdot hr)}{H_{CO2\_cp}(0.011, T_{Liq} \cdot K) \cdot mmHg \cdot \left(Coefficient(Inputs)_{16} \cdots + Coefficient(Inputs)_{18} - Coefficient(Inputs)_{19} \cdot R7\right)}$$

$$kla1 \leftarrow kla1 \cdot \left(1 + 0.1 \cdot \frac{Input_{i,25} - pCO22}{Input_{i,25}}\right)$$

$result_{i,0} \leftarrow eTime$

$result_{i,1} \leftarrow kla1$

$result_{i,2} \leftarrow OUR1 \cdot \frac{L \cdot hr}{mmol}$

$result_{i,3} \leftarrow pCO21$

$result_{i,4} \leftarrow OUR2 \cdot \frac{L \cdot hr}{mmol}$

$result_{i,5} \leftarrow pCO22$

$result_{i,6} \leftarrow Input_{i,25}$

$result_{i,7} \leftarrow Input_{i,24}$

$result_{i,8} \leftarrow temp1$

$result_{i,9} \leftarrow P\_V$

$result_{i,10} \leftarrow kla2$

$result_{i,11} \leftarrow q\_sup2$

$result_{i,12} \leftarrow qtotal2$

for $i \in 0..7$ otherwise

$result_{1,i} \leftarrow 0$

$result_{0,0} \leftarrow$ "Time"

$result_{0,1} \leftarrow$ "kla"

$result_{0,2} \leftarrow$ "OUR"

$result_{0,3} \leftarrow$ "CO2"

$result_{0,4} \leftarrow$ "OUR_fit"

$result_{0,5} \leftarrow$ "CO2_fit"

$result_{0,6} \leftarrow$ "CO2_meas"

$result_{0,7} \leftarrow$ "VCC"

result

Outcome := M3(Data) (9.3 − 5)

The results from the M3 matrix is divided into vectors for the different process parameters.

$$\text{OUR\_calc} := \text{submatrix}\left(\text{Outcome}^{\langle 2\rangle}, 1, \text{rows}(\text{Outcome}) - 1, 0, 0\right) \quad (9.3-6)$$

$$\text{VCC\_meas} := \text{submatrix}\left(\text{Outcome}^{\langle 7\rangle}, 1, \text{rows}(\text{Outcome}) - 1, 0, 0\right) \quad (9.3-7)$$

$$\text{Run\_duration} := \text{submatrix}\left(\text{Outcome}^{\langle 0\rangle}, 1, \text{rows}(\text{Outcome}) - 1, 0, 0\right) \quad (9.3-8)$$

$$\text{OUR\_fit} := \text{submatrix}\left(\text{Outcome}^{\langle 4\rangle}, 1, \text{rows}(\text{Outcome}) - 1, 0, 0\right) \quad (9.3-9)$$

$$\text{CO2\_meas} := \text{submatrix}\left(\text{Outcome}^{\langle 6\rangle}, 4, \text{rows}(\text{Outcome}) - 1, 0, 0\right) \quad (9.3-10)$$

$$\text{CO2\_calc} := \text{submatrix}\left(\text{Outcome}^{\langle 3\rangle}, 4, \text{rows}(\text{Outcome}) - 1, 0, 0\right) \quad (9.3-11)$$

$$\text{lactate} := \frac{\text{submatrix}\left[(\text{Data})^{\langle 26\rangle}, 1, \text{rows}(\text{Data}) - 1, 0, 0\right]}{90.08} \cdot 1000 \quad (9.3-12)$$

$$\text{ammonia} := \text{submatrix}\left[(\text{Data})^{\langle 27\rangle}, 1, \text{rows}(\text{Data}) - 1, 0, 0\right] \quad (9.3-13)$$

Continuous functions for the oxygen uptake rate, the lactate accumulation and ammonia accumulation are determined by fitting to seventh order polynomials.

$$\text{OUR\_proc(elapsed\_time)} := \left| \begin{array}{l} X \leftarrow \text{Run\_duration} \\ Y \leftarrow \text{OUR\_calc} \\ S \leftarrow \text{regress}(X, Y, 6) \\ \text{result} \leftarrow \text{interp}(S, X, Y, \text{elapsed\_time}) \\ \text{result} \leftarrow 0 \ \text{ if } \ \text{result} < 0 \end{array} \right. \quad (9.3-14)$$

$$\text{lact\_proc(elapsed\_time)} := \left| \begin{array}{l} X \leftarrow \text{Run\_duration} \\ Y \leftarrow \text{lactate} \\ S \leftarrow \text{regress}(X, Y, 6) \\ \text{result} \leftarrow \text{interp}(S, X, Y, \text{elapsed\_time}) \end{array} \right. \quad (9.3-15)$$

$$\text{amm\_proc(elapsed\_time)} := \left| \begin{array}{l} X \leftarrow \text{Run\_duration} \\ Y \leftarrow \text{ammonia} \\ S \leftarrow \text{regress}(X, Y, 6) \\ \text{result} \leftarrow \text{interp}(S, X, Y, \text{elapsed\_time}) \end{array} \right. \quad (9.3-16)$$

Representative Run Calculation

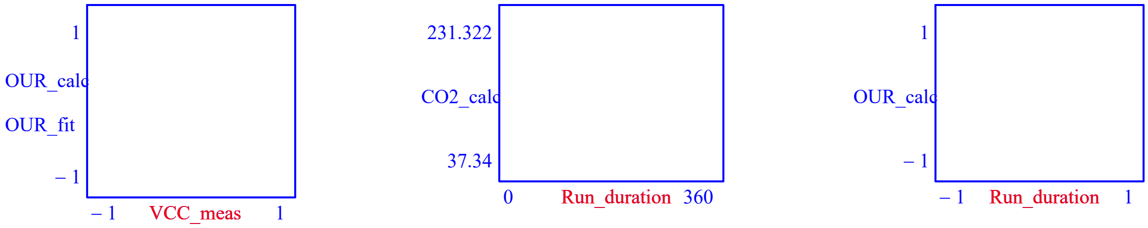


Figure 9.3-1: Plot of calculated oxygen uptake rate versus cell count, comparison of calculated dissolved carbon dioxide versus measured carbon dioxide, and calculated oxygen uptake rate versus run duration.

9.4 Part 4. Process Prediction

The process prediction utilizes the information about the medium, bioreactor, and process performance to determine the resulting pH and dissolved carbon dioxide. The user can specify different gas flow rates, and different cascade controls for dissolved oxygen control. The model will then forecast the gas flows and base addition necessary for pH and dissolved oxygen control. The model will also forecast the resulting dissolved carbon dioxide and pH.

Utilize Model Part 4	DO NOT Run Part 4 Run Part 4

Table 9.4-1: Decision to run 9.4 Part 4 for forecasting the resulting pCO2 accumulation from the dissolved oxygen and pH control in a vessel.

BIOREACTOR PARAMETERS

Vessel Volume [ ] L

Vessel Diameter [ ] m

Impeller Characteristics

"Power Number"	"Diameter"	"Volume"
0	0	0
0	0	0

Number of Spargers

Zero
One
Two

Spargers Gas Transfer Coefficients

"Coefficient"	"alpha"	"beta"
0	0	0
0	0	0

Surface Aeration Gas Transfer Coefficients

"Int"	0
"V"	0
"N"	0
"S"	0
"VN"	0
"VS"	0
"NS"	0
"V2"	0
"N2"	0
"S2"	0

$P_{atm} := 12.2psi$

Table 9.4-2: Properties of the bioreactor used for the proposed run.

Assume that the respiration quotient (the ratio of the moles of carbon dioxide produced per mole of oxygen consumed) is equal to 1.

$$RQ := 1 \qquad (9.4-1)$$

Enter the saline equivalence of the representative run.

Saline [0.011]

Table 9.4-3: Saline equivalence of medium. Used to determine the oxygen and carbon dioxide gas solubility.

Operating Parameters

Dissolved Oxygen Control Cascade

The sequence of controlled parameters to maintain the dissolved oxygen set point.

- Air Flow
- Agitation
- O2 Enrichment

- None
- Air Flow
- Agitation

- None
- Air Flow
- Agitation

- None
- Air Flow
- Agitation

Table 9.4-4: Select the cascade order for dissolved oxygen control. The top most selection option is the first controllable parameter. When that has hit its upper limit, then the second selected parameter would be adjusted to maintain the dissolved oxygen control. The bottom most selection option is the active control parameter after the middle parameter has hit its upper limit.

Cascade Matrix Build

The following program converts the selections from the above list boxes to a vector called CASCADE with six string instances. The first instance is the string "Empty", and the last instance is also the string "Empty." The other instances correspond to values selected in the above list box. The CASCADE vector is used to determine which is the controlled parameter. If the current controlled parameter is at the maximum output, and the dissolved oxygen is below set point, then the program will step through to the next array instance and change which is the controlling parameter for dissolved oxygen. If the value of the next instance is "Empty", then the program will not change the controlling parameter.

```
CASCADE :=  result_0 ← "Empty"                                              (9.4 – 2)
            result_1 ←  value ← "None"  if Cas1 = 0
                        value ← "Air"  if Cas1 = 1
                        value ← "Agitation"  if Cas1 = 2
                        value ← "O2 Enrichment"  if Cas1 = 3
                        value ← "Pressure"  if Cas1 = 4
                        value ← "Proportional"  if Cas1 = 5
                        value ← "Ratio Control"  if Cas1 = 6
                        value
            result_2 ←  value ← "None"  if Cas2 = 0
                        value ← "Air"  if Cas2 = 1
                        value ← "Agitation"  if Cas2 = 2
                        value ← "O2 Enrichment"  if Cas2 = 3
                        value ← "Pressure"  if Cas2 = 4
                        value ← "Proportional"  if Cas2 = 5
                        value ← "Ratio Control"  if Cas2 = 6
                        value
            result_3 ←  value ← "None"  if Cas3 = 0
                        value ← "Air"  if Cas3 = 1
                        value ← "Agitation"  if Cas3 = 2
                        value ← "O2 Enrichment"  if Cas3 = 3
                        value ← "Pressure"  if Cas3 = 4
                        value ← "Proportional"  if Cas3 = 5
                        value ← "Ratio Control"  if Cas3 = 6
                        value
            result_4 ←  value ← "None"  if Cas4 = 0
                        value ← "Air"  if Cas4 = 1
                        value ← "Agitation"  if Cas4 = 2
                        value ← "O2 Enrichment"  if Cas4 = 3
                        value ← "Pressure"  if Cas4 = 4
                        value ← "Proportional"  if Cas4 = 5
                        value ← "Ratio Control"  if Cas4 = 6
                        value
            result_5 ← "Empty"
            result
```

Cascade Matrix Build

DO Control

Overlay DO Control
Sparge 1 DO Control
Sparge 2 DO Control

Table 9.4-5: Select the gas stream used for dissolved oxygen control

Total Flow [1] SLPM

Table 9.4-6: Specify the total flow rate that would be used for when one of the cascade controls are specified as ratio control.

Proportional Constant (Air to O_2) [0]

Table 9.4-7: Specify the proportion of air to oxygen (0 to 1) that would be used when one of the cascade controls are specified as proportional control.

O_2 Split Ratio (Between Spargers) [100] %

Table 9.4-8: Specify the spilt of the oxygen flow between the primary (selected in Table 9.4-5) and secondary sparger.

CO_2 Addition for pH Control

- Overlay CO2 Addition
- Sparge 1 CO2 Addition
- Sparge 2 CO2 Addition

Min CO2 Addition (percent of total) [0] %

Table 9.4-9: Select the gas stream used for carbon dioxide addition for pH control

CO_2 Split Ratio [100] %

Table 9.4-10: Specify the spilt of the carbon dioxide flow between the primary (selected in Table 8.4-9) and secondary sparger.

Base Concentration [1] mol/L

Table 9.4-11: The concentration of carbonate used for pH control.

Air Sparge For pCO2 Control

- Off
- Sparge 1 pH Variant
- Sparge 2 pH Variant

Tuning Parameters qAir = ([ ] + [ ] x (pH_setpoint - pH)) x qO2

Maximum Air Flow		SLPM
Switch Over Time		hours from inoculation
Manual Air Flow After Switch		SLPM

Table 9.4-12: Automated pCO2 control algorithm implemented at the DSC and FMC. The air flow is directly proportionate to the oxygen flow. The air flow decreases as the culture becomes more basic to allow carbon dioxide to accumulate. The air flow increases as the culture becomes more acidic to remove carbon dioxide. The air flow will increase until the maximum air flow is reach (entered above). In the manufacturing production records, there may be instructions to turn the pH variant control off and set the air to a manual set point. This is modeled with the switch over time and manual air flow after switch fields above.

""	"Min"	"Actionable Min"	"Max"	"Cascade"	"Units"
"Volume"	""	""	""	""	"L"
"Pressure"	""	""	""	""	"psig"
"Agitation"	""	""	""	""	"rpm"
"Air Ovly"	""	""	""	""	"SLPM"
"O2 Ovly"	""	""	""	""	"SLPM"
"CO2 Ovly"	""	""	""	""	"SLPM"
"N2 Ovly"	""	""	""	""	"SLPM"
"Air Spg 1"	""	""	""	""	"SLPM"
"O2 Spg 1"	""	""	""	""	"SLPM"
"CO2 Spg 1"	""	""	""	""	"SLPM"
"N2 Spg 1"	""	""	""	""	"SLPM"
"Air Spg 2"	""	""	""	""	"SLPM"
"O2 Spg 2"	""	""	""	""	"SLPM"
"CO2 Spg 2"	""	""	""	""	"SLPM"
"N2 Spg 2"	""	""	""	""	"SLPM"
"Base"	""	""	""	""	...

Table 9.4-13: The minimum and maximum limits for the bioreactor control parameters. These values would be used to determine when a parameter would trigger a cascade or can longer be increased by the controller.

Proposed Process Conditions

"Process Time"	"Hours"					
"V"	"L"					
"Temp"	"C"					
"DO"	"%"					
"Pressure"	"psig"					
"Agitation"	"rpm"					
"pH Lower"	"pH units"					
"pH Upper"	"pH units"					
"Air Ovly"	"SLPM"					
"O2 Ovly"	"SLPM"					
"CO2 Ovly"	"SLPM"					
"N2 Ovly"	"SLPM"					
"Air Spg 1"	"SLPM"					
"O2 Spg 1"	"SLPM"					
"CO2 Spg 1"	"SLPM"					
"N2 Spg 1"	"SLPM"					
"Air Spg 2"	"SLPM"					
"O2 Spg 2"	"SLPM"					
"CO2 Spg 2"	"SLPM"					
"N2 Spg 2"	"SLPM"					
"Feed 1"	"% of Vi"					
"Feed 2"	"% of Vi"					
"Antifoam"	"L"					

Table 9.4-14: The proposed process conditions for the cell culture run. The expected oxygen uptake rate, lactate production, and ammonia production are obtained from section 9.3. (Note that section 9.3 needs to be active for section 9.4 to perform appropriate calculations.)

Enter the duration of the model (not to exceed the representative run data) and the time interval for the numeric calculations. Note that the more frequent the time interval, the more capable the forecast, but the longer the calculation time. Thirty minute intervals is typically a good compromise.

Model End Time		hours
Model Time Interval		minutes

Table 9.4-15: Duration and time interval to use for numeric calculations.

Medium pKa values

"K1"	-6.352
"K2"	-10.323
"Kh"	-2.59
"K1P"	-2.12
"K2P"	-7.21
"K3P"	-12.67
"K_HEPES"	-7.55
"K_glucose"	-9.41
"K_Lactate"	-3.86
"K1Ala"	-2.34
"K2Ala"	-9.69
"K_Ammonia"	-9.255
"K_Water"	-14
"Ka"	0
"Kb"	0

Table 9.4-16: pKa values for medium (use the basis medium values determined by using section 9.2).

Basis Medium Basal Concentration (mol/L)

"CO3"	0
"HCO3"	0
"H2CO3"	0
"PO4"	0
"HPO4"	0
"H2PO4"	0
"H3PO4"	0
"HEPES"	0
"Glucose"	0
"Lactate"	0
"Ammonia"	0
"Ala"	0
"OH"	0
"A_salts"	0
"B_salts"	0
"A_eq"	0

Table 9.4-17: Concentration of basal medium components for the basis medium (use the basis medium values determined by using section 9.2).

Proposed Concentrations (mol/L)

"CO3"	0
"HCO3"	0
"H2CO3"	0
"PO4"	0
"HPO4"	0
"H2PO4"	0
"H3PO4"	0
"HEPES"	0
"Glucose"	0
"Lactate"	0
"Ammonia"	0
"Ala"	0
"OH"	0
"A_salts"	0
"B_salts"	0
"A_eq"	0

Table 9.4-18: Concentration of basal medium components to be used in the model forecast. If this is the same medium as determined in the basis medium, then re-enter the same values in

Table 9.2.1-3, otherwise enter the values for the medium to be used.

Adjustable Coefficients

"B_ammonia"	0
"B_base"	0
"B_feed"	0
"B_lactate"	0
"B_ala"	0

Table 9.4-19: Fitted terms from the nonlinear medium model.

	Minimum	Maximum	
Air Flow for Contour Plot	0		SLPM
Oxygen Flow for Contour Plot	0		SLPM

Table 9.4-20: Minimum and maximum flow rates to use in contour plots of Figure 9.4-1. If these cells are blank, it will use the limits provided in Table 9.4-13 for the gas stream used for dissolved oxygen control.

Air Flow for Source for Contour Plot

Overlay
Sparge 1
Sparge 2

Table 9.4-21: Selected gas stream with variable air flow in contour plots.

Processing Time 70 hours

Table 9.4-22: Processing time to use for contour plots (Figure 9.4-1). This will apply the gas flow rates predicted at the entered time to provide how the ability to support oxygen uptake rate and the resulting accumulated carbon dioxide change with air and oxygen flow rates.

Calculation of OUR and pCO2 contour plots

$$Total := str2num(Total)\cdot SLPM \quad (9.4-1)$$

$$CO2split := \frac{str2num(CO2split)}{100} \quad (9.4-2)$$

$$Base\_Conc := str2num(Base\_Conc)\cdot\frac{mol}{L} \quad (9.4-3)$$

$$Proportional := str2num(Proportional) \quad (9.4-4)$$

$$DOsplit := \frac{str2num(DOsplit)}{100} \quad (9.4-5)$$

$$\mathrm{Input} := \left|\begin{array}{l} i \leftarrow 2 \\ \mathrm{eTime} \leftarrow 0 \;\; \text{if } \; \mathrm{strlen(ProcessTime)} = 0 \\ \mathrm{eTime} \leftarrow \mathrm{str2num(ProcessTime)} \;\; \text{otherwise} \\ \text{while } \; \mathrm{Proposed}_{0,i} < (\mathrm{eTime} \wedge i < \mathrm{cols(Proposed)} - 1) \\ \quad i \leftarrow i + 1 \\ i \leftarrow i - 1 \;\; \text{if } \; i > \mathrm{cols(Proposed)} - 1 \\ i \leftarrow i - 1 \;\; \text{if } \; \mathrm{Proposed}_{0,i} > \mathrm{eTime} \\ \mathrm{submatrix(Proposed}, 1, 19, i, i) \end{array}\right. \qquad (9.4-6)$$

$$D := \mathrm{str2num}(D) \cdot m \qquad (9.4-7)$$

$$V_0 := \mathrm{Proposed}_{1,2} \cdot L \qquad (9.4-8)$$

$$V_T := \mathrm{str2num}(V_T) \cdot L \qquad (9.4-9)$$

$$V_H(V_T, V_L) := V_T - V_L \qquad (9.4-10)$$

The coefficient matrix for equations 3.1-15, 3.1-16, 4.1.2-1, 4.1.2-2, and 4.1.3-2.is rebuilt to account for values that may be different in the proposed vessel than were used for the reference run.

$$\mathrm{Coefficient(Inputs)} := \left|\begin{array}{l} V_{Liq} \leftarrow \mathrm{Inputs}_0 \cdot L \\ T_{Liq} \leftarrow \mathrm{Inputs}_1 \cdot K \\ P_{HS} \leftarrow \mathrm{Inputs}_2 \cdot \mathrm{psi} \\ N_{rpm} \leftarrow \mathrm{Inputs}_3 \cdot \dfrac{1}{\mathrm{min}} \\ \mathrm{qair\_ovly} \leftarrow \mathrm{Inputs}_4 \cdot \mathrm{SLPM} \\ \mathrm{qO2\_ovly} \leftarrow \mathrm{Inputs}_5 \cdot \mathrm{SLPM} \\ \mathrm{qCO2\_ovly} \leftarrow \mathrm{Input}_6 \cdot \mathrm{SLPM} \\ \mathrm{qN2\_ovly} \leftarrow \mathrm{Inputs}_7 \cdot \mathrm{SLPM} \\ \mathrm{qair\_1} \leftarrow \mathrm{Inputs}_8 \cdot \mathrm{SLPM} \\ \mathrm{qO2\_1} \leftarrow \mathrm{Inputs}_9 \cdot \mathrm{SLPM} \\ \mathrm{qCO2\_1} \leftarrow \mathrm{Inputs}_{10} \cdot \mathrm{SLPM} \\ \mathrm{qN2\_1} \leftarrow \mathrm{Inputs}_{11} \cdot \mathrm{SLPM} \\ \mathrm{qair\_2} \leftarrow \mathrm{Inputs}_{12} \cdot \mathrm{SLPM} \\ \mathrm{qO2\_2} \leftarrow \mathrm{Inputs}_{13} \cdot \mathrm{SLPM} \\ \mathrm{qCO2\_2} \leftarrow \mathrm{Inputs}_{14} \cdot \mathrm{SLPM} \\ \mathrm{qN2\_2} \leftarrow \mathrm{Inputs}_{15} \cdot \mathrm{SLPM} \\ \mathrm{OUR} \leftarrow \mathrm{Inputs}_{16} \cdot \dfrac{\mathrm{mmol}}{L \cdot \mathrm{hr}} \end{array}\right. \qquad (9.4-11)$$

$kla_{surf} \leftarrow Inputs_{17} \cdot \frac{1}{hr}$

$kla1 \leftarrow Inputs_{18} \cdot \frac{1}{hr}$

$kla2 \leftarrow Inputs_{19} \cdot \frac{1}{hr}$

$saline \leftarrow Inputs_{20}$

$T_{HS} \leftarrow T_H(T_{Liq})$

$V_{HS} \leftarrow V_H(V_T, V_{Liq})$

$p_{Liq} \leftarrow p_L(V_{Liq}, D, p_{HS})$

$q_{ovly} \leftarrow q_{total}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly) + 0.00000000000000001 \cdot SLPM$

$q_{spg1} \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.00000000000000001 \cdot SLPM$

$q_{spg2} \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.000000000000000001 \cdot SLPM$

$xO2\_ovly \leftarrow x_{O2}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$

$xO2spg\_1 \leftarrow x_{O2}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$

$xO2spg\_2 \leftarrow x_{O2}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$

$xCO2\_ovly \leftarrow x_{CO2}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$

$xCO2spg\_1 \leftarrow x_{CO2}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$

$xCO2spg\_2 \leftarrow x_{CO2}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$

$H_{O2\_} \leftarrow H_{O2\_cp}(saline, T_{Liq})$

$H_{CO2\_} \leftarrow H_{CO2\_cp}(saline, T_{Liq})$

$$result_0 \leftarrow xO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_1 \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_2 \leftarrow xO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_3 \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$result_4 \leftarrow kla_{surf} \cdot hr$

$result_5 \leftarrow p_{HS} \cdot H_{O2\_} \cdot \frac{L}{mol}$

$result_6 \leftarrow OUR \cdot \frac{L \cdot hr}{mmol}$

$$result_7 \leftarrow xO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_8 \leftarrow \frac{1}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_9 \leftarrow \left(xO2spg\_1 \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot hr$$

$$result_{10} \leftarrow \left[\frac{1}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{11} \leftarrow \left(xO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{12} \leftarrow \frac{\left(q_{ovly} + q_{spg1} + q_{spg2}\right) \cdot T_{HS} \cdot p_{stp}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{13} \leftarrow p_{HS} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_{14} \leftarrow \frac{R_{gas} \cdot T_{HS} \cdot V_{Liq}}{p_{HS} \cdot V_{HS}} \cdot \frac{mol}{L}$$

$$result_{15} \leftarrow xCO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{16} \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_{17} \leftarrow xCO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{18} \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$$result_{19} \leftarrow 0.893 \cdot kla_{surf} \cdot hr$$

$$result_{20} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{21} \leftarrow xCO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{22} \leftarrow \frac{1}{H_{CO2\_} \cdot p_{Liq}} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{23} \leftarrow \left(xCO2spg\_1 \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot hr$$

$$result_{24} \leftarrow \left[\frac{1}{H_{CO2\_} \cdot p_{Liq}} \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{25} \leftarrow \left(xCO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{26} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{27} \leftarrow p_{Liq} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_{28} \leftarrow OUR \cdot \frac{L \cdot hr}{mmol}$$

result

This program returns the oxygen uptake rate that is supported by different air and oxygen flow rates through the controlling gas stream (overlay, sparge 1, or sparge 2).

OUR_(Control, Input, qAir, qO2) := (9.4 − 12)

```
V_Liq ← Input_0
T_Liq ← Input_1 + T_0C/K
DO ← Input_2
p_HS ← Input_3 + p_atm/psi
N_rpm ← Input_4
qair_ovly ← qAir  if  AIR_CONTOUR = 0
qair_ovly ← Input_7  otherwise
qO2_ovly ← qO2  if  Control = 0
qO2_ovly ← Input_8  otherwise
qN2_ovly ← Input_10
qCO2_ovly ← | str2num(Min_CO2)/(100 − str2num(Min_CO2)) ·(qair_ovly + qO2_ovly + qN2_ovly)  if  CO2_CONTROL = 0
            | Input_9  otherwise
if  Control = 1
    qO2_1 ← qO2
    qair_1 ← qAir  if  AIR_CONTOUR = 1
    otherwise
        if  AIR_CO2_CONTROL = 1
            qair_1 ← str2num(AIR_CO2_GAIN) qO2_1
            qair_1 ← str2num(AIR_CO2_MAX)  if  qair_1 > str2num(AIR_CO2_MAX)
            qair_1 ← Input_11  if  qair_1 < Input_11
        qair_1 ← Input_11  otherwise
    qair_2 ← qAir  if  AIR_CONTOUR = 2
    otherwise
        if  AIR_CO2_CONTROL = 2
            qair_2 ← str2num(AIR_CO2_GAIN) qO2_1
            qair_2 ← str2num(AIR_CO2_MAX)  if  qair_2 > str2num(AIR_CO2_MAX)
            qair_2 ← Input_15  if  qair_2 < Input_15
        qair_2 ← Input_15  otherwise
qO2_1 ← Input_12  otherwise
qN2_1 ← Input_14
qCO2_1 ← | str2num(Min_CO2)/(100 − str2num(Min_CO2)) ·(qair_1 + qO2_1 + qN2_1)  if  CO2_CONTROL = 1
         | Input_13  otherwise
if  Control = 2
    qO2_2 ← qO2
    qair_1 ← qAir  if  AIR_CONTOUR = 1
    otherwise
```

```
if AIR_CO2_CONTROL = 1
    qair_1 ← str2num(AIR_CO2_GAIN) qO2_2
    qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
    qair_1 ← Input_11  if qair_1 < Input_11
qair_1 ← Input_11  otherwise
qair_2 ← qAir  if AIR_CONTOUR = 2
otherwise
  if AIR_CO2_CONTROL = 2
      qair_2 ← str2num(AIR_CO2_GAIN) qO2_2
      qair_2 ← str2num(AIR_CO2_MAX)  if qair_2 > str2num(AIR_CO2_MAX)
      qair_2 ← Input_15  if qair_2 < Input_15
  qair_2 ← Input_15  otherwise
qO2_2 ← Input_16  otherwise
qN2_2 ← Input_18
```

$$qCO2\_2 \leftarrow \begin{cases} \frac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_2 + qO2\_2 + qN2\_2) & \text{if } CO2\_CONTROL = 2 \\ Input_{17} & \text{otherwise} \end{cases}$$

$OUR \leftarrow 1$

$qtotalovly \leftarrow q_{total}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly) + 0.0000000000000001 \cdot SIUnitsOf(qair\_ovly)$

$qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.0000000000000001 \cdot SIUnitsOf(qair\_1)$

$qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.0000000000000001 \cdot SIUnitsOf(qair\_2)$

$kla\_surf \leftarrow 0$

$Power \leftarrow 0 \cdot W$

for $i \in 1..rows(Impeller) - 1$

$$Power \leftarrow Power + Impeller_{i,0} \cdot \left(\frac{N_{rpm}}{min}\right)^3 \cdot (Impeller_{i,1} \cdot m)^5 \frac{gm}{cm^3} \quad \text{if } Impeller_{i,2} \cdot L < V_{Liq} \cdot L$$

$$P\_V \leftarrow \frac{Power}{V_{Liq} \cdot L} \cdot \frac{m^3}{W} \quad \text{if } V_{Liq} \cdot L > 0 \cdot L$$

$P\_V \leftarrow 0$ otherwise

$$q\_sup1 \leftarrow \frac{qtotal1 \cdot SLPM}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$A1 \leftarrow Sparge_{1,0}$

$alpha1 \leftarrow Sparge_{1,1}$

$beta1 \leftarrow Sparge_{1,2}$

$kla1 \leftarrow A1 P\_V^{alpha1} \cdot q\_sup1^{beta1}$

$$q\_sup2 \leftarrow \frac{qtotal2 \cdot SLPM}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$A2 \leftarrow Sparge_{2,0}$

$alpha2 \leftarrow Sparge_{2,1}$

$beta2 \leftarrow Sparge_{2,2}$

$kla2 \leftarrow A2 \cdot P\_V^{alpha2} \cdot q\_sup2^{beta2}$

$$kla\_surf \leftarrow Surface_{0,1} + Surface_{1,1} \cdot V_{Liq} \ldots$$
$$+ Surface_{2,1} \cdot N_{rpm} \cdot \frac{min}{s} \ldots$$
$$+ Surface_{3,1}(qtotal1 + qtotal2) \ldots$$
$$+ Surface_{4,1} \cdot \left(V_{Liq} \cdot N_{rpm} \cdot \frac{min}{s}\right) \ldots$$
$$+ Surface_{5,1} \cdot \left[V_{Liq} \cdot (qtotal1 + qtotal2)\right] \ldots$$
$$+ Surface_{6,1} \cdot \left[N_{rpm} \cdot \frac{min}{s} \cdot (qtotal1 + qtotal2)\right] \ldots$$
$$+ Surface_{8,1} \cdot \left(N_{rpm} \cdot \frac{min}{s}\right)^2 \ldots$$
$$+ Surface_{9,1} \cdot (qtotal1 + qtotal2)^2 \ldots$$
$$+ Surface_{7,1} \cdot V_{Liq}^{\ 2}$$

$kla\_surf \leftarrow 0 \ \text{ if } \ kla\_surf < 0$

$$Inputs \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ kla\_surf \\ kla1 \\ kla2 \\ 0.011 \end{pmatrix}$$

$$T1 \leftarrow Coefficient(Inputs)_0 \cdot \left(\frac{mol}{L \cdot hr}\right) - \frac{DO}{100} \cdot 0.21 \cdot Coefficient(Inputs)_{27} \cdot \frac{mol}{L} \cdot Coefficient(Inputs)_1 \cdot \frac{1}{hr}$$

$$T2 \leftarrow Coefficient(Inputs)_2 \cdot \left(\frac{}{L\cdot hr}\right) - \frac{}{100} \cdot 0.21 \cdot Coefficient(Inputs)_{27} \cdot \frac{}{L} \cdot Coefficient(Inputs)_3 \cdot \frac{}{hr}$$

$$T3 \leftarrow \frac{\begin{gathered}Coefficient(Inputs)_7 \ldots \\ + Coefficient(Inputs)_8 \cdot \frac{DO}{100} \cdot 0.21 \cdot Coefficient(Inputs)_{27} + Coefficient(Inputs)_9 \ldots \\ + Coefficient(Inputs)_{10} \cdot \frac{DO}{100} \cdot 0.21 \cdot Coefficient(Inputs)_{27} + Coefficient(Inputs)_{11} \ldots \\ + Coefficient(Inputs)_4 \cdot \frac{DO}{100} \cdot 0.21 \cdot Coefficient(Inputs)_{27} \cdot Coefficient(Inputs)_{14}\end{gathered}}{Coefficient(Inputs)_{12} + Coefficient(Inputs)_5 \cdot Coefficient(Inputs)_4 \cdot Coefficient(Inputs)_{14}}$$

$$result \leftarrow \left[\begin{gathered}T1 \ldots \\ + T2 \ldots \\ + \frac{Coefficient(Inputs)_4}{hr} \cdot \left(\begin{gathered}T3 \cdot Coefficient(Inputs)_5 \cdot \frac{mol}{L} \ldots \\ + -\frac{DO}{100} \cdot 0.21 \cdot Coefficient(Inputs)_{27} \cdot \frac{mol}{L}\end{gathered}\right)\end{gathered}\right] \cdot \frac{L\cdot hr}{mmol}$$

This program returns the dissolved carbon dioxide that results from different air and oxygen flow rates through the controlling gas stream (overlay, sparge 1, or sparge 2).

$CO2\_(Control, Input, qAir, qO2) :=$ (9.4 – 13)

$$V_{Liq} \leftarrow Input_0$$

$$T_{Liq} \leftarrow Input_1 + \frac{T_{0C}}{K}$$

$$DO \leftarrow Input_2$$

$$p_{HS} \leftarrow Input_3 + \frac{p_{atm}}{psi}$$

$$N_{rpm} \leftarrow Input_4$$

$qair\_ovly \leftarrow qAir$ if $AIR\_CONTOUR = 0$

$qair\_ovly \leftarrow Input_7$ otherwise

$qO2\_ovly \leftarrow qO2$ if $Control = 0$

$qO2\_ovly \leftarrow Input_8$ otherwise

$qN2\_ovly \leftarrow Input_{10}$

$qCO2\_ovly \leftarrow$ $\frac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_ovly + qO2\_ovly + qN2\_ovly)$ if $CO2\_CONTROL = 0$

$Input_9$ otherwise

if $Control = 1$

$qO2\_1 \leftarrow qO2$

$qair\_1 \leftarrow qAir$ if $AIR\_CONTOUR = 1$

otherwise

if $AIR\_CO2\_CONTROL = 1$

$qair\_1 \leftarrow str2num(AIR\_CO2\_GAIN)\, qO2\_1$

$qair\_1 \leftarrow str2num(AIR\_CO2\_MAX)$ if $qair\_1 > str2num(AIR\_CO2\_MAX)$

$qair\_1 \leftarrow Input_{11}$ if $qair\_1 < Input_{11}$

$qair\_1 \leftarrow Input$ otherwise

$qair\_2 \leftarrow qAir \;\; \text{if} \;\; AIR\_CONTOUR = 2$

otherwise

if $AIR\_CO2\_CONTROL = 2$

$qair\_2 \leftarrow str2num(AIR\_CO2\_GAIN)\, qO2\_1$

$qair\_2 \leftarrow str2num(AIR\_CO2\_MAX) \;\; \text{if} \;\; qair\_2 > str2num(AIR\_CO2\_MAX)$

$qair\_2 \leftarrow Input_{15} \;\; \text{if} \;\; qair\_2 < Input_{15}$

$qair\_2 \leftarrow Input_{15} \;\; \text{otherwise}$

$qO2\_1 \leftarrow Input_{12} \;\; \text{otherwise}$

$qN2\_1 \leftarrow Input_{14}$

$$qCO2\_1 \leftarrow \begin{cases} \dfrac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_1 + qO2\_1 + qN2\_1) & \text{if } CO2\_CONTROL = 1 \\ Input_{13} & \text{otherwise} \end{cases}$$

if $Control = 2$

$qO2\_2 \leftarrow qO2$

$qair\_1 \leftarrow qAir \;\; \text{if} \;\; AIR\_CONTOUR = 1$

otherwise

if $AIR\_CO2\_CONTROL = 1$

$qair\_1 \leftarrow str2num(AIR\_CO2\_GAIN)\, qO2\_2$

$qair\_1 \leftarrow str2num(AIR\_CO2\_MAX) \;\; \text{if} \;\; qair\_1 > str2num(AIR\_CO2\_MAX)$

$qair\_1 \leftarrow Input_{11} \;\; \text{if} \;\; qair\_1 < Input_{11}$

$qair\_1 \leftarrow Input_{11} \;\; \text{otherwise}$

$qair\_2 \leftarrow qAir \;\; \text{if} \;\; AIR\_CONTOUR = 2$

otherwise

if $AIR\_CO2\_CONTROL = 2$

$qair\_2 \leftarrow str2num(AIR\_CO2\_GAIN)\, qO2\_2$

$qair\_2 \leftarrow str2num(AIR\_CO2\_MAX) \;\; \text{if} \;\; qair\_2 > str2num(AIR\_CO2\_MAX)$

$qair\_2 \leftarrow Input_{15} \;\; \text{if} \;\; qair\_2 < Input_{15}$

$qair\_2 \leftarrow Input_{15} \;\; \text{otherwise}$

$qO2\_2 \leftarrow Input_{16} \;\; \text{otherwise}$

$qN2\_2 \leftarrow Input_{18}$

$$qCO2\_2 \leftarrow \begin{cases} \dfrac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_2 + qO2\_2 + qN2\_2) & \text{if } CO2\_CONTROL = 2 \\ Input_{17} & \text{otherwise} \end{cases}$$

$OUR \leftarrow 1$

$qtotalovly \leftarrow q_{total}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly) + 0.00000000000001 \cdot SIUnitsOf(qair\_ovly)$

$qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.00000000000001 \cdot SIUnitsOf(qair\_1)$

$qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.00000000000001 \cdot SIUnitsOf(qair\_2)$

$kla\_surf \leftarrow Surface_{0,1} + Surface_{1,1} \cdot V_{Liq} \cdots$

$+ Surface_{2,1} \cdot N_{rpm} \cdot \frac{min}{} \cdots$

$$+\,Surface_{3,1}(qtotal1 + qtotal2)\ \ldots$$
$$+\,Surface_{4,1}\cdot\left(V_{Liq}\cdot N_{rpm}\cdot\frac{min}{s}\right)\ \ldots$$
$$+\,Surface_{5,1}\cdot\left[V_{Liq}\cdot(qtotal1 + qtotal2)\right]\ \ldots$$
$$+\,Surface_{6,1}\cdot\left[N_{rpm}\cdot\frac{min}{s}\cdot(qtotal1 + qtotal2)\right]\ \ldots$$
$$+\,Surface_{8,1}\cdot\left(N_{rpm}\cdot\frac{min}{s}\right)^2\ \ldots$$
$$+\,Surface_{9,1}\cdot(qtotal1 + qtotal2)^2\ \ldots$$
$$+\,Surface_{7,1}\cdot V_{Liq}{}^2$$

$kla\_surf \leftarrow 0\ \text{ if }\ kla\_surf < 0$

$Power \leftarrow 0\cdot W$

for $i \in 1..rows(Impeller) - 1$

$$Power \leftarrow Power + Impeller_{i,0}\cdot\left(\frac{N_{rpm}}{min}\right)^3\cdot\left(Impeller_{i,1}\cdot m\right)^5\frac{gm}{cm^3}\ \text{ if }\ Impeller_{i,2}\cdot L < V_{Liq}\cdot L$$

$$P\_V \leftarrow \frac{Power}{V_{Liq}\cdot L}\cdot\frac{m^3}{W}\ \text{ if }\ V_{Liq}\cdot L > 0\cdot L$$

$$q\_sup1 \leftarrow \frac{qtotal1\cdot SLPM}{\pi\cdot\left(\frac{D}{2}\right)^2}\cdot\frac{s}{m}$$

$A1 \leftarrow Sparge_{1,0}$

$alpha1 \leftarrow Sparge_{1,1}$

$beta1 \leftarrow Sparge_{1,2}$

$kla1 \leftarrow A1\cdot P\_V^{alpha1}\cdot q\_sup1^{beta1}$

$$q\_sup2 \leftarrow \frac{qtotal2\cdot SLPM}{\pi\cdot\left(\frac{D}{2}\right)^2}\cdot\frac{s}{m}$$

$A2 \leftarrow Sparge_{2,0}$

$alpha2 \leftarrow Sparge_{2,1}$

$beta2 \leftarrow Sparge_{2,2}$

$kla2 \leftarrow A2\cdot P\_V^{alpha2}\cdot q\_sup2^{beta2}$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \end{pmatrix}$$

$$\text{Inputs} \leftarrow \begin{pmatrix} qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ kla\_surf \\ kla1 \\ kla2 \\ 0.011 \end{pmatrix}$$

$$OUR \leftarrow OUR\_(Control, Input, qAir, qO2)\cdot\frac{mmol}{L\cdot hr}$$

$$T1 \leftarrow Coefficient(Inputs)_{15}\cdot\frac{mol}{L} + Coefficient(Inputs)_{17}\cdot\frac{mol}{L}$$

$$T2 \leftarrow \frac{\left(Coefficient(Inputs)_{21} + Coefficient(Inputs)_{23} + Coefficient(Inputs)_{25}\right)}{Coefficient(Inputs)_{12} + Coefficient(Inputs)_{4}\cdot 0.893\cdot Coefficient(Inputs)_{20}\cdot Coefficient(Inputs)_{14}}$$

$$T3 \leftarrow \left[\frac{\begin{pmatrix} Coefficient(Inputs)_{22}\ \ldots \\ + Coefficient(Inputs)_{24}\ \ldots \\ + Coefficient(Inputs)_{4}\cdot 0.893\cdot Coefficient(Inputs)_{14} \end{pmatrix}\cdot Coefficient(Inputs)_{20}}{Coefficient(Inputs)_{12} + Coefficient(Inputs)_{4}\cdot 0.893\cdot Coefficient(Inputs)_{20}\cdot Coefficient(Inputs)_{14}} - 1\right]$$

$$result \leftarrow \frac{\begin{bmatrix} T1\ \ldots \\ + Coefficient(Inputs)_{19}\cdot\left(T2\cdot Coefficient(Inputs)_{20}\cdot\frac{mol}{L}\right)\ \ldots \\ + RQ\,OUR\cdot hr \end{bmatrix}}{H_{CO2\_cp}\left(0.011, T_{Liq}\cdot K\right)\cdot mmHg\cdot\begin{pmatrix} Coefficient(Inputs)_{16}\ \ldots \\ + Coefficient(Inputs)_{18}\ \ldots \\ + -Coefficient(Inputs)_{19}\cdot T3 \end{pmatrix}}$$

These two programs create the response surface for the oxygen uptake rate and the dissolved carbon dioxide if the user selected section 9.4 to be active.

$$OURmesh(qO2, qAir) := \begin{array}{|l} result \leftarrow 0 \;\; \text{if} \;\; Part4 = 0 \\ result \leftarrow OUR\_(DO\_CONTROL, Input, qAir, qO2) \;\; \text{otherwise} \\ result \end{array} \qquad (9.4 - 14)$$

$$CO2mesh(qO2, qAir) := \begin{array}{|l} result \leftarrow 0 \;\; \text{if} \;\; Part4 = 0 \\ result \leftarrow CO2\_(DO\_CONTROL, Input, qAir, qO2) \;\; \text{otherwise} \\ result \end{array} \qquad (9.4 - 15)$$

$$\mathrm{Air\_min} := \left| \begin{array}{l} \text{if } \ \mathrm{str2num(AirMin)} = 0 \\ \quad \left| \begin{array}{l} \mathrm{result} \leftarrow \mathrm{Limits}_{4,1} \ \text{ if } \ \mathrm{DO\_CONTROL} = 0 \\ \text{otherwise} \\ \quad \left| \begin{array}{l} \mathrm{result} \leftarrow \mathrm{Limits}_{8,1} \ \text{ if } \ \mathrm{DO\_CONTROL} = 1 \\ \mathrm{result} \leftarrow \mathrm{Limits}_{12,1} \ \text{ otherwise} \end{array} \right. \end{array} \right. \\ \mathrm{result} \leftarrow \mathrm{str2num(AirMin)} \ \text{ otherwise} \end{array} \right. \qquad (9.4 - 16)$$

$$\mathrm{Air\_max} := \left| \begin{array}{l} \text{if } \ \mathrm{str2num(AirMax)} = 0 \\ \quad \left| \begin{array}{l} \mathrm{result} \leftarrow \mathrm{Limits}_{4,3} \ \text{ if } \ \mathrm{DO\_CONTROL} = 0 \\ \text{otherwise} \\ \quad \left| \begin{array}{l} \mathrm{result} \leftarrow \mathrm{Limits}_{8,3} \ \text{ if } \ \mathrm{DO\_CONTROL} = 1 \\ \mathrm{result} \leftarrow \mathrm{Limits}_{12,3} \ \text{ otherwise} \end{array} \right. \end{array} \right. \\ \mathrm{result} \leftarrow \mathrm{str2num(AirMax)} \ \text{ otherwise} \end{array} \right. \qquad (9.4 - 17)$$

$$\mathrm{O2\_min} := \left| \begin{array}{l} \text{if } \ \mathrm{str2num(O2Min)} = 0 \\ \quad \left| \begin{array}{l} \mathrm{result} \leftarrow \mathrm{Limits}_{5,1} \ \text{ if } \ \mathrm{DO\_CONTROL} = 0 \\ \text{otherwise} \\ \quad \left| \begin{array}{l} \mathrm{result} \leftarrow \mathrm{Limits}_{9,1} \ \text{ if } \ \mathrm{DO\_CONTROL} = 1 \\ \mathrm{result} \leftarrow \mathrm{Limits}_{13,1} \ \text{ otherwise} \end{array} \right. \end{array} \right. \\ \mathrm{result} \leftarrow \mathrm{str2num(O2Min)} \ \text{ otherwise} \end{array} \right. \qquad (9.4 - 18)$$

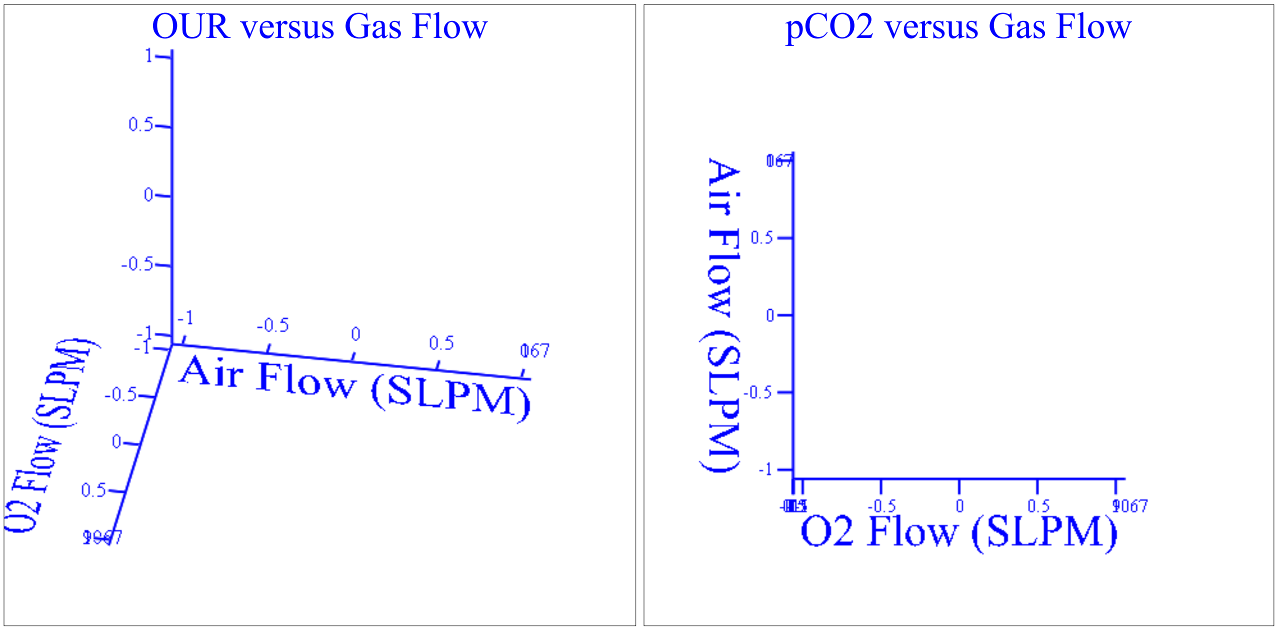
$$\mathrm{O2\_max} := \left| \begin{array}{l} \text{if } \ \mathrm{str2num(O2Max)} = 0 \\ \quad \left| \begin{array}{l} \mathrm{result} \leftarrow \mathrm{Limits}_{5,3} \ \text{ if } \ \mathrm{DO\_CONTROL} = 0 \\ \text{otherwise} \\ \quad \left| \begin{array}{l} \mathrm{result} \leftarrow \mathrm{Limits}_{9,3} \ \text{ if } \ \mathrm{DO\_CONTROL} = 1 \\ \mathrm{result} \leftarrow \mathrm{Limits}_{13,3} \ \text{ otherwise} \end{array} \right. \end{array} \right. \\ \mathrm{result} \leftarrow \mathrm{str2num(O2Max)} \ \text{ otherwise} \end{array} \right. \qquad (9.4 - 19)$$

$$\mathrm{OUR} := \mathrm{CreateMesh(OURmesh, O2\_min, O2\_max, Air\_min, Air\_max, 100, 50)} \qquad (9.4 - 20)$$

$$\mathrm{CO2} := \mathrm{CreateMesh(CO2mesh, O2\_min, O2\_max, Air\_min, Air\_max, 100, 50)} \qquad (9.4 - 21)$$

$$
\begin{aligned}
GasFlows := \; & result_{0,1} \leftarrow \text{"Air"} \\
& result_{0,2} \leftarrow \text{"Oxygen"} \\
& result_{0,3} \leftarrow \text{"Carbon Dioxide"} \\
& result_{1,0} \leftarrow \text{"Overlay"} \\
& result_{2,0} \leftarrow \text{"Sparger 1"} \\
& result_{3,0} \leftarrow \text{"Sparger 2"} \\
& result_{1,1} \leftarrow Proposed_{8,2} \\
& result_{1,2} \leftarrow Proposed_{9,2} \\
& result_{1,3} \leftarrow Proposed_{10,2} \\
& result_{2,1} \leftarrow Proposed_{12,2} \\
& result_{2,2} \leftarrow Proposed_{13,2} \\
& result_{2,3} \leftarrow Proposed_{14,2} \\
& result_{3,1} \leftarrow Proposed_{16,2} \\
& result_{3,2} \leftarrow Proposed_{17,2} \\
& result_{3,3} \leftarrow Proposed_{18,2} \\
& result_{1,2} \leftarrow \text{"X\_values"} \;\; \text{if } DO\_CONTROL = 0 \\
& result_{2,2} \leftarrow \text{"X\_values"} \;\; \text{if } DO\_CONTROL = 1 \\
& result_{3,2} \leftarrow \text{"X\_values"} \;\; \text{if } DO\_CONTROL = 2 \\
& result_{1,1} \leftarrow \text{"Y\_values"} \;\; \text{if } AIR\_CONTOUR = 0 \\
& result_{2,1} \leftarrow \text{"Y\_values"} \;\; \text{if } AIR\_CONTOUR = 1 \\
& result_{3,1} \leftarrow \text{"Y\_values"} \;\; \text{if } AIR\_CONTOUR = 2 \\
& result
\end{aligned}
\qquad (9.4-22)
$$

Calculation of OUR and pCO2 contour plots



GasFlows = ▮

Figure 9.4-1: Contour plots for the oxygen uptake rate and resulting carbon dioxide accumulation for the vessel with the specified air, oxygen, and carbon dioxide flow rates through the overlay, sparger 1, and sparger 2 (time zero settings for the bioreactor in table 9.4-13).

Process Model Numerical Calculations

Convert the text fields for model end time and model interval time to numeric values.

$$Time\_end := str2num(Time\_end)\cdot hr \qquad (9.4-23)$$

$$Time\_int := str2num(Time\_int)\cdot min \qquad (9.4-24)$$

This program calculates the oxygen uptake rate that can be supported by the process conditions. The purpose of this calculation is to allow for a numeric iteration to select the process conditions that meet the demand for oxygen.

$$
\begin{aligned}
\text{OURcalc}(\text{processVariables}, \text{Antifoam}) := \; & V_{Liq} \leftarrow \text{processVariables}_0 \cdot L \qquad (9.4-25) \\
& T_{Liq} \leftarrow \text{processVariables}_1 \cdot K + T_{0C} \\
& p_{HS} \leftarrow \text{processVariables}_2 \cdot psi + p_{atm} \\
& Nrpm \leftarrow \text{processVariables}_3 \cdot \frac{1}{min} \\
& qair\_ovly \leftarrow \text{processVariables}_4 \cdot SLPM \\
& qO2\_ovly \leftarrow \text{processVariables}_5 \cdot SLPM \\
& qCO2\_ovly \leftarrow \text{processVariables}_6 \cdot SLPM \\
& qN2\_ovly \leftarrow \text{processVariables}_7 \cdot SLPM \\
& qair\_1 \leftarrow \text{processVariables}_8 \cdot SLPM \\
& qO2\_1 \leftarrow \text{processVariables}_9 \cdot SLPM \\
& qCO2\_1 \leftarrow \text{processVariables}_{10} \cdot SLPM \\
& qN2\_1 \leftarrow \text{processVariables}_{11} \cdot SLPM \\
& qair\_2 \leftarrow \text{processVariables}_{12} \cdot SLPM \\
& qO2\_2 \leftarrow \text{processVariables}_{13} \cdot SLPM \\
& qCO2\_2 \leftarrow \text{processVariables}_{14} \cdot SLPM \\
& qN2\_2 \leftarrow \text{processVariables}_{15} \cdot SLPM \\
& V_{FEED} \leftarrow \text{processVariables}_{20} \cdot L \\
& DO \leftarrow \frac{\text{processVariables}_{21}}{100} \\
& pH \leftarrow \text{processVariables}_{22} \\
& pH_{Upper} \leftarrow \text{processVariables}_{23} \\
& pH_{Lower} \leftarrow \text{processVariables}_{24} \\
& adj \leftarrow \text{processVariables}_{25} \\
& qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) \\
& qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) \\
& M \leftarrow 0.5 + 0.5 \cdot \exp(-1 \cdot \text{Antifoam}) \\
& kla\_surf \leftarrow M \cdot \Big[kla\_surf \leftarrow Surface_{0,1} + Surface_{1,1} \cdot V_{Liq} \cdot \frac{1}{L} \; ... \\
& \qquad + Surface_{2,1} \cdot Nrpm \cdot min \cdot \frac{min}{s} \; ... \\
& \qquad + Surface_{3,1} (qtotal1 + qtotal2) \cdot \frac{1}{SLPM} \; ... \\
& \qquad + Surface_{4,1} \cdot \left(V_{Liq} \cdot Nrpm \cdot \frac{min}{L} \cdot \frac{min}{s} \right) \; ... \\
& \qquad + Surface_{5,1} \cdot \left[V_{Liq} \cdot (qtotal1 + qtotal2) \right] \cdot \frac{1}{L \cdot SLPM} \; ... \\
& \qquad + Surface_{6,1} \cdot \left[Nrpm \cdot min \cdot \frac{min}{s} \cdot (qtotal1 + qtotal2) \right] \cdot \frac{1}{SLPM} \; ... \\
& \qquad + Surface_{8,1} \cdot \left(Nrpm \cdot min \cdot \frac{min}{s} \right)^2 \; ... \\
& \qquad + Surface_{9,1} \cdot (qtotal1 + qtotal2)^2 \cdot \frac{1}{SLPM^2} \; ...
\end{aligned}
$$

$$+ \text{Surface}_{7,1} \cdot V_{Liq}^{2} \cdot \frac{1}{L^2}$$

$\text{kla\_surf} \leftarrow 0 \cdot \frac{1}{hr}$ if $\text{kla\_surf} < 0 \cdot \frac{1}{hr}$

$\text{Power} \leftarrow 0 \cdot W$

for $i \in 1 .. \text{rows}(\text{Impeller}) - 1$

if $\text{Impeller}_{i,2} \cdot L < V_{Liq}$

$\text{Power} \leftarrow \text{Power} + \text{Impeller}_{i,0} \cdot (\text{Nrpm})^3 \cdot \left(\text{Impeller}_{i,1} \cdot m\right)^5 \frac{gm}{cm^3}$

$\text{temp} \leftarrow 1$

$\text{P\_V} \leftarrow \frac{\text{Power}}{V_{Liq}} \cdot \frac{m^3}{W}$ if $V_{Liq} > 0 \cdot L$

$\text{P\_V} \leftarrow 0$ otherwise

$$\text{q\_sup1} \leftarrow \frac{\text{qtotal1}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$\text{A1} \leftarrow \text{Sparge}_{1,0}$

$\text{alpha1} \leftarrow \text{Sparge}_{1,1}$

$\text{beta1} \leftarrow \text{Sparge}_{1,2}$

$\text{kla1} \leftarrow M \cdot A1 \cdot \text{P\_V}^{\text{alpha1}} \cdot \text{q\_sup1}^{\text{beta1}}$

$$\text{q\_sup2} \leftarrow \frac{\text{qtotal2}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$\text{A2} \leftarrow \text{Sparge}_{2,0}$

$\text{alpha2} \leftarrow \text{Sparge}_{2,1}$

$\text{beta2} \leftarrow \text{Sparge}_{2,2}$

$\text{kla2} \leftarrow M \cdot A2 \cdot \text{P\_V}^{\text{alpha2}} \cdot \text{q\_sup2}^{\text{beta2}}$

$$\begin{pmatrix} \frac{V_{Liq}}{L} \\ \frac{T_{Liq}}{K} \\ \frac{p_{HS}}{psi} \\ \text{Nrpm} \cdot \text{min} \\ \frac{\text{qair\_ovly}}{SLPM} \\ \frac{\text{qO2\_ovly}}{SLPM} \\ \frac{\text{qCO2\_ovly}}{SLPM} \\ \frac{\text{qN2\_ovly}}{SLPM} \end{pmatrix}$$

$$\text{Inputs} \leftarrow \begin{pmatrix} \frac{\text{qair\_1}}{\text{SLPM}} \\ \frac{\text{qO2\_1}}{\text{SLPM}} \\ \frac{\text{qCO2\_1}}{\text{SLPM}} \\ \frac{\text{qN2\_1}}{\text{SLPM}} \\ \frac{\text{qair\_2}}{\text{SLPM}} \\ \frac{\text{qO2\_2}}{\text{SLPM}} \\ \frac{\text{qCO2\_2}}{\text{SLPM}} \\ \frac{\text{qN2\_2}}{\text{SLPM}} \\ 0 \\ \text{kla\_surf} \\ \text{kla1} \\ \text{kla2} \\ 0.011 \end{pmatrix}$$

$$T1 \leftarrow \text{Coefficient(Inputs)}_0 \cdot \left(\frac{\text{mol}}{\text{L}\cdot\text{hr}}\right) - \text{DO}\cdot 0.21\cdot\text{Coefficient(Inputs)}_{27}\cdot\frac{\text{mol}}{\text{L}}\cdot\text{Coefficient(Inputs)}_1\cdot\frac{1}{\text{hr}}$$

$$T2 \leftarrow \text{Coefficient(Inputs)}_2 \cdot \left(\frac{\text{mol}}{\text{L}\cdot\text{hr}}\right) - \text{DO}\cdot 0.21\cdot\text{Coefficient(Inputs)}_{27}\cdot\frac{\text{mol}}{\text{L}}\cdot\text{Coefficient(Inputs)}_3\cdot\frac{1}{\text{hr}}$$

$$T3 \leftarrow \frac{\begin{pmatrix} \text{Coefficient(Inputs)}_7 \ldots \\ + \text{Coefficient(Inputs)}_8\cdot\text{DO}\cdot 0.21\cdot\text{Coefficient(Inputs)}_{27} + \text{Coefficient(Inputs)}_9 \ldots \\ + \text{Coefficient(Inputs)}_{10}\cdot\text{DO}\cdot 0.21\cdot\text{Coefficient(Inputs)}_{27} + \text{Coefficient(Inputs)}_{11} \ldots \\ + \text{Coefficient(Inputs)}_4\cdot\text{DO}\cdot 0.21\cdot\text{Coefficient(Inputs)}_{27}\cdot\text{Coefficient(Inputs)}_{14} \end{pmatrix}}{\text{Coefficient(Inputs)}_{12} + \text{Coefficient(Inputs)}_{13}\cdot\text{Coefficient(Inputs)}_4\cdot\text{Coefficient(Inputs)}_{14}}$$

$$\text{result} \leftarrow \left[\begin{array}{l} T1 + T2 \ldots \\ + \frac{\text{Coefficient(Inputs)}_4}{\text{hr}}\cdot\begin{pmatrix} T3\cdot\text{Coefficient(Inputs)}_5\cdot\frac{\text{mol}}{\text{L}} \ldots \\ + -\text{DO}\cdot 0.21\cdot\text{Coefficient(Inputs)}_{27}\cdot\frac{\text{mol}}{\text{L}} \end{pmatrix} \end{array}\right]$$

This program calculates how the process controller will cascade through the control elements to meet the dissolved oxygen set point.

$$
\begin{aligned}
\mathrm{DOcontrol(processVariables, Antifoam, DOsp)} := \;& V_{Liq} \leftarrow \mathrm{processVariables}_0 \cdot L \\
& T_{Liq} \leftarrow \mathrm{processVariables}_1 \cdot K + T_{0C} \\
& p_{HS} \leftarrow \mathrm{processVariables}_2 \cdot psi + p_{atm} \\
& Nrpm \leftarrow \mathrm{processVariables}_3 \cdot \frac{1}{min} \\
& qair\_ovly \leftarrow \mathrm{processVariables}_4 \cdot SLPM \\
& qO2\_ovly \leftarrow \mathrm{processVariables}_5 \cdot SLPM \\
& qCO2\_ovly \leftarrow \mathrm{processVariables}_6 \cdot SLPM \\
& qN2\_ovly \leftarrow \mathrm{processVariables}_7 \cdot SLPM \\
& qair\_1 \leftarrow \mathrm{processVariables}_8 \cdot SLPM \\
& qO2\_1 \leftarrow \mathrm{processVariables}_9 \cdot SLPM \\
& qCO2\_1 \leftarrow \mathrm{processVariables}_{10} \cdot SLPM \\
& qN2\_1 \leftarrow \mathrm{processVariables}_{11} \cdot SLPM \\
& qair\_2 \leftarrow \mathrm{processVariables}_{12} \cdot SLPM \\
& qO2\_2 \leftarrow \mathrm{processVariables}_{13} \cdot SLPM \\
& qCO2\_2 \leftarrow \mathrm{processVariables}_{14} \cdot SLPM \\
& qN2\_2 \leftarrow \mathrm{processVariables}_{15} \cdot SLPM \\
& OUR \leftarrow \mathrm{processVariables}_{16} \cdot \frac{mmol}{L \cdot hr} \\
& lactate \leftarrow \mathrm{processVariables}_{17} \cdot \frac{mmol}{L} \\
& ammonia \leftarrow \mathrm{processVariables}_{18} \cdot \frac{mmol}{L} \\
& V_{Base} \leftarrow \mathrm{processVariables}_{19} \cdot L \\
& V_{FEED} \leftarrow \mathrm{processVariables}_{20} \cdot L \\
& DO \leftarrow DOsp \\
& \mathrm{processVariables}_{21} \leftarrow DOsp \cdot 100 \\
& pH \leftarrow \mathrm{processVariables}_{22} \\
& pH_{Upper} \leftarrow \mathrm{processVariables}_{23} \\
& pH_{Lower} \leftarrow \mathrm{processVariables}_{24} \\
& adj \leftarrow \mathrm{processVariables}_{25} \\
& MIN\_p_{HS} \leftarrow Limits_{2,1} \cdot psi + p_{atm} \\
& MAX\_p_{HS} \leftarrow Limits_{2,3} \cdot psi + p_{atm} \\
& MIN\_Nrpm \leftarrow Limits_{3,1} \cdot \frac{1}{min} \\
& MAX\_Nrpm \leftarrow Limits_{3,3} \cdot \frac{1}{min} \\
& MIN\_qair\_ovly \leftarrow Limits_{4,1} \cdot SLPM \\
& MAX\_qair\_ovly \leftarrow Limits_{4,3} \cdot SLPM
\end{aligned}
\qquad (9.4-26)
$$

$MIN\_qO2\_ovly \leftarrow Limits_{5,1} \cdot SLPM$

$MAX\_qO2\_ovly \leftarrow Limits_{5,3} \cdot SLPM$

$MIN\_qCO2\_ovly \leftarrow Limits_{6,1} \cdot SLPM$

$MAX\_qCO2\_ovly \leftarrow Limits_{6,3} \cdot SLPM$

$MIN\_qN2\_ovly \leftarrow Limits_{7,1} \cdot SLPM$

$MAX\_qN2\_ovly \leftarrow Limits_{7,3} \cdot SLPM$

$MIN\_qair\_1 \leftarrow Limits_{8,1} \cdot SLPM$

$MAX\_qair\_1 \leftarrow Limits_{8,3} \cdot SLPM$

$MIN\_qO2\_1 \leftarrow Limits_{9,1} \cdot SLPM$

$MAX\_qO2\_1 \leftarrow Limits_{9,3} \cdot SLPM$

$MIN\_qCO2\_1 \leftarrow Limits_{10,1} \cdot SLPM$

$MAX\_qCO2\_1 \leftarrow Limits_{10,3} \cdot SLPM$

$MIN\_qN2\_1 \leftarrow Limits_{11,1} \cdot SLPM$

$MAX\_qN2\_1 \leftarrow Limits_{11,3} \cdot SLPM$

$MIN\_qair\_2 \leftarrow Limits_{12,1} \cdot SLPM$

$MAX\_qair\_2 \leftarrow Limits_{12,3} \cdot SLPM$

$MIN\_qO2\_2 \leftarrow Limits_{13,1} \cdot SLPM$

$MAX\_qO2\_2 \leftarrow Limits_{13,3} \cdot SLPM$

$MIN\_qCO2\_2 \leftarrow Limits_{14,1} \cdot SLPM$

$MAX\_qCO2\_2 \leftarrow Limits_{14,3} \cdot SLPM$

$MIN\_qN2\_2 \leftarrow Limits_{15,1} \cdot SLPM$

$MAX\_qN2\_2 \leftarrow Limits_{15,3} \cdot SLPM$

$Step \leftarrow 1$

$Count \leftarrow 1$

$$OUR_{calc} \leftarrow Re\left[OURcalc(processVariables, Antifoam) + \left(\frac{-1}{10^{10}}\right)^{0.5} \cdot \frac{mmol}{L \cdot hr}\right]$$

$$\text{while } Count < 5 \vee \left[Count < 500 \wedge \left[\left(\frac{OUR}{OUR_{calc}}\right) > 1.02 \vee \left(\frac{OUR}{OUR_{calc}}\right) < 0.98\right]\right]$$

$Count \leftarrow Count + 1$

$\text{if } CASCADE_{Step} = \text{"Agitation"}$

$$Nrpm \leftarrow Nrpm \cdot \left[\left(\frac{OUR}{OUR_{calc}}\right)^{\frac{1}{3}}\right]$$

$\text{if } Nrpm > MAX\_Nrpm$

$Nrpm \leftarrow MAX\_Nrpm$

$Step \leftarrow Step + 1 \ \text{ if } CASCADE_{Step+1} \neq \text{"None"}$

$\text{if } Nrpm < MIN\_Nrpm$

$Nrpm \leftarrow MIN\_Nrpm$

$Step \leftarrow Step - 1 \ \text{ if } \ CASCADE_{Step-1} \neq \text{"None"}$

$\text{if } \ CASCADE_{Step} = \text{"Air"}$

$\text{if } \ DO\_CONTROL = 0$

$qair\_ovly \leftarrow qair\_ovly \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$\text{if } \ qair\_ovly > MAX\_qair\_ovly$

$qair\_ovly \leftarrow MAX\_qair\_ovly$

$Step \leftarrow Step + 1 \ \text{ if } \ CASCADE_{Step+1} \neq \text{"None"}$

$\text{if } \ qair\_ovly < MIN\_qair\_ovly$

$qair\_ovly \leftarrow MIN\_qair\_ovly$

$Step \leftarrow Step - 1 \ \text{ if } \ CASCADE_{Step-1} \neq \text{"None"}$

$\text{if } \ DO\_CONTROL = 1$

$qair\_1 \leftarrow qair\_1 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$\text{if } \ qair\_1 > MAX\_qair\_1$

$qair\_1 \leftarrow MAX\_qair\_1$

$Step \leftarrow Step + 1 \ \text{ if } \ CASCADE_{Step+1} \neq \text{"None"}$

$\text{if } \ qair\_1 < MIN\_qair\_1$

$qair\_1 \leftarrow MIN\_qair\_1$

$Step \leftarrow Step - 1 \ \text{ if } \ CASCADE_{Step-1} \neq \text{"None"}$

$\text{if } \ DO\_CONTROL = 2$

$qair\_2 \leftarrow qair\_2 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$\text{if } \ qair\_2 > MAX\_qair\_2$

$qair\_2 \leftarrow MAX\_qair\_2$

$Step \leftarrow Step + 1 \ \text{ if } \ CASCADE_{Step+1} \neq \text{"None"}$

$\text{if } \ qair\_2 < MIN\_qair\_2$

$qair\_2 \leftarrow MIN\_qair\_2$

$Step \leftarrow Step - 1 \ \text{ if } \ CASCADE_{Step-1} \neq \text{"None"}$

$\text{if } \ DO\_CONTROL = 3$

$qair\_1 \leftarrow qair\_1 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$qair\_2 \leftarrow qair\_1 \cdot (DOsplit - 1)$

$\text{if } \ qair\_1 > MAX\_qair\_1$

$qair\_1 \leftarrow MAX\_qair\_1$

$Step \leftarrow Step + 1 \ \text{ if } \ CASCADE_{Step+1} \neq \text{"None"}$

$\text{if } \ qair\_1 < MIN\_qair\_1$

$qair\_1 \leftarrow MIN\_qair\_1$

```
            Step ← Step − 1  if  CASCADE_(Step−1) ≠ "None"
  if  CASCADE_Step = "O2 Enrichment"
    if  DO_CONTROL = 0
      qO2_ovly ← qO2_ovly·(OUR / OUR_calc)
      if  qO2_ovly > MAX_qO2_ovly
        qO2_ovly ← MAX_qO2_ovly
        Step ← Step + 1  if  CASCADE_(Step+1) ≠ "None"
      if  qO2_ovly < MIN_qO2_ovly
        qO2_ovly ← MIN_qO2_ovly
        Step ← Step − 1  if  CASCADE_(Step−1) ≠ "None"
    if  DO_CONTROL = 1
      qO2_1 ← qO2_1·(OUR / OUR_calc)
      if  qO2_1 > MAX_qO2_1
        qO2_1 ← MAX_qO2_1
        Step ← Step + 1  if  CASCADE_(Step+1) ≠ "None"
      if  qO2_1 < 0.01·MIN_qO2_1
        qO2_1 ← MIN_qO2_1
        Step ← Step − 1  if  CASCADE_(Step−1) ≠ "None"
      if  AIR_CO2_CONTROL = 1
        Gain ← str2num(AIR_CO2_GAIN)
        qair_1 ← str2num(AIR_CO2_MAN) SLPM  if  adj = 100
        otherwise
          multiplier ← Gain + adj
          qair_1 ← multiplier·qO2_1
          Max ← str2num(AIR_CO2_MAX)·SLPM
          qair_1 ← Max  if  qair_1 > Max
          qair_1 ← MAX_qair_1  if  qair_1 > MAX_qair_1
          qair_1 ← MIN_qair_1  if  qair_1 < MIN_qair_1
      if  AIR_CO2_CONTROL = 2
        Gain ← str2num(AIR_CO2_GAIN)
        qair_2 ← str2num(AIR_CO2_MAN) SLPM  if  adj = 100
        otherwise
          multiplier ← Gain + adj
          qair_2 ← (multiplier)·qO2_1
          Max ← str2num(AIR_CO2_MAX)·SLPM
          qair_2 ← Max  if  qair_2 > Max
          qair_2 ← MAX_qair_2  if  qair_2 > MAX_qair_2
          qair_2 ← MIN_qair_2  if  qair_2 < MIN_qair_2
```

```
if DO_CONTROL = 2
    qO2_2 ← qO2_2·(OUR / OUR_calc)
    if qO2_2 > MAX_qO2_2
        qO2_2 ← MAX_qO2_2
        Step ← Step + 1  if CASCADE_(Step+1) ≠ "None"
    if qO2_2 < MIN_qO2_2
        qO2_2 ← MIN_qO2_2
        Step ← Step − 1  if CASCADE_(Step−1) ≠ "None"
    if AIR_CO2_CONTROL = 1
        Gain ← str2num(AIR_CO2_GAIN)
        qair_1 ← str2num(AIR_CO2_MAN) SLPM  if adj = 100
        otherwise
            multiplier ← Gain + adj
            qair_1 ← (multiplier)·qO2_2
            Max ← str2num(AIR_CO2_MAX)·SLPM
            qair_1 ← Max  if qair_1 > Max
            qair_1 ← MAX_qair_1  if qair_1 > MAX_qair_1
            qair_1 ← MIN_qair_1  if qair_1 < MIN_qair_1
    if AIR_CO2_CONTROL = 2
        Gain ← str2num(AIR_CO2_GAIN)
        qair_2 ← str2num(AIR_CO2_MAN) SLPM  if adj = 100
        otherwise
            multiplier ← Gain + adj
            qair_2 ← multiplier·qO2_2
            Max ← str2num(AIR_CO2_MAX)·SLPM
            qair_2 ← Max  if qair_2 > Max
            qair_2 ← MAX_qair_2  if qair_2 > MAX_qair_2
            qair_2 ← MIN_qair_2  if qair_2 < MIN_qair_2
if DO_CONTROL = 3
    qO2_1 ← qO2_1·(OUR / OUR_calc)
    if qO2_1 > MAX_qO2_1
        qO2_1 ← MAX_qO2_1
        Step ← Step + 1  if CASCADE_(Step+1) ≠ "None"
    if qO2_1 < MIN_qO2_1
        qO2_1 ← MIN_qO2_1
        Step ← Step − 1  if CASCADE_(Step−1) ≠ "None"
    qO2_2 ← qO2_1·(DOsplit − 1)
    if AIR_CO2_CONTROL = 1
        Gain ← str2num(AIR_CO2_GAIN)
```

```
qair_1 ← str2num(AIR_CO2_MAN) SLPM  if adj = 100
otherwise
    multiplier ← Gain + adj
    qair_1 ← (multiplier)·(qO2_1 + qO2_2)
    Max ← str2num(AIR_CO2_MAX)·SLPM
    qair_1 ← Max  if qair_1 > Max
    qair_1 ← MAX_qair_1  if qair_1 > MAX_qair_1
    qair_1 ← MIN_qair_1  if qair_1 < MIN_qair_1
if AIR_CO2_CONTROL = 2
    Gain ← str2num(AIR_CO2_GAIN)
    qair_2 ← str2num(AIR_CO2_MAN) SLPM  if adj = 100
    otherwise
        multiplier ← Gain + adj
        qair_2 ← (multiplier)·(qO2_1 + qO2_2)
        Max ← str2num(AIR_CO2_MAX)·SLPM
        qair_2 ← Max  if qair_2 > Max
        qair_2 ← MAX_qair_2  if qair_2 > MAX_qair_2
        qair_2 ← MIN_qair_2  if qair_2 < MIN_qair_2
if CASCADE_Step = "Pressure"
    p_HS ← p_HS·(OUR / OUR_calc)
    if p_HS > MAX_p_HS
        p_HS ← MAX_p_HS
        Step ← Step + 1  if CASCADE_Step+1 ≠ "None"
    if p_HS < MIN_p_HS
        p_HS ← MIN_p_HS
        Step ← Step − 1  if CASCADE_Step−1 ≠ "None"
if CASCADE_Step = "Proportional"
    if DO_CONTROL = 0
        qO2_ovly ← qO2_ovly·(OUR / OUR_calc)
        qair_ovly ← Proportional·qO2_ovly
        if qair_ovly > MAX_qair_ovly
            qair_ovly ← MAX_qair_ovly
            Step ← Step + 1  if CASCADE_Step+1 ≠ "None"
        if qair_ovly < MIN_qair_ovly
            qair_ovly ← MIN_qair_ovly
            Step ← Step − 1  if CASCADE_Step−1 ≠ "None"
    if DO_CONTROL = 1
```

$qO2\_1 \leftarrow qO2\_1 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$qair\_1 \leftarrow Proportional \cdot qO2\_1$

if $qair\_1 > MAX\_qair\_1$

$qair\_1 \leftarrow MAX\_qair\_1$

$Step \leftarrow Step + 1$ if $CASCADE_{Step+1} \neq$ "None"

$qair\_1 \leftarrow MIN\_qair\_1$ if $qair\_1 < MIN\_qair\_1$

if $DO\_CONTROL = 2$

$qO2\_2 \leftarrow qO2\_2 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$qO2\_2 \leftarrow Proportional \cdot qair\_2$

if $qair\_2 > MAX\_qair\_2$

$qair\_2 \leftarrow MAX\_qair\_2$

$Step \leftarrow Step + 1$ if $CASCADE_{Step+1} \neq$ "None"

if $qair\_2 < MIN\_qair\_2$

$qair\_2 \leftarrow MIN\_qair\_2$

$Step \leftarrow Step - 1$ if $CASCADE_{Step-1} \neq$ "None"

if $DO\_CONTROL = 3$

$qO2\_1 \leftarrow qO2\_1 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$qair\_1 \leftarrow Proportional \cdot qO2\_1$

$qO2\_2 \leftarrow qO2\_1 \cdot (DOsplit - 1)$

$qair\_2 \leftarrow Proportional \cdot qO2\_2$

if $qair\_1 > MAX\_qair\_1$

$qair\_1 \leftarrow MAX\_qair\_1$

$Step \leftarrow Step + 1$ if $CASCADE_{Step+1} \neq$ "None"

if $qair\_1 < MIN\_qair\_1$

$qair\_1 \leftarrow MIN\_qair\_1$

$Step \leftarrow Step - 1$ if $CASCADE_{Step-1} \neq$ "None"

if $CASCADE_{Step}$ = "Ratio Control"

if $DO\_CONTROL = 0$

$qO2\_ovly \leftarrow qO2\_ovly \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$qair\_ovly \leftarrow Total - qO2\_ovly$

if $qO2\_ovly > MAX\_qO2\_ovly$

$qO2\_ovly \leftarrow MAX\_qO2\_ovly$

$Step \leftarrow Step + 1$ if $CASCADE_{Step+1} \neq$ "None"

if $qO2\_ovly < MIN\_qO2\_ovly$

$qO2\_ovly \leftarrow MIN\_qO2\_ovly$

$Step \leftarrow Step - 1 \;\; \text{if} \;\; CASCADE_{Step-1} \neq \text{"None"}$

$qair\_ovly \leftarrow Total - qO2\_ovly$

$\text{if} \;\; DO\_CONTROL = 1$

$qO2\_1 \leftarrow qO2\_1 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$\text{if} \;\; qO2\_1 > MAX\_qO2\_1$

$qO2\_1 \leftarrow MAX\_qO2\_1$

$Step \leftarrow Step + 1 \;\; \text{if} \;\; CASCADE_{Step+1} \neq \text{"None"}$

$\text{if} \;\; qO2\_1 < MIN\_qO2\_1$

$qO2\_1 \leftarrow MIN\_qO2\_1$

$Step \leftarrow Step - 1 \;\; \text{if} \;\; CASCADE_{Step-1} \neq \text{"None"}$

$qair\_1 \leftarrow Total - qO2\_1$

$\text{if} \;\; DO\_CONTROL = 2$

$qO2\_2 \leftarrow qO2\_2 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$\text{if} \;\; qO2\_2 > MAX\_qO2\_2$

$qO2\_2 \leftarrow MAX\_qO2\_2$

$Step \leftarrow Step + 1 \;\; \text{if} \;\; CASCADE_{Step+1} \neq \text{"None"}$

$\text{if} \;\; qO2\_2 < MIN\_qO2\_2$

$qO2\_2 \leftarrow MIN\_qO2\_2$

$Step \leftarrow Step - 1 \;\; \text{if} \;\; CASCADE_{Step-1} \neq \text{"None"}$

$qair\_2 \leftarrow Total - qO2\_2$

$\text{if} \;\; DO\_CONTROL = 3$

$qO2\_1 \leftarrow qO2\_1 \cdot \left(\frac{OUR}{OUR_{calc}}\right)$

$\text{if} \;\; qO2\_1 > MAX\_qO2\_1$

$qO2\_1 \leftarrow MAX\_qO2\_1$

$Step \leftarrow Step + 1 \;\; \text{if} \;\; CASCADE_{Step+1} \neq \text{"None"}$

$\text{if} \;\; qO2\_1 < MIN\_qO2\_1$

$qO2\_1 \leftarrow MIN\_qO2\_1$

$Step \leftarrow Step - 1 \;\; \text{if} \;\; CASCADE_{Step-1} \neq \text{"None"}$

$qair\_1 \leftarrow Total - qO2\_1$

$qair\_2 \leftarrow qair\_1 \cdot (DOsplit - 1)$

$qO2\_2 \leftarrow qO2\_1 \cdot (DOsplit - 1)$

$$\left(\begin{array}{c} \frac{V_{Liq}}{L} \\ \frac{T_{Liq} - T_{0C}}{K} \\ \frac{p_{HS} - p_{atm}}{psi} \end{array} \right.$$

$$
\text{processVariables} \leftarrow \begin{pmatrix} \overline{\text{psi}} \\ \text{Nrpm}\cdot\text{min} \\ \frac{\text{qair\_ovly}}{\text{SLPM}} \\ \frac{\text{qO2\_ovly}}{\text{SLPM}} \\ \frac{\text{qCO2\_ovly}}{\text{SLPM}} \\ \frac{\text{qN2\_ovly}}{\text{SLPM}} \\ \frac{\text{qair\_1}}{\text{SLPM}} \\ \frac{\text{qO2\_1}}{\text{SLPM}} \\ \frac{\text{qCO2\_1}}{\text{SLPM}} \\ \frac{\text{qN2\_1}}{\text{SLPM}} \\ \frac{\text{qair\_2}}{\text{SLPM}} \\ \frac{\text{qO2\_2}}{\text{SLPM}} \\ \frac{\text{qCO2\_2}}{\text{SLPM}} \\ \frac{\text{qN2\_2}}{\text{SLPM}} \\ \frac{\text{OUR}}{\frac{\text{mmol}}{\text{L}\cdot\text{hr}}} \\ \frac{\text{lactate}}{\frac{\text{mmol}}{\text{L}}} \\ \frac{\text{ammonia}}{\frac{\text{mmol}}{\text{L}}} \\ \frac{V_{Base}}{\text{L}} \\ \frac{V_{FEED}}{\text{L}} \\ \text{DO}\cdot 100 \\ \text{pH} \\ \text{pH}_{Upper} \\ \text{pH}_{Lower} \\ \text{adj} \end{pmatrix}
$$

$$
\text{OUR}_{calc} \leftarrow \text{Re}\left[\text{OURcalc}(\text{processVariables}, \text{Antifoam}) + \left(\frac{-1}{10^{10}}\right)^{0.5} \cdot \frac{\text{mmol}}{\text{L}\cdot\text{hr}}\right]
$$

$$\text{if } \left(\frac{OUR}{OUR_{calc}}\right) > 1.02 \vee \left(\frac{OUR}{OUR_{calc}}\right) < 0.98$$

$$temp \leftarrow 1$$

$$\text{for } i \in 1..5$$

$$OUR1 \leftarrow OUR_{calc}$$

$$DO2 \leftarrow 1.1DO$$

$$processVariables_{21} \leftarrow DO2 \cdot 100$$

$$OUR2 \leftarrow Re\left[OURcalc(processVariables, Antifoam) + \left(\frac{-1}{10^{10}}\right)^{0.5} \cdot \frac{mmol}{L \cdot hr}\right]$$

$$D1 \leftarrow 0.0001 \cdot \frac{mol}{L \cdot hr} \quad \text{if } (OUR1 - OUR) - (OUR2 - OUR) = 0 \cdot \frac{mol}{L \cdot hr}$$

$$D1 \leftarrow (OUR1 - OUR) - (OUR2 - OUR) \quad \text{otherwise}$$

$$DO \leftarrow \frac{(OUR1 - OUR) \cdot DO2 - (OUR2 - OUR) \cdot DO1}{D1}$$

$$processVariables_{21} \leftarrow DO \cdot 100$$

$$OUR_{calc} \leftarrow Re\left[OURcalc(processVariables, Antifoam) + \left(\frac{-1}{10^{10}}\right)^{0.5} \cdot \frac{mmol}{L \cdot hr}\right]$$

$$DO1 \leftarrow DO$$

$$result \leftarrow processVariables$$

$$result$$

This program calculates the dissolved carbon dioxide that is in equilibrium with the gas flow. This allows for a numeric iteration to determine the resulting pH and determine whether the pH controller will take no action, add carbon dioxide, or add base.

$$
\begin{aligned}
\text{gasBalanceCO2(processVariables, Antifoam)} := \;& V_{Liq} \leftarrow \text{processVariables}_0 \cdot L \qquad (9.4-27)\\
& T_{Liq} \leftarrow \text{processVariables}_1 \cdot K + T_{0C}\\
& p_{HS} \leftarrow \text{processVariables}_2 \cdot psi + p_{atm}\\
& Nrpm \leftarrow \text{processVariables}_3 \cdot \frac{1}{min}\\
& qair\_ovly \leftarrow \text{processVariables}_4 \cdot SLPM\\
& qO2\_ovly \leftarrow \text{processVariables}_5 \cdot SLPM\\
& qCO2\_ovly \leftarrow \text{processVariables}_6 \cdot SLPM\\
& qN2\_ovly \leftarrow \text{processVariables}_7 \cdot SLPM\\
& qair\_1 \leftarrow \text{processVariables}_8 \cdot SLPM\\
& qO2\_1 \leftarrow \text{processVariables}_9 \cdot SLPM\\
& qCO2\_1 \leftarrow \text{processVariables}_{10} \cdot SLPM\\
& qN2\_1 \leftarrow \text{processVariables}_{11} \cdot SLPM\\
& qair\_2 \leftarrow \text{processVariables}_{12} \cdot SLPM\\
& qO2\_2 \leftarrow \text{processVariables}_{13} \cdot SLPM\\
& qCO2\_2 \leftarrow \text{processVariables}_{14} \cdot SLPM\\
& qN2\_2 \leftarrow \text{processVariables}_{15} \cdot SLPM\\
& OUR \leftarrow \text{processVariables}_{16} \cdot \frac{mmol}{L \cdot hr}\\
& lactate \leftarrow \text{processVariables}_{17} \cdot \frac{mmol}{L}\\
& ammonia \leftarrow \text{processVariables}_{18} \cdot \frac{mmol}{L}\\
& V_{Base} \leftarrow \text{processVariables}_{19} \cdot L\\
& V_{FEED} \leftarrow \text{processVariables}_{20} \cdot L\\
& DO \leftarrow \frac{\text{processVariables}_{21}}{100}\\
& pH \leftarrow \text{processVariables}_{22}\\
& pH_{Upper} \leftarrow \text{processVariables}_{23}\\
& pH_{Lower} \leftarrow \text{processVariables}_{24}\\
& qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)\\
& qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)\\
& M \leftarrow 0.5 + 0.5 \cdot \exp(-1 \cdot Antifoam)\\
& kla\_surf \leftarrow M \cdot \Big[kla\_surf \leftarrow Surface_{0,1} + Surface_{1,1} \cdot V_{Liq} \cdot \frac{1}{L} \ldots\\
& \qquad + Surface_{2,1} \cdot Nrpm \cdot min \cdot \frac{min}{s} \ldots\\
& \qquad + Surface_{3,1} (qtotal1 + qtotal2) \cdot \frac{1}{SLPM} \ldots\\
& \qquad + Surface_{4,1} \cdot \left(V_{Liq} \cdot Nrpm \cdot min \cdot \frac{min}{s} \right) \cdot \frac{1}{L} \ldots\\
& \qquad + Surface_{5,1} \cdot \left[V_{Liq} \cdot (qtotal1 + qtotal2) \right] \cdot \frac{1}{\ldots} \ldots
\end{aligned}
$$

$$+ Surface_{6,1}\cdot\left[Nrpm\cdot min\cdot\frac{min}{s}\cdot(qtotal1 + qtotal2)\right]\cdot\frac{1}{SLPM} \ldots$$

$$+ Surface_{8,1}\cdot\left(Nrpm\cdot min\cdot\frac{min}{s}\right)^2 \ldots$$

$$+ Surface_{9,1}\cdot(qtotal1 + qtotal2)^2\cdot\frac{1}{SLPM^2} \ldots$$

$$+ Surface_{7,1}\cdot V_{Liq}^{\ 2}\cdot\frac{1}{L^2}$$

$kla\_surf \leftarrow 0 \ \text{ if } \ kla\_surf < 0$

$Power \leftarrow 0\cdot W$

$\text{for } \ i \in 1..rows(Impeller) - 1$

$\text{if } \ Impeller_{i,2}\cdot L < V_{Liq}$

$$Power \leftarrow Power + Impeller_{i,0}\cdot(Nrpm)^3\cdot\left(Impeller_{i,1}\cdot m\right)^5\frac{gm}{cm^3}$$

$temp \leftarrow 1$

$$P\_V \leftarrow \frac{Power}{V_{Liq}}\cdot\frac{m^3}{W} \ \text{ if } \ V_{Liq} > 0\cdot L$$

$P\_V \leftarrow 0 \ \text{ otherwise}$

$$q\_sup1 \leftarrow \frac{qtotal1}{\pi\cdot\left(\frac{D}{2}\right)^2}\cdot\frac{s}{m}$$

$A1 \leftarrow Sparge_{1,0}$

$alpha1 \leftarrow Sparge_{1,1}$

$beta1 \leftarrow Sparge_{1,2}$

$kla1 \leftarrow M\cdot A1\cdot P\_V^{alpha1}\cdot q\_sup1^{beta1}$

$$q\_sup2 \leftarrow \frac{qtotal2}{\pi\cdot\left(\frac{D}{2}\right)^2}\cdot\frac{s}{m}$$

$A2 \leftarrow Sparge_{2,0}$

$alpha2 \leftarrow Sparge_{2,1}$

$beta2 \leftarrow Sparge_{2,2}$

$kla2 \leftarrow M\cdot A2\cdot P\_V^{alpha2}\cdot q\_sup2^{beta2}$

$$\begin{pmatrix} \frac{V_{Liq}}{L} \\ \frac{T_{Liq}}{K} \\ \frac{p_{HS}}{psi} \\ Nrpm\cdot min \\ \frac{qair\_ovly}{SLPM} \\ \frac{qO2\_ovly}{SLPM} \end{pmatrix}$$

$$\text{Inputs} \leftarrow \begin{pmatrix} \vdots \\ \frac{qCO2\_ovly}{SLPM} \\ \frac{qN2\_ovly}{SLPM} \\ \frac{qair\_1}{SLPM} \\ \frac{qO2\_1}{SLPM} \\ \frac{qCO2\_1}{SLPM} \\ \frac{qN2\_1}{SLPM} \\ \frac{qair\_2}{SLPM} \\ \frac{qO2\_2}{SLPM} \\ \frac{qCO2\_2}{SLPM} \\ \frac{qN2\_2}{SLPM} \\ \frac{OUR}{\frac{mmol}{L \cdot hr}} \\ kla\_surf \\ kla1 \\ kla2 \\ 0.011 \end{pmatrix}$$

$$T1 \leftarrow \text{Coefficient(Inputs)}_{15} \cdot \frac{mol}{L} + \text{Coefficient(Inputs)}_{17} \cdot \frac{mol}{L}$$

$$T2 \leftarrow \frac{\left(\text{Coefficient(Inputs)}_{21} \ldots + \text{Coefficient(Inputs)}_{23} \ldots + \text{Coefficient(Inputs)}_{25}\right) \cdot \left(\text{Coefficient(Inputs)}_{20} \cdot \frac{mol}{L}\right)}{\text{Coefficient(Inputs)}_{12} \ldots + \text{Coefficient(Inputs)}_{4} \cdot 0.893 \cdot \text{Coefficient(Inputs)}_{26} \cdot \text{Coefficient(Inputs)}_{14}}$$

$$T3 \leftarrow \left[\frac{\left(\text{Coefficient(Inputs)}_{22} \ldots + \text{Coefficient(Inputs)}_{24} \ldots + \text{Coefficient(Inputs)}_{4} \cdot 0.893 \cdot \text{Coefficient(Inputs)}_{14}\right) \cdot \text{Coefficient(Inputs)}_{20}}{\text{Coefficient(Inputs)}_{12} \ldots + \text{Coefficient(Inputs)}_{4} \cdot 0.893 \cdot \text{Coefficient(Inputs)}_{26} \cdot \text{Coefficient(Inputs)}_{14}} - 1\right]$$

$$\text{result} \leftarrow \frac{\left(T1 + \text{Coefficient(Inputs)}_{19} \cdot T2 + RQ\,OUR \cdot hr\right)}{H_{CO2\_cp}\left(0.011, T_{Liq}\right) \cdot mmHg \cdot \left(\text{Coefficient(Inputs)}_{16} \ldots + \text{Coefficient(Inputs)}_{18} \ldots + -\text{Coefficient(Inputs)}_{19} \cdot T3\right)}$$

$$\text{result}$$

The following equations load the medium properties for the basis medium and the proposed medium.

$$\begin{pmatrix} CO3_{basis} \\ HCO3_{basis} \\ H2CO3_{basis} \\ PO4_{basis} \\ HPO4_{basis} \\ H2PO4_{basis} \\ H3PO4_{basis} \\ HEPES_{basis} \\ Gluc \\ Lact \\ Amm \\ Ala \\ OH_{basis} \\ Salts_{A\_basis} \\ Salts_{B\_basis} \\ A_{eq\_basis} \end{pmatrix} := Conc^{\langle 1 \rangle} \qquad (9.4-28)$$

$$\begin{pmatrix} K_1 \\ K_2 \\ K_h \\ K_{1P} \\ K_{2P} \\ K_{3P} \\ K_{HEPES} \\ K_{Gluc} \\ K_{lact} \\ K_{1Ala} \\ K_{2Ala} \\ K_{amm} \\ K_w \\ K_a \\ K_b \end{pmatrix} := Kvalues^{\langle 1 \rangle} \qquad (9.4-29)$$

$$\begin{pmatrix} B_{amm} \\ B_{base} \\ B_{feed} \\ B_{lact} \\ B_{ala} \end{pmatrix} := Coeff^{\langle 1 \rangle} \qquad (9.4-30)$$

$$PO4\_basis := \begin{pmatrix} PO4_{basis} & HPO4_{basis} & H2PO4_{basis} & H3PO4_{basis} \end{pmatrix} \qquad (9.4-31)$$

$$carbonate\_basis := \begin{pmatrix} CO3_{basis} & HCO3_{basis} & H2CO3_{basis} \end{pmatrix} \qquad (9.4-32)$$

$$HEPES\_basis := HEPES_{basis} \qquad (9.4-33)$$

$$\begin{pmatrix} CO3_i \\ HCO3_i \\ H2CO3_i \\ PO4_i \\ HPO4_i \\ H2PO4_i \\ H3PO4_i \\ HEPES \\ Gluc \\ Lact \\ Amm \\ Ala \\ OH_i \end{pmatrix} := Conc2^{\langle 1 \rangle} \qquad (9.4-34)$$

$$PO4\_i := \begin{pmatrix} PO4_i & HPO4_i & H2PO4_i & H3PO4_i \end{pmatrix} \qquad (9.4-35)$$

$$carbonate\_i := \begin{pmatrix} CO3_i & HCO3_i & H2CO3_i \end{pmatrix} \qquad (9.4-36)$$

The equilibrium between the carbon dioxide and the pH is calculated as follows:

(9.4 − 37)

$$
\begin{aligned}
pCO2(pH, Temp) := \;& G1 \leftarrow \frac{\Phi\_carbonate(pH, carbonate\_i)}{\Phi\_carbonate(pH, carbonate\_basis)} \\
& G2 \leftarrow \frac{\Phi\_total\left(pH, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_{A\_basis}, Salts_{B\_basis}\right)}{\Phi\_total\left(pH, carbonate\_i, PO4\_i, HEPES, Salts_{A\_basis}, Salts_{B\_basis}\right)} \\
& result \leftarrow \left[\frac{\left[\dfrac{10^{-2 \cdot pH}}{\left[2 \times 10^{K_1+K_2} \ldots + 2 \times 10^{\left(K_1+K_2+K_h\right)} \ldots + 10^{\left(K_1-pH\right)} \ldots + 10^{\left(K_1+K_h-pH\right)}\right]} \cdot \left[HCO3_i \cdot \dfrac{mol}{L} + 2CO3_i \cdot \dfrac{mol}{L} \ldots + G1 \cdot G2 \cdot \left[10^{-pH} \cdot \dfrac{mole}{L} \ldots + OH_i \cdot \dfrac{mol}{L} \ldots + -\dfrac{10^{-14}}{10^{-pH}} \cdot \dfrac{mole}{L} \ldots + -\Delta H\_PO4(pH, PO4\_i) \ldots + -\dfrac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-pH}} \cdot HEPES \cdot \dfrac{mol}{L} \ldots + \dfrac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-pH}} \cdot (Gluc) \cdot \dfrac{mol}{L} \ldots + -\dfrac{10^{K_a}}{10^{K_a} + 10^{-pH}} \cdot Salts_{A\_basis} \cdot \dfrac{mol}{L} \ldots + \dfrac{10^{-pH}}{10^{K_b} + 10^{-pH}} \cdot Salts_{B\_basis} \cdot \dfrac{mol}{L} \ldots + -A_{eq\_basis} \cdot \dfrac{mol}{L}\right]\right]\right]}{H_{CO2\_cp}(str2num(saline), Temp) \cdot mmHg}\right]
\end{aligned}
$$

This program is calculating the resulting pH due to the dissolved carbon dioxide that would be in equilibrium with the gas stream, the medium buffering, and the addition of base, feeds, and metabolites. This value is used to determine what action the pH controller would take to maintain the pH within the target range.

$$
\text{Resulting\_pH}(\text{procVariables}, \text{CO2\_gasBal}) := \left|
\begin{aligned}
& V_{Liq} \leftarrow \text{procVariables}_0 \cdot L \\
& T_{Liq} \leftarrow \text{procVariables}_1 \cdot K + T_{0C} \\
& p_{HS} \leftarrow \text{procVariables}_2 \cdot \text{psi} + p_{atm} \\
& \text{Nrpm} \leftarrow \text{procVariables}_3 \cdot \frac{1}{\text{min}} \\
& \text{qair\_ovly} \leftarrow \text{procVariables}_4 \cdot \text{SLPM} \\
& \text{qO2\_ovly} \leftarrow \text{procVariables}_5 \cdot \text{SLPM} \\
& \text{qCO2\_ovly} \leftarrow \text{procVariables}_6 \cdot \text{SLPM} \\
& \text{qN2\_ovly} \leftarrow \text{procVariables}_7 \cdot \text{SLPM} \\
& \text{qair\_1} \leftarrow \text{procVariables}_8 \cdot \text{SLPM} \\
& \text{qO2\_1} \leftarrow \text{procVariables}_9 \cdot \text{SLPM} \\
& \text{qCO2\_1} \leftarrow \text{procVariables}_{10} \cdot \text{SLPM} \\
& \text{qN2\_1} \leftarrow \text{procVariables}_{11} \cdot \text{SLPM} \\
& \text{qair\_2} \leftarrow \text{procVariables}_{12} \cdot \text{SLPM} \\
& \text{qO2\_2} \leftarrow \text{procVariables}_{13} \cdot \text{SLPM} \\
& \text{qCO2\_2} \leftarrow \text{procVariables}_{14} \cdot \text{SLPM} \\
& \text{qN2\_2} \leftarrow \text{procVariables}_{15} \cdot \text{SLPM} \\
& \text{OUR} \leftarrow \text{procVariables}_{16} \cdot \frac{\text{mmol}}{L \cdot \text{hr}} \\
& \text{lactate} \leftarrow \text{procVariables}_{17} \cdot \frac{\text{mmol}}{L} \\
& \text{ammonia} \leftarrow \text{procVariables}_{18} \cdot \frac{\text{mmol}}{L} \\
& V_{Base} \leftarrow \text{procVariables}_{19} \cdot L \\
& V_{FEED} \leftarrow \text{procVariables}_{20} \cdot L \\
& \text{DO} \leftarrow \frac{\text{procVariables}_{21}}{100} \\
& \text{pH} \leftarrow \text{procVariables}_{22} \\
& \text{pH}_{Upper} \leftarrow \text{procVariables}_{23} \\
& \text{pH}_{Lower} \leftarrow \text{procVariables}_{24} \\
& \text{CO2\_gasBal} \leftarrow \frac{\text{CO2\_gasBal}}{\exp\left[0.04375 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)\right]} \\
& \text{pH}_{Upper} \leftarrow \frac{\text{pH}_{Upper} - 0.033 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)}{1 - 0.0065 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)} \\
& \text{pH}_{Lower} \leftarrow \frac{\text{pH}_{Lower} - 0.033 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)}{1 - 0.0065 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)}
\end{aligned}
\right. \qquad (9.4 - 38)
$$

$$pH_{est} \leftarrow \frac{pH_{Lower} + pH_{Upper}}{2}$$

for $i \in 1..10$

$$carbonate\_i \leftarrow \left(CO3_i + B_{base} \cdot \frac{V_{Base}}{V_{Liq}} \cdot Base\_Conc \cdot \frac{L}{mol} \quad HCO3_i \quad H2CO3_i \right)$$

$$G1 \leftarrow \frac{\Phi\_carbonate(pH_{est}, carbonate\_i)}{\Phi\_carbonate(pH_{est}, carbonate\_basis)}$$

$$G2 \leftarrow \frac{\Phi\_total(pH_{est}, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_{A\_basis}, Salts_{B\_basis})}{\Phi\_total(pH_{est}, carbonate\_i, PO4\_i, HEPES, Salts_{A\_basis}, Salts_{B\_basis})}$$

$$\left[\frac{10^{-2 \cdot pH_{est}}}{\left[\begin{array}{l} 2 \times 10^{K_1+K_2} \ldots \\ + 2 \times 10^{(K_1+K_2+K_h)} \ldots \\ + 10^{(K_1-pH_{est})} \ldots \\ + 10^{(K_1+K_h-pH_{est})} \end{array} \right]} \cdot \left[\begin{array}{l} HCO3_i \cdot \frac{mol}{L} \ldots \\ + 2\left(\begin{array}{l} CO3_i \ldots \\ + B_{base} \cdot \frac{V_{Base}}{V_{Liq}} \cdot Base\_Conc \cdot \frac{L}{mol} \end{array} \right) \cdot \frac{mol}{L} \ldots \\ + G1 \cdot G2 \cdot \left[\begin{array}{l} 10^{-pH_{est}} \cdot \frac{mole}{L} \ldots \\ + B_{feed} \cdot \frac{V_{FEED}}{V_0} \cdot \frac{mol}{L} \ldots \\ + OH_i \cdot \frac{mol}{L} \ldots \\ + -\frac{10^{-14}}{10^{-pH_{est}}} \cdot \frac{mole}{L} \ldots \\ + -\Delta H\_PO4(pH_{est}, PO4\_i) \ldots \\ + -\frac{10^{K_{HEPES}} \cdot \left(HEPES \cdot \frac{mol}{L} \right)}{10^{K_{HEPES}} + 10^{-pH_{est}}} \ldots \\ + -\frac{10^{K_a} \cdot \left(Salts_{A\_basis} \cdot \frac{mol}{L} \right)}{10^{K_a} + 10^{-pH_{est}}} \ldots \\ + \frac{10^{-pH_{est}} \cdot \left(Salts_{B\_basis} \cdot \frac{mol}{L} \right)}{10^{K_b} + 10^{-pH_{est}}} \ldots \\ + -A_{eq\_basis} \cdot \frac{mol}{L} \ldots \\ + \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-pH_{est}}} \cdot (Gluc) \cdot \frac{mol}{L} \ldots \\ + \frac{-B_{lact} \cdot 10^{K_{lact}}}{10^{K_{lact}} + 10^{-pH_{est}}} \cdot lactate \ldots \\ + \frac{B_{amm} \cdot 10^{-pH_{est}}}{10^{K_{amm}} + 10^{-pH_{est}}} \cdot ammonia \ldots \\ + -\left(\frac{\begin{array}{l} 2 \cdot 10^{K_{1Ala}+K_{2Ala}} \ldots \\ + 10^{K_{1Ala}-pH_{est}} \end{array}}{10^{K_{1Ala}+K_{2Ala}} \ldots} \right) \cdot B_{ala} \cdot lactate \end{array} \right. \end{array} \right.$$

$$CO2\_med \leftarrow \frac{\left[\quad \left[\quad \left[\left(+ 10^{K_{1Ala}-pH_{est}} \ldots + 10^{-2pH_{est}} \right) \right] \right] \right]}{H_{CO2\_cp}\left(str2num(saline), T_{Liq}\right) \cdot mmHg}$$

$$pH_{est} \leftarrow pH_{est} + 0.1 \cdot \left(\frac{CO2\_med - CO2\_gasBal}{CO2\_med} \right)$$

$$pH_{est} \leftarrow 5 \ \text{ if } \ pH_{est} < 5$$

$$pH_{est} \leftarrow 8 \ \text{ if } \ pH_{est} > 8$$

$$pH_{est} \leftarrow pH_{est} + \left[-0.0147 + 0.0065 \cdot \left(7.400 - pH_{est} \right) \right] \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37 \right)$$

$$\text{return} \leftarrow pH_{est}$$

This program calculates the volume of base added to the culture to keep the pH within the target range.

$$AddBase(procVariables, CO2\_gasBal) := \quad (9.4 - 39)$$

$$V_{Liq} \leftarrow procVariables_0 \cdot L$$

$$T_{Liq} \leftarrow procVariables_1 \cdot K + T_{0C}$$

$$p_{HS} \leftarrow procVariables_2 \cdot psi + p_{atm}$$

$$Nrpm \leftarrow procVariables_3 \cdot \frac{1}{min}$$

$$qair\_ovly \leftarrow procVariables_4 \cdot SLPM$$

$$qO2\_ovly \leftarrow procVariables_5 \cdot SLPM$$

$$qCO2\_ovly \leftarrow procVariables_6 \cdot SLPM$$

$$qN2\_ovly \leftarrow procVariables_7 \cdot SLPM$$

$$qair\_1 \leftarrow procVariables_8 \cdot SLPM$$

$$qO2\_1 \leftarrow procVariables_9 \cdot SLPM$$

$$qCO2\_1 \leftarrow procVariables_{10} \cdot SLPM$$

$$qN2\_1 \leftarrow procVariables_{11} \cdot SLPM$$

$$qair\_2 \leftarrow procVariables_{12} \cdot SLPM$$

$$qO2\_2 \leftarrow procVariables_{13} \cdot SLPM$$

$$qCO2\_2 \leftarrow procVariables_{14} \cdot SLPM$$

$$qN2\_2 \leftarrow procVariables_{15} \cdot SLPM$$

$$OUR \leftarrow procVariables_{16} \cdot \frac{mmol}{L \cdot hr}$$

$$lactate \leftarrow procVariables_{17} \cdot \frac{mmol}{L}$$

$$ammonia \leftarrow procVariables_{18} \cdot \frac{mmol}{L}$$

$$V_{Base} \leftarrow procVariables_{19} \cdot L$$

$$V_{FEED} \leftarrow procVariables_{20} \cdot L$$

$$DO \leftarrow \frac{procVariables_{21}}{100}$$

100

$pH \leftarrow procVariables_{22}$

$pH_{Upper} \leftarrow procVariables_{23}$

$pH_{Lower} \leftarrow procVariables_{24}$

$$CO2\_gasBal \leftarrow \frac{CO2\_gasBal}{\exp\left[0.04375\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)\right]}$$

$$pH_{Lower} \leftarrow \frac{pH_{Lower}-0.033\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)}{1-0.0065\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)}$$

$temp \leftarrow V_{Base}$

$temp2 \leftarrow temp$

for $i \in 1..10$

$$temp \leftarrow \frac{temp+temp2}{2}$$

$$carbonate\_i \leftarrow \left(CO3_i + B_{base}\cdot\frac{temp}{V_{Liq}}\cdot Base\_Conc\cdot\frac{L}{mol} \quad HCO3_i \quad H2CO3_i\right)$$

$$G1 \leftarrow \frac{\Phi\_carbonate(pH_{Lower}, carbonate\_i)}{\Phi\_carbonate(pH_{Lower}, carbonate\_basis)}$$

$$G2 \leftarrow \frac{\Phi\_total(pH_{Lower}, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_{A\_basis}, Salts_{B\_basis})}{\Phi\_total(pH_{Lower}, carbonate\_i, PO4\_i, HEPES, Salts_{A\_basis}, Salts_{B\_basis})}$$

$$\Bigg[\Bigg[\frac{CO2\_gasBal\cdot H_{CO2\_cp}(0.011, T_{Liq})\cdot mmHg}{10^{-2\cdot pH_{Lower}}\left[2\times10^{K_1+K_2} \ldots + 2\times10^{(K_1+K_2+K_h)} \ldots + 10^{(K_1-pH_{Lower})} \ldots + 10^{(K_1+K_h-pH_{Lower})}\right]} \ldots$$

$$+ -\Bigg[HCO3_i\cdot\frac{mol}{L} \ldots$$

$$+ 2CO3_i\cdot\frac{mol}{L} \ldots$$

$$+ G1\cdot G2\cdot\Bigg[10^{-pH_{Lower}}\cdot\frac{mole}{L} \ldots$$

$$+ B_{feed}\cdot\frac{V_{FEED}}{V_0}\cdot\frac{mol}{L} \ldots$$

$$+ OH_i\cdot\frac{mol}{L} \ldots$$

$$+ -\frac{10^{-14}}{10^{-pH_{Lower}}}\cdot\frac{mole}{L} \ldots$$

$$+ -\Delta H\_PO4(pH_{Lower}, PO4\_i) \ldots$$

$$+ -\frac{10^{K_{HEPES}}}{10^{K_{HEPES}} + 10^{-pH_{Lower}}}\cdot HEPES\cdot\frac{mol}{L} \ldots$$

$$+ -\frac{10^{K_a}}{10^{K_a} + 10^{-pH_{Lower}}}\cdot Salts_{A\_basis}\cdot\frac{mol}{L} \ldots$$

$$10 \quad + 10$$

$$+ \frac{10^{-pH_{Lower}}}{10^{K_b} + 10^{-pH_{Lower}}} \cdot Salts_{B\_basis} \cdot \frac{mol}{L} \ldots$$

$$+ -A_{eq\_basis} \cdot \frac{mol}{L} \ldots$$

$$+ \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-pH_{Lower}}} \cdot (Gluc) \cdot \frac{mol}{L} \ldots$$

$$+ \frac{-B_{lact} \cdot 10^{K_{lact}}}{10^{K_{lact}} + 10^{-pH_{Lower}}} \cdot lactate \ldots$$

$$+ \frac{B_{amm} \cdot 10^{-pH_{Lower}}}{10^{K_{amm}} + 10^{-pH_{Lower}}} \cdot ammonia \ldots$$

$$+ -\left(\frac{2 \cdot 10^{K_{1Ala}+K_{2Ala}} \ldots + 10^{K_{1Ala}-pH_{Lower}}}{10^{K_{1Ala}+K_{2Ala}} \ldots + 10^{K_{1Ala}-pH_{Lower}} \ldots + 10^{-2pH_{Lower}}} \right) \cdot B_{ala} \cdot lactate$$

$$temp2 \leftarrow \frac{[\ldots]}{\frac{2B_{base} \cdot Base\_Conc}{V_{Liq}}}$$

$$result \leftarrow temp$$

This program calculates the amount of carbon dioxide added through the controlling gas stream to maintain the culture within the target pH range.

AddCO2(procVariables, CO2_gasBal, Antifoam) := (9.4 – 40)

$V_{Liq} \leftarrow procVariables_0 \cdot L$

$T_{Liq} \leftarrow procVariables_1 \cdot K + T_{0C}$

$p_{HS} \leftarrow procVariables_2 \cdot psi + p_{atm}$

$Nrpm \leftarrow procVariables_3 \cdot \frac{1}{min}$

$qair\_ovly \leftarrow procVariables_4 \cdot SLPM$

$qO2\_ovly \leftarrow procVariables_5 \cdot SLPM$

$qCO2\_ovly \leftarrow procVariables_6 \cdot SLPM$

$qN2\_ovly \leftarrow procVariables_7 \cdot SLPM$

$qair\_1 \leftarrow procVariables_8 \cdot SLPM$

$qO2\_1 \leftarrow procVariables_9 \cdot SLPM$

$qCO2\_1 \leftarrow procVariables_{10} \cdot SLPM$

$qN2\_1 \leftarrow procVariables_{11} \cdot SLPM$

$qair\_2 \leftarrow procVariables_{12} \cdot SLPM$

$qO2\_2 \leftarrow procVariables_{13} \cdot SLPM$

$qCO2\_2 \leftarrow procVariables_{14} \cdot SLPM$

$qN2\_2 \leftarrow procVariables_{15} \cdot SLPM$

$OUR \leftarrow procVariables_{16} \cdot \frac{mmol}{L \cdot hr}$

$lactate \leftarrow procVariables_{17} \cdot \frac{mmol}{L}$

$ammonia \leftarrow procVariables_{18} \cdot \frac{mmol}{L}$

$V_{Base} \leftarrow procVariables_{19} \cdot L$

$V_{FEED} \leftarrow procVariables_{20} \cdot L$

$DO \leftarrow \frac{procVariables_{21}}{100}$

$pH \leftarrow procVariables_{22}$

$pH_{Upper} \leftarrow procVariables_{23}$

$pH_{Lower} \leftarrow procVariables_{24}$

$adj \leftarrow procVariables_{25}$

$MIN\_qCO2\_ovly \leftarrow Limits_{6,1} \cdot SLPM$

$MAX\_qCO2\_ovly \leftarrow Limits_{6,3} \cdot SLPM$

$MIN\_qCO2\_1 \leftarrow Limits_{10,1} \cdot SLPM$

$MAX\_qCO2\_1 \leftarrow Limits_{10,3} \cdot SLPM$

$MIN\_qCO2\_2 \leftarrow Limits_{14,1} \cdot SLPM$

$MAX\_qCO2\_2 \leftarrow Limits_{14,3} \cdot SLPM$

$pH_{LL} \leftarrow -0.033 \cdot \left(\frac{T_{Liq} - T_{0C}}{} - 37 \right)$

$$pHcalc \leftarrow \frac{pH_{Upper} \cdots 0.0035 \cdots \left(\frac{\cdots}{K} \cdots 37 \right)}{1 - 0.0065 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37 \right)}$$

$$carbonate\_i \leftarrow \left(CO3_i + B_{base} \cdot \frac{V_{Base}}{V_{Liq}} \cdot Base\_Conc \cdot \frac{L}{mol} \quad HCO3_i \quad H2CO3_i \right)$$

$$G1 \leftarrow \frac{\Phi\_carbonate(pHcalc, carbonate\_i)}{\Phi\_carbonate(pHcalc, carbonate\_basis)}$$

$$G2 \leftarrow \frac{\Phi\_total\left(pHcalc, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_{A\_basis}, Salts_{B\_basis}\right)}{\Phi\_total\left(pHcalc, carbonate\_i, PO4\_i, HEPES, Salts_{A\_basis}, Salts_{B\_basis}\right)}$$

$$\frac{10^{-2 \cdot pHcalc}}{\left[\begin{array}{l} 2 \times 10^{K_1+K_2} \ldots \\ + 2 \times 10^{(K_1+K_2+K_h)} \ldots \\ + 10^{(K_1-pHcalc)} \ldots \\ + 10^{(K_1+K_h-pHcalc)} \end{array} \right]} \cdot \left[\begin{array}{l} HCO3_i \cdot \frac{mol}{L} \ldots \\ + 2\left(\begin{array}{l} CO3_i \ldots \\ + B_{base} \cdot \frac{V_{Base}}{V_{Liq}} \cdot Base\_Conc \cdot \frac{L}{mol} \end{array} \right) \cdot \frac{mol}{L} \ldots \\ + G1 \cdot G2 \cdot \left[\begin{array}{l} 10^{-pHcalc} \frac{mole}{L} \ldots \\ + B_{base} \cdot \frac{V_{Base}}{V_{Liq}} \cdot Base\_Conc \ldots \\ + B_{feed} \cdot \frac{V_{FEED}}{V_0} \cdot \frac{mol}{L} \ldots \\ + OH_i \cdot \frac{mol}{L} \ldots \\ + -\frac{10^{-14}}{10^{-pHcalc}} \cdot \frac{mole}{L} \ldots \\ + -\Delta H\_PO4(pHcalc, PO4\_i) \ldots \\ + -\frac{10^{K_{HEPES}} \cdot \left(HEPES \cdot \frac{mol}{L} \right)}{10^{K_{HEPES}} + 10^{-pHcalc}} \ldots \\ + -\frac{10^{K_a} \cdot \left(Salts_{A\_basis} \cdot \frac{mol}{L} \right)}{10^{K_a} + 10^{-pHcalc}} \ldots \\ + \frac{10^{-pHcalc} \cdot \left(Salts_{B\_basis} \cdot \frac{mol}{L} \right)}{10^{K_b} + 10^{-pHcalc}} \ldots \\ + -A_{eq\_basis} \cdot \frac{mol}{L} \ldots \\ + \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-pHcalc}} \cdot (Gluc) \cdot \frac{mol}{L} \ldots \\ + \frac{-B_{lact} \cdot 10^{K_{lact}}}{10^{K_{lact}} + 10^{-pHcalc}} \cdot lactate \ldots \\ + \frac{B_{amm} \cdot 10^{-pHcalc}}{10^{K_{amm}} + 10^{-pHcalc}} \cdot ammonia \ldots \\ + -\left(\frac{\begin{array}{l} 2 \cdot 10^{K_{1Ala}+K_{2Ala}} \ldots \\ + 10^{K_{1Ala}-pHcalc} \end{array}}{\begin{array}{l} 10^{K_{1Ala}+K_{2Ala}} \ldots \\ \cdots \end{array}} \right) \cdot B_{ala} \cdot lactate \end{array} \right. \end{array} \right.$$

$$+10^{K_{1Ala}-pHcalc} \ldots$$
$$+10^{-2pHcalc}$$

$$CO2\_med \leftarrow \frac{\cdots}{H_{CO2\_cp}(0.011, T_{Liq})\cdot mmHg}$$

$$CO2\_med \leftarrow Re\left[CO2\_med\cdot exp\left[0.04375\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)\right]+\left(\frac{-1}{10^{10}}\right)^{0.5}\right]$$

for $i \in 1..10$

if CO2_CONTROL = 0

$$qCO2\_ovly \leftarrow Re\left[qCO2\_ovly\cdot\frac{CO2\_med}{CO2\_gasBal}+\left(\frac{-1}{10^{10}}\right)^{0.5}\cdot SLPM\right]$$

$$minFlow \leftarrow \frac{str2num(Min\_CO2)}{100-str2num(Min\_CO2)}\cdot(qair\_ovly+qO2\_ovly+qN2\_ovly)$$

qCO2_ovly ← minFlow if qCO2_ovly < minFlow

qCO2_ovly ← MIN_qCO2_ovly if qCO2_ovly < MIN_qCO2_ovly

qCO2_ovly ← MAX_qCO2_ovly if qCO2_ovly > MAX_qCO2_ovly

if CO2_CONTROL = 1

$$qCO2\_1 \leftarrow Re\left[qCO2\_1\cdot\frac{CO2\_med}{CO2\_gasBal}+\left(\frac{-1}{10^{10}}\right)^{0.5}\cdot SLPM\right]$$

$$minFlow \leftarrow \frac{str2num(Min\_CO2)}{100-str2num(Min\_CO2)}\cdot(qair\_1+qO2\_1+qN2\_1)$$

qCO2_1 ← minFlow if qCO2_1 < minFlow

qCO2_1 ← MIN_qCO2_1 if qCO2_1 < MIN_qCO2_1

qCO2_1 ← MAX_qCO2_1 if qCO2_1 > MAX_qCO2_1

if CO2_CONTROL = 2

$$qCO2\_2 \leftarrow Re\left[qCO2\_2\cdot\frac{CO2\_med}{CO2\_gasBal}+\left(\frac{-1}{10^{10}}\right)^{0.5}\cdot SLPM\right]$$

$$minFlow \leftarrow \frac{str2num(Min\_CO2)}{100-str2num(Min\_CO2)}\cdot(qair\_2+qO2\_2+qN2\_2)$$

qCO2_2 ← minFlow if qCO2_2 < minFlow

qCO2_2 ← MIN_qCO2_2 if qCO2_2 < MIN_qCO2_2

qCO2_2 ← MAX_qCO2_2 if qCO2_2 > MAX_qCO2_2

qCO2_2 ← MIN_qCO2_2

if CO2_CONTROL = 3

$$qCO2\_1 \leftarrow Re\left[qCO2\_1\cdot\frac{CO2\_med}{CO2\_gasBal}+\left(\frac{-1}{10^{10}}\right)^{0.5}\cdot SLPM\right]$$

qCO2_1 ← MIN_qCO2_1 if qCO2_1 < MIN_qCO2_1

qCO2_1 ← MAX_qCO2_1 if qCO2_1 > MAX_qCO2_1

qCO2_2 ← qCO2_1·CO2split

qCO2_2 ← MIN_qCO2_2 if qCO2_2 < MIN_qCO2_2

qCO2_2 ← MAX_qCO2_2 if qCO2_2 > MAX_qCO2_2

$$\left(\frac{V_{Liq}}{L}\right.$$

$$
procVariables \leftarrow \left| \begin{array}{c}
\frac{T_{Liq} - T_{0C}}{K} \\
\frac{p_{HS} - p_{atm}}{psi} \\
Nrpm \cdot min \\
\frac{qair\_ovly}{SLPM} \\
\frac{qO2\_ovly}{SLPM} \\
\frac{qCO2\_ovly}{SLPM} \\
\frac{qN2\_ovly}{SLPM} \\
\frac{qair\_1}{SLPM} \\
\frac{qO2\_1}{SLPM} \\
\frac{qCO2\_1}{SLPM} \\
\frac{qN2\_1}{SLPM} \\
\frac{qair\_2}{SLPM} \\
\frac{qO2\_2}{SLPM} \\
\frac{qCO2\_2}{SLPM} \\
\frac{qN2\_2}{SLPM} \\
\frac{OUR}{\frac{mmol}{L \cdot hr}} \\
\frac{lactate}{\frac{mmol}{L}} \\
\frac{ammonia}{\frac{mmol}{L}} \\
\frac{V_{Base}}{L} \\
\frac{V_{FEED}}{L} \\
DO \cdot 100 \\
pH \\
pH_{Upper} \\
pH_{Lower}
\end{array} \right.
$$

$$\left(\quad adj \quad \right)$$

$$CO2\_gasBal \leftarrow Re\left[gasBalanceCO2(procVariables, Antifoam) + \left(\frac{-1}{10^{10}}\right)^{0.5}\right]$$

$$CO2\_gasBal \leftarrow 1 \ \text{ if } \ CO2\_gasBal < 1$$

$$procVariables \leftarrow \begin{pmatrix} \frac{V_{Liq}}{L} \\ \frac{T_{Liq} - T_{0C}}{K} \\ \frac{p_{HS} - p_{atm}}{psi} \\ Nrpm \cdot min \\ \frac{qair\_ovly}{SLPM} \\ \frac{qO2\_ovly}{SLPM} \\ \frac{qCO2\_ovly}{SLPM} \\ \frac{qN2\_ovly}{SLPM} \\ \frac{qair\_1}{SLPM} \\ \frac{qO2\_1}{SLPM} \\ \frac{qCO2\_1}{SLPM} \\ \frac{qN2\_1}{SLPM} \\ \frac{qair\_2}{SLPM} \\ \frac{qO2\_2}{SLPM} \\ \frac{qCO2\_2}{SLPM} \\ \frac{qN2\_2}{SLPM} \\ \frac{OUR}{\frac{mmol}{L \cdot hr}} \\ \frac{lactate}{\frac{mmol}{L}} \\ \frac{ammonia}{\frac{mmol}{L}} \\ \frac{V_{Base}}{L} \end{pmatrix}$$

$$\begin{pmatrix} \frac{V_{FEED}}{L} \\ CO2\_med \\ pH \\ pH_{Upper} \\ pH_{Lower} \\ adj \end{pmatrix}$$

$result \leftarrow procVariables$

This model steps through the logic presented in Figure 5-1. The resulting process control parameters and process outcomes are captured in the matrix Model. The program is only performed when the user selects section 9.4 to be active.

$$\text{Model} := \begin{array}{|l} V_{Base} \leftarrow 0L \\ V_{FEED} \leftarrow 0 \cdot L \\ \text{Antifoam} \leftarrow 0 \\ \text{eTime} \leftarrow 0 \\ \text{count\_N} \leftarrow 0 \\ \text{while } \text{eTime} \le \frac{\text{Time\_end}}{\text{hr}} \\ \quad \begin{array}{|l} \text{ProcessColumn} \leftarrow 2 \\ \text{while } \text{Proposed}_{0,\text{ProcessColumn}} \le \text{eTime} \\ \quad \text{ProcessColumn} \leftarrow \text{ProcessColumn} + 1 \\ \text{ProcessColumn} \leftarrow \text{ProcessColumn} - 1 \\ T_{Liq} \leftarrow \text{Proposed}_{2,\text{ProcessColumn}} \cdot K + T_{0C} \\ p_{HS} \leftarrow \text{Proposed}_{4,\text{ProcessColumn}} \cdot psi + p_{atm} \\ \text{Nrpm} \leftarrow \text{Proposed}_{5,\text{ProcessColumn}} \cdot \frac{1}{min} \\ \text{qair\_ovly} \leftarrow \text{Proposed}_{8,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qO2\_ovly} \leftarrow \text{Proposed}_{9,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qN2\_ovly} \leftarrow \text{Proposed}_{11,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qCO2\_ovly} \leftarrow \text{Proposed}_{10,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qair\_1} \leftarrow \text{Proposed}_{12,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qO2\_1} \leftarrow \text{Proposed}_{13,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qN2\_1} \leftarrow \text{Proposed}_{15,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qCO2\_1} \leftarrow \text{Proposed}_{14,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qair\_2} \leftarrow \text{Proposed}_{16,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qO2\_2} \leftarrow \text{Proposed}_{17,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qN2\_2} \leftarrow \text{Proposed}_{19,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \\ \text{qCO2\_2} \leftarrow \text{Proposed}_{18,\text{ProcessColumn}} \cdot SLPM + 0.00001 \cdot SLPM \end{array} \end{array} \quad \text{if Part4} = 1 \qquad (9.4-41)$$

$$OUR \leftarrow OUR\_proc(eTime)\cdot\frac{mmol}{L\cdot hr}$$

$$lact \leftarrow lact\_proc(eTime)\cdot\frac{mmol}{L}$$

$$ammonia \leftarrow amm\_proc(eTime)\cdot\frac{mmol}{L}$$

$$V_{FEED} \leftarrow Proposed_{20,ProcessColumn}\cdot V_0 + V_{FEED} \text{ if } Proposed_{0,ProcessColumn} > eTime - \frac{Time\_int}{hr}$$

$$V_{Liq} \leftarrow V_0 + V_{FEED}$$

$$DO \leftarrow \frac{Proposed_{3,ProcessColumn}}{100}$$

$$pH_{Upper} \leftarrow Proposed_{7,ProcessColumn}$$

$$pH_{Lower} \leftarrow Proposed_{6,ProcessColumn}$$

$$pH \leftarrow 0.5pH_{Upper} + 0.5pH_{Lower}$$

$$adj \leftarrow str2num(AIR\_CO2\_ADJ)\cdot\left(\frac{pH_{Upper} + pH_{Lower}}{2} - pH\right)$$

$$Antifoam \leftarrow Proposed_{22,ProcessColumn} + Antifoam\cdot\exp\left(\frac{-Time\_int}{62.4\cdot hr}\right)$$

$$procVariables \leftarrow \begin{pmatrix} \frac{V_{Liq}}{L} \\ \frac{T_{Liq} - T_{0C}}{K} \\ \frac{p_{HS} - p_{atm}}{psi} \\ Nrpm\cdot min \\ \frac{qair\_ovly}{SLPM} \\ \frac{qO2\_ovly}{SLPM} \\ \frac{qCO2\_ovly}{SLPM} \\ \frac{qN2\_ovly}{SLPM} \\ \frac{qair\_1}{SLPM} \\ \frac{qO2\_1}{SLPM} \\ \frac{qCO2\_1}{SLPM} \\ \frac{qN2\_1}{SLPM} \\ \frac{qair\_2}{SLPM} \\ \frac{qO2\_2}{SLPM} \\ qCO2\_2 \end{pmatrix}$$

$$\begin{pmatrix} \overline{\text{SLPM}} \\ \frac{\text{qN2\_2}}{\text{SLPM}} \\ \frac{\text{OUR}}{\frac{\text{mmol}}{\text{L}\cdot\text{hr}}} \\ \frac{\text{lact}}{\frac{\text{mmol}}{\text{L}}} \\ \frac{\text{ammonia}}{\frac{\text{mmol}}{\text{L}}} \\ \frac{V_{Base}}{\text{L}} \\ \frac{V_{FEED}}{\text{L}} \\ \text{DO}\cdot 100 \\ \text{pH} \\ \text{pH}_{Upper} \\ \text{pH}_{Lower} \\ \text{adj} \end{pmatrix}$$

preSetPoint ← procVariables if ProcessColumn = 2

preValues ← procVariables if count_N = 0

for i ∈ 1.. 10

$$\text{procVariables} \leftarrow \text{DOcontrol}\left(\text{procVariables}, \text{Antifoam}, \frac{\text{Proposed}_{3,\text{ProcessColumn}}}{100}\right)$$

if CO2_CONTROL = 0

$$\text{minFlow} \leftarrow \frac{\text{str2num(Min\_CO2)}}{100 - \text{str2num(Min\_CO2)}} \cdot \left(\text{procVariables}_4 + \text{procVariables}_5 + \text{procVariables}_7\right) \cdot \text{SLPM}$$

$$\text{procVariables}_6 \leftarrow \frac{\text{minFlow}}{\text{SLPM}} \quad \text{if procVariables}_6 < \frac{\text{minFlow}}{\text{SLPM}}$$

if CO2_CONTROL = 1

$$\text{minFlow} \leftarrow \frac{\text{str2num(Min\_CO2)}}{100 - \text{str2num(Min\_CO2)}} \cdot \left(\text{procVariables}_8 + \text{procVariables}_9 + \text{procVariables}_{11}\right) \cdot \text{SLPM}$$

$$\text{procVariables}_{10} \leftarrow \frac{\text{minFlow}}{\text{SLPM}} \quad \text{if procVariables}_{10} < \frac{\text{minFlow}}{\text{SLPM}}$$

if CO2_CONTROL = 2

$$\text{minFlow} \leftarrow \frac{\text{str2num(Min\_CO2)}}{100 - \text{str2num(Min\_CO2)}} \cdot \left(\text{procVariables}_{12} + \text{procVariables}_{13} + \text{procVariables}_{15}\right) \cdot \text{SLPM}$$

$$\text{procVariables}_{14} \leftarrow \frac{\text{minFlow}}{\text{SLPM}} \quad \text{if procVariables}_{14} < \frac{\text{minFlow}}{\text{SLPM}}$$

if CO2_CONTROL = 3

$$\text{minFlow} \leftarrow \frac{\text{str2num(Min\_CO2)}}{100 - \text{str2num(Min\_CO2)}} \cdot \left(\text{qair\_2} + \text{qO2\_2} + \text{qN2\_2}\right)$$

qCO2_ovly ← minFlow if qCO2_ovly < 0minFlow

$$\text{CO2\_gasBal} \leftarrow \text{Re}\left[\text{gasBalanceCO2(procVariables, Antifoam)} + \left(\frac{-1}{10^{10}}\right)^{0.5}\right]$$

$CO2\_gasBal \leftarrow 1 \;\text{ if }\; CO2\_gasBal < 1$

$CO2temp \leftarrow CO2\_gasBal$

$pH \leftarrow Re\left[Resulting\_pH(procVariables, CO2\_gasBal) + \left(\frac{-1}{10^{10}}\right)^{0.5}\right]$

$adj \leftarrow str2num(AIR\_CO2\_ADJ)\cdot\left(\frac{pH_{Upper} + pH_{Lower}}{2} - pH\right)$

$adj \leftarrow 100 \;\text{ if }\; eTime > str2num(AIR\_CO2\_SWITCH)$

$procVariables_{22} \leftarrow pH$

$procVariables_{25} \leftarrow adj$

if $pH < pH_{Lower}$

if $adj \neq 100$

$adj \leftarrow str2num(AIR\_CO2\_ADJ)\cdot\left(\frac{pH_{Upper} + pH_{Lower}}{2} - pH_{Lower}\right)$

$procVariables_{25} \leftarrow adj$

$procVariables \leftarrow DOcontrol\left(procVariables, Antifoam, \frac{Proposed_{3, ProcessColumn}}{100}\right)$

$V_{Base} \leftarrow AddBase(procVariables, CO2\_gasBal)$

$V_{Liq} \leftarrow V_{Liq} + V_{Base}$

$procVariables_0 \leftarrow \frac{V_{Liq}}{L}$

if $pH > pH_{Upper}$

$procVariables_{22} \leftarrow pH_{Upper}$

for $i \in 1..10$

$DOtemp \leftarrow \frac{procVariables_{21}}{1}$

if $adj \neq 100$

$adj \leftarrow str2num(AIR\_CO2\_ADJ)\cdot\left(\frac{pH_{Upper} + pH_{Lower}}{2} - pH_{Upper}\right)$

$procVariables_{25} \leftarrow adj$

$procVariables \leftarrow Re\left[AddCO2(procVariables, CO2\_gasBal, Antifoam) + \left(\frac{-1}{10^{10}}\right)^{0.5}\right]$

$CO2temp \leftarrow procVariables_{21}$

$procVariables_{21} \leftarrow DOtemp$

$procVariables \leftarrow DOcontrol\left(procVariables, Antifoam, \frac{Proposed_{3, ProcessColumn}}{100}\right)$

$$\begin{pmatrix} eTime \\ procVariables_0 \\ procVariables_1 \\ procVariables_2 \\ procVariables_3 \end{pmatrix}$$

$$
\begin{aligned}
&\quad ReturnValues_{count\_N} \leftarrow \begin{pmatrix} procVariables_4 \\ procVariables_5 \\ procVariables_6 \\ procVariables_7 \\ procVariables_8 \\ procVariables_9 \\ procVariables_{10} \\ procVariables_{11} \\ procVariables_{12} \\ procVariables_{13} \\ procVariables_{14} \\ procVariables_{15} \\ procVariables_{16} \\ procVariables_{17} \\ procVariables_{18} \\ procVariables_{19} \\ procVariables_{20} \\ procVariables_{21} \\ CO2temp \\ procVariables_{22} \\ procVariables_{23} \\ procVariables_{24} \\ procVariables_{25} \end{pmatrix} \\
&\quad preValues \leftarrow ReturnValues_{count\_N} \\
&\quad count\_N \leftarrow count\_N + 1 \\
&\quad eTime \leftarrow eTime + \frac{Time\_int}{hr} \\
&otherwise \\
&\quad \text{for } i \in 0..24 \\
&\quad\quad Mtx_i \leftarrow 0 \\
&\quad ReturnValues_0 \leftarrow Mtx \\
&ReturnValues
\end{aligned}
$$

The outcome from the model is converted to vectors to allow for graphing in the plots below.

$$
eTime := \begin{aligned} &\text{for } i \in 0..\,rows(Model) - 1 \\ &\quad result_i \leftarrow \left(Model_i\right)_0 \\ &result \end{aligned} \qquad (9.4-42)
$$

$$Vol := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_1 \\ result \end{array} \right. \tag{9.4 – 43}$$

$$Temperature := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_2 \\ result \end{array} \right. \tag{9.4 – 44}$$

$$Pressure := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_3 \\ result \end{array} \right. \tag{9.4 – 45}$$

$$NRPM := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_4 \\ result \end{array} \right. \tag{9.4 – 46}$$

$$Air\_Ovly := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_5 \\ result \end{array} \right. \tag{9.4 – 47}$$

$$O2\_Ovly := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_6 \\ result \end{array} \right. \tag{9.4 – 48}$$

$$CO2\_Ovly := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_7 \\ result \end{array} \right. \tag{9.4 – 49}$$

$$N2\_Ovly := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_8 \\ result \end{array} \right. \tag{9.4 – 50}$$

$$Air\_Spg1 := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_9 \\ result \end{array} \right. \tag{9.4 – 51}$$

$$O2\_Spg1 := \left| \begin{array}{l} \text{for } i \in 0..\,\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{10} \\ result \end{array} \right. \tag{9.4 – 52}$$

$$CO2\_Spg1 := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{11} \\ result \end{array} \right. \qquad (9.4-53)$$

$$N2\_Spg1 := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{12} \\ result \end{array} \right. \qquad (9.4-54)$$

$$Air\_Spg2 := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{13} \\ result \end{array} \right. \qquad (9.4-55)$$

$$O2\_Spg2 := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{14} \\ result \end{array} \right. \qquad (9.4-56)$$

$$CO2\_Spg2 := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{15} \\ result \end{array} \right. \qquad (9.4-57)$$

$$N2\_Spg2 := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{16} \\ result \end{array} \right. \qquad (9.4-58)$$

$$OUR := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{17} \\ result \end{array} \right. \qquad (9.4-59)$$

$$Lactate := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \dfrac{\left(Model_i\right)_{18}}{1000} \cdot 90.08 \\ result \end{array} \right. \qquad (9.4-60)$$

$$Ammonia := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{19} \\ result \end{array} \right. \qquad (9.4-61)$$

$$V_{BASE} := \left| \begin{array}{l} \text{for } i \in 0..\text{rows}(Model) - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{20} \\ result \end{array} \right. \qquad (9.4-62)$$

$$V_{FEED} := \left| \begin{array}{l} \text{for } i \in 0..\text{rows(Model)} - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{21} \\ result \end{array} \right. \tag{9.4 - 63}$$

$$DO := \left| \begin{array}{l} \text{for } i \in 0..\text{rows(Model)} - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{22} \\ result \end{array} \right. \tag{9.4 - 64}$$

$$pCO2 := \left| \begin{array}{l} \text{for } i \in 0..\text{rows(Model)} - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{23} \\ result \end{array} \right. \tag{9.4 - 65}$$

$$pH := \left| \begin{array}{l} \text{for } i \in 0..\text{rows(Model)} - 1 \\ \quad result_i \leftarrow \left(Model_i\right)_{24} \\ result \end{array} \right. \tag{9.4 - 66}$$

Process Model Numerical Calculations

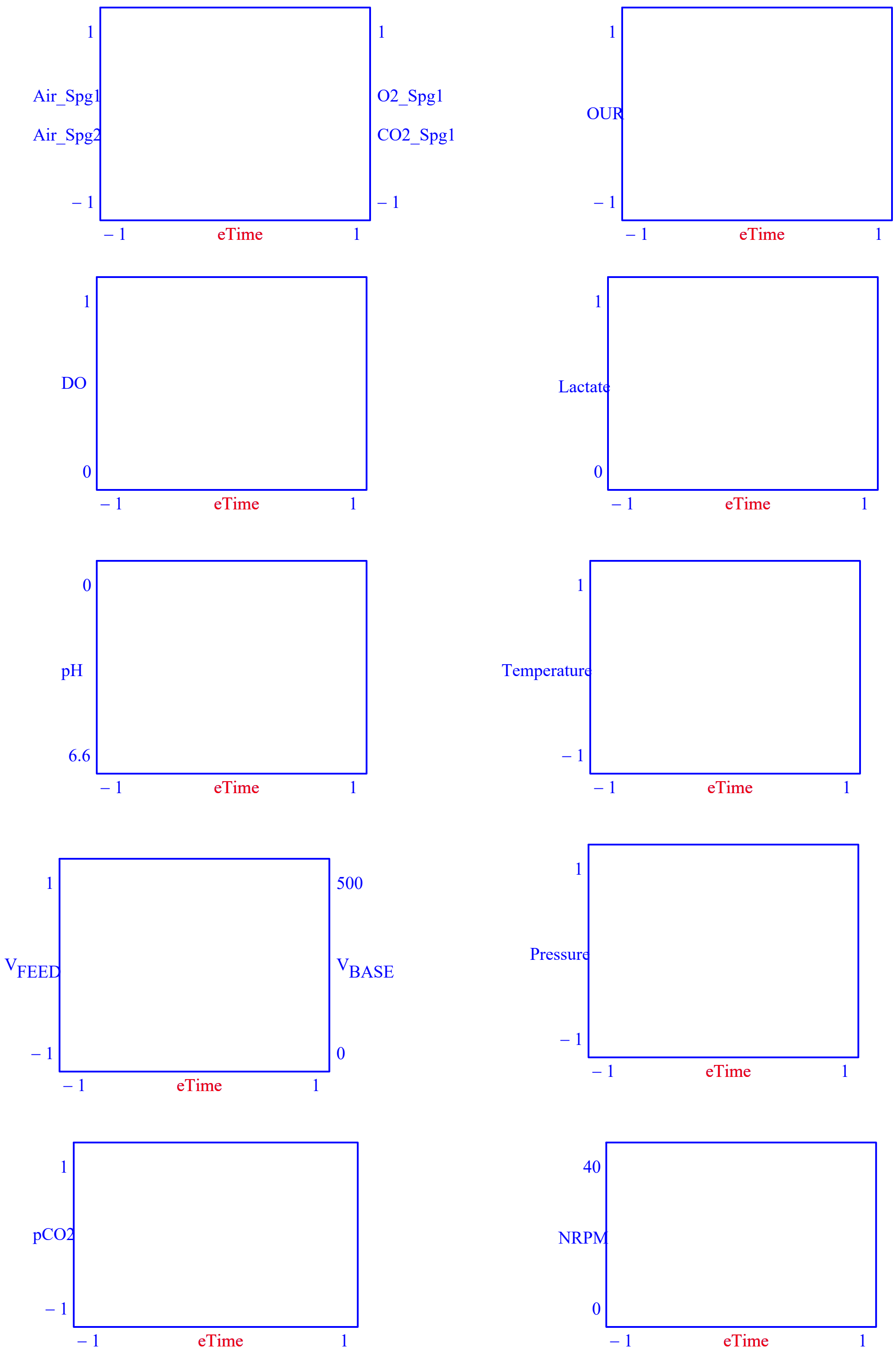


Figure 9.4-2: Forecasted process operations to meet representative run demands.

9.5 Part 5. Dynamic Event Modeling

The dynamic event model utilizes the information about the medium, bioreactor, and process performance to determine the resulting dissolved oxygen and dissolved carbon dioxide over the course of a dynamic event. The user can specify different gas flow rates at different time points and model the resulting solution for the series of ordinary differential equations.

Utilize Model Part 5

DO NOT Run Part 5
Run Part 5

Table 9.5-1: Decision to run 9.5 Part 5 for forecasting the resulting pCO2 accumulation from dynamic conditions in a vessel.

BIOREACTOR PARAMETERS

Vessel Volume L

Vessel Diameter m

Impeller Characteristics

"Power Number"	"Diameter"	"Volume"
0	0	0
0	0	0

Number of Spargers

Zero
One
Two

Spargers Gas Transfer Coefficients

"Coefficient"	"alpha"	"beta"
0	0	0
0	0	0

Surface Aeration Gas Transfer Coefficients

"Int"	0
"V"	0
"N"	0
"S"	0
"VN"	0
"VS"	0
"NS"	0
"V2"	0
"N2"	0
"S2"	0

Saline 0.011

$P_{atm} := 14.7psi$

Table 9.5-2: Properties of the bioreactor used for the dynamic evaluation run.

Medium pKa values

"K1"	-6.352
"K2"	-10.323
"Kh"	-2.59
"K1P"	-2.12
"K2P"	-7.21
"K3P"	-12.67
"K_HEPES"	-7.55
"K_glucose"	-9.41
"K_Lactate"	-3.86
"K1Ala"	-2.34
"K2Ala"	-9.69
"K_Ammonia"	-9.255
"K_Water"	-14
"Ka"	0
"Kb"	0

Table 9.5-3: pKa values for medium (use the basis medium values determined by using section 9.2).

Basis Medium Basal Concentration (mol/L)

"CO3"	0
"HCO3"	0
"H2CO3"	0
"PO4"	0
"HPO4"	0
"H2PO4"	0
"H3PO4"	0
"HEPES"	0
"Glucose"	0
"Lactate"	0
"Ammonia"	0
"Ala"	0
"OH"	0
"A_salts"	0
"B_salts"	0
"A_eq"	0

Table 9.5-4: Concentration of basal medium components for the basis medium (use the basis medium values determined by using section 9.2).

Proposed Concentrations (mol/L)

"CO3"	0
"HCO3"	0
"H2CO3"	0
"PO4"	0
"HPO4"	0
"H2PO4"	0
"H3PO4"	0
"HEPES"	0
"Glucose"	0
"Lactate"	0
"Ammonia"	0
"Ala"	0
"OH"	0
"A_salts"	0
"B_salts"	0
"A_eq"	0

Table 9.5-5: Concentration of basal medium components to be used in the model forecast. If this

is the same medium as determined in the basis medium, the re-enter the same values from Table 9.2.1-3, otherwise enter the values for the medium to be used.

Adjustable Coefficients

"B_ammonia"	0
"B_base"	0
"B_feed"	0
"B_lactate"	0
"B_ala"	0

Table 9.5-6: Fitted terms from the nonlinear medium model from Table 9.2.3-3.

This model needs to assume an initial steady-state condition to calculate the head space composition, the liquid composition, and the oxygen uptake rate. The values listed in table 9.5-7 are used to determine the steady-state condition prior to the disturbance.

Initial Conditions

"Process Time"	"Hours"	
"V"	"L"	
"Temp"	"C"	
"DO"	"%"	
"Pressure"	"psig"	
"Agitation"	"rpm"	
"pH Lower"	"pH units"	
"pH Upper"	"pH units"	
"Air Ovly"	"SLPM"	
"O2 Ovly"	"SLPM"	
"CO2 Ovly"	"SLPM"	
"N2 Ovly"	"SLPM"	
"Air Spg 1"	"SLPM"	
"O2 Spg 1"	"SLPM"	
"CO2 Spg 1"	"SLPM"	
"N2 Spg 1"	"SLPM"	
"Air Spg 2"	"SLPM"	
"O2 Spg 2"	"SLPM"	
"CO2 Spg 2"	"SLPM"	
"N2 Spg 2"	"SLPM"	
"Feed 1"	"L"	
"Feed 2"	"L"	
"Antifoam"	"L"	
"Base Added"	"L"	
"Lactate"	"mmol/L"	
"Ammonia"	"mmol/L"	
"Glucose"	"mmol/L"	

Table 9.5-7: Conditions when bioreactor was at steady-state (prior to the dynamic disturbance).

The dynamic conditions will be determined by step changes in the controlled parameters as listed in the following table. The model will assume that the parameter is constant until there is a time stamp with a new value. At that point, it will calculate how the headspace composition and dissolved gasses will will respond to the step change. The time stamped values for the dynamic conditions are presented in table 9.5-4. For each parameter, there are two rows. One row is for the time stamp, and the second row is for the process value.

"V"	"Discrete"				
"V"	"L"				
"Temp"	"Discrete"				
"Temp"	"C"				
"DO"	"Discrete"				
"DO"	"%"				
"Pressure"	"Discrete"				
"Pressure"	"psig"				
"Agitation"	"Discrete"				
"Agitation"	"rpm"				
"pH Lower"	"Discrete"				
"pH Lower"	"pH units"				
"pH Upper"	"Discrete"				
"pH Upper"	"pH units"				
"Air Ovly"	"Discrete"				
"Air Ovly"	"SLPM"				
"O2 Ovly"	"Discrete"				
"O2 Ovly"	"SLPM"				
"CO2 Ovly"	"Discrete"				
"CO2 Ovly"	"SLPM"				
"N2 Ovly"	"Discrete"				
"N2 Ovly"	"SLPM"				
"Air Spg 1"	"Discrete"				
"Air Spg 1"	"SLPM"				
"O2 Spg 1"	"Discrete"				
"O2 Spg 1"	"SLPM"				
"CO2 Spg 1"	"Discrete"				
"CO2 Spg 1"	"SLPM"				
"N2 Spg 1"	"Discrete"				
"N2 Spg 1"	"SLPM"				
"Air Spg 2"	"Discrete"				
"Air Spg 2"	"SLPM"				
"O2 Spg 2"	"Discrete"				
"O2 Spg 2"	"SLPM"				
"CO2 Spg 2"	"Discrete"				
"CO2 Spg 2"	"SLPM"				
"N2 Spg 2"	"Discrete"				
"N2 Spg 2"	"SLPM"				
"Feed 1"	"Discrete"				
"Feed 1"	"L"				
"Feed 2"	"Discrete"				
"Feed 2"	"L"				
"Antifoam"	"Discrete"				
"Antifoam"	"L"				
"Base Added"	"Discrete"				
"Base Added"	"L"				
"Lactate"	"Interpolate"				
"Lactate"	"mmol/L"				
"Ammonia"	"Interpolate"				

"Ammonia"	"mmol/L"				
"Glucose"	"Interpolate"				
"Glucose"	"mmol/L"				

Table 9.5-8: Time dependent parameter values for the dynamic event.

Model End Time [ ] hours

Model Time Interval [ ] minutes

Dynamic Calculations

Convert the text fields into numerical values.

$$V_T := \text{str2num}(V_T)\cdot L \qquad (9.5-1)$$

$$D := \text{str2num}(D)\cdot m \qquad (9.5-2)$$

$$\text{TimeEnd} := \text{str2num}(\text{Time\_end})\cdot hr \qquad (9.5-3)$$

$$\text{TimeInt} := \text{str2num}(\text{Time\_int})\cdot min \qquad (9.5-4)$$

$$\begin{pmatrix} K_1 \\ K_2 \\ K_b \\ K_{1P} \\ K_{2P} \\ K_{3P} \\ K_{HEPES} \\ K_{Gluc} \\ K_{lact} \\ K_{1Ala} \\ K_{2Ala} \\ K_{amm} \\ K_w \\ K_a \\ K_b \end{pmatrix} := \text{Kvalues}^{\langle 1 \rangle} \qquad (9.5-5)$$

$$\begin{pmatrix} CO3_{basis} \\ HCO3_{basis} \\ H2CO3_{basis} \\ PO4_{basis} \\ HPO4_{basis} \\ H2PO4_{basis} \\ H3PO4_{basis} \\ HEPES_{basis} \\ Gluc_{basis} \\ Lact \\ Amm \\ Ala \\ OH_{basis} \\ Salts_{A\_basis} \\ Salts_{B\_basis} \\ A_{eq\_basis} \end{pmatrix} := Conc^{\langle 1 \rangle} \qquad (9.5-6)$$

$$\begin{pmatrix} B_{amm} \\ B_{base} \\ B_{feed} \\ B_{lact} \\ B_{ala} \end{pmatrix} := Coeff^{\langle 1 \rangle} \qquad (9.5-7)$$

$$PO4\_basis := \left(PO4_{basis} \;\; HPO4_{basis} \;\; H2PO4_{basis} \;\; H3PO4_{basis} \right) \qquad (9.5-8)$$

$$carbonate\_basis := \left(CO3_{basis} \;\; HCO3_{basis} \;\; H2CO3_{basis} \right) \qquad (9.5-9)$$

$$HEPES\_basis := HEPES_{basis} \qquad (9.5-10)$$

$$\begin{pmatrix} CO3_i \\ HCO3_i \\ H2CO3_i \\ PO4_i \\ HPO4_i \\ H2PO4_i \\ H3PO4_i \\ HEPES \\ Gluc \\ Lact \\ Amm \\ Ala \\ OH_i \end{pmatrix} := Conc2^{\langle 1 \rangle} \qquad (9.5-11)$$

$$PO4\_i := \left(PO4_i \;\; HPO4_i \;\; H2PO4_i \;\; H3PO4_i \right) \qquad (9.5-12)$$

$$carbonate\_i := \left(CO3_i \;\; HCO3_i \;\; H2CO3_i \right) \qquad (9.5-13)$$

The following equation will examine the values in the Dynamic matrix (table 9.5-8) to determine the elapsed times when a parameter goes through a step change. The times are listed in a matrix TimeShifts.

(9.5 – 14)

```
TimeShifts := | count ← 0
              | for i ∈ 0..25
              |   for j ∈ 2..cols(Dynamic) − 1
              |     | if count = 0
              |     |   | result_count ← Dynamic_{i·2,j}
              |     |   | count ← count + 1
              |     | otherwise
              |     |   | k ← 0
              |     |   | while k < count − 1 ∧ result_k < Dynamic_{i·2,j}
              |     |   |   k ← k + 1
              |     |   | if k = count
              |     |   |   | result_count ← Dynamic_{i·2,j}
              |     |   |   | count ← count + 1
              |     |   | if result_k ≠ Dynamic_{i·2,j}   otherwise
              |     |   |   | for k2 ∈ count − 1..k
              |     |   |   |   result_{k2+1} ← result_{k2}
              |     |   |   | result_k ← Dynamic_{i·2,j}
              |     |   |   | count ← count + 1
              | result ← submatrix(result, 0, rows(result) − 2, 0, 0)
```

From the listed values in the Dynamic matrix (table 9.5-8), the values between the list time stamps are determined by either treating each as a discrete step change, or by interpolating a constant slope.

$$\begin{aligned}
\text{DynamicValue}(i,\text{time}) := \; & j \leftarrow 3 \\
& \text{while } j < \text{cols}(\text{Dynamic}) - 1 \wedge \text{Dynamic}_{i,j} \le \text{time} \\
& \quad j \leftarrow j + 1 \\
& \text{result} \leftarrow \text{Dynamic}_{i+1,j-1} \;\; \text{if } \text{Dynamic}_{i,j} = 0 \\
& \text{otherwise} \\
& \quad \text{result} \leftarrow \frac{\text{time} - \text{Dynamic}_{i,j-1}}{\text{Dynamic}_{i,j} - \text{Dynamic}_{i,j-1}}\left(\text{Dynamic}_{i+1,j} - \text{Dynamic}_{i+1,j-1}\right) \ldots \;\; \text{if } \text{Dynamic}_{i,1} = \text{"Interpolate"} \\
& \qquad + \text{Dynamic}_{i+1,j-1} \\
& \quad \text{if } \text{Dynamic}_{i,1} = \text{"Discrete"} \\
& \qquad \text{result} \leftarrow \text{Dynamic}_{i+1,j} \;\; \text{if } \text{Dynamic}_{i,j} \le \text{time} \\
& \qquad \text{result} \leftarrow \text{Dynamic}_{i+1,j-1} \;\; \text{otherwise} \\
& \text{result}
\end{aligned} \qquad (9.5-15)$$

The coefficient matrix for equations 3.1-15, 3.1-16, 4.1.2-1, 4.1.2-2, and 4.1.3-2 is rebuilt to account for values that may be different in the dynamic model than were used for the reference standard.

$$\begin{aligned}
\text{Coefficient}(\text{Inputs}) := \; & V_{Liq} \leftarrow \text{Inputs}_0 \cdot L \\
& T_{Liq} \leftarrow \text{Inputs}_1 \cdot K \\
& p_{HS} \leftarrow \text{Inputs}_2 \cdot \text{psi} \\
& N_{rpm} \leftarrow \text{Inputs}_3 \cdot \frac{1}{\text{min}} \\
& \text{qair_ovly} \leftarrow \text{Inputs}_4 \cdot \text{SLPM} \\
& \text{qO2_ovly} \leftarrow \text{Inputs}_5 \cdot \text{SLPM} \\
& \text{qCO2_ovly} \leftarrow \text{Inputs}_6 \cdot \text{SLPM} \\
& \text{qN2_ovly} \leftarrow \text{Inputs}_7 \cdot \text{SLPM} \\
& \text{qair_1} \leftarrow \text{Inputs}_8 \cdot \text{SLPM} \\
& \text{qO2_1} \leftarrow \text{Inputs}_9 \cdot \text{SLPM} \\
& \text{qCO2_1} \leftarrow \text{Inputs}_{10} \cdot \text{SLPM} \\
& \text{qN2_1} \leftarrow \text{Inputs}_{11} \cdot \text{SLPM} \\
& \text{qair_2} \leftarrow \text{Inputs}_{12} \cdot \text{SLPM} \\
& \text{qO2_2} \leftarrow \text{Inputs}_{13} \cdot \text{SLPM} \\
& \text{qCO2_2} \leftarrow \text{Inputs}_{14} \cdot \text{SLPM} \\
& \text{qN2_2} \leftarrow \text{Inputs}_{15} \cdot \text{SLPM} \\
& \text{OUR} \leftarrow \text{Inputs}_{16} \cdot \frac{\text{mmol}}{L \cdot \text{hr}} \\
& \text{kla}_{surf} \leftarrow \text{Inputs}_{17} \cdot \frac{1}{\text{hr}}
\end{aligned} \qquad (9.5-16)$$

$$kla1 \leftarrow Inputs_{18} \cdot \frac{1}{hr}$$

$$kla2 \leftarrow Inputs_{19} \cdot \frac{1}{hr}$$

$$saline \leftarrow Inputs_{20}$$

$$T_{HS} \leftarrow T_H\left(T_{Liq}\right)$$

$$V_{HS} \leftarrow V_H\left(V_T, V_{Liq}\right)$$

$$p_{Liq} \leftarrow p_L\left(V_{Liq}, D, p_{HS}\right)$$

$$q_{ovly} \leftarrow q_{total}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly) + 0.0000000000000001 \cdot SLPM$$

$$q_{spg1} \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.0000000000000001 \cdot SLPM$$

$$q_{spg2} \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.00000000000000001 \cdot SLPM$$

$$xO2\_ovly \leftarrow x_{O2}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$$

$$xO2spg\_1 \leftarrow x_{O2}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$$

$$xO2spg\_2 \leftarrow x_{O2}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$$

$$xCO2\_ovly \leftarrow x_{CO2}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$$

$$xCO2spg\_1 \leftarrow x_{CO2}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$$

$$xCO2spg\_2 \leftarrow x_{CO2}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$$

$$H_{O2\_} \leftarrow H_{O2\_cp}\left(saline, T_{Liq}\right)$$

$$H_{CO2\_} \leftarrow H_{CO2\_cp}\left(saline, T_{Liq}\right)$$

$$result_0 \leftarrow xO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_1 \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_2 \leftarrow xO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_3 \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$$result_4 \leftarrow kla_{surf} \cdot hr$$

$$result_5 \leftarrow p_{HS} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_6 \leftarrow OUR \cdot \frac{L \cdot hr}{mmol}$$

$$result_7 \leftarrow xO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_8 \leftarrow \frac{1}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_9 \leftarrow \left(xO2spg\_1 \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot hr$$

$$result_{10} \leftarrow \left[\frac{1}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{11} \leftarrow \left(xO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{[illegible]}\right) \cdot \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{[illegible]}\right)\right] \cdot hr$$

$$result_{11} \leftarrow \left(xO2spg\_2 \cdot \frac{\quad}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot exp\left[-\left(\frac{\quad}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{12} \leftarrow \frac{\left(q_{ovly} + q_{spg1} + q_{spg2}\right) \cdot T_{HS} \cdot p_{stp}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{13} \leftarrow p_{HS} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_{14} \leftarrow \frac{R_{gas} \cdot T_{HS} \cdot V_{Liq}}{p_{HS} \cdot V_{HS}} \cdot \frac{mol}{L}$$

$$result_{15} \leftarrow xCO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{16} \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_{17} \leftarrow xCO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{18} \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$$result_{19} \leftarrow 0.893 \cdot kla_{surf} \cdot hr$$

$$result_{20} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{21} \leftarrow xCO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{22} \leftarrow \frac{1}{H_{CO2\_} \cdot \rho_{Liq}} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{23} \leftarrow \left(xCO2spg\_1 \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot hr$$

$$result_{24} \leftarrow \left[\frac{1}{H_{CO2\_} \cdot \rho_{Liq}} \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{25} \leftarrow \left(xCO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{26} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{27} \leftarrow \rho_{Liq} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_{28} \leftarrow OUR \cdot \frac{L \cdot hr}{mmol}$$

$$result_{29} \leftarrow xO2\_ovly \cdot q_{ovly} \cdot \frac{hr}{L}$$

$$result_{30} \leftarrow q_{spg1} \cdot \frac{hr}{L}$$

$$result_{31} \leftarrow xO2spg\_1 \cdot exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot q_{spg1} \cdot \frac{hr}{L}$$

$$result_{32} \leftarrow exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot \rho_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot q_{spg1} \cdot \frac{hr}{L}$$

$$result_{33} \leftarrow q_{spg2} \cdot \frac{hr}{L}$$

$$result_{34} \leftarrow xO2spg\_2 \cdot \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot q_{spg2} \cdot \frac{hr}{L}$$

$$result_{35} \leftarrow \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot q_{spg2} \cdot \frac{hr}{L}$$

$$result_{36} \leftarrow kla_{surf} \cdot \frac{R_{gas} \cdot T_{q\_ref}}{p_{stp}} \cdot V_{Liq} \cdot \left(H_{O2\_} \cdot p_{Liq}\right) \cdot \frac{hr}{L}$$

$$result_{37} \leftarrow q_{ovly} \cdot \frac{hr}{L}$$

$$result_{38} \leftarrow xCO2\_ovly \cdot q_{ovly} \cdot \frac{hr}{L}$$

$$result_{39} \leftarrow xCO2spg\_1 \cdot \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot q_{spg1} \cdot \frac{hr}{L}$$

$$result_{40} \leftarrow \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot q_{spg1} \cdot \frac{hr}{L}$$

$$result_{41} \leftarrow xCO2spg\_2 \cdot \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot q_{spg2} \cdot \frac{hr}{L}$$

$$result_{42} \leftarrow \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot q_{spg2} \cdot \frac{hr}{L}$$

$$result_{43} \leftarrow 0.893 \cdot kla_{surf} \cdot \frac{R_{gas} \cdot T_{q\_ref}}{p_{stp}} \cdot V_{Liq} \cdot \left(H_{CO2\_} \cdot p_{Liq}\right) \cdot \frac{hr}{L}$$

$$result$$

The initial values for the oxygen uptake rate, the dissolved carbon dioxide concentration, the oxygen ratio in the headspace, and the carbon dioxide ratio in the headspace are determined from the operating conditions listed in the Initial matrix (table 9.5-7).

$$\text{InitialValues} := \left| \begin{array}{l}
V_{Liq} \leftarrow \text{InitialM}_{1,2} \cdot L \\
T_{Liq} \leftarrow \text{InitialM}_{2,2} \cdot K + T_{0C} \\
p_{HS} \leftarrow \text{InitialM}_{4,2} \cdot psi + p_{atm} \\
\text{Nrpm} \leftarrow \text{InitialM}_{5,2} \cdot \frac{1}{min} \\
\text{qair_ovly} \leftarrow \text{InitialM}_{8,2} \cdot \text{SLPM} \\
\text{qO2_ovly} \leftarrow \text{InitialM}_{9,2} \cdot \text{SLPM} \\
\text{qCO2_ovly} \leftarrow \text{InitialM}_{10,2} \cdot \text{SLPM} \\
\text{qN2_ovly} \leftarrow \text{InitialM}_{11,2} \cdot \text{SLPM} \\
\text{qair_1} \leftarrow \text{InitialM}_{12,2} \cdot \text{SLPM} \\
\text{qO2_1} \leftarrow \text{InitialM}_{13,2} \cdot \text{SLPM} \\
\text{qCO2_1} \leftarrow \text{InitialM}_{14,2} \cdot \text{SLPM} \\
\text{qN2_1} \leftarrow \text{InitialM}_{15,2} \cdot \text{SLPM} \\
\text{qair_2} \leftarrow \text{InitialM}_{16,2} \cdot \text{SLPM} \\
\text{qO2_2} \leftarrow \text{InitialM}_{17,2} \cdot \text{SLPM} \\
\text{qCO2_2} \leftarrow \text{InitialM}_{18,2} \cdot \text{SLPM} \\
\text{qN2_2} \leftarrow \text{InitialM}_{19,2} \cdot \text{SLPM} \\
\text{Antifoam} \leftarrow \text{InitialM}_{22,2} \\
DO \leftarrow \frac{\text{InitialM}_{3,2}}{100} \\
\text{qtotal1} \leftarrow q_{total}(\text{qair_1}, \text{qO2_1}, \text{qCO2_1}, \text{qN2_1}) \\
\text{qtotal2} \leftarrow q_{total}(\text{qair_2}, \text{qO2_2}, \text{qCO2_2}, \text{qN2_2}) \\
M \leftarrow 0.5 + 0.5 \cdot \exp(-1 \cdot \text{Antifoam}) \\
M \leftarrow 1 \\
\text{kla_surf} \leftarrow M \cdot \left[\begin{array}{l}
\text{Surface}_{0,1} + \text{Surface}_{1,1} \cdot V_{Liq} \cdot \frac{1}{L} \ldots \\
+ \text{Surface}_{2,1} \cdot \text{Nrpm} \cdot min \cdot \frac{min}{s} \ldots \\
+ \text{Surface}_{3,1} (\text{qtotal1} + \text{qtotal2}) \cdot \frac{1}{\text{SLPM}} \ldots \\
+ \text{Surface}_{4,1} \cdot \left(V_{Liq} \cdot \text{Nrpm} \cdot min \cdot \frac{min}{s} \right) \cdot \frac{1}{L} \ldots \\
+ \text{Surface}_{5,1} \cdot \left[V_{Liq} \cdot (\text{qtotal1} + \text{qtotal2}) \right] \cdot \frac{1}{L \cdot \text{SLPM}} \ldots \\
+ \text{Surface}_{6,1} \cdot \left[\text{Nrpm} \cdot min \cdot \frac{min}{s} \cdot (\text{qtotal1} + \text{qtotal2}) \right] \cdot \frac{1}{\text{SLPM}} \ldots \\
+ \text{Surface}_{8,1} \cdot \left(\text{Nrpm} \cdot min \cdot \frac{min}{s} \right)^2 \ldots \\
+ \text{Surface}_{9,1} \cdot (\text{qtotal1} + \text{qtotal2})^2 \cdot \frac{1}{\text{SLPM}^2} \ldots \\
+ \text{Surface}_{7,1} \cdot V_{Liq}{}^2 \cdot \frac{1}{L^2}
\end{array} \right] \\
\text{kla_surf} \leftarrow 0 \cdot \frac{1}{hr} \quad \text{if } \text{kla_surf} < 0 \cdot \frac{1}{hr}
\end{array} \right. \qquad (9.5-17)$$

$$\text{Power} \leftarrow 0 \cdot W$$

for $i \in 1..\text{rows}(\text{Impeller}) - 1$

if $\text{Impeller}_{i,2} \cdot L < V_{Liq}$

$$\text{Power} \leftarrow \text{Power} + \text{Impeller}_{i,0} \cdot (\text{Nrpm})^3 \cdot \left(\text{Impeller}_{i,1} \cdot m\right)^5 \frac{gm}{cm^3}$$

$$\text{temp} \leftarrow 1$$

$$\text{P\_V} \leftarrow \frac{\text{Power}}{V_{Liq}} \cdot \frac{m^3}{W} \quad \text{if } V_{Liq} > 0 \cdot L$$

$$\text{P\_V} \leftarrow 0 \quad \text{otherwise}$$

$$\text{q\_sup1} \leftarrow \frac{\text{qtotal1}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$$\text{A1} \leftarrow \text{Sparge}_{1,0}$$

$$\text{alpha1} \leftarrow \text{Sparge}_{1,1}$$

$$\text{beta1} \leftarrow \text{Sparge}_{1,2}$$

$$\text{kla1} \leftarrow M \cdot \text{A1} \cdot \text{P\_V}^{\text{alpha1}} \cdot \text{q\_sup1}^{\text{beta1}}$$

$$\text{q\_sup2} \leftarrow \frac{\text{qtotal2}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$$\text{A2} \leftarrow \text{Sparge}_{2,0}$$

$$\text{alpha2} \leftarrow \text{Sparge}_{2,1}$$

$$\text{beta2} \leftarrow \text{Sparge}_{2,2}$$

$$\text{kla2} \leftarrow M \cdot \text{A2} \cdot \text{P\_V}^{\text{alpha2}} \cdot \text{q\_sup2}^{\text{beta2}}$$

$$\text{Inputs} \leftarrow \begin{pmatrix} \frac{V_{Liq}}{L} \\ \frac{T_{Liq}}{K} \\ \frac{P_{HS}}{psi} \\ \text{Nrpm} \cdot \text{min} \\ \frac{\text{qair\_ovly}}{SLPM} \\ \frac{\text{qO2\_ovly}}{SLPM} \\ \frac{\text{qCO2\_ovly}}{SLPM} \\ \frac{\text{qN2\_ovly}}{SLPM} \\ \frac{\text{qair\_1}}{SLPM} \\ \frac{\text{qO2\_1}}{SLPM} \\ \text{qCO2\_1} \ldots \end{pmatrix}$$

$$\begin{pmatrix} \frac{qCO2\_1}{SLPM} \\ \frac{qN2\_1}{SLPM} \\ \frac{qair\_2}{SLPM} \\ \frac{qO2\_2}{SLPM} \\ \frac{qCO2\_2}{SLPM} \\ \frac{qN2\_2}{SLPM} \\ 0 \\ kla\_surf \\ kla1 \\ kla2 \\ 0.011 \end{pmatrix}$$

$$T1 \leftarrow Coefficient(Inputs)_0 \cdot \left(\frac{mol}{L \cdot hr}\right) - DO \cdot 0.21 \cdot Coefficient(Inputs)_{27} \cdot \frac{mol}{L} \cdot Coefficient(Inputs)_1 \cdot \frac{1}{hr}$$

$$T2 \leftarrow Coefficient(Inputs)_2 \cdot \left(\frac{mol}{L \cdot hr}\right) - DO \cdot 0.21 \cdot Coefficient(Inputs)_{27} \cdot \frac{mol}{L} \cdot Coefficient(Inputs)_3 \cdot \frac{1}{hr}$$

$$T3 \leftarrow \frac{\begin{pmatrix} Coefficient(Inputs)_7 \ldots \\ + Coefficient(Inputs)_8 \cdot DO \cdot 0.21 \cdot Coefficient(Inputs)_{27} + Coefficient(Inputs)_9 \ldots \\ + Coefficient(Inputs)_{10} \cdot DO \cdot 0.21 \cdot Coefficient(Inputs)_{27} + Coefficient(Inputs)_{11} \ldots \\ + Coefficient(Inputs)_4 \cdot DO \cdot 0.21 \cdot Coefficient(Inputs)_{27} \cdot Coefficient(Inputs)_{14} \end{pmatrix}}{Coefficient(Inputs)_{12} + Coefficient(Inputs)_{13} \cdot Coefficient(Inputs)_4 \cdot Coefficient(Inputs)_{14}}$$

$$result_0 \leftarrow \left[T1 + T2 \ldots + \frac{Coefficient(Inputs)_4}{hr} \cdot \begin{pmatrix} T3 \cdot Coefficient(Inputs)_5 \cdot \frac{mol}{L} \ldots \\ + -DO \cdot 0.21 \cdot Coefficient(Inputs)_{27} \cdot \frac{mol}{L} \end{pmatrix} \right] \cdot L \cdot \frac{hr}{mmol}$$

$$T4 \leftarrow Coefficient(Inputs)_{15} \cdot \frac{mol}{L} + Coefficient(Inputs)_{17} \cdot \frac{mol}{L}$$

$$T5 \leftarrow \frac{\begin{pmatrix} Coefficient(Inputs)_{21} \ldots \\ + Coefficient(Inputs)_{23} \ldots \\ + Coefficient(Inputs)_{25} \end{pmatrix} \cdot \left(Coefficient(Inputs)_{20} \cdot \frac{mol}{L} \right)}{\begin{matrix} Coefficient(Inputs)_{12} \ldots \\ + Coefficient(Inputs)_4 \cdot 0.893 \cdot Coefficient(Inputs)_{26} \cdot Coefficient(Inputs)_{14} \end{matrix}}$$

$$T6 \leftarrow \left[\frac{\begin{pmatrix} Coefficient(Inputs)_{22} \ldots \\ + Coefficient(Inputs)_{24} \ldots \\ + Coefficient(Inputs)_4 \cdot 0.893 \cdot Coefficient(Inputs)_{14} \end{pmatrix} \cdot Coefficient(Inputs)_{20}}{\begin{matrix} Coefficient(Inputs)_{12} \ldots \\ + Coefficient(Inputs)_4 \cdot 0.893 \cdot Coefficient(Inputs)_{26} \cdot Coefficient(Inputs)_{14} \end{matrix}} - 1 \right]$$

$$result_1 \leftarrow \frac{\left(T4 + Coefficient(Inputs)_{19} \cdot T5 + RQ result_0 \frac{mmol}{L \cdot hr} \cdot hr \right)}{H_{CO2\_cp}(0.011, T_{Liq}) \cdot mmHg \cdot (Coefficient(Inputs)_{16} \ldots}$$

$$\left(\ldots + \text{Coefficient(Inputs)}_{18} \ldots + -\text{Coefficient(Inputs)}_{19}\cdot T6 + 000.0000001 \right)$$

$$\text{result}_2 \leftarrow \frac{\left[\begin{array}{l}\text{Coefficient(Inputs)}_{29} \ldots \\ + DO\cdot 0.21\cdot\text{Coefficient(Inputs)}_{30} + \text{Coefficient(Inputs)}_{31} - DO\cdot 0.21\cdot\text{Coefficient(Inputs)}_{32} \ldots \\ + DO\cdot 0.21\cdot\text{Coefficient(Inputs)}_{33} + \left(\text{Coefficient(Inputs)}_{34} - DO\cdot 0.21\cdot\text{Coefficient(Inputs)}_{35}\right) \ldots \\ + \text{Coefficient(Inputs)}_{36}\cdot DO\cdot 0.21\end{array}\right]}{\left[\left(\text{Coefficient(Inputs)}_{37} + \text{Coefficient(Inputs)}_{30} + \text{Coefficient(Inputs)}_{33}\right) + \text{Coefficient(Inputs)}_{36}\cdot\frac{p_{HS}}{p_L\left(V_{Liq}, D, p_{HS}\right)}\right]}$$

$$\text{result}_3 \leftarrow \frac{\left[\begin{array}{l}\text{Coefficient(Inputs)}_{38} \ldots \\ + \frac{\text{result}_2\cdot mmHg}{p_L\left(V_{Liq}, D, p_{HS}\right)}\cdot\text{Coefficient(Inputs)}_{30} + \text{Coefficient(Inputs)}_{39} - \frac{\text{result}_2\cdot mmHg}{p_L\left(V_{Liq}, D, p_{HS}\right)}\cdot\text{Coefficient(Inputs)}_{40} \ldots \\ + \frac{\text{result}_2\cdot mmHg}{p_L\left(V_{Liq}, D, p_{HS}\right)}\cdot\text{Coefficient(Inputs)}_{33} + \left(\text{Coefficient(Inputs)}_{41} - \frac{\text{result}_2\cdot mmHg}{p_L\left(V_{Liq}, D, p_{HS}\right)}\cdot\text{Coefficient(Inputs)}_{42}\right) \ldots \\ + \text{Coefficient(Inputs)}_{43}\cdot\frac{\text{result}_1\cdot mmHg}{p_L\left(V_{Liq}, D, p_{HS}\right)}\end{array}\right]}{\left[\left(\text{Coefficient(Inputs)}_{37} + \text{Coefficient(Inputs)}_{30} + \text{Coefficient(Inputs)}_{33}\right) + \text{Coefficient(Inputs)}_{43}\cdot\frac{p_{HS}}{p_L\left(V_{Liq}, D, p_{HS}\right)}\right]}$$

result

The following solver block allows for the five differential equations for the dissolved oxygen, the dissolved carbon dioxide, the oxygen ratio in the headspace, the carbon dioxide ratio in the headspace and the dissolved oxygen probe to be solved simutaneously.

Given

$$O2(0) = \text{Initial}_0 \qquad (9.5-18)$$

$$x_{O2\_hs}(0) = \text{Initial}_1 \qquad (9.5-19)$$

$$O_e(0) = \text{Initial}_2 \qquad (9.5-20)$$

$$CO2(0) = \text{Initial}_3 \qquad (9.5-21)$$

$$x_{CO2\_hs}(0) = \text{Initial}_4 \qquad (9.5-22)$$

$$\frac{d}{dt}O2(t) = \left[\begin{array}{l}\text{Coefficient(Inputs)}_0\cdot\left(\frac{mol}{L\cdot hr}\right) - O2(t)\cdot\frac{mol}{L}\cdot\text{Coefficient(Inputs)}_1\cdot\frac{1}{hr} \ldots \\ + \text{Coefficient(Inputs)}_2\cdot\left(\frac{mol}{L\cdot hr}\right) - O2(t)\cdot\frac{mol}{L}\cdot\text{Coefficient(Inputs)}_3\cdot\frac{1}{hr} \ldots \\ + \frac{\text{Coefficient(Inputs)}_4}{hr}\cdot\left(x_{O2\_hs}(t)\cdot\text{Coefficient(Inputs)}_5\cdot\frac{mol}{L} - O2(t)\cdot\frac{mol}{L}\right) \ldots \\ + -\text{Coefficient(Inputs)}_6\cdot\frac{mmol}{L\cdot hr}\end{array}\right]\cdot\frac{hr\cdot L}{mol} \qquad (9.5-23)$$

$$\frac{d}{dt}CO2(t) = \begin{bmatrix} \text{Coefficient(Inputs)}_{15}\cdot\frac{mol}{L\cdot hr} - CO2(t)\cdot\frac{mol}{L}\cdot\text{Coefficient(Inputs)}_{16}\cdot\frac{1}{hr}\ \ldots \\ + \text{Coefficient(Inputs)}_{17}\cdot\frac{mol}{L\cdot hr} - CO2(t)\cdot\frac{mol}{L}\cdot\text{Coefficient(Inputs)}_{18}\cdot\frac{1}{hr}\ \ldots \\ + \frac{\text{Coefficient(Inputs)}_{19}}{hr}\cdot\left(x_{CO2\_hs}(t)\cdot\text{Coefficient(Inputs)}_{20}\cdot\frac{mol}{L} - CO2(t)\cdot\frac{mol}{L}\right)\ \ldots \\ + RQ\,\text{Coefficient(Inputs)}_{6}\cdot\frac{mmol}{L\cdot hr} \end{bmatrix}\cdot\frac{hr\cdot L}{mol} \qquad (9.5-24)$$

$$\frac{d}{dt}x_{O2\_hs}(t) = \begin{bmatrix} \text{Coefficient(Inputs)}_{7}\cdot\frac{1}{hr}\ \ldots \\ + \text{Coefficient(Inputs)}_{8}\cdot\frac{L}{hr\cdot mol}\cdot\left(O2(t)\cdot\frac{mol}{L}\right) + \text{Coefficient(Inputs)}_{9}\cdot\frac{1}{hr}\ \ldots \\ + \text{Coefficient(Inputs)}_{10}\cdot\frac{L}{hr\cdot mol}\cdot\left(O2(t)\cdot\frac{mol}{L}\right) + \text{Coefficient(Inputs)}_{11}\cdot\frac{1}{hr}\ \ldots \\ + -x_{O2\_hs}(t)\cdot\text{Coefficient(Inputs)}_{12}\cdot\frac{1}{hr}\ \ldots \\ + \frac{-\text{Coefficient(Inputs)}_{4}}{hr}\cdot\left(x_{O2\_hs}(t)\cdot\text{Coefficient(Inputs)}_{5}\cdot\frac{mol}{L} - O2(t)\cdot\frac{mol}{L}\right)\cdot\text{Coefficient(Inputs)}_{14}\cdot\frac{L}{mol} \end{bmatrix}\cdot hr \qquad (9.5-25)$$

$$\frac{d}{dt}x_{CO2\_hs}(t) = \begin{bmatrix} \text{Coefficient(Inputs)}_{21}\cdot\frac{1}{hr}\ \ldots \\ + \text{Coefficient(Inputs)}_{22}\cdot\frac{L}{mol\cdot hr}\cdot\left(CO2(t)\cdot\frac{mol}{L}\right) + \text{Coefficient(Inputs)}_{23}\cdot\frac{1}{hr}\ \ldots \\ + \text{Coefficient(Inputs)}_{24}\cdot\frac{L}{mol\cdot hr}\cdot\left(CO2(t)\cdot\frac{mol}{L}\right) + \text{Coefficient(Inputs)}_{25}\cdot\frac{1}{hr}\ \ldots \\ + -x_{CO2\_hs}(t)\cdot\text{Coefficient(Inputs)}_{12}\cdot\frac{1}{hr}\ \ldots \\ + \frac{-\text{Coefficient(Inputs)}_{19}}{hr}\cdot\left(x_{CO2\_hs}(t)\cdot\text{Coefficient(Inputs)}_{20}\cdot\frac{mol}{L} - CO2(t)\cdot\frac{mol}{L}\right)\cdot\text{Coefficient(Inputs)}_{14}\cdot\frac{L}{mol} \end{bmatrix}\cdot hr \qquad (9.5-26)$$

$$\frac{d}{dt}O_e(t) = \frac{hr}{\tau(Nrpm)}\cdot\left(\frac{\frac{100}{0.21}\cdot O2(t)\cdot\frac{mol}{L}}{\text{Coefficient(Inputs)}_{27}\cdot\frac{mol}{L}} - O_e(t)\right) \qquad (9.5-27)$$

$$\text{Result(Inputs, Initial, DataPoints)} := \text{Odesolve}\left[\begin{pmatrix} O2 \\ x_{O2\_hs} \\ O_e \\ CO2 \\ x_{CO2\_hs} \end{pmatrix}, t, 10, \text{DataPoints}\right] \qquad (9.5-28)$$

The dynamic model is calculated by first priming the initial matrix with the oxygen uptake rate, oxygen ratio in the headspace, dissolved oxygen concentration, carbon dioxide ratio in the headspace and dissolved carbon dioxide concentration. The time dependent values for the process conditions are then uploaded, and the dynamic equations are solved. This is done until a time stamp when a process change is noted. At that point, the initial matrix is set to the values that satisfy the differential equations and the process is repeated.

DynamicModel :=

$$eTime \leftarrow 0 \qquad (9.5-29)$$

$$j \leftarrow 0$$

$$count\_N \leftarrow 0$$

$$V_{FEED} \leftarrow 0 \cdot L$$

$$DO\_0 \leftarrow \frac{Dynamic_{5,2}}{100}$$

$$OUR\_0 \leftarrow InitialValues_0 \cdot \frac{mmol}{L \cdot hr}$$

$$CO2\_0 \leftarrow InitialValues_1$$

$$xO2HS\_0 \leftarrow InitialValues_2$$

$$xCO2HS\_0 \leftarrow InitialValues_3$$

$$Initial \leftarrow \begin{pmatrix} \frac{Dynamic_{5,2} \cdot 0.21}{100} \cdot H_{O2\_cp}\left(0.011, Dynamic_{3,2} \cdot K + T_{0C}\right) \cdot p_L\left(Dynamic_{1,2} \cdot L, D, Dynamic_{7,2} \cdot psi + p_{atm}\right) \cdot \frac{L}{mol} \\ InitialValues_2 \\ Dynamic_{5,2} \\ InitialValues_1 \cdot mmHg \cdot H_{CO2\_cp}\left(0.011, Dynamic_{3,2} \cdot K + T_{0C}\right) \cdot \frac{L}{mol} \\ InitialValues_3 \end{pmatrix}$$

$$result_{count\_N} \leftarrow \begin{pmatrix} 0 \\ Dynamic_{5,2} \\ InitialValues_2 \\ Dynamic_{5,2} \\ InitialValues_1 \\ InitialValues_3 \end{pmatrix}$$

$$\text{while } eTime \le \frac{TimeEnd}{hr} \qquad \text{if } Part5 = 1$$

$$TimeZero \leftarrow eTime$$

$$V_{Liq} \leftarrow DynamicValue(0, eTime) \cdot L$$

$$T_{Liq} \leftarrow DynamicValue(2, eTime) \cdot K + T_{0C}$$

$$DO \leftarrow \frac{DynamicValue(4, eTime)}{100}$$

$$p_{HS} \leftarrow DynamicValue(6, eTime) \cdot psi + p_{atm}$$

$$Nrpm \leftarrow DynamicValue(8, eTime) \cdot \frac{1}{min}$$

$$qair\_ovly \leftarrow DynamicValue(14, eTime) \cdot SLPM$$

$$qO2\_ovly \leftarrow DynamicValue(16, eTime) \cdot SLPM$$

$$qCO2\_ovly \leftarrow DynamicValue(18, eTime) \cdot SLPM$$

$$qN2\_ovly \leftarrow DynamicValue(20, eTime) \cdot SLPM$$

$$qair\_1 \leftarrow DynamicValue(22, eTime) \cdot SLPM$$

$$qO2\_1 \leftarrow DynamicValue(24, eTime) \cdot SLPM$$

$$qCO2\_1 \leftarrow DynamicValue(26, eTime) \cdot SLPM$$

$qN2\_1 \leftarrow \mathrm{DynamicValue}(28, eTime) \cdot SLPM$

$qair\_2 \leftarrow \mathrm{DynamicValue}(30, eTime) \cdot SLPM$

$qO2\_2 \leftarrow \mathrm{DynamicValue}(32, eTime) \cdot SLPM$

$qCO2\_2 \leftarrow \mathrm{DynamicValue}(34, eTime) \cdot SLPM$

$qN2\_2 \leftarrow \mathrm{DynamicValue}(36, eTime) \cdot SLPM$

$ammonia \leftarrow 0 \cdot \frac{mmol}{L}$

$lact \leftarrow 0 \cdot \frac{mmol}{L}$

$V_{FEED} \leftarrow \mathrm{DynamicValue}(38, eTime) \cdot L + \mathrm{DynamicValue}(40, eTime) \cdot L + V_{FEED}$

$pH_{Upper} \leftarrow \mathrm{DynamicValue}(12, eTime)$

$pH_{Lower} \leftarrow \mathrm{DynamicValue}(10, eTime)$

$pH \leftarrow 0.5 pH_{Upper} + 0.5 pH_{Lower}$

$Antifoam \leftarrow \mathrm{DynamicValue}(42, eTime) + Antifoam \cdot \exp\left(\frac{\frac{-TimeInt}{hr}}{62.4}\right)$

$qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.00000000000001 \cdot SLPM$

$qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.00000000000001 \cdot SLPM$

$M \leftarrow 0.5 + 0.5 \cdot \exp(-1 \cdot Antifoam)$

$M \leftarrow 1$

$$
\begin{aligned}
kla\_surf \leftarrow M \cdot \Bigg[& Surface_{0,1} + Surface_{1,1} \cdot V_{Liq} \cdot 1 \frac{1}{L} \ldots \\
& + Surface_{2,1} \cdot Nrpm \cdot min \cdot \frac{min}{s} \ldots \\
& + Surface_{3,1} (qtotal1 + qtotal2) \cdot \frac{1}{SLPM} \ldots \\
& + Surface_{4,1} \cdot \left(V_{Liq} \cdot Nrpm \cdot min \cdot \frac{min}{s}\right) \cdot \frac{1}{L} \ldots \\
& + Surface_{5,1} \cdot \left[V_{Liq} \cdot (qtotal1 + qtotal2)\right] \cdot \frac{1}{L \cdot SLPM} \ldots \\
& + Surface_{6,1} \cdot \left[Nrpm \cdot min \cdot \frac{min}{s} \cdot (qtotal1 + qtotal2) \cdot \frac{1}{SLPM}\right] \ldots \\
& + Surface_{8,1} \cdot \left(Nrpm \cdot min \cdot \frac{min}{s}\right)^2 \ldots \\
& + Surface_{9,1} \cdot (qtotal1 + qtotal2)^2 \cdot \frac{1}{SLPM^2} \ldots \\
& + Surface_{7,1} \cdot V_{Liq}{}^2 \cdot \frac{1}{L^2} \Bigg]
\end{aligned}
$$

$kla\_surf \leftarrow 0 \cdot \frac{1}{hr} \quad \text{if } kla\_surf < 0 \cdot \frac{1}{hr}$

$Power \leftarrow 0 \cdot W$

$\text{for } i \in 1 .. \mathrm{rows}(Impeller) - 1$

$\quad \text{if } Impeller_{i,2} \cdot L < V_{Liq}$

$\qquad Power \leftarrow Power + Impeller_{i,0} \cdot (Nrpm)^3 \cdot (Impeller_{i,1} \cdot m)^5 \frac{gm}{cm^3}$

$\qquad temp \leftarrow 1$

$P\_V \leftarrow \frac{Power}{V_{Liq}} \cdot \frac{m^3}{W} \quad \text{if } V_{Liq} > 0 \cdot L$

$_{Liq}$

$P\_V \leftarrow 0 \quad \text{otherwise}$

$$q\_sup1 \leftarrow \frac{qtotal1}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$A1 \leftarrow Sparge_{1,0}$

$alpha1 \leftarrow Sparge_{1,1}$

$beta1 \leftarrow Sparge_{1,2}$

$kla1 \leftarrow M \cdot A1 \cdot P\_V^{alpha1} \cdot q\_sup1^{beta1}$

$$q\_sup2 \leftarrow \frac{qtotal2}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$A2 \leftarrow Sparge_{2,0}$

$alpha2 \leftarrow Sparge_{2,1}$

$beta2 \leftarrow Sparge_{2,2}$

$kla2 \leftarrow M \cdot A2 \cdot P\_V^{alpha2} \cdot q\_sup2^{beta2}$

$$Inputs \leftarrow \begin{bmatrix} \frac{V_{Liq}}{L} \\ \frac{T_{Liq}}{K} \\ \frac{p_{HS}}{psi} \\ Nrpm \cdot min \\ \frac{qair\_ovly}{SLPM} \\ \frac{qO2\_ovly}{SLPM} \\ \frac{qCO2\_ovly}{SLPM} \\ \frac{qN2\_ovly}{SLPM} \\ \frac{qair\_1}{SLPM} \\ \frac{qO2\_1}{SLPM} \\ \frac{qCO2\_1}{SLPM} \\ \frac{qN2\_1}{SLPM} \\ \frac{qair\_2}{SLPM} \\ \frac{qO2\_2}{SLPM} \end{bmatrix}$$

$$\begin{bmatrix} \frac{qCO2\_2}{SLPM} \\ \frac{qN2\_2}{SLPM} \\ OUR\_0 \cdot \left(\frac{L \cdot hr}{mmol}\right) \\ kla\_surf \\ kla1 \\ kla2 \\ 0.011 \end{bmatrix}$$

$O2calc \leftarrow Result(Inputs, Initial, Points)_0$

$xO2calc \leftarrow Result(Inputs, Initial, Points)_1$

$DOcalc \leftarrow Result(Inputs, Initial, Points)_2$

$CO2calc \leftarrow Result(Inputs, Initial, Points)_3$

$xCO2calc \leftarrow Result(Inputs, Initial, Points)_4$

$k \leftarrow 1$

while $eTime + \frac{TimeInt}{hr} < TimeShifts_j$

$count\_N \leftarrow count\_N + 1$

$eTime \leftarrow eTime + \frac{TimeInt}{hr}$

$$result_{count\_N} \leftarrow \begin{pmatrix} eTime \\ \frac{O2calc\left(k \cdot \frac{TimeInt}{hr}\right) \cdot \frac{mol}{L}}{H_{O2\_cp}(0.011, T_{Liq}) \cdot p_L(V_{Liq}, D, p_{HS})} \cdot \frac{100}{0.21} \\ xO2calc\left(k \cdot \frac{TimeInt}{hr}\right) \\ DOcalc\left(k \cdot \frac{TimeInt}{hr}\right) \\ \frac{CO2calc\left(k \cdot \frac{TimeInt}{hr}\right) \cdot \frac{mol}{L} \cdot \frac{1}{mmHg}}{H_{CO2\_cp}(0.011, T_{Liq})} \\ xCO2calc\left(k \cdot \frac{TimeInt}{hr}\right) \end{pmatrix}$$

$k \leftarrow k + 1$

if $j < rows(TimeShifts) - 1$

$count\_N \leftarrow count\_N + 1$

$eTime \leftarrow TimeShifts_j$

$$result_{count\_N} \leftarrow \begin{pmatrix} eTime \\ \frac{O2calc\left(TimeShifts_j - TimeZero\right) \cdot \frac{mol}{L}}{H_{O2\_cp}(0.011, T_{Liq}) \cdot p_L(V_{Liq}, D, p_{HS})} \cdot \frac{100}{0.21} \\ xO2calc\left(TimeShifts_j - TimeZero\right) \\ DOcalc\left(TimeShifts_j - TimeZero\right) \\ CO2calc\left(TimeShifts_j - TimeZero\right) \cdot \frac{mol}{L} \cdot \frac{1}{mmHg} \end{pmatrix}$$

$$
\begin{aligned}
&\quad\quad \left|\begin{array}{l}
\quad \frac{\quad\quad\quad\quad L \;\; mmHg}{H_{CO2\_cp}(0.011, T_{Liq})} \\
\quad xCO2calc\left(TimeShifts_j - TimeZero\right)
\end{array}\right) \\
&\quad\quad Initial \leftarrow \begin{pmatrix} O2calc\left(TimeShifts_j - TimeZero\right) \\ xO2calc\left(TimeShifts_j - TimeZero\right) \\ DOcalc\left(TimeShifts_j - TimeZero\right) \\ CO2calc\left(TimeShifts_j - TimeZero\right) \\ xCO2calc\left(TimeShifts_j - TimeZero\right) \end{pmatrix} \\
&\quad\quad j \leftarrow j + 1 \\
&result
\end{aligned}
$$

The values from the dynamic model matrix are used to build arrays for the elapsed time, the dissolved oxygen reading and the dissolved carbon dioxide partial pressure.

$$
ElapsedTime := \left|\begin{array}{l} \text{for } i \in 0..rows(DynamicModel) - 1 \\ \quad result_i \leftarrow \left(DynamicModel_i\right)_0 \\ result \end{array}\right. \qquad (9.5 - 30)
$$

$$
DynamicDO := \left|\begin{array}{l} \text{for } i \in 0..rows(DynamicModel) - 1 \\ \quad \left|\begin{array}{l} result_i \leftarrow \left(DynamicModel_i\right)_3 \\ result_i \leftarrow 0 \;\; \text{if } \; result_i < 0 \end{array}\right. \\ result \end{array}\right. \qquad (9.5 - 31)
$$

$$
DynamicCO2 := \left|\begin{array}{l} \text{for } i \in 0..rows(DynamicModel) - 1 \\ \quad result_i \leftarrow \left(DynamicModel_i\right)_4 \\ result \end{array}\right. \qquad (9.5 - 32)
$$

The dynamic model for pH is calculated by numerically solving the for the medium dissolved carbon dioxide that is is in equilibrium with an estimate pH. The pH is adjusted iteratively until the medium dissolved carbon dioxide that is in equilibrium with the adjusted pH is the dissolved carbon dioxide that satisfy the gas transport mass balance equations.

$$\text{Dynamic\_pH} := \quad V_{FEED} \leftarrow 0L \qquad (9.5-33)$$

$$\text{for } i \in 0..\text{rows}(\text{ElapsedTime})-1$$

$$V_{Liq} \leftarrow \text{DynamicValue}\left(0, \text{ElapsedTime}_i\right)\cdot L$$

$$T_{Liq} \leftarrow \text{DynamicValue}\left(2, \text{ElapsedTime}_i\right)\cdot K + T_{0C}$$

$$pH_{Upper} \leftarrow \text{DynamicValue}\left(12, \text{ElapsedTime}_i\right)$$

$$pH_{Lower} \leftarrow \text{DynamicValue}\left(10, \text{ElapsedTime}_i\right)$$

$$V_{FEED} \leftarrow \text{DynamicValue}\left(38, \text{ElapsedTime}_i\right)\cdot L + \text{DynamicValue}\left(40, \text{ElapsedTime}_i\right)\cdot L + V_{FEED}$$

$$V_{Base} \leftarrow \text{DynamicValue}\left(44, \text{ElapsedTime}_i\right)\cdot L$$

$$\text{lactate} \leftarrow \text{DynamicValue}\left(46, \text{ElapsedTime}_i\right)\cdot \frac{mmol}{L}$$

$$\text{ammonia} \leftarrow \text{DynamicValue}\left(48, \text{ElapsedTime}_i\right)\cdot \frac{mmol}{L}$$

$$\text{glucose} \leftarrow \text{DynamicValue}\left(50, \text{ElapsedTime}_i\right)\cdot \frac{mmol}{L}$$

$$\text{CO2\_gasBal} \leftarrow \frac{\text{DynamicCO2}_i}{\exp\left[0.04375\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)\right]}$$

$$pH_{Upper} \leftarrow \frac{pH_{Upper} - 0.033\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)}{1-0.0065\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)}$$

$$pH_{Lower} \leftarrow \frac{pH_{Lower} - 0.033\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)}{1-0.0065\cdot\left(\frac{T_{Liq}-T_{0C}}{K}-37\right)}$$

$$pH_{est} \leftarrow \frac{pH_{Lower}+pH_{Upper}}{2}$$

$$\text{for } j \in 1..10$$

$$\text{carbonate\_i} \leftarrow \left(CO3_i + B_{base}\cdot\frac{V_{Base}}{V_{Liq}}\cdot \text{Base\_Conc}\cdot\frac{L}{mol} \quad HCO3_i \quad H2CO3_i\right)$$

$$G1 \leftarrow \frac{\Phi\text{\_carbonate}\left(pH_{est}, \text{carbonate\_i}\right)}{\Phi\text{\_carbonate}\left(pH_{est}, \text{carbonate\_basis}\right)}$$

$$G2 \leftarrow \frac{\Phi\text{\_total}\left(pH_{est}, \text{carbonate\_basis}, \text{PO4\_basis}, \text{HEPES\_basis}, \text{Salts}_{A\_basis}, \text{Salts}_{B\_basis}\right)}{\Phi\text{\_total}\left(pH_{est}, \text{carbonate\_i}, \text{PO4\_i}, \text{HEPES}, \text{Salts}_{A\_basis}, \text{Salts}_{B\_basis}\right)}$$

$$\left[\frac{10^{-2\cdot pH_{est}}}{\left[\begin{array}{l}2\times 10^{K_1+K_2}\ \ldots\\ +2\times 10^{\left(K_1+K_2+K_h\right)}\ \ldots\\ +10^{\left(K_1-pH_{est}\right)}\ \ldots\\ +10^{\left(K_1+K_h-pH_{est}\right)}\end{array}\right]}\cdot\left[\begin{array}{l}HCO3_i\cdot\frac{mol}{L}\ \ldots\\ +2\left(\begin{array}{l}CO3_i\ \ldots\\ +\text{Coeff}_{1,1}\cdot\frac{V_{Base}}{V_{Liq}}\cdot\text{Base\_Conc}\cdot\frac{L}{mol}\end{array}\right)\cdot\frac{mol}{L}\ \ldots\\ +G1\cdot G2\cdot\left[\begin{array}{l}10^{-pH_{est}}\cdot\frac{mole}{L}\ \ldots\\ +\text{Coeff}_{2,1}\cdot\frac{V_{FEED}}{V_0}\cdot\frac{mol}{L}\ \ldots\\ +OH_i\cdot\frac{mol}{}\ \ldots\end{array}\right.\end{array}\right.\right.$$

$$\mathrm{L}$$

$$+ -\frac{10^{-14}}{10^{-pH_{est}}}\cdot\frac{mole}{L}\ \ldots$$

$$+ -\Delta H\_PO4(pH_{est}, PO4\_i)\ \ldots$$

$$+ -\frac{10^{K_{HEPES}}\cdot\left(HEPES\cdot\frac{mol}{L}\right)}{10^{K_{HEPES}} + 10^{-pH_{est}}}\ \ldots$$

$$+ -\frac{10^{K_a}\cdot\left(Salts_{A\_basis}\cdot\frac{mol}{L}\right)}{10^{K_a} + 10^{-pH_{est}}}\ \ldots$$

$$+ \frac{10^{-pH_{est}}\cdot\left(Salts_{B\_basis}\cdot\frac{mol}{L}\right)}{10^{K_b} + 10^{-pH_{est}}}\ \ldots$$

$$+ -A_{eq\_basis}\cdot\frac{mol}{L}\ \ldots$$

$$+ \frac{-B_{lact}\cdot 10^{K_{lact}}}{10^{K_{lact}} + 10^{-pH_{est}}}\cdot lactate\ \ldots$$

$$+ \frac{B_{amm}\cdot 10^{-pH_{est}}}{10^{K_{amm}} + 10^{-pH_{est}}}\cdot ammonia\ \ldots$$

$$+ \frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-pH_{est}}}\cdot glucose\ \ldots$$

$$+ -\left(\frac{2\cdot 10^{K_{1Ala}+K_{2Ala}}\ \ldots + 10^{K_{1Ala}-pH_{est}}}{10^{K_{1Ala}+K_{2Ala}}\ \ldots + 10^{K_{1Ala}-pH_{est}}\ \ldots + 10^{-2pH_{est}}}\right)\cdot B_{ala}\cdot lactate$$

$$CO2\_med \leftarrow \frac{[\ldots]}{H_{CO2\_cp}(str2num(saline), T_{Liq})\cdot mmHg}$$

$$pH_{est} \leftarrow pH_{est} + 0.1\cdot\left(\frac{CO2\_med - CO2\_gasBal}{CO2\_med}\right)$$

$$pH_{est} \leftarrow 5\ \text{ if }\ pH_{est} < 5$$

$$pH_{est} \leftarrow 9\ \text{ if }\ pH_{est} > 9$$

$$pH_{est} \leftarrow pH_{est} + [-0.0147 + 0.0065\cdot(7.400 - pH_{est})]\cdot\left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)$$

$$result_i \leftarrow pH_{est}$$

result

Dynamic Calculations

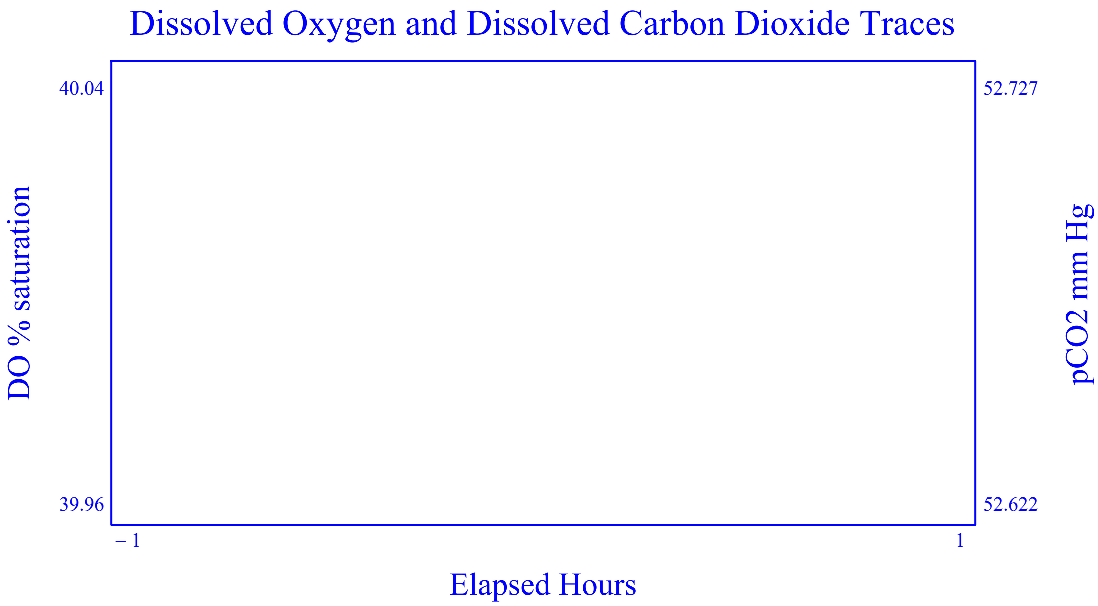


Figure 9.5-1: Dissolved oxygen and dissolved carbon dioxide levels resulting from transient events noted in Table 9.5-8.

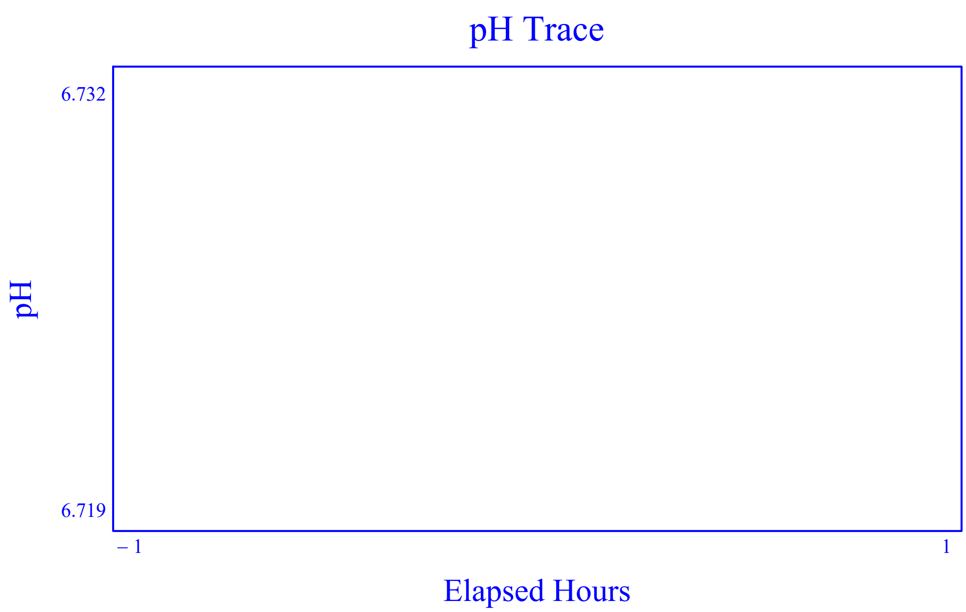


Figure 9.5-2: pH resulting from transient events noted in Table 9.5-8.

9.6 Part 6. PI Loop Tuning by Direct Synthesis Method

A direct synthesis method for the controller loop tuning can be performed by leveraging the gas transport model for the bioreactor. This section will calculate the gain and integral for the tuner that satisfies the methods described in section 5.3.

Utilize Model Part 6	DO NOT Run Part 6 Run Part 6

Table 9.6-1: Decision to run 9.6 Part 6 for determining the gain and integral terms for the controller.

MEDIUM PROPERTIES

The medium buffering capacity will impact the relationship between the carbon dioxide, base and pH for the pH control. The medium properties are entered here for section 9.6 to calculate the appropriate gain for carbon dioxide add and base add.

Medium pKa values

"K1"	-6.352
"K2"	-10.323
"Kh"	-2.59
"K1P"	-2.12
"K2P"	-7.21
"K3P"	-12.67
"K_HEPES"	-7.55
"K_glucose"	-9.41
"K_Lactate"	-3.86
"K1Ala"	-2.34
"K2Ala"	-9.69
"K_Ammonia"	-9.255
"K_Water"	-14
"Ka"	0
"Kb"	0

Table 9.6-1.1: pKa values for medium (use the basis medium values determined by using section 9.2).

Basis Medium Basal Concentration (mol/L)

"CO3"	0
"HCO3"	0
"H2CO3"	0
"PO4"	0
"HPO4"	0
"H2PO4"	0
"H3PO4"	0
"HEPES"	0
"Glucose"	0
"Lactate"	0
"Ammonia"	0
"Ala"	0
"OH"	0
"A_salts"	0
"B_salts"	0
"A_eq"	0

Table 9.6-1.2: Concentration of basal medium components for the basis medium (use the basis medium values determined by using section 9.2).

Proposed Concentrations (mol/L)

"CO3"	0
"HCO3"	0
"H2CO3"	0
"PO4"	0
"HPO4"	0
"H2PO4"	0
"H3PO4"	0
"HEPES"	0
"Glucose"	0
"Lactate"	0
"Ammonia"	0
"Ala"	0
"OH"	0
"A_salts"	0
"B_salts"	0
"A_eq"	0

Table 9.6-1.3: Concentration of basal medium components to be used in the model forecast. If this is the same medium as determined in the basis medium, then re-enter the same values from Table 9.2.1-3, otherwise enter the values for the medium to be used.

Adjustable Coefficients

"B_ammonia"	0
"B_base"	0
"B_feed"	0
"B_lactate"	0
"B_ala"	0

Table 9.6-1.4: Fitted terms from the nonlinear medium model from Table 9.2.3-3.

BIOREACTOR PARAMETERS

Vessel Volume [ ] L

Vessel Diameter [ ] m

Impeller Characteristics

"Power Number"	"Diameter"	"Volume"
0	0	0
0	0	0

Number of Spargers

Zero
One
Two

Spargers Gas Transfer Coefficients

"Coefficient"	"alpha"	"beta"
0	0	0
0	0	0

Surface Aeration Gas Transfer Coefficients

"Int"	0
"V"	0
"N"	0
"S"	0
"VN"	0
"VS"	0
"NS"	0
"V2"	0
"N2"	0
"S2"	0

Saline 0.011

$P_{atm} := 12.2psi$

Table 9.6-2: Properties of the bioreactor used for the dissolved oxygen loop tuning.

Operating Parameters

Dissolved Oxygen Control Cascade

The sequence of controlled parameters to maintain the dissolved oxygen set point.

Air Flow
Agitation
O2 Enrichment

None
Air Flow
Agitation

None
Air Flow
Agitation

None
Air Flow
Agitation

Table 9.6-3: Select the cascade order for dissolved oxygen control. The top most selection option is the first controllable parameter. When that has hit its upper limit, then the second selected parameter would be adjusted to maintain the dissolved oxygen control. The bottom most selection option is the active control parameter after the middle parameter has hit its upper limit.

Cascade Matrix Build

The following program converts the selections from the above list boxes to a vector called CASCADE with six string instances. The first instance is the string "Empty", and the last instance is also the string "Empty." The other instances correspond to values selected in the above list box. The CASCADE vector is used to determine which is the controlled parameter. If the current controlled parameter is at the maximum output, and the dissolved oxygen is below set point, then the program will step through to the next array instance and change which is the controlling parameter for dissolved oxygen. If the value of the next instance is "Empty", then the program will not change the controlling parameter.

$$RQ := 1 \qquad (9.6-1)$$

```
CASCADE := | result_0 ← "Empty"                                                    (9.6 – 2)
           | result_1 ← | value ← "None"  if Cas1 = 0
           |            | value ← "Air"  if Cas1 = 1
           |            | value ← "Agitation"  if Cas1 = 2
           |            | value ← "O2 Enrichment"  if Cas1 = 3
           |            | value ← "Pressure"  if Cas1 = 4
           |            | value ← "Proportional"  if Cas1 = 5
           |            | value ← "Ratio Control"  if Cas1 = 6
           |            | value
           | result_2 ← | value ← "None"  if Cas2 = 0
           |            | value ← "Air"  if Cas2 = 1
           |            | value ← "Agitation"  if Cas2 = 2
           |            | value ← "O2 Enrichment"  if Cas2 = 3
           |            | value ← "Pressure"  if Cas2 = 4
           |            | value ← "Proportional"  if Cas2 = 5
           |            | value ← "Ratio Control"  if Cas2 = 6
           |            | value
           | result_3 ← | value ← "None"  if Cas3 = 0
           |            | value ← "Air"  if Cas3 = 1
           |            | value ← "Agitation"  if Cas3 = 2
           |            | value ← "O2 Enrichment"  if Cas3 = 3
           |            | value ← "Pressure"  if Cas3 = 4
           |            | value ← "Proportional"  if Cas3 = 5
           |            | value ← "Ratio Control"  if Cas3 = 6
           |            | value
           | result_4 ← | value ← "None"  if Cas4 = 0
           |            | value ← "Air"  if Cas4 = 1
           |            | value ← "Agitation"  if Cas4 = 2
           |            | value ← "O2 Enrichment"  if Cas4 = 3
           |            | value ← "Pressure"  if Cas4 = 4
           |            | value ← "Proportional"  if Cas4 = 5
           |            | value ← "Ratio Control"  if Cas4 = 6
           |            | value
           | result_5 ← "Empty"
           | result
```

Cascade Matrix Build

DO Control

Overlay DO Control
Sparge 1 DO Control
Sparge 2 DO Control

Table 9.6-4: Select the gas stream used for dissolved oxygen control.

Total Flow	1	SLPM

Table 9.6-5: Specify the total flow rate that would be used when one of the cascade controls are specified as ratio control.

Proportional Constant (Air to O2)	1

Table 9.6-6: Specify the proportion of air to oxygen (0 to 1) that would be used when one of the cascade controls are specified as proportional control.

O2 Split Ratio (Between Spargers)	100	%

Table 9.6-7: Specify the spilt of the oxygen flow between the primary (selected in Table 9.6-3) and secondary sparger.

CO2 Addition for pH Control	Overlay CO2 Addition Sparge 1 CO2 Addition Sparge 2 CO2 Addition	
Min CO2 Addition (percent of total)	0	%

Table 9.6-8: Select the gas stream used for carbon dioxide addition for pH control.

CO2 Split Ratio	100	%

Table 9.6-9: Specify the spilt of the carbon dioxide flow between the primary (selected in Table 9.6-8) and secondary sparger.

Base Concentration		mol/L
Air Sparge For pCO2 Control	Off Sparge 1 pH Variant Sparge 2 pH Variant	
Tuning Parameters	qAir = (___ + ___ x (pH_setpoint - pH)) x qO2	
Maximum Air Flow		SLPM
Switch Over Time		hours from inoculation
Manual Air Flow After Switch		SLPM

Table 9.6-10: The concentration of carbonate used for pH control and Automated pCO2 control algorithm.

Process Conditions

"Process Time"	"Hours"				
"V"	"L"				
"Temp"	"C"				
"DO"	"%"				
"Pressure"	"psig"				
"Agitation"	"rpm"				
"pH Lower"	"pH units"				
"pH Upper"	"pH units"				
"Air Ovly"	"SLPM"				
"O2 Ovly"	"SLPM"				
"CO2 Ovly"	"SLPM"				
"N2 Ovly"	"SLPM"				
"Air Spg 1"	"SLPM"				
"O2 Spg 1"	"SLPM"				
"CO2 Spg 1"	"SLPM"				
"N2 Spg 1"	"SLPM"				
"Air Spg 2"	"SLPM"				
"O2 Spg 2"	"SLPM"				
"CO2 Spg 2"	"SLPM"				
"N2 Spg 2"	"SLPM"				
"Feed 1"	"L"				
"Feed 2"	"L"				
"Antifoam"	"L"				
"Base Added"	"L"				
"Lactate"	"mmol/L"				
"Ammonia"	"mmol/L"				
"Glucose"	"mmol/L"				

Process Limits

""	"Min"	"Actionable Min"	"Max"	"Cascade"	"Units"
"Volume"	""	""	""	""	"L"
"Pressure"	""	""	""	""	"psig"
"Agitation"	""	""	""	""	"rpm"
"Air Ovly"	""	""	""	""	"SLPM"
"O2 Ovly"	""	""	""	""	"SLPM"
"CO2 Ovly"	""	""	""	""	"SLPM"
"N2 Ovly"	""	""	""	""	"SLPM"
"Air Spg 1"	""	""	""	""	"SLPM"
"O2 Spg 1"	""	""	""	""	"SLPM"
"CO2 Spg 1"	""	""	""	""	"SLPM"
"N2 Spg 1"	""	""	""	""	"SLPM"
"Air Spg 2"	""	""	""	""	...

Table 9.6-11: The process conditions and process limits used for dissolved oxygen controller loop tuning. The minimum and maximum limits for the bioreactor control parameters. These values would be used to determine when a parameter would trigger a cascade or can longer be increased by the controller.

Some controllers are capable of using a variable gain based on the oxygen flow meter output. This is to account that the gain for the system is changing with the different gas transfer coefficients. The variable gain is described by a polygon. The polygon has a maximum output (qMax) and a specified number of increments between 0 and 100% output. The coefficients of the polygon are relative to the gain identified for a specified gas flow rate (The specified gas flow rate is defined as the Basis in equation 9.6-4). The qMax, the increments, and the basis flow rate are specified for calculating the polygon of a variable gain controller.

$$qMaxO2 := 250 \qquad (9.6-3)$$

$$Basis := 25 \qquad (9.6-4)$$

$$qMaxCO2 := 250 \qquad (9.6-5)$$

$$MaxPump := 0.01$$

$$Inc := 10 \qquad (9.6-6)$$

The time lag (in seconds) for the system and the time constant for the controller is provided to solve equation 5.3-1, 5.3-3, and 5.3-4.

$$\theta := 10 \qquad (9.6-7)$$

$$\tau c := 120 \qquad (9.6-8)$$

Mixing time constant for base addition and pH probe response time

"Theta_CO2"	0
"Theta_Base"	0
"CO2 Add Loop Time Constant"	0
"Base Loop Time Constant"	0
"Vessel Mix Time"	0

Table 9.6-11.1: Time constants used for the pH control through CO2 add and base add. Values are listed with units of seconds.

$$TimeConstants := TimeConstants^{\langle 1 \rangle} \qquad (9.6-9)$$

	Minimum	Maximum	
Air Flow for Contour Plot	0		SLPM
Oxygen Flow for Contour Plot	0		SLPM

Table 9.6-12: Minimum and maximum flow rates to use in contour plots of Figure 9.6-1. If these cells are blank, it will use the limits provided in Table 9.4-13 for the gas stream used for dissolved oxygen control.

Air Flow for Source for Contour Plot	Overlay Sparge 1 Sparge 2

Table 9.6-13: Selected gas stream with variable air flow in contour plots.

Processing Time		hours

Table 9.6-14: Processing time to use for contour plots (Figure 9.6-1).

Direct Synthesis Calculations

The values captured in the text fields above are converted to numerical values.

$$Total := str2num(Total) \cdot SLPM \qquad (9.6-12)$$

$$CO2split := \frac{str2num(CO2split)}{100} \qquad (9.6-13)$$

$$Base\_Conc := str2num(Base\_Conc) \cdot \frac{mol}{L} \qquad (9.6-14)$$

$$Proportional := str2num(Proportional) \qquad (9.6-15)$$

$$DOsplit := \frac{str2num(DOsplit)}{100} \qquad (9.6-16)$$

$$\begin{aligned} Input := \; & i \leftarrow 2 \\ & eTime \leftarrow 0 \text{ if } strlen(ProcessTime) = 0 \\ & eTime \leftarrow str2num(ProcessTime) \text{ otherwise} \\ & \text{while } ProcessData_{0,i} < eTime \wedge i < cols(ProcessData) - 1 \\ & \quad i \leftarrow i + 1 \\ & i \leftarrow i - 1 \text{ if } i > cols(ProcessData) - 1 \\ & i \leftarrow i - 1 \text{ if } ProcessData_{0,i} > eTime \\ & submatrix(ProcessData, 1, rows(ProcessData) - 1, i, i) \end{aligned} \qquad (9.6-17)$$

$$D := str2num(D) \cdot m \qquad (9.6-18)$$

$$V_T := str2num(V_T) \cdot L \qquad (9.6-19)$$

The coefficient matrix for equations 3.1-15, 3.1-16, 4.1.2-1, 4.1.2-2, and 4.1.3-2.is rebuilt to account for values that may be different in the proposed vessel than were used for the gold standard run.

Coefficient(Inputs) := (9.6 – 20)

$V_{Liq} \leftarrow Inputs_0 \cdot L$

$T_{Liq} \leftarrow Inputs_1 \cdot K$

$DO \leftarrow Inputs_2$

$p_{HS} \leftarrow Inputs_3 \cdot psi$

$N_{rpm} \leftarrow Inputs_4 \cdot \frac{1}{min}$

$qair\_ovly \leftarrow Inputs_5 \cdot SLPM$

$qO2\_ovly \leftarrow Inputs_6 \cdot SLPM$

$qCO2\_ovly \leftarrow Inputs_7 \cdot SLPM$

$qN2\_ovly \leftarrow Inputs_8 \cdot SLPM$

$qair\_1 \leftarrow Inputs_9 \cdot SLPM$

$qO2\_1 \leftarrow Inputs_{10} \cdot SLPM$

$qCO2\_1 \leftarrow Inputs_{11} \cdot SLPM$

$qN2\_1 \leftarrow Inputs_{12} \cdot SLPM$

$qair\_2 \leftarrow Inputs_{13} \cdot SLPM$

$qO2\_2 \leftarrow Inputs_{14} \cdot SLPM$

$qCO2\_2 \leftarrow Inputs_{15} \cdot SLPM$

$qN2\_2 \leftarrow Inputs_{16} \cdot SLPM$

$OUR \leftarrow Inputs_{17} \cdot \frac{mmol}{L \cdot hr}$

$kla\_surf \leftarrow Inputs_{18} \cdot \frac{1}{hr}$

$kla1 \leftarrow Inputs_{19} \cdot \frac{1}{hr}$

$kla2 \leftarrow Inputs_{20} \cdot \frac{1}{hr}$

$saline \leftarrow str2num(saline)$

$T_{HS} \leftarrow T_H(T_{Liq})$

$V_{HS} \leftarrow V_H(V_T, V_{Liq})$

$p_{Liq} \leftarrow p_L(V_{Liq}, D, p_{HS})$

$q_{ovly} \leftarrow q_{total}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly) + 0.00000000000001 \cdot SIUnitsOf(qair\_ovly)$

$q_{spg1} \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.00000000000001 \cdot SIUnitsOf(qair\_1)$

$q_{spg2} \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.000000000000001 \cdot SIUnitsOf(qair\_2)$

$xO2\_ovly \leftarrow x_{O2}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$

$xO2spg\_1 \leftarrow x_{O2}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$

$xO2spg\_2 \leftarrow x_{O2}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$

$xCO2\_ovly \leftarrow x_{CO2}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly)$

$$xCO2spg\_1 \leftarrow x_{CO2}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$$

$$xCO2spg\_2 \leftarrow x_{CO2}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$$

$$H_{O2\_} \leftarrow H_{O2\_cp}(saline, T_{Liq})$$

$$H_{CO2\_} \leftarrow H_{CO2\_cp}(saline, T_{Liq})$$

$$result_0 \leftarrow xO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_1 \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_2 \leftarrow xO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_3 \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$$result_4 \leftarrow kla\_surf \cdot hr$$

$$result_5 \leftarrow p_{HS} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_6 \leftarrow OUR \cdot \frac{L \cdot hr}{mmol}$$

$$result_7 \leftarrow xO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_8 \leftarrow \frac{1}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_9 \leftarrow \left(xO2spg\_1 \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{kla1 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot hr$$

$$result_{10} \leftarrow \left[\frac{1}{H_{O2\_} \cdot p_{Liq}} \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{11} \leftarrow \left(xO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{kla2 \cdot H_{O2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{12} \leftarrow \frac{(q_{ovly} + q_{spg1} + q_{spg2}) \cdot T_{HS} \cdot p_{stp}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{13} \leftarrow p_{HS} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_{14} \leftarrow \frac{R_{gas} \cdot T_{HS} \cdot V_{Liq}}{p_{HS} \cdot V_{HS}} \cdot \frac{mol}{L}$$

$$result_{15} \leftarrow xCO2spg\_1 \cdot \frac{p_{stp} \cdot q_{spg1}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{16} \leftarrow \frac{p_{stp} \cdot q_{spg1}}{H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot hr$$

$$result_{17} \leftarrow xCO2spg\_2 \cdot \frac{p_{stp} \cdot q_{spg2}}{V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot \frac{L \cdot hr}{mol}$$

$$result_{18} \leftarrow \frac{p_{stp} \cdot q_{spg2}}{H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right] \cdot hr$$

$$result \leftarrow 0.893 \cdot kla\_surf \cdot hr$$

$$result_{19} \leftarrow 0.893 \cdot kla\_surf \cdot hr$$

$$result_{20} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{21} \leftarrow xCO2\_ovly \cdot \frac{q_{ovly} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot hr$$

$$result_{22} \leftarrow \frac{1}{H_{CO2\_} \cdot p_{Liq}} \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{23} \leftarrow \left(xCO2spg\_1 \cdot \frac{q_{spg1} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{0.893 \cdot kla1 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg1}}\right)\right] \cdot hr$$

$$result_{24} \leftarrow \left[\frac{1}{H_{CO2\_} \cdot p_{Liq}} \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}} \cdot \left[1 - \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right]\right]\right] \cdot \frac{mol}{L} \cdot hr$$

$$result_{25} \leftarrow \left(xCO2spg\_2 \cdot \frac{q_{spg2} \cdot p_{stp} \cdot T_{HS}}{V_{HS} \cdot T_{q\_ref} \cdot p_{HS}}\right) \cdot \exp\left[-\left(\frac{0.893 \cdot kla2 \cdot H_{CO2\_} \cdot p_{Liq} \cdot V_{Liq} \cdot R_{gas} \cdot T_{q\_ref}}{p_{stp} \cdot q_{spg2}}\right)\right] \cdot hr$$

$$result_{26} \leftarrow p_{HS} \cdot H_{CO2\_} \cdot \frac{L}{mol}$$

$$result_{27} \leftarrow p_{Liq} \cdot H_{O2\_} \cdot \frac{L}{mol}$$

$$result_{28} \leftarrow OUR \cdot \frac{L \cdot hr}{mmol}$$

result

This program returns the oxygen uptake rate that is supported by different air and oxygen flow rates through the controlling gas stream (overlay, sparge 1, or sparge 2).

(9.6 − 21)

```
OUR_(Control, Input, qAir, qO2) := | V_Liq ← Input_0
                                   | T_Liq ← Input_1 + 273.15
                                   | DO ← Input_2
                                   | p_HS ← Input_3 + p_atm/psi
                                   | N_rpm ← Input_4
                                   | qair_ovly ← qAir  if  AIR_CONTOUR = 0
                                   | qair_ovly ← Input_7  otherwise
                                   | qO2_ovly ← qO2  if  Control = 0
                                   | qO2_ovly ← Input_8  otherwise
                                   | qN2_ovly ← Input_10
                                   | qCO2_ovly ← | max[Input_9, str2num(Min_CO2)/(100 − str2num(Min_CO2)) · (qair_ovly ... + qO2_ovly ... + qN2_ovly)]  if  CO2_CONTROL = 0
                                   |             | Input_9  otherwise
                                   | if  Control = 1
                                   |   | qO2_1 ← qO2
                                   |   | qair_1 ← qAir  if  AIR_CONTOUR = 1
                                   |   | otherwise
                                   |   |   | if  AIR_CO2_CONTROL = 1
                                   |   |   |   | qair_1 ← str2num(AIR_CO2_GAIN) qO2_1
                                   |   |   |   | qair_1 ← str2num(AIR_CO2_MAX)  if  qair_1 > str2num(AIR_CO2_MAX)
                                   |   |   |   | qair_1 ← Input_11  if  qair_1 < Input_11
                                   |   |   | qair_1 ← Input_11  otherwise
                                   |   | qair_2 ← qAir  if  AIR_CONTOUR = 2
                                   |   | otherwise
                                   |   |   | if  AIR_CO2_CONTROL = 2
                                   |   |   |   | qair_2 ← str2num(AIR_CO2_GAIN) qO2_1
                                   |   |   |   | qair_2 ← str2num(AIR_CO2_MAX)  if  qair_2 > str2num(AIR_CO2_MAX)
                                   |   |   |   | qair_2 ← Input_15  if  qair_2 < Input_15
                                   |   |   | qair_2 ← Input_15  otherwise
                                   | qO2_1 ← Input_12  otherwise
                                   | qN2_1 ← Input_14
                                   | qCO2_1 ← | max[Input_13, str2num(Min_CO2)/(100 − str2num(Min_CO2)) · (qair_1 ... + qO2_1 ... + qN2_1)]  if  CO2_CONTROL = 1
                                   |          | Input_13  otherwise
                                   | if  Control = 2
                                   |   | qO2_2 ← qO2
```

qair_1 ← qAir if AIR_CONTOUR = 1

otherwise

if AIR_CO2_CONTROL = 1

qair_1 ← str2num(AIR_CO2_GAIN) qO2_2

qair_1 ← str2num(AIR_CO2_MAX) if qair_1 > str2num(AIR_CO2_MAX)

qair_1 ← $Input_{11}$ if qair_1 < $Input_{11}$

qair_1 ← $Input_{11}$ otherwise

qair_2 ← qAir if AIR_CONTOUR = 2

otherwise

if AIR_CO2_CONTROL = 2

qair_2 ← str2num(AIR_CO2_GAIN) qO2_2

qair_2 ← str2num(AIR_CO2_MAX) if qair_2 > str2num(AIR_CO2_MAX)

qair_2 ← $Input_{15}$ if qair_2 < $Input_{15}$

qair_2 ← $Input_{15}$ otherwise

qO2_2 ← $Input_{16}$ otherwise

qN2_2 ← $Input_{18}$

$$qCO2\_2 \leftarrow \begin{cases} \max\left[Input_{17}, \dfrac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_2 + qO2\_2 + qN2\_2)\right] & \text{if } CO2\_CONTROL = 2 \\ Input_{17} & \text{otherwise} \end{cases}$$

OUR ← 1

$qtotalovly \leftarrow q_{total}(qair\_ovly, qO2\_ovly, qCO2\_ovly, qN2\_ovly) + 0.0000000000000001 \cdot SIUnitsOf(qair\_ovly)$

$qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.0000000000000001 \cdot SIUnitsOf(qair\_1)$

$qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.0000000000000001 \cdot SIUnitsOf(qair\_2)$

kla_surf ← 0

Power ← 0·W

for i ∈ 1 .. rows(Impeller) − 1

$$Power \leftarrow Power + Impeller_{i,0} \cdot \left(\frac{N_{rpm}}{min}\right)^3 \cdot (Impeller_{i,1} \cdot m)^5 \frac{gm}{cm^3} \quad \text{if } Impeller_{i,2} \cdot L < V_{Liq} \cdot L$$

$$P\_V \leftarrow \frac{Power}{V_{Liq} \cdot L} \cdot \frac{m^3}{W} \quad \text{if } V_{Liq} \cdot L > 0 \cdot L$$

P_V ← 0 otherwise

$$q\_sup1 \leftarrow \frac{qtotal1 \cdot SLPM}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$A1 \leftarrow Sparge_{1,0}$

$alpha1 \leftarrow Sparge_{1,1}$

$beta1 \leftarrow Sparge_{1,2}$

$kla1 \leftarrow A1 P\_V^{alpha1} \cdot q\_sup1^{beta1}$

$$kla1 \leftarrow A1 \cdot P\_V \quad \cdot q\_sup1$$

$$q\_sup2 \leftarrow \frac{qtotal2 \cdot SLPM}{\pi \cdot \left(\frac{D}{2}\right)^{2}} \cdot \frac{s}{m}$$

$$A2 \leftarrow Sparge_{2,0}$$

$$alpha2 \leftarrow Sparge_{2,1}$$

$$beta2 \leftarrow Sparge_{2,2}$$

$$kla2 \leftarrow A2 \cdot P\_V^{alpha2} \cdot q\_sup2^{beta2}$$

$$\begin{aligned} kla\_surf \leftarrow\ & Surface_{0,1} + Surface_{1,1} \cdot V_{Liq} \ldots \\ & + Surface_{2,1} \cdot N_{rpm} \cdot \frac{min}{s} \ldots \\ & + Surface_{3,1} (qtotal1 + qtotal2) \ldots \\ & + Surface_{4,1} \cdot \left(V_{Liq} \cdot N_{rpm} \cdot \frac{min}{s}\right) \ldots \\ & + Surface_{5,1} \cdot \left[V_{Liq} \cdot (qtotal1 + qtotal2)\right] \ldots \\ & + Surface_{6,1} \cdot \left[N_{rpm} \cdot \frac{min}{s} \cdot (qtotal1 + qtotal2)\right] \ldots \\ & + Surface_{8,1} \cdot \left(N_{rpm} \cdot \frac{min}{s}\right)^{2} \ldots \\ & + Surface_{9,1} \cdot (qtotal1 + qtotal2)^{2} \ldots \\ & + Surface_{7,1} \cdot V_{Liq}^{\ 2} \end{aligned}$$

$$kla\_surf \leftarrow 0 \quad \text{if} \quad kla\_surf < 0$$

$$Inputs \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ DO \\ p_{HS} \\ N_{rpm} \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ kla\_surf \\ kla1 \end{pmatrix}$$

$$\begin{pmatrix} \text{kla2} \\ 0.011 \end{pmatrix}$$

$$T1 \leftarrow \text{Coefficient(Inputs)}_0 \cdot \left(\frac{\text{mol}}{\text{L}\cdot\text{hr}}\right) - \frac{DO}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} \cdot \frac{\text{mol}}{\text{L}} \cdot \text{Coefficient(Inputs)}_1 \cdot \frac{1}{\text{hr}}$$

$$T2 \leftarrow \text{Coefficient(Inputs)}_2 \cdot \left(\frac{\text{mol}}{\text{L}\cdot\text{hr}}\right) - \frac{DO}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} \cdot \frac{\text{mol}}{\text{L}} \cdot \text{Coefficient(Inputs)}_3 \cdot \frac{1}{\text{hr}}$$

$$T3 \leftarrow \frac{\begin{gathered}\text{Coefficient(Inputs)}_7 \ldots \\ + \text{Coefficient(Inputs)}_8 \cdot \frac{DO}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} + \text{Coefficient(Inputs)}_9 \ldots \\ + \text{Coefficient(Inputs)}_{10} \cdot \frac{DO}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} + \text{Coefficient(Inputs)}_{11} \ldots \\ + \text{Coefficient(Inputs)}_4 \cdot \frac{DO}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} \cdot \text{Coefficient(Inputs)}_{14}\end{gathered}}{\text{Coefficient(Inputs)}_{12} + \text{Coefficient(Inputs)}_5 \cdot \text{Coefficient(Inputs)}_4 \cdot \text{Coefficient(Inputs)}_{14}}$$

$$\text{result} \leftarrow \left[\begin{gathered} T1 \ldots \\ + T2 \ldots \\ + \frac{\text{Coefficient(Inputs)}_4}{\text{hr}} \cdot \left(\begin{gathered} T3 \cdot \text{Coefficient(Inputs)}_5 \cdot \frac{\text{mol}}{\text{L}} \ldots \\ + -\frac{DO}{100} \cdot 0.21 \cdot \text{Coefficient(Inputs)}_{27} \cdot \frac{\text{mol}}{\text{L}} \end{gathered}\right) \end{gathered}\right] \cdot \frac{\text{L}\cdot\text{hr}}{\text{mmol}}$$

The gain for the system is calculated by determining the step change in the dissolved oxygen that would result from a step change in the oxygen flow rate.

(9.6 – 22)

```
Kc(Control, Input, qAir, qO2, Θ, τc, qMax) := | DO ← Input_2
                                              | result1 ← OUR_(Control, Input, qAir, qO2)
                                              | DO2 ← Input_2 + 5
                                              | result2 ← OUR_(Control, Input, qAir, qO2 + 0.01·qMax)
                                              | for i ∈ 0..100
                                              |   | DO2 ← DO2 + 0.5DO2·(result2 − result1)/result1
                                              |   | Input_2 ← DO2
                                              |   | result2 ← OUR_(Control, Input, qAir, qO2 + 0.01·qMax)
                                              | K ← (DO2 − DO)/(0.01·qMax)
                                              | V_Liq ← Input_0
                                              | T_Liq ← Input_1 + 273.15
                                              | N_rpm ← Input_4
                                              | qair_ovly ← qAir  if AIR_CONTOUR = 0
                                              | qair_ovly ← Input_7  otherwise
                                              | qO2_ovly ← qO2  if Control = 0
                                              | qO2_ovly ← Input_8  otherwise
                                              | qN2_ovly ← Input_10
                                              | qCO2_ovly ← | max[Input_9, str2num(Min_CO2)/(100 − str2num(Min_CO2))·(qair_ovly ... + qO2_ovly ... + qN2_ovly)]  if CO2_CONTROL = 0
                                              |             | Input_9  otherwise
                                              | if Control = 1
                                              |   | qO2_1 ← qO2
                                              |   | qair_1 ← qAir  if AIR_CONTOUR = 1
                                              |   | otherwise
                                              |   |   | if AIR_CO2_CONTROL = 1
                                              |   |   |   | qair_1 ← str2num(AIR_CO2_GAIN) qO2_1
                                              |   |   |   | qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
                                              |   |   |   | qair_1 ← Input_11  if qair_1 < Input_11
                                              |   |   | qair_1 ← Input_11  otherwise
                                              |   | qair_2 ← qAir  if AIR_CONTOUR = 2
                                              |   | otherwise
                                              |   |   | if AIR_CO2_CONTROL = 2
                                              |   |   |   | qair_2 ← str2num(AIR_CO2_GAIN) qO2_1
                                              |   |   |   | qair_2 ← str2num(AIR_CO2_MAX)  if qair_2 > str2num(AIR_CO2_MAX)
                                              |   |   |   | qair_2 ← Input_15  if qair_2 < Input_15
                                              |   |   | qair_2 ← Input_15  otherwise
                                              | qO2_1 ← Input_12  otherwise
```

$qN2\_1 \leftarrow Input_{14}$

$qCO2\_1 \leftarrow \begin{cases} \max\left[Input_{13}, \dfrac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_1 + qO2\_1 + qN2\_1)\right] & \text{if } CO2\_CONTROL = 1 \\ Input_{13} & \text{otherwise} \end{cases}$

if Control = 2

$qO2\_2 \leftarrow qO2$

$qair\_1 \leftarrow qAir$ if $AIR\_CONTOUR = 1$

otherwise

if $AIR\_CO2\_CONTROL = 1$

$qair\_1 \leftarrow str2num(AIR\_CO2\_GAIN)\, qO2\_2$

$qair\_1 \leftarrow str2num(AIR\_CO2\_MAX)$ if $qair\_1 > str2num(AIR\_CO2\_MAX)$

$qair\_1 \leftarrow Input_{11}$ if $qair\_1 < Input_{11}$

$qair\_1 \leftarrow Input_{11}$ otherwise

$qair\_2 \leftarrow qAir$ if $AIR\_CONTOUR = 2$

otherwise

if $AIR\_CO2\_CONTROL = 2$

$qair\_2 \leftarrow str2num(AIR\_CO2\_GAIN)\, qO2\_2$

$qair\_2 \leftarrow str2num(AIR\_CO2\_MAX)$ if $qair\_2 > str2num(AIR\_CO2\_MAX)$

$qair\_2 \leftarrow Input_{15}$ if $qair\_2 < Input_{15}$

$qair\_2 \leftarrow Input_{15}$ otherwise

$qO2\_2 \leftarrow Input_{16}$ otherwise

$qN2\_2 \leftarrow Input_{18}$

$qCO2\_2 \leftarrow \begin{cases} \max\left[Input_{17}, \dfrac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_2 + qO2\_2 + qN2\_2)\right] & \text{if } CO2\_CONTROL = 2 \\ Input_{17} & \text{otherwise} \end{cases}$

$qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1) + 0.0000000000000001 \cdot SIUnitsOf(qair\_1)$

$qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2) + 0.0000000000000001 \cdot SIUnitsOf(qair\_2)$

$kla\_surf \leftarrow 0$

$Power \leftarrow 0 \cdot W$

for $i \in 1 .. rows(Impeller) - 1$

$Power \leftarrow Power + Impeller_{i,0} \cdot \left(\dfrac{N_{rpm}}{min}\right)^3 \cdot (Impeller_{i,1} \cdot m)^5 \dfrac{gm}{cm^3}$ if $Impeller_{i,2} \cdot L < V_{Liq} \cdot L$

$P\_V \leftarrow \dfrac{Power}{V_{Liq} \cdot L} \cdot \dfrac{m^3}{W}$ if $V_{Liq} \cdot L > 0 \cdot L$

$P\_V \leftarrow 0$ otherwise

$q\_sup1 \leftarrow \dfrac{qtotal1 \cdot SLPM}{(D)^2} \cdot \dfrac{s}{m}$

$$\pi\cdot\left(\frac{\;}{2}\right)$$

$$A1 \leftarrow Sparge_{1,0}$$

$$alpha1 \leftarrow Sparge_{1,1}$$

$$beta1 \leftarrow Sparge_{1,2}$$

$$kla1 \leftarrow A1P\_V^{alpha1}\cdot q\_sup1^{beta1}$$

$$q\_sup2 \leftarrow \frac{qtotal2\cdot SLPM}{\pi\cdot\left(\frac{D}{2}\right)^2}\cdot\frac{s}{m}$$

$$A2 \leftarrow Sparge_{2,0}$$

$$alpha2 \leftarrow Sparge_{2,1}$$

$$beta2 \leftarrow Sparge_{2,2}$$

$$kla2 \leftarrow A2\cdot P\_V^{alpha2}\cdot q\_sup2^{beta2}$$

$$\begin{aligned} kla\_surf \leftarrow\ & Surface_{0,1} + Surface_{1,1}\cdot V_{Liq} \ldots \\ & + Surface_{2,1}\cdot N_{rpm}\cdot\frac{min}{s} \ldots \\ & + Surface_{3,1}(qtotal1 + qtotal2) \ldots \\ & + Surface_{4,1}\cdot\left(V_{Liq}\cdot N_{rpm}\cdot\frac{min}{s}\right) \ldots \\ & + Surface_{5,1}\cdot\left[V_{Liq}\cdot(qtotal1 + qtotal2)\right] \ldots \\ & + Surface_{6,1}\cdot\left[N_{rpm}\cdot\frac{min}{s}\cdot(qtotal1 + qtotal2)\right] \ldots \\ & + Surface_{8,1}\cdot\left(N_{rpm}\cdot\frac{min}{s}\right)^2 \ldots \\ & + Surface_{9,1}\cdot(qtotal1 + qtotal2)^2 \ldots \\ & + Surface_{7,1}\cdot V_{Liq}^{\;2} \end{aligned}$$

$$kla\_surf \leftarrow 0 \quad \text{if } kla\_surf < 0$$

$$\tau \leftarrow \frac{3600}{kla\_surf} \quad \text{if } Control = 0$$

$$\tau \leftarrow \frac{3600}{kla1} \quad \text{if } Control = 1$$

$$\tau \leftarrow \frac{3600}{kla2} \quad \text{if } Control = 2$$

$$result_0 \leftarrow \frac{1}{K}\cdot\frac{\tau}{\Theta + \tau c}$$

$$result_1 \leftarrow \frac{1}{K}\cdot\frac{\tau^2 + \tau\cdot\Theta - (\tau c - \tau)^2}{(\Theta + \tau c)^2}$$

$$result$$

The integral time constant is determined from the controlling gas transfer coefficient.

```
ts(Control, Input, qAir, qO2) :=                                                    (9.6 – 23)
  | V_Liq ← Input_0
  | T_Liq ← Input_1 + 273.15
  | DO ← Input_2
  | N_rpm ← Input_4
  | if Control = 0
  |   | qair_ovly ← qAir
  |   | qO2_ovly ← qO2
  | otherwise
  |   | qair_ovly ← Input_7
  |   | qO2_ovly ← Input_8
  | qN2_ovly ← Input_10
  | qCO2_ovly ← | str2num(Min_CO2) / (100 − str2num(Min_CO2)) · (qair_ovly ...
  |             |        + qO2_ovly ...
  |             |        + qN2_ovly)   if CO2_CONTROL = 0
  |             | Input_9  otherwise
  | if Control = 1
  |   | qO2_1 ← qO2
  |   | if AIR_CO2_CONTROL = 1
  |   |   | qair_1 ← str2num(AIR_CO2_GAIN)·qO2_1
  |   |   | qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
  |   |   | qair_1 ← Input_11  if qair_1 < Input_11
  |   | qair_1 ← Input_11  otherwise
  |   | if AIR_CO2_CONTROL = 2
  |   |   | qair_2 ← str2num(AIR_CO2_GAIN)·qO2_1
  |   |   | qair_2 ← str2num(AIR_CO2_MAX)  if qair_2 > str2num(AIR_CO2_MAX)
  |   |   | qair_2 ← Input_15  if qair_2 < Input_15
  |   | qair_2 ← Input_15  otherwise
  | qO2_1 ← Input_12  otherwise
  | qN2_1 ← Input_14
  | qCO2_1 ← | str2num(Min_CO2) / (100 − str2num(Min_CO2)) · (qair_1 + qO2_1 + qN2_1)  if CO2_CONTROL = 1
  |          | Input_13  otherwise
  | if Control = 2
  |   | qO2_2 ← qO2
  |   | if AIR_CO2_CONTROL = 1
  |   |   | qair_1 ← str2num(AIR_CO2_GAIN) qO2_2
  |   |   | qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
  |   |   | qair_1 ← Input_11  if qair_1 < Input_11
  |   | qair_1 ← Input_11  otherwise
```

if AIR_CO2_CONTROL = 2

$qair\_2 \leftarrow str2num(AIR\_CO2\_GAIN)\,qO2\_2$

$qair\_2 \leftarrow str2num(AIR\_CO2\_MAX) \quad \text{if } qair\_2 > str2num(AIR\_CO2\_MAX)$

$qair\_2 \leftarrow Input_{15} \quad \text{if } qair\_2 < Input_{15}$

$qair\_2 \leftarrow Input_{15} \quad \text{otherwise}$

$qO2\_2 \leftarrow Input_{16} \quad \text{otherwise}$

$qN2\_2 \leftarrow Input_{18}$

$$qCO2\_2 \leftarrow \begin{cases} \dfrac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_2 + qO2\_2 + qN2\_2) & \text{if } CO2\_CONTROL = 2 \\ Input_{17} & \text{otherwise} \end{cases}$$

$qtotal1 \leftarrow q_{total}(qair\_1, qO2\_1, qCO2\_1, qN2\_1)$

$qtotal2 \leftarrow q_{total}(qair\_2, qO2\_2, qCO2\_2, qN2\_2)$

$$\begin{aligned} kla\_surf \leftarrow\ & Surface_{0,1} + Surface_{1,1} \cdot V_{Liq} \ldots \\ & + Surface_{2,1} \cdot N_{rpm} \cdot \frac{min}{s} \ldots \\ & + Surface_{3,1}(qtotal1 + qtotal2) \ldots \\ & + Surface_{4,1} \cdot \left(V_{Liq} \cdot N_{rpm} \cdot \frac{min}{s}\right) \ldots \\ & + Surface_{5,1} \cdot \left[V_{Liq} \cdot (qtotal1 + qtotal2)\right] \ldots \\ & + Surface_{6,1} \cdot \left[N_{rpm} \cdot \frac{min}{s} \cdot (qtotal1 + qtotal2)\right] \ldots \\ & + Surface_{8,1} \cdot \left(N_{rpm} \cdot \frac{min}{s}\right)^2 \ldots \\ & + Surface_{9,1} \cdot (qtotal1 + qtotal2)^2 \ldots \\ & + Surface_{7,1} \cdot V_{Liq}^{\;2} \end{aligned}$$

$Power \leftarrow 0 \cdot W$

for $i \in 1 .. rows(Impeller) - 1$

$$Power \leftarrow Power + Impeller_{i,0} \cdot \left(\frac{N_{rpm}}{min}\right)^3 \cdot \left(Impeller_{i,1} \cdot m\right)^5 \frac{gm}{cm^3} \quad \text{if } Impeller_{i,2} \cdot L < V_{Liq} \cdot L$$

$$P\_V \leftarrow \frac{Power}{V_{Liq} \cdot L} \cdot \frac{m^3}{W} \quad \text{if } V_{Liq} \cdot L > 0 \cdot L$$

$P\_V \leftarrow 0 \quad \text{otherwise}$

$$q\_sup1 \leftarrow \frac{qtotal1 \cdot SLPM}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$A1 \leftarrow Sparge_{1,0}$

$alpha1 \leftarrow Sparge_{1,1}$

$beta1 \leftarrow Sparge_{1,2}$

$kla1 \leftarrow A1\,P\_V^{alpha1} \cdot q\_sup1^{beta1}$

$$q\_sup2 \leftarrow \frac{qtotal2 \cdot SLPM}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$$
\begin{aligned}
&A2 \leftarrow Sparge_{2,0}\\
&alpha2 \leftarrow Sparge_{2,1}\\
&beta2 \leftarrow Sparge_{2,2}\\
&kla2 \leftarrow A2 \cdot P\_V^{alpha2} \cdot q\_sup2^{beta2}\\
&\tau \leftarrow \frac{3600}{kla\_surf} \quad \text{if } Control = 0\\
&\tau \leftarrow \frac{3600}{kla1} \quad \text{if } Control = 1\\
&\tau \leftarrow \frac{3600}{kla2} \quad \text{if } Control = 2\\
&result \leftarrow \tau
\end{aligned}
$$

This program returns the dissolved carbon dioxide that results from different air and oxygen flow rates through the controlling gas stream (overlay, sparge 1, or sparge 2).

$$
\begin{aligned}
CO2\_(Control, Input, qAir, qO2) := \;&V_{Liq} \leftarrow Input_0\\
&T_{Liq} \leftarrow Input_1 + 273.15\\
&DO \leftarrow Input_2\\
&P_{HS} \leftarrow Input_3 + \frac{P_{atm}}{psi}\\
&N_{rpm} \leftarrow Input_4\\
&qair\_ovly \leftarrow qAir \quad \text{if } AIR\_CONTOUR = 0\\
&qair\_ovly \leftarrow Input_7 \quad \text{otherwise}\\
&qO2\_ovly \leftarrow qO2 \quad \text{if } Control = 0\\
&qO2\_ovly \leftarrow Input_8 \quad \text{otherwise}\\
&qN2\_ovly \leftarrow Input_{10}\\
&qCO2\_ovly \leftarrow \left|\begin{array}{l} \max\left[Input_9, \frac{str2num(Min\_CO2)}{100 - str2num(Min\_CO2)} \cdot (qair\_ovly \ldots + qO2\_ovly \ldots + qN2\_ovly)\right] \quad \text{if } CO2\_CONTROL = 0\\ Input_9 \quad \text{otherwise}\end{array}\right.\\
&\text{if } Control = 1\\
&\quad qO2\_1 \leftarrow qO2\\
&\quad qair\_1 \leftarrow qAir \quad \text{if } AIR\_CONTOUR = 1\\
&\quad \text{otherwise}\\
&\quad\quad \text{if } AIR\_CO2\_CONTROL = 1\\
&\quad\quad\quad qair\_1 \leftarrow str2num(AIR\_CO2\_GAIN)\, qO2\_1\\
&\quad\quad\quad qair\_1 \leftarrow str2num(AIR\_CO2\_MAX) \quad \text{if } qair\_1 > str2num(AIR\_CO2\_MAX)\\
&\quad\quad\quad qair\_1 \leftarrow Input_{11} \quad \text{if } qair\_1 < Input_{11}\\
&\quad\quad qair\_1 \leftarrow Input_{11} \quad \text{otherwise}
\end{aligned}
\qquad (9.6-24)
$$

```
    qair_2 ← qAir  if  AIR_CONTOUR = 2
    otherwise
        if  AIR_CO2_CONTROL = 2
            qair_2 ← str2num(AIR_CO2_GAIN) qO2_1
            qair_2 ← str2num(AIR_CO2_MAX)  if  qair_2 > str2num(AIR_CO2_MAX)
            qair_2 ← Input_15  if  qair_2 < Input_15
        qair_2 ← Input_15  otherwise
qO2_1 ← Input_12  otherwise
qN2_1 ← Input_14
qCO2_1 ← | max[ Input_13 , str2num(Min_CO2) / (100 − str2num(Min_CO2)) · (qair_1 ... + qO2_1 ... + qN2_1) ]  if  CO2_CONTROL = 1
         | Input_13  otherwise
if  Control = 2
    qO2_2 ← qO2
    qair_1 ← qAir  if  AIR_CONTOUR = 1
    otherwise
        if  AIR_CO2_CONTROL = 1
            qair_1 ← str2num(AIR_CO2_GAIN) qO2_2
            qair_1 ← str2num(AIR_CO2_MAX)  if  qair_1 > str2num(AIR_CO2_MAX)
            qair_1 ← Input_11  if  qair_1 < Input_11
        qair_1 ← Input_11  otherwise
    qair_2 ← qAir  if  AIR_CONTOUR = 2
    otherwise
        if  AIR_CO2_CONTROL = 2
            qair_2 ← str2num(AIR_CO2_GAIN) qO2_2
            qair_2 ← str2num(AIR_CO2_MAX)  if  qair_2 > str2num(AIR_CO2_MAX)
            qair_2 ← Input_15  if  qair_2 < Input_15
        qair_2 ← Input_15  otherwise
qO2_2 ← Input_16  otherwise
qN2_2 ← Input_18
qCO2_2 ← | max[ Input_17 , str2num(Min_CO2) / (100 − str2num(Min_CO2)) · (qair_2 ... + qO2_2 ... + qN2_2) ]  if  CO2_CONTROL = 2
         | Input_17  otherwise
qtotalovly ← q_total(qair_ovly, qO2_ovly, qCO2_ovly, qN2_ovly) + 0.00000000000001·SIUnitsOf(qair_ovly)
qtotal1 ← q_total(qair_1, qO2_1, qCO2_1, qN2_1) + 0.00000000000001·SIUnitsOf(qair_1)
qtotal2 ← q_total(qair_2, qO2_2, qCO2_2, qN2_2) + 0.00000000000001·SIUnitsOf(qair_2)
kla_surf ← Surface_0,1 + Surface_1,1·V_Liq ...
```

$$+ \text{Surface}_{2,1} \cdot N_{rpm} \cdot \frac{\text{min}}{\text{s}} \ldots$$
$$+ \text{Surface}_{3,1}(\text{qtotal1} + \text{qtotal2}) \ldots$$
$$+ \text{Surface}_{4,1} \cdot \left(V_{Liq} \cdot N_{rpm} \cdot \frac{\text{min}}{\text{s}}\right) \ldots$$
$$+ \text{Surface}_{5,1} \cdot \left[V_{Liq} \cdot (\text{qtotal1} + \text{qtotal2})\right] \ldots$$
$$+ \text{Surface}_{6,1} \cdot \left[N_{rpm} \cdot \frac{\text{min}}{\text{s}} \cdot (\text{qtotal1} + \text{qtotal2})\right] \ldots$$
$$+ \text{Surface}_{8,1} \cdot \left(N_{rpm} \cdot \frac{\text{min}}{\text{s}}\right)^2 \ldots$$
$$+ \text{Surface}_{9,1} \cdot (\text{qtotal1} + \text{qtotal2})^2 \ldots$$
$$+ \text{Surface}_{7,1} \cdot V_{Liq}^{\ 2}$$

$\text{kla\_surf} \leftarrow 0 \ \text{ if } \ \text{kla\_surf} < 0$

$\text{Power} \leftarrow 0 \cdot W$

$\text{for } \ i \in 1 .. \text{rows(Impeller)} - 1$

$$\text{Power} \leftarrow \text{Power} + \text{Impeller}_{i,0} \cdot \left(\frac{N_{rpm}}{\text{min}}\right)^3 \cdot \left(\text{Impeller}_{i,1} \cdot m\right)^5 \frac{gm}{cm^3} \ \text{ if } \ \text{Impeller}_{i,2} \cdot L < V_{Liq} \cdot L$$

$$\text{P\_V} \leftarrow \frac{\text{Power}}{V_{Liq} \cdot L} \cdot \frac{m^3}{W} \ \text{ if } \ V_{Liq} \cdot L > 0 \cdot L$$

$$\text{q\_sup1} \leftarrow \frac{\text{qtotal1} \cdot \text{SLPM}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$\text{A1} \leftarrow \text{Sparge}_{1,0}$

$\text{alpha1} \leftarrow \text{Sparge}_{1,1}$

$\text{beta1} \leftarrow \text{Sparge}_{1,2}$

$\text{kla1} \leftarrow \text{A1} \cdot \text{P\_V}^{\text{alpha1}} \cdot \text{q\_sup1}^{\text{beta1}}$

$$\text{q\_sup2} \leftarrow \frac{\text{qtotal2} \cdot \text{SLPM}}{\pi \cdot \left(\frac{D}{2}\right)^2} \cdot \frac{s}{m}$$

$\text{A2} \leftarrow \text{Sparge}_{2,0}$

$\text{alpha2} \leftarrow \text{Sparge}_{2,1}$

$\text{beta2} \leftarrow \text{Sparge}_{2,2}$

$\text{kla2} \leftarrow \text{A2} \cdot \text{P\_V}^{\text{alpha2}} \cdot \text{q\_sup2}^{\text{beta2}}$

$\text{OUR} \leftarrow \text{OUR\_(Control, Input, qAir, qO2)}$

$$\begin{pmatrix} V_{Liq} \\ T_{Liq} \\ DO \\ p_{HS} \\ N_{rpm} \\ \text{qair\_ovly} \\ \text{qO2\_ovly} \\ \ldots \end{pmatrix}$$

$$\text{Inputs} \leftarrow \begin{pmatrix} \text{qCO2\_ovly} \\ \text{qN2\_ovly} \\ \text{qair\_1} \\ \text{qO2\_1} \\ \text{qCO2\_1} \\ \text{qN2\_1} \\ \text{qair\_2} \\ \text{qO2\_2} \\ \text{qCO2\_2} \\ \text{qN2\_2} \\ \text{OUR} \\ \text{kla\_surf} \\ \text{kla1} \\ \text{kla2} \\ 0.011 \end{pmatrix}$$

$$T1 \leftarrow \text{Coefficient(Inputs)}_{15}\cdot\frac{mol}{L} + \text{Coefficient(Inputs)}_{17}\cdot\frac{mol}{L}$$

$$T2 \leftarrow \frac{\left(\text{Coefficient(Inputs)}_{21} + \text{Coefficient(Inputs)}_{23} + \text{Coefficient(Inputs)}_{25}\right)}{\text{Coefficient(Inputs)}_{12} + \text{Coefficient(Inputs)}_{4}\cdot 0.893\cdot\text{Coefficient(Inputs)}_{20}\cdot\text{Coefficient(Inputs)}_{14}}$$

$$T3 \leftarrow \left[\frac{\left(\text{Coefficient(Inputs)}_{22} \ldots + \text{Coefficient(Inputs)}_{24} \ldots + \text{Coefficient(Inputs)}_{4}\cdot 0.893\cdot\text{Coefficient(Inputs)}_{14}\right)\cdot\text{Coefficient(Inputs)}_{20}}{\text{Coefficient(Inputs)}_{12} + \text{Coefficient(Inputs)}_{4}\cdot 0.893\cdot\text{Coefficient(Inputs)}_{20}\cdot\text{Coefficient(Inputs)}_{14}} - 1\right]$$

$$\text{result} \leftarrow \frac{\left[T1 \ldots + \text{Coefficient(Inputs)}_{19}\cdot\left(T2\cdot\text{Coefficient(Inputs)}_{20}\cdot\frac{mol}{L}\right) \ldots + RQ\,OUR\cdot\frac{mmol}{L\cdot hr}\cdot hr\right]}{H_{CO2\_cp}\left(0.011, T_{Liq}\cdot K\right)\cdot mmHg\cdot\left(\text{Coefficient(Inputs)}_{16} \ldots + \text{Coefficient(Inputs)}_{18} \ldots + -\text{Coefficient(Inputs)}_{19}\cdot T3\right)}$$

These two programs create the response surface for the oxygen uptake rate and the dissolved carbon dioxide if the user selected section 9.6 to be active.

$$\text{OURmesh(qO2, qAir)} := \left| \begin{array}{l} \text{result} \leftarrow 0 \ \text{ if } \ \text{Part6} = 0 \\ \text{result} \leftarrow \text{OUR\_(DO\_CONTROL, Input, qAir, qO2)} \ \text{ otherwise} \\ \text{result} \end{array} \right. \qquad (9.6-25)$$

$$\text{CO2mesh(qO2, qAir)} := \left| \begin{array}{l} \text{result} \leftarrow 0 \ \text{ if } \ \text{Part6} = 0 \\ \text{result} \leftarrow \text{CO2\_(DO\_CONTROL, Input, qAir, qO2)} \ \text{ otherwise} \\ \text{result} \end{array} \right. \qquad (9.6-26)$$

```
Air_min := | if str2num(AirMin) = 0                                           (9.6 – 27)
           |   | result ← Limits_{4,1}  if DO_CONTROL = 0
           |   | otherwise
           |   |   | result ← Limits_{8,1}  if DO_CONTROL = 1
           |   |   | result ← Limits_{12,1}  otherwise
           | result ← str2num(AirMin)  otherwise
```

```
Air_max := | if str2num(AirMax) = 0                                           (9.6 – 28)
           |   | result ← Limits_{4,3}  if DO_CONTROL = 0
           |   | otherwise
           |   |   | result ← Limits_{8,3}  if DO_CONTROL = 1
           |   |   | result ← Limits_{12,3}  otherwise
           | result ← str2num(AirMax)  otherwise
```

```
O2_min := | if str2num(O2Min) = 0                                             (9.6 – 29)
          |   | result ← Limits_{5,1}  if DO_CONTROL = 0
          |   | otherwise
          |   |   | result ← Limits_{9,1}  if DO_CONTROL = 1
          |   |   | result ← Limits_{13,1}  otherwise
          | result ← str2num(O2Min)  otherwise
```

```
O2_max := | if str2num(O2Max) = 0                                             (9.6 – 30)
          |   | result ← Limits_{5,3}  if DO_CONTROL = 0
          |   | otherwise
          |   |   | result ← Limits_{9,3}  if DO_CONTROL = 1
          |   |   | result ← Limits_{13,3}  otherwise
          | result ← str2num(O2Max)  otherwise
```

$$OUR := CreateMesh(OURmesh, O2\_min, O2\_max, Air\_min, Air\_max, 100, 50) \tag{9.6 – 31}$$

$$CO2 := CreateMesh(CO2mesh, O2\_min, O2\_max, Air\_min, Air\_max, 100, 50) \tag{9.6 – 32}$$

$$
\begin{aligned}
GasFlows := \quad & result_{0,1} \leftarrow \text{"Air"} \\
& result_{0,2} \leftarrow \text{"Oxygen"} \\
& result_{0,3} \leftarrow \text{"Carbon Dioxide"} \\
& result_{1,0} \leftarrow \text{"Overlay"} \\
& result_{2,0} \leftarrow \text{"Sparger 1"} \\
& result_{3,0} \leftarrow \text{"Sparger 2"} \\
& result_{1,1} \leftarrow ProcessData_{8,2} \\
& result_{1,2} \leftarrow ProcessData_{9,2} \\
& result_{1,3} \leftarrow ProcessData_{10,2} \\
& result_{2,1} \leftarrow ProcessData_{12,2} \\
& result_{2,2} \leftarrow ProcessData_{13,2} \\
& result_{2,3} \leftarrow ProcessData_{14,2} \\
& result_{3,1} \leftarrow ProcessData_{16,2} \\
& result_{3,2} \leftarrow ProcessData_{17,2} \\
& result_{3,3} \leftarrow ProcessData_{18,2} \\
& result_{1,2} \leftarrow \text{"X\_values"} \quad \text{if } DO\_CONTROL = 0 \\
& result_{2,2} \leftarrow \text{"X\_values"} \quad \text{if } DO\_CONTROL = 1 \\
& result_{3,2} \leftarrow \text{"X\_values"} \quad \text{if } DO\_CONTROL = 2 \\
& result_{1,1} \leftarrow \text{"Y\_values"} \quad \text{if } AIR\_CONTOUR = 0 \\
& result_{2,1} \leftarrow \text{"Y\_values"} \quad \text{if } AIR\_CONTOUR = 1 \\
& result_{3,1} \leftarrow \text{"Y\_values"} \quad \text{if } AIR\_CONTOUR = 2 \\
& result
\end{aligned}
\qquad (9.6-33)
$$

The matrix of gain and time integral is calculated from the variable gain system.

(9.6 – 34)

```
Tuning(θ, τc, qMax, Inc) :=
  for i ∈ 0..6      if Part6 = 0
     result[0,i] ← 0
  for i ∈ 0..Inc                                        otherwise
     result[i,0] ← (i/Inc)·qMax
     if DO_CONTROL = 1
        qO2_1 ← result[i,0]
        qair_1 ← Input[11]
        qair_2 ← Input[15]
        air_flow ← qair_1
        if AIR_CO2_CONTROL = 1
           qair_1 ← str2num(AIR_CO2_GAIN) qO2_1
           qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
           qair_1 ← Input[11]  if qair_1 < Input[11]
           air_flow ← qair_1
        if AIR_CO2_CONTROL = 2
           qair_2 ← str2num(AIR_CO2_GAIN) qO2_1
           qair_2 ← str2num(AIR_CO2_MAX)  if qair_2 > str2num(AIR_CO2_MAX)
           qair_2 ← Input[15]  if qair_2 < Input[15]
           air_flow ← qair_2
     if DO_CONTROL = 2
        qO2_2 ← result[i,0]
        qair_1 ← Input[11]
        qair_2 ← Input[15]
        air_flow ← qair_2
        if AIR_CO2_CONTROL = 1
           qair_1 ← str2num(AIR_CO2_GAIN) qO2_2
           qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
           qair_1 ← Input[11]  if qair_1 < Input[11]
           airBasis ← Input[11]  if airBasis < Input[11]
           air_flow ← qair_1
        if AIR_CO2_CONTROL = 2
           qair_2 ← str2num(AIR_CO2_GAIN) qO2_2
           qair_2 ← str2num(AIR_CO2_MAX)  if qair_2 > str2num(AIR_CO2_MAX)
           qair_2 ← Input[15]  if qair_2 < Input[15]
           air_flow ← qair_2
     result[i,1] ← Kc(DO_CONTROL, Input, air_flow, (i·qMax)/Inc, θ, τc, qMax)[0]·(100/qMax)
     result[i,2] ← Kc(DO_CONTROL, Input, air_flow, (i·qMax)/Inc, θ, τc, qMax)[1]·(100/qMax)
```

$$\mathrm{result}_{i,3} \leftarrow \frac{100}{\mathrm{result}_{i,2}}$$

$$\mathrm{result}_{i,4} \leftarrow \mathrm{ts}\left(\mathrm{DO\_CONTROL}, \mathrm{Input}, \mathrm{air\_flow}, \frac{i \cdot \mathrm{qMax}}{\mathrm{Inc}}\right)$$

$$\mathrm{result}_{i,5} \leftarrow \frac{\left[\left(\mathrm{ts}\left(\mathrm{DO\_CONTROL}, \mathrm{Input}, \mathrm{air\_flow}, \frac{i \cdot \mathrm{qMax}}{\mathrm{Inc}}\right)\right)^2 \ldots + \theta \cdot \mathrm{ts}\left(\mathrm{DO\_CONTROL}, \mathrm{Input}, \mathrm{air\_flow}, \frac{i \cdot \mathrm{qMax}}{\mathrm{Inc}}\right) \ldots + -\left(\tau c - \mathrm{ts}\left(\mathrm{DO\_CONTROL}, \mathrm{Input}, \mathrm{air\_flow}, \frac{i \cdot \mathrm{qMax}}{\mathrm{Inc}}\right)\right)^2\right]}{\left(\mathrm{ts}\left(\mathrm{DO\_CONTROL}, \mathrm{Input}, \mathrm{air\_flow}, \frac{i \cdot \mathrm{qMax}}{\mathrm{Inc}}\right) + \theta\right)}$$

$$\mathrm{result}_{i,6} \leftarrow 4 \cdot (\tau c)$$

result

The gain for the system is calculated.

```
GAIN(Basis, θ, τc, qMax) := | if DO_CONTROL = 1                                              (9.6 – 35)
                            |   | qO2_1 ← Basis
                            |   | qair_1 ← Input_11
                            |   | qair_2 ← Input_15
                            |   | air_flow ← qair_1
                            |   | if AIR_CO2_CONTROL = 1
                            |   |   | qair_1 ← str2num(AIR_CO2_GAIN) qO2_1
                            |   |   | qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
                            |   |   | qair_1 ← Input_11  if qair_1 < Input_11
                            |   |   | air_flow ← qair_1
                            |   | if AIR_CO2_CONTROL = 2
                            |   |   | qair_2 ← str2num(AIR_CO2_GAIN) qO2_1
                            |   |   | qair_2 ← str2num(AIR_CO2_MAX)  if qair_2 > str2num(AIR_CO2_MAX)
                            |   |   | qair_2 ← Input_15  if qair_2 < Input_15
                            |   |   | air_flow ← qair_2
                            | if DO_CONTROL = 2
                            |   | qO2_2 ← Basis
                            |   | qair_1 ← Input_11
                            |   | qair_2 ← Input_15
                            |   | air_flow ← qair_2
                            |   | if AIR_CO2_CONTROL = 1
                            |   |   | qair_1 ← str2num(AIR_CO2_GAIN) qO2_2
                            |   |   | qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
                            |   |   | qair_1 ← Input_11  if qair_1 < Input_11
```

$$\begin{array}{l} \quad \quad air\_flow \leftarrow qair\_1 \\ \quad \text{if } AIR\_CO2\_CONTROL = 2 \\ \quad \quad qair\_2 \leftarrow str2num(AIR\_CO2\_GAIN)\, qO2\_2 \\ \quad \quad qair\_2 \leftarrow str2num(AIR\_CO2\_MAX) \;\; \text{if } qair\_2 > str2num(AIR\_CO2\_MAX) \\ \quad \quad qair\_2 \leftarrow Input_{15} \;\; \text{if } qair\_2 < Input_{15} \\ \quad \quad air\_flow \leftarrow qair\_2 \\ Kc(DO\_CONTROL, Input, air\_flow, Basis, \theta, \tau c, qMax)_1 \cdot \frac{100}{qMax} \end{array}$$

The polygon is expressed as the x and y array to produce the graph in figure 9.6-2

$$x := Tuning(\theta, \tau c, qMaxO2, Inc)^{\langle 0 \rangle} \tag{9.6 – 36}$$

$$y := Tuning(\theta, \tau c, qMaxO2, Inc)^{\langle 2 \rangle} \tag{9.6 – 37}$$

The average time integral for the controller is calculated by the following.

$$avg\_ti(\theta, \tau c, qMax, Increments) := \left| \begin{array}{l} M \leftarrow Tuning(\theta, \tau c, qMax, Inc)^{\langle 5 \rangle} \\ mean(M) \end{array} \right. \tag{9.6 – 38}$$

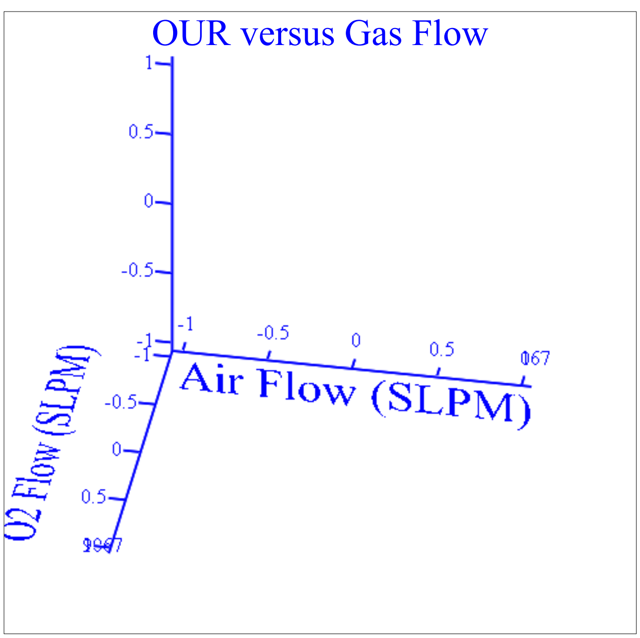
Alternatively, the polygon for the output adjustment is calculated by the following.

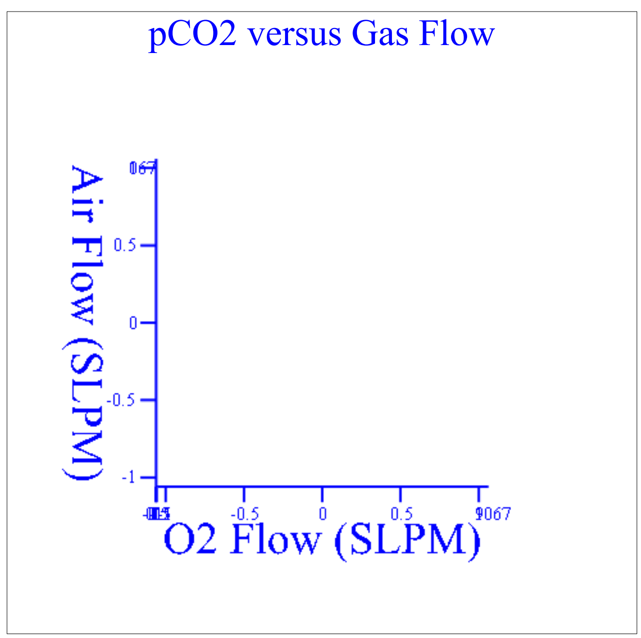
$$air := \left| \begin{array}{l} result \leftarrow Input_7 \;\; \text{if } AIR\_CONTOUR = 0 \\ result \leftarrow Input_{11} \;\; \text{if } AIR\_CONTOUR = 1 \\ result \leftarrow Input_{15} \;\; \text{if } AIR\_CONTOUR = 2 \\ result \end{array} \right. \tag{9.6 – 39}$$

(9.6 – 40)

$$
\text{Polygon}(\text{Basis}, \text{air}, \theta, \tau c, \text{qMax}, \text{Increments}) :=
\left|
\begin{array}{l}
\text{for } i \in 0..2 \quad \text{if } \text{Part6} = 0 \\
\quad \text{result}_{0,i} \leftarrow 0 \\
\text{otherwise} \\
\quad \left|
\begin{array}{l}
\text{result}_{0,0} \leftarrow 0 \\
\text{result}_{0,1} \leftarrow 0 \\
\text{result}_{0,2} \leftarrow \dfrac{(\text{ts}(\text{DO\_CONTROL}, \text{Input}, \text{air}, 0))^2 \ldots + \theta \cdot \text{ts}(\text{DO\_CONTROL}, \text{Input}, \text{air}, 0) \ldots + -(\tau c - \text{ts}(\text{DO\_CONTROL}, \text{Input}, \text{air}, 0))^2}{\text{ts}(\text{DO\_CONTROL}, \text{Input}, \text{air}, 0) + \theta} \\
\text{for } i \in 1..\text{Increments} \\
\quad \left|
\begin{array}{l}
\text{result}_{i,0} \leftarrow \dfrac{i}{\text{Increments}} \cdot \text{qMax} \\
\text{result}_{i,1} \leftarrow \text{result}_{i-1,1} + \dfrac{\text{Kc}(\text{DO\_CONTROL}, \text{Input}, \text{air}, \text{Basis}, \theta, \tau c, \text{qMax})_1}{\text{Kc}\left(\text{DO\_CONTROL}, \text{Input}, \text{air}, \dfrac{i \cdot \text{qMax}}{\text{Increments}}, \theta, \tau c, \text{qMax}\right)_1} \cdot \dfrac{\text{qMax}}{\text{Increments}} \\
\text{result}_{i,2} \leftarrow \dfrac{\left(\text{ts}\left(\text{DO\_CONTROL}, \text{Input}, \text{air}, \dfrac{i \cdot \text{qMax}}{\text{Inc}}\right)\right)^2 \ldots + \theta \cdot \text{ts}\left(\text{DO\_CONTROL}, \text{Input}, \text{air}, \dfrac{i \cdot \text{qMax}}{\text{Inc}}\right) \ldots + -\left(\tau c - \text{ts}\left(\text{DO\_CONTROL}, \text{Input}, \text{air}, \dfrac{i \cdot \text{qMax}}{\text{Inc}}\right)\right)^2}{\text{ts}\left(\text{DO\_CONTROL}, \text{Input}, \text{air}, \dfrac{i \cdot \text{qMax}}{\text{Inc}}\right) + \theta}
\end{array}
\right.
\end{array}
\right. \\
\text{result}
\end{array}
\right.
$$

Direct Synthesis Calculations





GasFlows = ▮

Figure 9.6-1: Contour plots for the oxygen uptake rate and resulting carbon dioxide accumulation for the vessel with the specified air, oxygen, and carbon dioxide flow rates through the overlay sparger 1, and sparger 2 (time zero settings for the bioreactor in table 9.6-11).

9.6.1 Dissolved Oxygen PI Loop Tuning Parameters

The resulting controller gain (1/Kc) for the system (at the basis flow rate) is calculated to be as follows:

$$\mathrm{GAIN}(\mathrm{Basis}, \theta, \tau c, \mathrm{qMaxO2}) = ▮ \qquad (9.6.1-1)$$

Note that for some controllers, the proportional band is used for the gain. In those cases, the following would be used for the PI controller.

$$\frac{100}{\mathrm{GAIN}(\mathrm{Basis}, \theta, \tau c, \mathrm{qMaxO2})} = ▮ \qquad (9.6.1-2)$$

The average integral time is calculated to be as follows:

$$\mathrm{avg\_ti}(\theta, \tau c, \mathrm{qMaxO2}, \mathrm{Inc}) = ▮ \qquad (9.6.1-3)$$

The controller gain and time integral is calculated with different oxygen flow rates as follows:

"O2 Flow"	"GAIN"	DSd GAIN"	"PB"	"1/kla"	"DS_d TI"	"4 Tc"

Tuning(θ, τc, qMaxO2, Inc) = ▮

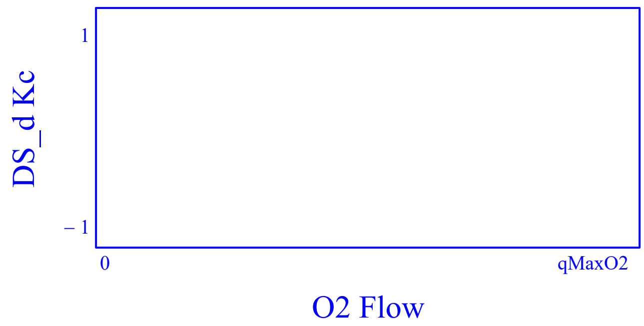


Figure 9.6.1-2: DO control loop variable gain controller.

Alternatively, instead of modifying the gain and integral, one could adjust the output to reflect the changing system gain with respect to the gas flow rate.

CV CV_adj Ti

$\mathrm{Polygon}(0.4\cdot qMaxO2, air, \theta, \tau c, qMaxO2, Inc) = ▮$

$\mathrm{Kc}(DO\_CONTROL, Input, air, 0.4\cdot qMaxO2, \theta, \tau c, qMaxO2)_1 = ▮$

Figure 9.6.1-3: Oxygen flow meter adjustment to account of variable gain.

9.6.2 pH PI Loop Tuning Parameters

The pH control loop is normally under a gain only controller with dead-band. This leads to an on-off controller with the addition of carbon dioxide and base. Depending on what the process controller is trying to achieve for dissolved oxygen control, this can result in the two loops fighting each other. Below are contour plots for the gain and integrals for the pCO2 addition and base addition if a PI control loop were to be implemented.

pH Direct Synthesis Calculations

The following equations allow for solving the system gain and time constant for the pH control loop,

$$\text{Resulting\_pH}(\text{procVariables}, \text{CO2\_gasBal}) := \begin{array}{|l} V_{Liq} \leftarrow \text{procVariables}_0 \cdot L \\ T_{Liq} \leftarrow \text{procVariables}_1 \cdot K + T_{0C} \\ p_{HS} \leftarrow \text{procVariables}_2 \cdot psi + p_{atm} \\ Nrpm \leftarrow \text{procVariables}_3 \cdot \frac{1}{min} \\ \text{qair\_ovly} \leftarrow \text{procVariables}_4 \cdot SLPM \\ \text{qO2\_ovly} \leftarrow \text{procVariables}_5 \cdot SLPM \\ \text{qCO2\_ovly} \leftarrow \text{procVariables}_6 \cdot SLPM \\ \text{qN2\_ovly} \leftarrow \text{procVariables}_7 \cdot SLPM \\ \text{qair\_1} \leftarrow \text{procVariables}_8 \cdot SLPM \\ \text{qO2\_1} \leftarrow \text{procVariables}_9 \cdot SLPM \\ \text{qCO2\_1} \leftarrow \text{procVariables}_{10} \cdot SLPM \\ \text{qN2\_1} \leftarrow \text{procVariables}_{11} \cdot SLPM \\ \text{qair\_2} \leftarrow \text{procVariables}_{12} \cdot SLPM \\ \text{qO2\_2} \leftarrow \text{procVariables}_{13} \cdot SLPM \\ \text{qCO2\_2} \leftarrow \text{procVariables}_{14} \cdot SLPM \\ \text{qN2\_2} \leftarrow \text{procVariables}_{15} \cdot SLPM \\ OUR \leftarrow \text{procVariables}_{16} \cdot \frac{mmol}{L \cdot hr} \\ lactate \leftarrow \text{procVariables}_{17} \cdot \frac{mmol}{L} \\ ammonia \leftarrow \text{procVariables}_{18} \cdot \frac{mmol}{L} \\ glucose \leftarrow \text{procVariables}_{19} \cdot \frac{mmol}{L} \\ V_{Base} \leftarrow \text{procVariables}_{20} \cdot L \\ V_{FEED} \leftarrow \text{procVariables}_{21} \cdot L \\ DO \leftarrow \frac{\text{procVariables}_{22}}{100} \\ pH \leftarrow \text{procVariables}_{23} \\ pH_{Upper} \leftarrow \text{procVariables}_{25} \\ pH_{Lower} \leftarrow \text{procVariables}_{24} \\ \text{CO2\_gasBal} \leftarrow \dfrac{\text{CO2\_gasBal}}{\exp\left[0.04375 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)\right]} \\ pH_{Upper} \leftarrow \dfrac{pH_{Upper} - 0.033 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)}{1 - 0.0065 \cdot \left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)} \end{array} \qquad (9.6.2-1)$$

$$pH_{Lower} \leftarrow \frac{pH_{Lower} - 0.033\cdot\left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)}{1 - 0.0065\cdot\left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)}$$

$$pH_{est} \leftarrow \frac{pH_{Lower} + pH_{Upper}}{2}$$

for $i \in 1..10$

$$carbonate\_i \leftarrow \left(CO3_i + B_{base}\cdot\frac{V_{Base}}{V_{Liq}}\cdot Base\_Conc\cdot\frac{L}{mol} \quad HCO3_i \quad H2CO3_i\right)$$

$$G1 \leftarrow \frac{\Phi\_carbonate(pH_{est}, carbonate\_i)}{\Phi\_carbonate(pH_{est}, carbonate\_basis)}$$

$$G2 \leftarrow \frac{\Phi\_total(pH_{est}, carbonate\_basis, PO4\_basis, HEPES\_basis, Salts_{A\_basis}, Salts_{B\_basis})}{\Phi\_total(pH_{est}, carbonate\_i, PO4\_i, HEPES, Salts_{A\_basis}, Salts_{B\_basis})}$$

$$\left[\frac{10^{-2\cdot pH_{est}}}{\left[\begin{array}{l}2\times 10^{K_1+K_2} \ldots \\ +2\times 10^{(K_1+K_2+K_h)} \ldots \\ +10^{(K_1-pH_{est})} \ldots \\ +10^{(K_1+K_h-pH_{est})}\end{array}\right]}\cdot\left[\begin{array}{l}HCO3_i\cdot\frac{mol}{L} \ldots \\ +2\left(\begin{array}{l}CO3_i \ldots \\ +B_{base}\cdot\frac{V_{Base}}{V_{Liq}}\cdot Base\_Conc\cdot\frac{L}{mol}\end{array}\right)\cdot\frac{mol}{L} \ldots \\ +G1\cdot G2\cdot\left[\begin{array}{l}10^{-pH_{est}}\cdot\frac{mole}{L} \ldots \\ +B_{feed}\cdot\frac{V_{FEED}}{V_0}\cdot\frac{mol}{L} \ldots \\ +OH_i\cdot\frac{mol}{L} \ldots \\ +-\frac{10^{-14}}{10^{-pH_{est}}}\cdot\frac{mole}{L} \ldots \\ +-\Delta H\_PO4(pH_{est}, PO4\_i) \ldots \\ +-\frac{10^{K_{HEPES}}\cdot\left(HEPES\cdot\frac{mol}{L}\right)}{10^{K_{HEPES}} + 10^{-pH_{est}}} \ldots \\ +-\frac{10^{K_a}\cdot\left(Salts_{A\_basis}\cdot\frac{mol}{L}\right)}{10^{K_a} + 10^{-pH_{est}}} \ldots \\ +\frac{10^{-pH_{est}}\cdot\left(Salts_{B\_basis}\cdot\frac{mol}{L}\right)}{10^{K_b} + 10^{-pH_{est}}} \ldots \\ +-A_{eq\_basis}\cdot\frac{mol}{L} \ldots \\ +\frac{-10^{K_{Gluc}}}{10^{K_{Gluc}} + 10^{-pH_{est}}}\cdot glucose \ldots \\ +\frac{-B_{lact}\cdot 10^{K_{lact}}}{10^{K_{lact}} + 10^{-pH_{est}}}\cdot lactate \ldots \\ +\frac{B_{amm}\cdot 10^{-pH_{est}}}{10^{K_{amm}} + 10^{-pH_{est}}}\cdot ammonia \ldots \\ \left(\quad K_{1\ldots} + K_{2\ldots} \quad \right)\end{array}\right.\end{array}\right.\right.$$

$$CO2\_med \leftarrow \frac{\left[\cdots + -\left(\frac{2\cdot 10^{K_{1Ala}+K_{2Ala}} \ldots + 10^{K_{1Ala}-pH_{est}}}{10^{K_{1Ala}+K_{2Ala}} \ldots + 10^{K_{1Ala}-pH_{est}} \ldots + 10^{-2pH_{est}}} \right)\cdot B_{ala}\cdot lactate \right]}{H_{CO2\_cp}(str2num(saline), T_{Liq})\cdot mmHg}$$

$$pH_{est} \leftarrow pH_{est} + 0.1\cdot\left(\frac{CO2\_med - CO2\_gasBal}{CO2\_med}\right)$$

$$pH_{est} \leftarrow 5 \ \text{ if } \ pH_{est} < 5$$

$$pH_{est} \leftarrow 8 \ \text{ if } \ pH_{est} > 8$$

$$pH_{est} \leftarrow pH_{est} + \left[-0.0147 + 0.0065\cdot(7.400 - pH_{est})\right]\cdot\left(\frac{T_{Liq} - T_{0C}}{K} - 37\right)$$

$$return \leftarrow pH_{est}$$

(9.6.2 – 2)

$KpH(Control, Input, qAir, qO2, TimeConstants, qMax) :=$

$$V_{Liq} \leftarrow Input_0$$
$$T_{Liq} \leftarrow Input_1$$
$$p_{HS} \leftarrow Input_3$$
$$Nrpm \leftarrow Input_4$$
$$qair\_ovly \leftarrow Input_7$$
$$qO2\_ovly \leftarrow Input_8$$
$$qCO2\_ovly \leftarrow Input_9$$
$$qN2\_ovly \leftarrow Input_{10}$$
$$qair\_1 \leftarrow Input_{11}$$
$$qO2\_1 \leftarrow Input_{12}$$
$$qCO2\_1 \leftarrow Input_{13}$$
$$qN2\_1 \leftarrow Input_{14}$$
$$qair\_2 \leftarrow Input_{15}$$
$$qO2\_2 \leftarrow Input_{16}$$
$$qCO2\_2 \leftarrow Input_{17}$$
$$qN2\_2 \leftarrow Input_{18}$$
$$OUR \leftarrow OUR\_(Control, Input, qAir, qO2)$$
$$lactate \leftarrow Input_{23}$$
$$ammonia \leftarrow Input_{24}$$
$$glucose \leftarrow Input_{25}$$
$$\theta CO2 \leftarrow TimeConstants_0$$
$$\theta Base \leftarrow TimeConstants_1$$

$\tau c\_pH\_CO2 \leftarrow TimeConstants_2$

$\tau c\_pH\_Base \leftarrow TimeConstants_3$

$\tau Mix \leftarrow TimeConstants_4$

$$endcolumn \leftarrow \left| \begin{array}{l} i \leftarrow 2 \\ eTime \leftarrow 0 \ \text{ if } \ strlen(ProcessTime) = 0 \\ eTime \leftarrow str2num(ProcessTime) \ \text{ otherwise} \\ \text{while } \ ProcessData_{0,i} < eTime \wedge i < cols(ProcessData) - 1 \\ \quad i \leftarrow i + 1 \\ i \leftarrow i - 1 \ \text{ if } \ i > cols(ProcessData) - 1 \\ i \leftarrow i - 1 \ \text{ if } \ ProcessData_{0,i} > eTime \end{array} \right.$$

$$V_{Base} \leftarrow \left| \begin{array}{l} sum \leftarrow 0 \\ \text{for } \ i \in 2..endcolumn \\ \quad sum \leftarrow sum + Input_{22} \\ sum \end{array} \right.$$

$$V_{FEED} \leftarrow \left| \begin{array}{l} sum \leftarrow 0 \\ \text{for } \ i \in 2..endcolumn \\ \quad sum \leftarrow sum + Input_{19} + Input_{20} \\ sum \cdot Input_0 \end{array} \right.$$

$DO \leftarrow Input_2$

$$pH \leftarrow \frac{Input_5 + Input_6}{2}$$

$pH_{Upper} \leftarrow Input_6$

$pH_{Lower} \leftarrow Input_5$

$adj \leftarrow str2num(AIR\_CO2\_ADJ)$

$$pHInputs \leftarrow \begin{pmatrix} V_{Liq} \\ T_{Liq} \\ p_{HS} \\ Nrpm \\ qair\_ovly \\ qO2\_ovly \\ qCO2\_ovly \\ qN2\_ovly \\ qair\_1 \\ qO2\_1 \\ qCO2\_1 \\ qN2\_1 \\ qair\_2 \\ qO2\_2 \\ qCO2\_2 \\ qN2\_2 \\ OUR \\ lactate \end{pmatrix}$$

$$\begin{pmatrix} \text{lactate} \\ \text{ammonia} \\ \text{glucose} \\ V_{Base} \\ V_{FEED} \\ DO \\ pH \\ pH_{Upper} \\ pH_{Lower} \\ adj \end{pmatrix}$$

$CO2\_1 \leftarrow CO2\_(Control, Input, qAir, qO2)$

$CO2targetU \leftarrow CO2\_1$

$CO2targetL \leftarrow CO2\_1$

$comparison \leftarrow \dfrac{Resulting\_pH(pHInputs, CO2targetU)}{pH_{Upper}}$

while $comparison > 1.00001 \lor comparison < 0.99999$

  $CO2targetU \leftarrow comparison \cdot CO2targetU$

  $comparison \leftarrow \dfrac{Resulting\_pH(pHInputs, CO2targetU)}{pH_{Upper}}$

$temp9 \leftarrow Input_9$

$temp13 \leftarrow Input_{13}$

$temp17 \leftarrow Input_{17}$

$pHtemp6 \leftarrow pHInputs_6$

$pHtemp10 \leftarrow pHInputs_{10}$

$pHtemp14 \leftarrow pHInputs_{14}$

$CO2\_1 \leftarrow CO2\_(Control, Input, qAir, qO2)$

while $\left(\dfrac{CO2targetU}{CO2\_1}\right) > 1.0001$

  $CO2\_1 \leftarrow CO2\_(Control, Input, qAir, qO2)$

  if $CO2\_CONTROL = 0$

    $Input_9 \leftarrow Re\left[Input_9 \cdot \dfrac{CO2targetU}{CO2\_1} \ldots + \left(\dfrac{-1}{10^{10}}\right)^{0.5}\right]$

    $Input_9 \leftarrow 0.0001 \text{ if } Input_9 = 0$

  if $CO2\_CONTROL = 1$

    $Input_{13} \leftarrow Re\left[Input_{13} \cdot \dfrac{CO2targetU}{CO2\_1} \ldots + \left(\dfrac{-1}{10^{10}}\right)^{0.5}\right]$

    $Input_{13} \leftarrow 0.0001 \text{ if } Input_{13} = 0$

  if $CO2\_CONTROL = 2$

    $Input_{17} \leftarrow Re\left[Input_{17} \cdot \dfrac{CO2targetU}{CO2\_1} \ldots\right]$

$$+\left(\frac{-1}{10^{10}}\right)^{0.5}$$

$$\text{Input}_{17} \leftarrow 0.0001 \quad \text{if } \text{Input}_{17} = 0$$

$$\text{if } \frac{\text{CO2targetU}}{\text{CO2\_1}} < 0.9999$$

$$\text{pHtemp20} \leftarrow \text{pHInputs}_{20}$$

$$\text{while } \frac{\text{pH}_{\text{Upper}}}{\text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_1})} > 1.00001$$

$$\text{pHInputs}_{20} \leftarrow \frac{\text{pH}_{\text{Upper}}}{\text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_1})} \cdot \text{pHInputs}_{20}$$

$$\text{pHInputs}_{20} \leftarrow 0.0001 \cdot V_{\text{Liq}} \quad \text{if } \text{pHInputs}_{20} = 0$$

$$\text{if } \text{CO2\_CONTROL} = 0$$

$$\text{Input}_9 \leftarrow \text{Input}_9 + 0.005\text{qMax}$$

$$\text{pHInputs}_6 \leftarrow \text{pHInputs}_6 + 0.005\text{qMax}$$

$$\text{if } \text{CO2\_CONTROL} = 1$$

$$\text{Input}_{13} \leftarrow \text{Input}_{13} + 0.005\text{qMax}$$

$$\text{pHInputs}_{10} \leftarrow \text{pHInputs}_{10} + 0.005 \cdot \text{qMax}$$

$$\text{if } \text{CO2\_CONTROL} = 2$$

$$\text{Input}_{17} \leftarrow \text{Input}_{17} + 0.005\text{qMax}$$

$$\text{pHInputs}_{14} \leftarrow \text{pHInputs}_{14} + 0.005\text{qMax}$$

$$\text{CO2\_2} \leftarrow \text{CO2\_}(\text{Control}, \text{Input}, \text{qAir}, \text{qO2})$$

$$\text{pH\_2} \leftarrow \text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_2})$$

$$\text{Input}_9 \leftarrow \text{temp9}$$

$$\text{Input}_{13} \leftarrow \text{temp13}$$

$$\text{Input}_{17} \leftarrow \text{temp17}$$

$$\text{pHInputs}_6 \leftarrow \text{pHtemp6}$$

$$\text{pHInputs}_{10} \leftarrow \text{pHtemp10}$$

$$\text{pHInputs}_{14} \leftarrow \text{pHtemp14}$$

$$\text{pHInputs}_{20} \leftarrow \text{pHtemp20}$$

$$\text{tempBase} \leftarrow \text{pHInputs}_{20}$$

$$\text{CO2\_1} \leftarrow \text{CO2\_}(\text{Control}, \text{Input}, \text{qAir}, \text{qO2})$$

$$\text{while } \frac{\text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_1})}{\text{pH}_{\text{Lower}}} > 1.00001$$

$$\text{pHInputs}_{17} \leftarrow \frac{\text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_1})}{\text{pH}_{\text{Lower}}} \cdot \text{pHInputs}_{17}$$

$$\text{pHInputs}_{17} \leftarrow 0.0001 \quad \text{if } \text{pHInputs}_{17} = 0$$

$$\text{if } \frac{\text{pH}_{\text{Lower}}}{\text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_1})} > 1.00001$$

$$\text{comparison} \leftarrow \frac{\text{pH}_{\text{Lower}}}{\text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_1})}$$

$$\text{while comparison} > 1.00001$$

$$\text{pHInputs}_{20} \leftarrow \text{comparison}\cdot\text{pHInputs}_{20}$$

$$\text{pHInputs}_{20} \leftarrow 0.0001\cdot V_{Liq} \quad \text{if } \text{pHInputs}_{20} = 0$$

$$\text{comparison} \leftarrow \frac{pH_{Lower}}{\text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_1})}$$

$$\text{pHInputs}_{20} \leftarrow \text{pHInputs}_{20} + 0.0001\cdot V_{Liq}$$

$$\text{pH\_3} \leftarrow \text{Resulting\_pH}(\text{pHInputs}, \text{CO2\_1})$$

$$\text{Tsys} \leftarrow \frac{\text{ts}(\text{Control}, \text{Input}, \text{qAir}, \text{qO2})}{0.893}$$

$$\text{result}_0 \leftarrow \frac{-0.005\text{qMax}}{\text{pH\_2} - pH_{Upper}}\cdot\frac{\text{Tsys}}{(\theta\text{CO2} + \tau\text{c\_pH\_CO2})}$$

$$\text{result}_1 \leftarrow \frac{-0.005\text{qMax}}{\text{pH\_2} - pH_{Upper}}\cdot\frac{\left[(\text{Tsys})^2 + \text{Tsys}\cdot\theta\text{CO2} + -(\tau\text{c\_pH\_CO2} - \text{Tsys})^2\right]}{(\theta\text{CO2} + \tau\text{c\_pH\_CO2})^2}$$

$$\text{result}_2 \leftarrow \frac{0.0001\cdot V_{Liq}}{\text{pH\_3} - pH_{Lower}}\cdot\frac{\theta\text{Base}}{(\tau\text{Mix} + \tau\text{c\_pH\_Base})}$$

$$\text{result}_3 \leftarrow \frac{0.0001\cdot V_{Liq}}{\text{pH\_3} - pH_{Lower}}\cdot\frac{\left[\tau\text{Mix}^2 + \tau\text{Mix}\cdot\theta\text{Base} - (\tau\text{c\_pH\_Base} - \tau\text{Mix})^2\right]}{(\theta\text{Base} + \tau\text{c\_pH\_Base})^2}$$

$$\text{result}$$

The pH Gain is a function of the oxygen flow rate

```
pH_Tune(Input, Inc, TimeConstants, qM, qMCO2) :=                                  (9.6.2 − 3)
  θCO2 ← TimeConstants_0
  θBase ← TimeConstants_1
  τc_pH_CO2 ← TimeConstants_2
  τc_pH_Base ← TimeConstants_3
  τMix ← TimeConstants_4
  for i ∈ 0.. Inc
      result_{i,0} ← (i/Inc)·qM
      if DO_CONTROL = 1
          qO2_1 ← result_{i,0}
          qair_1 ← Input_11
          qair_2 ← Input_15
          air_flow ← qair_1
          if AIR_CO2_CONTROL = 1
              qair_1 ← str2num(AIR_CO2_GAIN) qO2_1
              qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
              qair_1 ← Input_11  if qair_1 < Input_11
              air_flow ← qair_1
          if AIR_CO2_CONTROL = 2
              qair_2 ← str2num(AIR_CO2_GAIN) qO2_1
              qair_2 ← str2num(AIR_CO2_MAX)  if qair_2 > str2num(AIR_CO2_MAX)
              qair_2 ← Input_15  if qair_2 < Input_15
              air_flow ← qair_2
      if DO_CONTROL = 2
          qO2_2 ← result_{i,0}
          qair_1 ← Input_11
          qair_2 ← Input_15
          air_flow ← qair_2
          if AIR_CO2_CONTROL = 1
              qair_1 ← str2num(AIR_CO2_GAIN) qO2_2
              qair_1 ← str2num(AIR_CO2_MAX)  if qair_1 > str2num(AIR_CO2_MAX)
              qair_1 ← Input_11  if qair_1 < Input_11
              airBasis ← Input_11  if airBasis < Input_11
              air_flow ← qair_1
          if AIR_CO2_CONTROL = 2
              qair_2 ← str2num(AIR_CO2_GAIN) qO2_2
              qair_2 ← str2num(AIR_CO2_MAX)  if qair_2 > str2num(AIR_CO2_MAX)
              qair_2 ← Input_15  if qair_2 < Input_15
              air_flow ← qair_2
```

$$\mathrm{CTRL} \leftarrow \mathrm{DO\_CONTROL}$$

$$\mathrm{Kc} \leftarrow \mathrm{KpH}\left(\mathrm{CTRL}, \mathrm{Input}, \mathrm{air\_flow}, \frac{i \cdot \mathrm{qM}}{\mathrm{Inc}}, \mathrm{TimeConstants}, \mathrm{qM}\right)$$

$$\mathrm{Tsys} \leftarrow \frac{\mathrm{ts}\left(\mathrm{CTRL}, \mathrm{Input}, \mathrm{air\_flow}, \frac{i \cdot \mathrm{qM}}{\mathrm{Inc}}\right)}{0.893}$$

$$\mathrm{result}_{i,1} \leftarrow \frac{\mathrm{Kc}_0}{\left(\frac{\mathrm{qMCO2}}{100}\right)}$$

$$\mathrm{result}_{i,2} \leftarrow \frac{\mathrm{Kc}_1}{\left(\frac{\mathrm{qMCO2}}{100}\right)}$$

$$\mathrm{result}_{i,3} \leftarrow \frac{100}{\mathrm{result}_{i,2}}$$

$$\mathrm{result}_{i,4} \leftarrow \mathrm{Tsys}$$

$$\mathrm{result}_{i,5} \leftarrow \frac{(\mathrm{Tsys})^2 \ldots + \theta\mathrm{CO2} \cdot \mathrm{Tsys} \ldots + -(\tau\mathrm{c\_pH\_CO2} - \mathrm{Tsys})^2}{\mathrm{Tsys} + \theta\mathrm{CO2}}$$

$$\mathrm{result}_{i,6} \leftarrow 4 \cdot (\tau\mathrm{c\_pH\_CO2})$$

$$\mathrm{result}_{i,7} \leftarrow \frac{\mathrm{Kc}_2}{\left(\frac{\mathrm{MaxPump}}{100}\right)}$$

$$\mathrm{result}_{i,8} \leftarrow \frac{\mathrm{Kc}_3}{\left(\frac{\mathrm{MaxPump}}{100}\right)}$$

$$\mathrm{result}_{i,9} \leftarrow \frac{100}{\mathrm{result}_{i,2}}$$

$$\mathrm{result}_{i,10} \leftarrow \frac{\tau\mathrm{Mix}}{\theta\mathrm{Base} + \tau\mathrm{c\_pH\_Base}}$$

$$\mathrm{result}_{i,11} \leftarrow \frac{(\tau\mathrm{Mix})^2 + \tau\mathrm{Mix} \cdot \theta\mathrm{Base} - (\tau\mathrm{c\_pH\_Base} - \tau\mathrm{Mix})^2}{\theta\mathrm{Base} + \tau\mathrm{Mix}}$$

$$\mathrm{result}_{i,12} \leftarrow 4 \cdot (\tau\mathrm{c\_pH\_Base})$$

$$\mathrm{result}$$

(9.6.2 – 4)

$$\mathrm{pH\_CO2GAINmesh}(\mathrm{qO2}, \mathrm{qAir}) := \left| \begin{array}{l} \mathrm{result} \leftarrow 0 \ \text{ if } \ \mathrm{Part6} = 0 \\ \mathrm{result} \leftarrow \mathrm{KpH}(\mathrm{DO\_CONTROL}, \mathrm{Input}, \mathrm{qAir}, \mathrm{qO2}, \mathrm{TimeConstants}, \mathrm{qMaxCO2})_1 \cdot \frac{100}{\mathrm{qMaxCO2}} \ \text{ otherwise} \\ \mathrm{result} \end{array} \right.$$

$$\text{pH\_CO2INTMesh}(qO2, qAir) := \left| \begin{array}{l} \text{result} \leftarrow 0 \ \text{ if } \ \text{Part6} = 0 \\ \text{otherwise} \\ \quad \left| \begin{array}{l} \theta CO2 \leftarrow \text{TimeConstants}_0 \\ \tau c\_pH\_CO2 \leftarrow \text{TimeConstants}_2 \\ \text{Tsys} \leftarrow \frac{1}{0.893} \cdot \text{ts}(\text{DO\_CONTROL}, \text{Input}, qAir, qO2) \\ \text{result} \leftarrow \frac{(\text{Tsys})^2 \ldots + \theta CO2 \cdot \text{Tsys} \ldots + -(\tau c\_pH\_CO2 - \text{Tsys})^2}{(\theta CO2 + \text{Tsys})} \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.6.2 – 5}$$

$$\text{pH\_BASEGAINMesh}(qO2, qAir) := \left| \begin{array}{l} \text{result} \leftarrow 0 \ \text{ if } \ \text{Part6} = 0 \\ \text{result} \leftarrow \text{KpH}(\text{DO\_CONTROL}, \text{Input}, qAir, qO2, \text{TimeConstants}, qMaxCO2)_3 \cdot \frac{100}{\text{MaxPump}} \ \text{ otherwise} \\ \text{result} \end{array} \right. \tag{9.6.2 – 6}$$

$$\text{CO2\_GAIN} := \text{CreateMesh}(\text{pH\_CO2GAINmesh}, 0.1, \text{O2\_max}, 0.1, \text{Air\_max}, 100, 50) \tag{9.6.2 – 7}$$

$$\text{CO2\_INT} := \text{CreateMesh}(\text{pH\_CO2INTMesh}, 0.1, \text{O2\_max}, 0.1, \text{Air\_max}, 100, 50) \tag{9.6.2 – 8}$$

$$\text{BASE\_GAIN} := \text{CreateMesh}(\text{pH\_BASEGAINMesh}, \text{O2\_min}, \text{O2\_max}, \text{Air\_min}, \text{Air\_max}, 100, 50) \tag{9.6.2 – 9}$$

$$\text{tCO2}(\text{Input}, \text{Inc}, \text{TimeConstants}, qMax, qMaxCO2) := \left| \begin{array}{l} \text{result} \leftarrow 0 \ \text{ if } \ \text{Part6} = 0 \\ \text{otherwise} \\ \quad \left| \begin{array}{l} \text{result} \leftarrow 0 \ \text{ if } \ \text{Part6} = 0 \\ M \leftarrow \text{pH\_Tune}(\text{Input}, \text{Inc}, \text{TimeConstants}, qMax, qMaxCO2) \\ \text{result} \leftarrow \text{submatrix}(M, 0, \text{rows}(M) - 1, 0, 6) \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.6.2 – 10}$$

$$\text{tBase}(\text{Input}, \text{Inc}, \text{TimeConstants}, qMax, qMaxCO2) := \left| \begin{array}{l} \text{result} \leftarrow 0 \ \text{ if } \ \text{Part6} = 0 \\ \text{otherwise} \\ \quad \left| \begin{array}{l} M \leftarrow \text{pH\_Tune}(\text{Input}, \text{Inc}, \text{TimeConstants}, qMax, qMaxCO2) \\ M1 \leftarrow M^{\langle 0 \rangle} \\ M2 \leftarrow \text{submatrix}(M, 0, \text{rows}(M) - 1, 7, 12) \\ \text{result} \leftarrow \text{augment}(M1, M2) \end{array} \right. \\ \text{result} \end{array} \right. \tag{9.6.2 – 11}$$

To help keep the equation and table on one page width, shortening the name for the maximum oxygen and carbon dioxide flow rates.

$$MO2 := qMaxO2 \tag{9.6.2 – 12}$$

$$MCO2 := qMaxCO2 \tag{9.6.2 – 13}$$

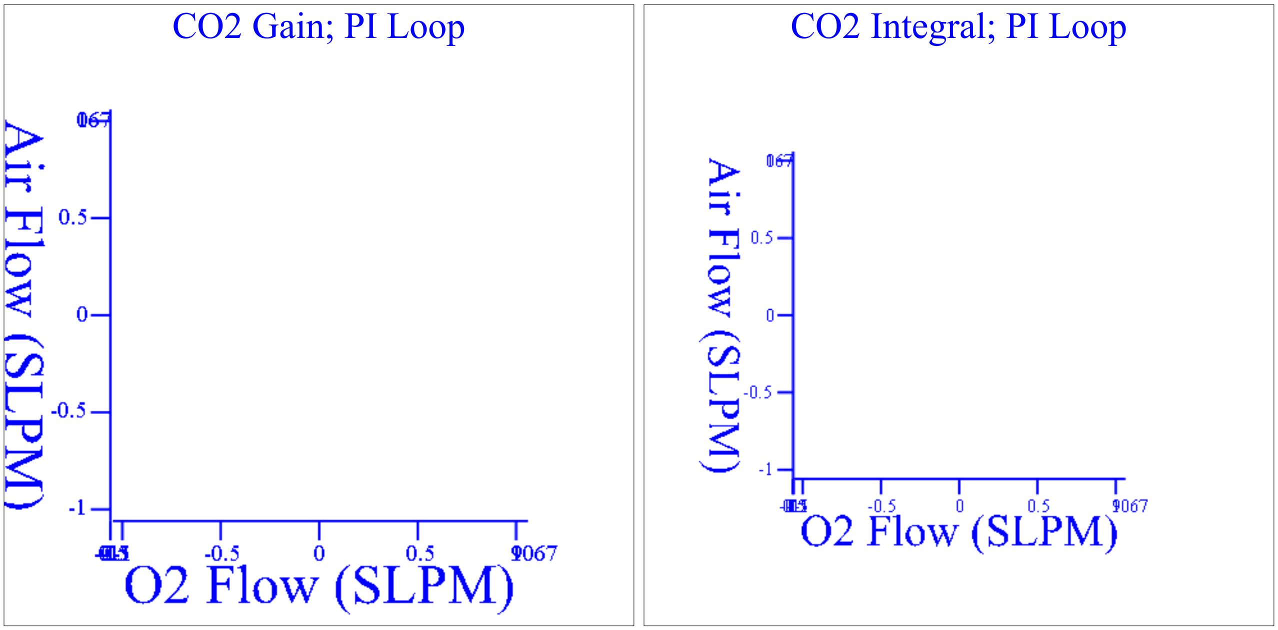


Figure 9.6.2-1: Contour plots for the carbon dioxide gain and integral for the vessel with the specified air, oxygen, and carbon dioxide flow rates through the overlay sparger 1, and sparger 2 (time specified for the bioreactor in table 9.6-14, settings for the bioreactor in table 9.6-11).

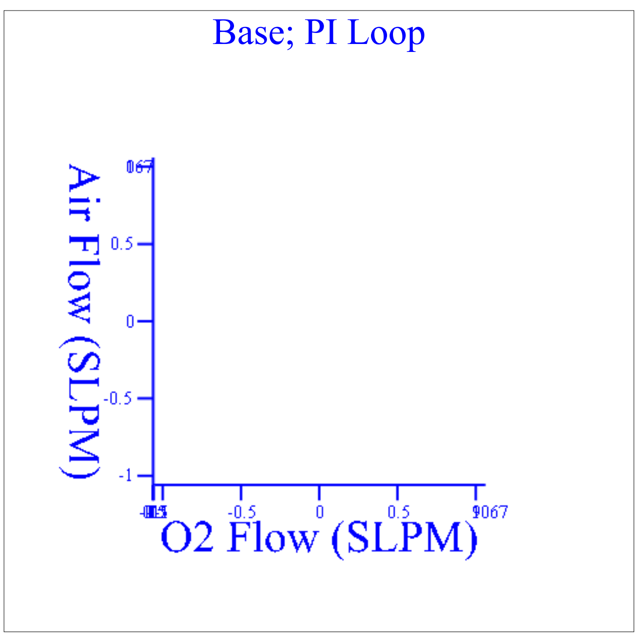


Figure 9.6.2-2: Contour plots for the base gain and assumed time constant for the vessel with the specified air, oxygen, and carbon dioxide flow rates through the overlay sparger 1, and sparger 2 (time specified for the run in table 9.6-14, settings for the bioreactor in table 9.6-11).

Add carbon dioxide for pH control

CO2_tune := tCO2(Input, Inc, TimeConstants, MO2, MCO2)

"O2 Flow"	"GAIN"	"DSd GAIN"	"PB"	"1/kla"	"DS_d TI"	"4 Tc"

CO2_tune = ▮

Table 9.6.2-1: Carbon dioxide add control loop variable gain controller.

Add base for pH control

Base_tune := tBase(Input, Inc, TimeConstants, MO2, MCO2)

"O2 Flow"	"GAIN"	"DSd GAIN"	"PB"	"1/kla"	"DS_d TI"	"4 Tc"

Base_tune = ▮

Table 9.6.2-2: Base add control loop variable gain controller.

Revision History

10. Revision History

Version ED001

New Document

Version ED002

Added glucose to the medium chemistry equilibrium equations to capture that the glucose is a weak acid. This impacts the following equations and tables: Equations 3.8.5-17, 3.8.5-18, 3.8.5-19, 3.8.6-1, 3.8.6-2, 9.2.1-1, 9.2.1-2, 9.2.1-6, 9.2.1-13, 9.2.1-14, 9.2.1-21, 9.2.2-1, 9.2.2-8, 9.2.3-1, 9.2.3-8, 9.2.3-13, 9.2.4-1, 9.2.4-4, 9.2.4-5, 9.4-28, 9.4-29, 9.4-34, 9.4-37, 9.4-38, 9.4-39, 9.4-40, 9.5-5, 9.5-6, 9.5-11, 9.5-33, 9.6-25, 9.6-26. Tables 9.2.1-2, 9.2.1-3, 9.2.2-1, 9.2.3-1, 9.4-16, 9.4-17, 9.4-18, 9.5-3, 9.5-4, 9.5-5, 9.5-7, 9.5-8, 9.6-11

Added equations for the direct synthesis with disturbance rejection calculation for the controller gain and controller integral calculations. This impacts the following equations, figures, and tables: Equations 5.3-4, 5.3-5, 9.6-9, 9.6-10, 9.6-11, 9.6-22, 9.6.2-2, 9.6.2-3. Figure 9.6.1-2. Added reference 21.

Added the ability to perform kla fitting without surface aeration for sparger 1, sparger 2, or both sparger 1 and sparge 2. (Table 9.1-10, Equations 9.1-54, 9.1-55, 9.1-66, 9.1-67, 9.1-68, and 9.1-69)

Added a default for the exp_num parameter to be set to -1 to avoid running the solver calculations until experiments are selected (Equations 9.1-54, 9.1-58, and 9.1-64).

Moved the definition of the klaInput matrix to be outside of the solver block (Equation 9.1-59).

Allow for the solver tolerance and convergence tolerance of the nonlinear solver to be based on the user selection of Table 9.1-12 (Equation 9.1-60 and 9.1-61).

Simplified how the constraints are defined for klaInput matrix (Equation 9.1-62 and 9.1-63).

Pass only one column into the Minerr calculation to improve the speed of the nonlinear solver (Equation 9.1-66).

Modified the vessel characterization section to allow for calibration values and delay time to be entered for each trial and experimental run. This impacts Table 9.1-5, Equations 9.1-54, 9.1-61.1 (trial loop to step through the columns of Exp_Data, vector InVtr being passed corrected in response to table 9.1-5), 9.1-61.3 (calAdjust), 9.1-53, 9.1-64 9.1-82 (change inputs for the ErrorSurf function).

Removed temperature terms from the kla surface parameters, removed third order terms, changed the units for agitation to inverse seconds to be consistent with the units table, and changed the order of the parameters to be consistent with the order presented in other tools (Tables 9.1-4, 9.1-17, 9.3-2, 9.4-2, 9.5-2, 9.6-2, Equations 9.1-90, 9.3-4, 9.4-12, 9.4-13, 9.4-25, 9.4-27, 9.5-17, 9.5-29, 9.6-21, 9.6-22, 9.6-23, and 9.6-24). Reduced the size of the Surface_solve matrix to reflect the removed terms (Equations 9.1-92 and 9.1-95).

Tightened the tolerance for convergence of the nonlinear solver (added Equations 9.1-92.1 and 9.1-92.2)

Clerical correction to figure 3.4.2-1 caption title.

Clerical corrections to which sections were referenced in the appendix. Section 9.1 Load Experimental Data: Corrected reference from section 7.1 to section 9.1. Refer to table 9.1-15 instead of 8.1-15 in caption Figure 9.1-1. Refer to figure 9.1-2 instead of figure 8.1-2 in caption Table 9.1-15.

Clerical correction to move the pH loop tuning calculations into section 9.6.2.

Added additional plots for data visualization. This impacts figure 9.1-1. Equation 9.1-78

Removed the T0 term fro the surface kla calculations because the temperature conversion is already captured in the program. This can impact the calculation for the surface kla. This impacts equations 9.1-90, 9.4-12, 9.6-21, 9.6-22, 9.6-23, 9.6-24

Included the second kla in the sum of square error calculation to simultaneously solve for two spargers. Equation 9.1-96.

Added the ability to have a minimum carbon dioxide flow rate relative to the total flow rate of the gas stream adding carbon dioxide. This impacts the following equations and tables: Equations 9.4-12, 9.4-40, 9.6-21, 9.6-22, 9.6-23, 9.6-24. Tables 9.4-9, 9.6-8.

Equation 9.4-26: Change check of CASCADE_step for "Proportional Control" to be a check for "Proportional" to be consistent with the drop down menu options. Within the Proportional CASCADE block, adjust the oxygen flow rate instead of the air flow rate (for DO_CONTROL 0, 1, and 2). Removed the step down in the CASCADE mode if less than minimum air flow rate.

Equation 9.5-14: Increased number in the outer for loop from 22 to 25 to account for changes in the input matrix.

Table 9.6-11: Added lactate, ammonia, and glucose concentration. Added a table for the process limits.

Added expressions that calculate air flow in equations 9.6-36 and 9.6-37.

Added atmospheric pressure as a variable that can be adjusted for calculating total pressure in tables 9.3-2, 9.4-2, 9.5-2, and 9.6-2 and equations 9.1-2.1, 9.1-14, 9.3-4, 9.4-12, 9.4-13, 9.4-25, 9.4-26, 9.4-27, 9.4-38, 9.4-39, 9.4-40, 9.4-41, 9.5-29, 9.6-21, 9.6-24, 9.6.2-1

Added experimental trial weighting to table 9.1-16 and equations 9.1-90, 9.1-96

Added expression to constrain numerical calculations for carbon dioxide to be positive in equations 9.4-40 and 9.4-41.

Added tables 9.6-1.1, 9.6-1.2, 9.6-1.3, and 9.6-1.4 so the medium properties can be entered into section 9.6.

Added table 9.6-11.1 to allow for the pH control time constants to be entered. Added the base loop time constant.

Replaced equation 9.6-9 with an expression to convert table 9.6-11.1 to a vector.

Removed equation 9.6-10 and 9.6-11 because these values are captured table 9.6-11.1.

Combined the control time constants inputs into an input vector TimeConstants for equations 9.6.2-2, 9.6.2-3, 9.6.2-10, and 9.6.2-11, and tables 9.6.2-1,9.6.2-2

Changed convergence for CO2 and pH in equation 9.6.2-2 to be while loops instead of for loops. Included the addition of lactate and base to allow for the pH balance at the upper and lower deadbands in equation 9.6.2-2.

Disabled equations 9.6.2-4, 9.6.2-5, 9.6.2-6, 9.6.2-7, 9.6.2-8, and 9.6.2-9. Moved the Figures 9.6.2-1 and 9.6.2-2 into the collapsed region since the equations producing these plots were disabled.

Revision History